Voids in morphology

Exploring "uninflectedness"

Edited by

Enrique L. Palancar

Sebastian Fedden

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Introduction to the book

Enrique L. Palancar & Sebastian Fedden

If we were to imagine an inflectional morphological system as a galactic universe with its matter and its governing laws, "uninflectedness" would mark the outer boundaries of this system, the "voids", the zones where the gravitational pull of inflection weakens and different structural principles come into play. Viewed in this analogical way, uninflectedness helps trace the limits of inflectional organization and offers valuable insights into how inflection operates and interacts with its periphery. Uninflectedness is thus a complex phenomenon that this book examines in depth through a comprehensive analysis of diverse languages and contexts, aiming to deepen our understanding of inflectional morphology.

In most languages in the world, it is common for the form of words to change to convey grammatical meaning. For example, in English, nouns typically change form to indicate plural (e.g., *men* instead of *man*, *cats* instead of *cat*). However, some nouns, such as SHEEP or DEER, do not change form even in plural contexts (e.g., we cannot say **these sheeps* or **these deers*, but *these sheep* or *these deer* are fine, and they mean more than one sheep or deer, respectively). Words like SHEEP and DEER are often referred to as "invariable". However, a more technical way to refer to them would be to treat them as "uninflected" (i.e., as not inflected) or even as "uninflecting", by referring to the fact that they do not inflect. Another way to characterize them is to treat them as "uninflectable" a term that highlights their inability to be inflected, that is, manifesting "uninflectability" as a grammatical property or characteristic. Authors often use the terms "uninflectedness" and "uninflectability" interchangeably, because in essence they refer to the same situation. However, one would perhaps like to be more precise and say that "uninflectability" refers to the property of being unable to inflect when there is the expectation that they should, while "uninflectedness" describes the broader phenomenon by presenting it as lack of inflection, whatever the interpretation one may give as to how or why it happened.

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Uninflectedness manifests in various ways across languages and the extent of its presence varies across inflectional systems. In English, uninflected nouns like SHEEP and DEER are somehow anomalies in the English lexicon (i.e., irregular nouns concerning plural formation), but in French, uninflected nouns are the norm, with only a few nouns inflecting for plural (e.g., *oeil/yeux* 'eye/eyes', *terminal/terminaux* 'terminal/terminals', etc.). Then, just as inflection is sensitive to word class, so is uninflectedness. While some word classes, like adverbs or interjections, commonly fall outside the realm of inflection, the lexicon of a given word class may be divided by uninflectedness. For example, in Basque there are only a handful of verbs which inflect for tense-aspect-mood and the person-number of their arguments, while most inflect periphrastically by auxiliaries (e.g., *da-kar-t* (ABS3SG-bring-ERG1SG) 'I bring it' vs. *erama-ten d-u-t* (take-VERBAL.NOUN ABS3SG-have-ERG1SG) 'I take it'). It may also affect entire paradigms or create splits within them, as seen with the noun MUZEUM 'museum' in Polish, which is uninflected in the singular, but which inflects normally in the plural.

Uninflectedness can also be observed in agreement. Verbs in Nakh-Dagestanian languages agree with their absolutive argument, but some fail to do so and thus fail to inflect. Systematic uninflectedness also exists. English adjectives whose stem has more than two syllables fail to inflect for comparative or superlative (e.g. *more/most interesting*). Similarly, Spanish adjectives with stems ending in /Xe#/ do not agree in gender; that is, they are uninflectable for gender (e.g. *la bici grande* 'the big bike'/*el coche grande* 'the big car' vs. *la bici roja* 'the red bike'/*el coche rojo* 'the red car'); raising the question of whether they should be considered as having a different paradigm. In such cases, we can predict the uninflectability of lexemes based on the phonological properties of their stems, suggesting that the phonology limits the lexeme's access to the morphophonological rules of inflection. This would explain why loanwords in many languages pose challenges to speakers inflecting them as native words, and why such words generally fall outside the realm of morphology and are only inflected once their phonology is adapted to the native system.

All these situations contrast with what Spencer (2020, this volume) calls "constructional uninflectedness", where words lose their inflectional properties in specific grammatical constructions. For example, in German, attributive adjectives inflect, but adjectives are uninflectable when used predicatively (e.g. *ein roter Stuhl* 'a red.NOM.SG.M chair' vs. *dieser Stuhl ist rot* 'this chair is red'). Even more strikingly, participial forms of verbs in French do not inflect for the gender of the object argument in perfective periphrastic forms with the auxiliary AVOIR 'have', but if reference to the object is anaphoric, like in headed relative clauses or constructions with fronted NPs, the participle agrees in gender (*e.g. j'ai pris la* *pomme* /jɛ **pʁi** la pɔm/ ‘I’ve taken the apple’, but *la pomme que j’ai prise* /la pɔm kə jɛ **pʁiz**/ ‘the apple I’ve taken’ or *la pomme, je l’ai prise* /la pɔm jə lɛ **pʁiz**/ ‘the apple, I’ve taken it’).

In this, uninflectedness contributes to understanding partial rule systems (Spencer 2020) and why languages might allow them at all over general rules. It also opens inquiries into its typological distribution across inflectional systems and its evolutionary dynamics. Given the cognitive load of partial rule systems, one might expect uninflectedness to diminish over time, with items becoming consistently inflected or uninflected, but as we show in the book (see Souag, this volume on Berber) uninflectedness can also be regarded as a stable feature through time in an entire language family. At the opposite end, increasing uninflectedness leads to the death of inflection, as it has happened to case in nouns in some Southern Slavic languages (see Baerman et al., this volume) or in French (see Esher, this volume).

This book explores uninflectedness comprehensively for the first time, assembling a collection of edited articles that examine the phenomenon theoretically and descriptively *with the purpose of building a better typologically-informed theory of inflection*. Detailed analyses of various languages highlight the diversity of uninflectedness, with a focus on Romance and Slavic languages due to the availability of large corpora for investigating the dynamic aspects of the phenomenon.

The volume consists of 13 chapters whose themes we have divided into five main areas:

I. Situating uninflectedness in the theoretical space of inflection:

—In his chapter, “The dog didn’t bark, the noun didn’t inflect: a typology of significant absences”, Greville G. Corbett argues that uninflectedness is significant because it is both unexpected and noteworthy. This prompts an exploration of why we expect inflection and why its absence matters. Corbett distinguishes uninflectedness from phenomena like syncretism and defectiveness, suggesting it exists on a continuum from fully inflected to uninflected, like the Polish noun *muzeum*, which is uninflected in the SINGULAR but inflected in the PLURAL or some Macedonian adjectives which are uninflected for GENDER but inflected for NUMBER. However, in syntax, uninflected items do not always fit seamlessly into expected slots and may be restricted in certain contexts. Uninflectedness varies across criteria, illustrated by the dramatic variations in Slavonic languages and the significant numbers of uninflecting items in Nakh-Dagestanian languages. Corbett calls for refined definitions and mapping of typological possibilities to fully appreciate the complexity and significance of uninflected items.

—Corbett's theoretical stance on uninflectedness is complemented by Andrew Spencer's chapter "Some concepts and consequences of uninflectedness," which raises three questions regarding uninflectability: First, as uninflectable lexemes do not resist derivation, transpositions or evaluative morphology. what counts as 'inflection'? Then there is the question about the relationship of uninflectedness to paradigm organization and Spencer asks if uninflectability is not just mass syncretism. The third issue discussed is the relationship to lexical insertion, where Spencer critiques the "Default Exponence Principle", which states that, by default, the form of any word undergoing lexical insertion is the lexical root, even in the case of a standardly inflecting lexeme. For uninflectable lexemes he proposes a model of lexical insertion using a coindexing mechanism over pairings of a lexeme's paradigm and syntactic terminals. For uninflecting lexemes, a virtual form/realized paradigm is defined without feature specification. Full constructional uninflectability occurs when syntax lacks expected morphosyntactic features, preventing pairing with the paradigm, thus treating the lexeme as uninflecting. Spencer's chapter highlights the complexity of uninflectedness, suggesting that it involves more than a simple absence of inflection, affecting derivation, transpositions, and evaluative morphology, and thus challenges existing morphological theories.

II. Conditions for uninflectedness:

—Ursula Doleschal identifies the lexeme classes most frequently subject to uninflectedness across several Slavic languages, including Czech, Polish, Russian, Serbian, Slovene, and Slovak. These classes predominantly comprise proper names, loanwords, and abbreviations, and most of her examples are nouns. The goal of Doleschal's study is to provide a comprehensive understanding of the factors influencing uninflectedness in Slavic noun classes by highlighting the varying degrees of integration of nouns within the inflectional systems of the different languages under study. Illustrating that each word class exhibits a continuum from a high degree of uninflectedness to nearly complete inflection, Doleschal argues that the absence of inflectional integration in nouns relates to essential features of declension classes, such as stem-ending, base form, and grammatical GENDER. These features can vary in restrictiveness, leading to differences among languages. For example, Russian displays a higher degree of uninflectedness in nouns compared to Slovene, but classes of FEMININE nouns tend to be more restrictive regarding stem structure and base form than those of MASCULINE nouns.

—Then Michele Loporcaro, in the chapter titled "Uninflectedness as a factor in agreement loss" argues that some of the Romance languages have gone a long way towards generalizing an isolating word structure, making uninflectedness the rule rather than the exception, at least in some areas of their grammars. The most widely known case is French nominal morphology, but a large subset of the Italo-Romance varieties – both north and south of Tuscany, whose dialects provided the basis for the standard language – parallel French in having reduced distinctions in inflection in general, in such a way that agreement has come to be signalled much less systematically than it used to be in Latin and still is in standard Italian. Elaborating on previous work, Loporcaro shows in which ways uninflectedness makes its way into these Romance varieties and in which ways it comes to interact with agreement. The main focus of Loporcaro's study is on the agreement by participles with the direct object, for which he shows that most Romance languages display four-cell paradigms which may show syncretisms or uninflectability, usually as a product of regular sound change.

III. Emergence of uninflectedness:

—In their co-authored Chapter, Matthew Baerman, Greville G. Corbett, Alexander Krasovitsky and Maria Kyuseva argue that ongoing loss of case in Serbian and Bulgarian dialects has caused the system to split into competing co-grammars, one where nouns inflect and one where they do not. This split between results in partial rules of different nature. The authors argue that three kinds of split are found, depending on the dialect (as illustrated from Bulgarian). Firstly, there is a split between morphological classes: while some retain case distinctions, others are uninflected. Secondly, classes that retain inflection are nevertheless affected by constructional uninflectedness. Thirdly, an otherwise inflecting morphological class may split on a semantic basis such as human vs. non-human. Asymmetry in morphological change leads to the rise of partial synchronic rules: morphological, syntactic or semantic. These rules persist over time, and change when a historical process moves to a new phase (for example, if uninflectedness spreads to a group within an inflected morphological class, e.g. to inanimates, splitting this class on a semantic basis). By examining these rules in contemporary dialects, and by arranging them chronologically, Baerman et al. show that one can uncover the fine-grained details of this historical process.

Louise Esher's chapter, "Loss of inflection in the diachrony of French nouns", is a case study that contributes a diachronic perspective on the development of uninflectedness and its interaction with other non-canonical phenomena. Esher argues that Modern French nouns have a "content paradigm"—the inventory of morphosyntactic feature sets required by syntax—with two cells, corresponding to the values SINGULAR and PLURAL of the feature NUMBER; this contrast is discernable through agreement patterns. Some nouns such as JOURNAL 'newspaper', have two distinct forms (*journal* vs. *journaux*) in the "realized paradigm"—the array of inflectional wordforms. However, nouns of the majority inflectional class, such as LIVRE 'book', display syncretism for NUMBER; as the realised paradigm has only a single form (*livre*(*s*)), such nouns may also be considered uninflectable. Esher shows that these patterns result from progressive "loss of inflection": Medieval French nouns had a content paradigm of four cells, with two values each for the features NUMBER and CASE. Loss of inflection occurs via a complex series of incremental changes, correlated with inflectional class, phonological shape and GENDER; the loss of CASE contrasts involves regular sound change, analogy and syntactic change, while the loss of NUMBER contrasts is principally due to sound change. While contrast in NUMBER is consistently retained longer than contrast in CASE, it is also noteworthy that, while the overall trend is towards reduction, change is not fast-paced as lexical items with and without given contrasts coexist over several centuries.

—Katja Friedewald studies the history of the French particle *voilà* 'here it is/there you go' in her chapter "French « voilà »: an uninflectable form arising from an inflecting verb". This particle is uninflecting, a situation that Friedewald argues is quite surprising with regard to its syntactic behavior because the various configurations where it occurs suggest that *voilà* actually behaves like a verb. In particular, *voilà* takes complements and it assigns accusative case to them, and it can even be embedded as the only element of a relative clause. From a diachronic point of view, the origins of the particle lie in the combination of the imperative form of the verb VEOIR 'see' from Old French with the locative adverb *là* 'there'. It was hence originally a multiword construction that shows certain similarities to constructional uninflectedness (Spencer, this volume). For Modern French, Friedewald argues that *voilà* cannot be regarded as the singular imperative form of VOIR 'see', but rather is an uninflected word since it does not have a PLURAL IMPERATIVE form, as it did in Medieval French. Corpus data suggest that after a period of parallel use, the SINGULAR IMPERATIVE form is the one usually used from the 16th c. onwards.

—In their chapter, "Emerging uninflectedness in French clipped verbs", Maria Copot, Ninoh A. Da Silva, Ahmed Beji, Arno Watiez and Olivier Bonami show that truncation in French nouns (*pneumatique* > *pneu*; *introduction* > *intro*) is well established and well documented, but recent years have seen the emergence of a parallel truncation process in verbs (*je me déconnecte* > *je me déco* 'I log off'). The combination of apocope and of the suffixal nature French conjugation provide an interesting new source of uninflectedness. The authors devise a computational method for retrieving truncated verbs from a corpus and analyse the retrieved forms in order to extract generalizations on the phenomenon. They first extract all verbal tokens in the corpus which are a substring of an inflected verb in the French lexicon (e.g. *déco* gets extracted since it is a substring of verbs such as *déco_nnecter*, *déco_rer*, *déco_der*, etc.). The full forms matched by the substring constitute potential matches for the corresponding full form. The best match is chosen by evaluating how well it fits the context, on the basis of methods from distributional semantics. Human evaluation reveals that the best match is indeed the correct corresponding full form in 88% of the cases. The conclusion of Copot et al.'s computational corpus study is that the phenomenon is productive, not limited to a subset of lexicalised truncated forms, nor to a restricted set of paradigm cells. The authors further argue that from the point of view of theoretical morphology, it is interesting to note that, while inflectional suffixes are fully lost in truncated verbs, stem allomorphy sometimes leads to the preservation of partial inflectional properties. For instance, among truncated forms of AVOIR 'have', a form such as *nous av* from *nous av_ons* or *nous av_ions* 'we have'/'we had' signal a present or imperfect, while *j'aur* from *j'aur_ai* or *j'aur_ais* 'I'll have/I'd have' signals a future or conditional. This raises interesting questions on the division of labour between stem allomorphy and affixal exponence in inflection systems, which the authors also discuss.

—The chapter "On the emergence of uninflectedness: The case of incipient verbal inflection dropping in present-day French" by Vincent Renner and Adam Renwick studies the adaptation of verbs from English to French inflectional paradigms resulting in the recent emergence of uninflected variants for a large number of such verbs (e.g., *je viens de 'play' cette vidéo...* 'I have just played this video...'). The authors' data come from a corpus of 50 million French tweets showing that the lack of inflection on verbs from English is a quantitatively remarkable fact, with thousands of daily attestations in 2021. Similarly, the distribution between inflected and uninflected infinitive forms in a sample of 50 verbs is quite striking as practically all verbs are predominantly uninflected. Renner and Renwick also argue that verbal uninflectedness in present-day French is not limited to borrowings from English. It appears repeatedly at the margins of the lexicon, in colloquial language of slang origin such as verlan and in borrowings from Romani, as well as in the newly emerged French-based contact languages of western Africa Nouchi and the so-called Camfranglais, as verbs, mostly borrowed from English, Cameroonian Pidgin English and Duala do not display inflection, especially in the infinitive and the past participle. The authors claim that while uninflectedness appears to be a marker of limited assimilation, flagging foreignness and peripherality, in the context of social media, uninflectedness seems to be used as a marker of textual genre and in-group membership.

—Anna Thornton and Paolo D'Achille address the development of uninflectedness over time in Italian nouns and adjectives, contributing to a typology of factors that allow them to predict whether an item is uninflectable. Even the factors that correlate with a noun's uninflectability increase and complexify over time: while in the 13^{th} century only nouns ending in a stressed vowel were uninflected, later other classes of nouns become (entirely or partially) uninflectable: nouns ending in unstressed *-e*, in *-i*, in a consonant, and finally MASCULINE nouns in *-a* and FEMININE nouns in *-o*, where a noun's GENDER plays a role in determining its uninflectability. These nouns are understood to have developed uninflectability because they contrast with the most system-adequate classes of inflectable nouns in Italian (MASCULINE nouns in *-o* and FEMININE nouns in *-a*). Thornton and D'Achille also study the behaviour of uninflectable Italian adjectives in diachrony for the first time. Adjectives, having a four-cell paradigm, have a potential for partial uninflectedness within a single paradigm. Preliminary results seem to point to a smaller rate of uninflectedness in Italian adjectives with respect to nouns; this possibly reflects a more pronounced necessity of having overt exponents of contextual inflectional feature values vs. inherent ones.

IV. Uninflectedness by specific word class:

—Jerzy Gaszewski's chapter, "Uninflectedness as a rule in Polish, an inflected language", examines the phenomenon of uninflected nouns in Polish, a language characterized by rich inflectional morphology involving distinction in CASE, NUMBER and declension classes. Uninflected nouns include stems with phonetic shapes that poorly match existing inflectional patterns (e.g., *kiwi, alibi, Peru*) and nouns denoting culturally distant concepts or foreign names (e.g., *karate, San Francisco, Sukarno*). However, usage can vary, as seen with the inflected noun *papaja* vs uninflected *mango*, despite the fact that both nouns refer to exotic fruits and fit productive declension classes. A particularly regular feature of Polish uninflected nouns is found in animate MASCULINE nouns used for female referents, such as common nouns (*profesor, minister, architekt*) and most surnames not ending in *-ska/-cka/-dzka*. This uninflected form signals FEMININE, although it is debated whether these forms and their inflected MASCULINE counterparts have become separate lexemes. These FEMININE forms have appeared in Polish at least since the 19^{th} century and have competed with normally inflected FEMININE forms derived with productive suffixes (e.g., *lekarka* for 'female medical doctor'). The choice between inflected and uninflected words for female referents has been ideologically charged, with debates in the early 20th c. re-emerging today. The current debate could potentially eliminate the uninflected pattern. However, only common nouns are subject to this controversy, not surnames, suggesting that the development of uninflected FEMININE forms may be influenced more by sociocultural factors than by linguistic systemic pressures.

—Viktor Köhlich's chapter examines the Japanese word class known as "rentaishi" (noun-modifying words), which has been largely overlooked in Western linguistics. Rentaishi words were introduced in the early 20th century to Japanese to align with adjectives in Western fusional languages but have language-specific and typological significance in the context of uninflectedness. Japanese features rich inflectional morphology for verbs and adjectives; in contrast, rentaishi words are characterized by their restriction to attributive positions, inability to inflect, and morphological diversity (e.g., *aru* 'a certain'; *rei-no* 'mere'), distinguishing them from major word classes based on morphology. Köhlich concludes by discussing whether rentaishi words should remain a distinct word class in Japanese or if these lexemes should be reclassified as uninflecting members of other word classes or even as part of the adjective group they were originally separated from.

V. Diachronic stability of uninflectedness:

—In his chapter, "The diachronic stability of uninflectedness in Berber," Lameen Souag argues that comparative data from varieties of Berber, an Afroasiatic language family indigenous to northern Africa, demonstrate that selective uninflectedness can remain stable over a period of approximately two millennia. He starts with the observation that most Berber varieties possess a robust system in which nouns are inflected for a feature known as "state." Typically, this involves two forms: the citation form (referred to as the "free state") and a marked form (traditionally misnamed the "construct state"). The exact distribution of these forms varies across different languages, occasionally interacting with information structure, but they are primarily used for postverbal nominatives and objects of prepositions. Souag shows that in the different Berber varieties, "state" inflection applies only to nouns with a gender-marked prefix, which a significant minority of nouns lack. In some instances, even nouns with such a prefix remain uninflected due to phonological reasons. Many (though not all) nouns borrowed from Arabic or Romance languages are uninflected for state, but this also applies to several inherited terms. Some of these terms, such as basic kinship terms, are unambiguously reconstructible as uninflected in proto-Berber. While there are plausible instances where state marking has been extended to previously uninflected nouns, no language has generalized this across the board to make all nouns inflected.

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Chapter 1

The dog didn’t bark, the noun didn’t inflect

Greville G. Corbett
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Uninflectability is notable because it defies our expectations and holds theoretical significance. I ask why inflection is typically expected and why its absence is important. Uninflectability is distinguished from related phenomena like syncretism and defectiveness. And rather than viewing it as an absolute, I identify a scale between fully inflected and uninflected items, where parts of a paradigm may remain uninflected. Using Canonical Typology, I identify four criteria which enable us to investigate uninflectability: (1) the extent of uninflectability across items, (2) the extent of uninflectability within individual items, where paradigms may be partially uninflected, (3) *incoming* conditions influencing uninflectability, and (4) *outgoing* requirements imposed by uninflectable items. The analysis draws on a wide range of languages, especially Slavonic and Nakh-Dagestanian languages, to refine definitions and explore the scope of uninflectability. Uninflectable items are more varied than most accounts allow for, and it is only when we map out the typological possibilities that we can appreciate their theoretical significance.[1]

1 Introduction: Unexpected absence

Inspector Gregory of Scotland Yard calls in Sherlock Holmes to advise on the disappearance of a racehorse:[2]

[1]In loving memory of Dr Julia Cissewski, linguist, primatologist, musician, activist.

[2]In *Silver Blaze* from *The Memoirs of Sherlock Holmes*, Arthur Conan Doyle, 1894.

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> "Is there any point to which you would wish to draw my attention?"
> "To the curious incident of the dog in the night-time."
> "The dog did nothing in the night-time."
> "That was the curious incident," remarked Sherlock Holmes.

There was an absence of the expected behaviour (barking), which Holmes points out as significant, and indeed leads to his solving the case. The interest of uninflectability is that our expectations are not met, and that this significant absence may reveal more than we could learn from an expected presence.

1.1 Expectations

In inflectional morphology, as in linguistics generally, we have names for the unexpected. We have "suppletion", "defectiveness", and so on, but we have no special term for items that fully meet our expectations ("regular" comes closest, but even here items may be regular in one respect and irregular in another). For uninflectability, it is the "inflectional expectation" (Spencer 2020: 146) that makes it nameworthy.

1.2 Intellectual housekeeping

Our expectations can be a starting point. To go further I look for clear definitions (and the attendant terms). I use *uninflected* for items which do not bear inflection. This is the superordinate term; within "uninflected", Spencer (2020: 142) makes a helpful distinction between "uninflecting" and "uninflectable". Words that belong to a class which, in the given language, are not associated with a paradigm of inflection are *uninflecting*. (We do not expect them to inflect, any more than we expect frogs to bark.) More interesting are words which belong to a class where we find inflection in a given language, yet they fail to inflect; these are termed *uninflectable*. (We do expect dogs to bark, and when the dog in *Silver Blaze* fails to bark, this is noteworthy.) We can specify these distinctions exactly, giving a logical end point to calibrate from, to yield a canonical typology (Section 2). I also discuss phenomena related to uninflectedness, since it is useful both to attempt to draw boundaries, and to recognize the interest of phenomena which resist classification.

1.3 Key data

Our data are taken from some thirty languages, with two families figuring prominently. First, the Slavonic languages provide several examples. These languages are promising in having complex inflectional morphology, and a research tradition on uninflectability.[3] The languages share many characteristics (which is valuable since it eliminates some variables) and yet they show interesting differences in terms of uninflectability. For instance, in all of them we expect nouns to inflect (for number, and in most languages for case too), but we also find instances of uninflectable nouns. As a first instance, consider the data in Table 1.

Table 1: *Kakadu* 'cockatoo' in a selection of Slavonic languages

	Russian	Ukrainian	Polish	Czech	Serbo-Croat
SINGULAR	kakadu	kakadu	kakadu	kakadu	kakadu
PLURAL	kakadu	kakadu	kakadu	kakadu-ové	kakadu-i

We see that in Russian, Ukrainian and Polish this noun is uninflectable, while inflecting in Czech and Serbo-Croat.[4] It is masculine in the languages in Table 1, except for Polish, where it is feminine. This noun points to a striking pattern: the languages form a 'great semicircle', running from Serbo-Croat in the southwest, where there are few uninflectable items, through Czech, Polish and Ukrainian, to Russian in the northeast, where there are many uninflectable items (Worth 1966: 11–12). The picture is interestingly nuanced, as I show in Section 3.2.

The second family which will be well represented is Nakh-Dagestanian. These languages prove significant since on the one hand they show a remarkable degree of inflection, and on the other there are numerous instances of the inability to inflect. This Archi example is a good way in:

[3]Note, for instance, Doleschal (2000, 2002a, 2002b, 2008, this volume). Uninflectedness figures relatively little in large surveys of the family, though Comrie & Corbett (1993) include several relevant points in descriptions of particular languages, and Sussex & Cubberley (2006: 250–251) have brief, helpful notes. From a rather different perspective, Maldjieva (1995) considers how the uninflecting items may be assigned a part of speech automatically, given that morphological indicators are lacking.

[4]I treat Serbo-Croat (hbs) as a pluricentric language, like German or English, with four standards: Bosnian (bos), Croatian (hrv), Montenegrin (cnr), and Serbian (srp) following the 2017 'Deklaracija o zajedničkom jeziku' (http://jezicinacionalizmi.com/deklaracija/). A linguistic overview can be found in Corbett & Browne (2018), and a good introduction to the complex issues of the sociolinguistic setting and language status is provided in Bugarski (2012, 2019).

(1) Archi (aqc, Chumakina & Bond 2016: 112)
D-ez Ajša **d**-ak:u.
SG.II-1SG.DAT Aisha(II)[SG.ABS] **SG.II**-see.PFV
'I saw Aisha (woman's name).' [with agreement]

In (1) the verb inflects, agreeing with the absolutive argument *Ajša* (a woman's name). Nouns also inflect (here the stem *Ajša* indicates the absolutive singular), and the pronoun *dez*, which is first singular dative, also inflects to agree in number (singular) and gender (II) with the absolutive argument. In fact, we find instances of inflection for all parts of speech. And yet, despite this prevalence of inflection, see example (2):

(2) D-ez Ajša kɬ'an.
SG.II-1SG.DAT Aisha(II)[SG.ABS] love
'I love Aisha.' [no agreement]

In (2) we have the same construction as in (1), namely a verb of emotion or perception, with the experiencer in the dative and the "object" in the absolutive. But in (2), the verb does not agree. This is a lexical matter: some verbs agree (as in (1)) and some do not (as in (2)), and this extends to verbs of other semantic and syntactic types, which again include inflectable and uninflectable members. The pronoun still agrees in (2), and more generally the syntax is "blind to the difference" (Chumakina & Bond 2016: 112). This may disturb some of our assumptions, and so I shall return to Archi in Section 3.1.2.

1.4 Outline

I first define a canonical inflecting lexeme (Section 2). From this baseline I set up four scales of non-canonicity. I take a canonical lexeme and compare outwards, measuring uninflectability *across* items (Section 3). Then I compare inwards, measuring uninflectability *within* items (Section 4). I analyse the context of uninflectability, looking at the conditions on it (the *incoming* conditions) in Section 5, followed by the external consequences (the *outgoing* impact) of uninflectability (Section 6). At a high level of generality these are the four theoretically possible dimensions of variability. I draw together the analyses in Section 7.

2 Canonical inflecting lexeme

I adopt a canonical approach (Section 2.1). I discuss relevant definitions (Section 2.2) and then consider the phenomena which border on uninflectedness (Section 2.3).

2.1 Canonical Typology

We wish to treat the phenomenon in a maximally inclusive way. Our method is well chosen:

> The canonical approach breaks down complex concepts in a way that clarifies where disagreements may lie between different linguists and theoretical frameworks
>
> (Nikolaeva 2013: 100; see also Round 2023 on this point)

In the canonical method, we analyse variable phenomena within and across languages to establish relevant scales (criteria). Then we push these scales to their logical end point, which gives us a baseline from which we can measure. This approach to measurement allows us "to conduct comparisons equally well within languages and between them" (Round & Corbett 2020). We identify more than one such scale, and this enables us to draw out the theoretical space to tease apart the clusters.

Canonical is a logical construct. It does not imply usual, normal, frequent, expected, unmarked or prototypical. As a comparison, zero Kelvin is not a frequent or indeed desirable temperature, but it is useful as an agreed baseline from which we measure; similarly, canonical is a helpful starting point, a useful exploratory tool, not a claim about what we shall find.[5] Among baselines, zero is favoured:

> ... nothings are usually extremes. They tend to sit at one end of a spectrum. And when scientists want to explore a phenomenon they look for an extreme version of it, because the factors are often easier to spot.
>
> Webb (2013: 2)

For uninflectability, the baseline from which we calibrate is canonical inflection. We can start from a well-grounded analysis of the morphosyntactic features and their values for the given system (Zaliznjak 1973; Corbett 2012: 73–88). The feature values established "multiply out", giving a paradigm with all the predicted paradigm cells. The lexical material is constant within lexemes and different across lexemes; conversely, inflectional material is different within lexemes and constant across lexemes. Thus every form of every lexeme will be distinct; and what can inflect will inflect (Corbett 2015: 150–151). Hence we obtain a viable baseline, namely the canonically inflecting item, from which we can compare outwards across items.

[5] For many examples of its usefulness see the Canonical Typology bibliography (permanent short-URL: http://tiny.cc/ctbib).

2.2 Definitions (base)

What then is the phenomenon which we wish to investigate (the "base" in Bond's 2013 view)? We can state this simply:

> Uninflectability is the phenomenon where a member of an inflecting part of speech[6] fails to inflect.

Compare Baerman et al. (2005: 27–35), and Spencer (2020: 142–143). Note that the phrase "inflecting part of speech" sets up the expectation, and we are interested here in instances where that expectation is not met. These may be individual items, or they may form a subclass. As noted earlier, Spencer distinguishes 'uninflectable' items, which do not inflect despite an expectation that they will, from 'uninflecting' items, which belong to non-inflecting classes, and hence there is no expectation that they will inflect (Spencer 2020: 142). Note that, within an inflecting class, we are interested if most items or even all but one item inflect, and equally if most do not. The statistical question is useful, but it is orthogonal to other criteria. Given that the class of items is in principle inflecting, the fact that inflection is not found consistently means there is a phenomenon worthy of our attention.

2.3 Closest related phenomena

Baerman et al. (2005: 27–35) are concerned with delimiting neutralization, syncretism and uninflectedness. They begin with two properties of canonical inflection. First, that feature values come together into fully distinguished combinations. A violation of this property is neutralization; here a distinction is not made in the inflectional morphology (for example, gender is not distinguished in the plural) and the syntax is also not sensitive to the distinction. The second property is that morphology always distinguishes syntactically relevant feature values. Violations of this property are uninflectedness, a total failure to distinguish relevant values, while syncretism (their main concern) represents a specific failure (for example, if dative and ablative case values are distinguished in the singular but not in the plural). To clarify the distinction, consider the data in Table 2.

[6] I refer to *part of speech*, as is found particularly in HPSG. In LFG "lexical category" is preferred, but 'lexical' brings unhelpful ambiguity, which is better avoided. And while some use 'lexical category' as a category of lexemes (equivalent to 'part of speech'), others use it more narrowly for part of speech with lexical meaning, for example, noun, verb, and adjective; this is the prevalent usage in Minimalism.

Table 2: Illustrative nouns from Russian[7]

		'boy'	'book'	'bone'	'bog'	'coat'
SINGULAR	NOM	mal′čik	knig-a	kost′	bolot-o	pal′to
	ACC	mal′čik-a	knig-u	kost′	bolot-o	pal′to
	GEN	mal′čik-a	knig-i	kost-i	bolot-a	pal′to
	DAT	mal′čik-u	knig-e	kost-i	bolot-u	pal′to
	LOC	mal′čik-e	knig-e	kost-i	bolot-e	pal′to
	INS	mal′čik-om	knig-oj	kost′-ju	bolot-om	pal′to
PLURAL	NOM	mal′čik-i	knig-i	kost-i	bolot-a	pal′to
	ACC	mal′čik-ov	knig-i	kost-i	bolot-a	pal′to
	GEN	mal′čik-ov	knig	kost-ej	bolot	pal′to
	DAT	mal′čik-am	knig-am	kostj-am	bolot-am	pal′to
	LOC	mal′čik-ax	knig-ax	kostj-ax	bolot-ax	pal′to
	INS	mal′čik-ami	knig-ami	kostj-ami	bolot-ami	pal′to

These nouns provide evidence to distinguish the core six case values of Russian. None of these nouns is fully canonical. We see several instances of syncretism. For example, *kost′* 'bone' and *boloto* 'bog' do not distinguish accusative from nominative in the singular or the plural, while *kniga* 'book' has this syncretism only in the plural. *Mal′čik* 'boy' has accusative-genitive syncretism in both numbers. Overall, *kost′* 'bone' has fewer distinct forms than the other inflectable nouns, yet clearly it inflects. With smaller paradigms, the syncretism-uninflectability border becomes less clear; for discussion, with respect to Italian, see (Thornton 2019a: 71–72, 90–91, and for the diachronic dimension see Loporcaro 2011). Also in Table 2 note the nominative singular of *mal′čik* 'boy', and the genitive plural of *kniga* 'book' and of *boloto* 'bog'. Here we find the zero form, the bare stem, what Stump (2017: 428n7) calls a "significative absence". There is no affixal inflection but, within the system, the morphosyntactic specification of forms with no overt inflection is as clearly identifiable as for the other forms; such forms do not count as uninflectable. In contrast, the noun *pal′to* 'coat' is uninflectable. This noun has a FULL PARADIGM, in the sense that if can appear in all syntactic slots, but the different morphosyntactic requirements are all met by the same form. This uninflectable noun is not distinct syntactically from inflectable

[7]These nouns illustrate the point we need, and are not fully representative of the system. For simplicity, stress alternations have not been included.

nouns; for instance, it controls number agreement just as normal inflecting nouns do. This is shown in these contrasting examples (3), appropriate for nominative or accusative cases, which occur several times in the Russian National Corpus (www.ruscorpora.ru):

(3) Russian

a. zimn-ee pal′to
winter-SG.N coat
'(a) winter coat'

b. zimn-ie pal′to
winter-PL coats
'winter coats'

Baerman et al. define uninflectedness as follows (2005: 33):

i. There is, in certain lexemes only, a loss of all values of a particular feature F found elsewhere in the language. This loss may depend on the presence of a particular combination of values of one or more other features (the context).

ii. Other syntactic objects distinguish values of feature F, either generally or in the given context, and feature F is therefore syntactically relevant.

The key point is that uninflectedness, and more specifically uninflectability, is about morphology being unresponsive to a feature that is syntactically relevant, and this definition allows for partial uninflectability.

The other 'boundary' to consider is that with defectiveness. In principle, the distinction between uninflectability and defectiveness is clear. Defectiveness means that the required form is missing and hence that an anticipated morphosyntactic specification cannot be realized (Baerman & Corbett 2010, Chuang et al. 2022, Sims 2023). Uninflectability is less dramatic: the morphosyntactic specification is not overtly realized, but the uninflected form is available and there is no resulting syntactic problem. Table 3 shows different possibilities.

For many Russian speakers, it is not possible to form the genitive singular of the noun *mečta* 'dream'. There is a gap here: the noun is defective. This contrasts with uninflectable *pal′to* 'coat', which has an unproblematic genitive plural, one which is like all its other forms. However, the contrast between defectiveness and uninflectability is not always so obvious. A noun may be uninflectable and also defective. We see this with *galife* 'riding breeches' (Table 3), which does not inflect, and controls only plural agreement, as in example (4):

Table 3: Defectiveness and uninflectability illustrated from Russian

		'dream'	'coat'	'riding breeches'	'scissors'
SINGULAR	NOM	mečt-a	pal′to	***	***
	ACC	mečt-u	pal′to	***	***
	GEN	mečt-y	pal′to	***	***
	DAT	mečt-e	pal′to	***	***
	LOC	mečt-e	pal′to	***	***
	INS	mečt-oj	pal′to	***	***
PLURAL	NOM	mečt-y	pal′to	galife	nožnic-y
	ACC	mečt-y	pal′to	galife	nožnic-y
	GEN	***	pal′to	galife	nožnic
	DAT	mečt-am	pal′to	galife	nožnic-am
	LOC	mečt-ax	pal′to	galife	nožnic-ami
	INS	mečt-ami	pal′to	galife	nožnic-ax

(4) Russian (Corbett 2019: 60)
v ser-yx galife
in grey-PL.LOC riding.breeches
'in grey riding breeches' (Ju. Trifonov, *Dom na naberežnoj* 1976)

This noun[8] is to be compared with nouns like *nožnicy* 'scissors' (Table 3), which is also a plurale tantum noun (hence defective in having no singular), but it inflects fully in the plural (Corbett 2019: 59–60). Given that defectiveness and uninflectability may intersect, we might still hope to maintain a clear distinction: uninflectability brings no syntactic problem while defectiveness causes the syntax to crash. But even here there are complications. Items may vary, being uninflectable or defective, according to other feature values, as I will show in Section 6.2.1 and Section 6.2.2 (and compare Nikolaev & Bermel 2022).

2.4 A specific issue with uninflectable nouns: gender assignment

Recall from Table 1 that *kakadu* 'cockatoo' in a selection of Slavonic languages varies according to whether it inflects, and according to its gender. There is a large issue here, but one restricted to nouns, namely gender assignment. It is

[8]See Dunn (1989) for further uninflectable pluralia tantum nouns.

important not to simplify: there is often a link between borrowings and uninflectability (but not all borrowings are uninflectable, and not all uninflectable nouns are borrowings); furthermore, not all uninflectable nouns need any additional specification as to gender. Indeed, the null hypothesis is that uninflectable nouns behave as ordinary nouns for the purposes of gender assignment. That is, assignment will be via semantic similarity, or formal similarity (whether morphological or phonological), see Fedden et al. (2025). However, the problem that prevents a noun from inflecting may equally be an issue for gender assignment. Thus, uninflectability here is important for what it can tell us about gender assignment, and in the case of borrowings, about the degree to which borrowed items are integrated. There is a considerable literature. The problem is acute in languages like those of the Slavonic family, where inflectional class is a strong predictor of gender (except when semantic assignment of animate nouns takes precedence). Within Slavonic, Russian is well studied, as in Murphy (2000), Wang (2014), Matushansky (2022), Chuprinko et al. (2023), Magomedova et al. (2024) and Timberlake (2004: 148–158), who provides a fine summary; for comparisons across the family, see Thomas (1983: 193–197) and Doleschal (2000: 48–151; this volume).

3 Extent of uninflectability across items

To make progress we need to measure, and we start by measuring across lexemes (Section 3.1). Then one may ask how we can specify which items inflect and which are uninflectable (Section 3.2).

3.1 Measuring across items

How many uninflectable items are there? In the canonical case, clearly none. We are interested in the extent of the difference between the canonical zero and what we find, and we have three types of comparison. The key type is within the part of speech, which sets up the expectation of inflectability (Section 3.1.1). We can also compare more generally within the language (Section 3.1.2) and even across languages (Section 3.1.3). We can then approach the question of which items fail to inflect (Section 3.2).

3.1.1 Comparison within the part of speech (the key comparison)

We start from the baseline expectation that behaviours will line up; specifically, members of a part of speech, defined in syntactic terms, will have the same morphological behaviour (Spencer 2005: 101; Corbett 2013). This is part of the general approach in Canonical Typology, namely that our criteria converge (Brown & Chumakina 2013: 9; Corbett & Fedden 2016: 527). This would mean that for our current purposes, within a given part of speech, all members inflect, or none do. We are interested in the instances where that expectation is not met. And, importantly, we are equally concerned with those examples where uninflectability is the minority behaviour and those where inflecting is unusual. We may be more familiar with the Russian-type situation, that is, where most nouns inflect, but some are uninflectable (like *pal′to* 'coat' in Table 3); however, the situation where a minority of items inflects, as in Table 4 below, is just as interesting. One reason for this is that – in most current theories of syntax – mechanisms for distributing and constraining feature values (for government and agreement, however these are modelled) apply 'blindly'; they do not 'peek ahead' to check whether the particular governee or agreement target has distinct forms for the feature values distributed. In other words, syntax is 'morphology-free' (Zwicky 1996: 301, discussed in Corbett 2009). Where only a (small) minority of the potential items is involved, this is relevant both for the functioning of the syntactic system and for its acquisition (see, for instance, Chumakina & Bond 2016: 111–117).

3.1.2 Comparison within the language (secondary)

There are languages where comparison across the parts of speech yields interesting results. For example, Serbo-Croat nominals typically retain inflection, yet numerals have largely lost it. This forms a surprising contrast to Russian, which has many uninflected nouns, yet retains inflection for numerals.

In Archi (introduced in Section 1.3) we find another unexpected contrast. On the one hand, we find inflection in all parts of speech. Example (5) illustrates part of the picture:

(5) Archi (aqc, Kibrik 1994: 349): inflection in different parts of speech

Buwa-mu	**b-ez**	dit:a<b>u	χ:^{w}alli
mother(II)-SG.ERG	**SG.III**-1SG.DAT	early<SG.III>	bread(III)[SG.ABS]

a<b>u.
make<SG.III>PFV
'Mother made bread for me early.'

In (5) we see case and number inflection on the noun *buwa* 'mother'. This example also shows other parts of speech inflecting for agreement (with the absolutive argument *χ:^{w}alli* 'bread'). Indeed there is agreement inflection found on all parts of speech apart from nouns (Chumakina & Corbett 2015: 94–95); in addition, for some parts of speech, there is inflection for other features too.[9] The contrast is that within these inflecting parts of speech there may be many items which do not inflect fully. This is clear from Table 4.

Table 4: Lexemes which inflect for agreement in the Archi dictionary (Chumakina & Corbett 2015: 95)[10]

	total	agreeing	% agreeing
verbs	1248	399	32.0
adverbs	383	13	3.4
postpositions	34	1	2.9
discourse clitics/particles	4	1	25.0

At this stage, the key point about Archi is the existence of parts of speech where full inflection is found with only a minority of items. I show another example in Mian in Section 4.2, particularly Table 8. It might be suggested that minority items which inflect (as in Table 4) could be treated as instances of overdifferentiation, but they are more "serious" than that. Overdifferentiation involves additional feature values, as when Kolami numerals show three gender values as opposed to the two values found with other parts of speech (Emeneau 1955: 56; Corbett 1991: 168). In other words, the feature gender is common to numerals and other parts of speech, but numerals have an additional gender value. In Table 4, it is a matter of inflecting for a feature as a whole, or not, which is a more radical difference than indicated by the term overdifferentiation. I return to Archi in Section 4.2, to take further the point about *full* inflection, since items can fail to show gender and number agreement, and still inflect for other features.

[9] It is worth noting that Archi has instances of "frivolous infixation", not determined by phonology (Chumakina & Corbett 2015).

[10] Pronouns are not included in Table 4, since we cannot simply count lexemes in this instance, as we shall see in Section 4.1. Chumakina (2023) reports on adverbs which inflect for orientation case markers (such as allative 'towards'); there are about ten such adverbs in Archi, but no adverb inflects for orientation case as well as for gender and number (Marina Chumakina, personal communication 13.11.2023).

3.1.3 Comparison cross-linguistically

We have general cross-linguistic (typological) expectations: we are not surprised to find verbs which inflect, while we may be relatively confident that conjunctions do not. And yet even this surprising possibility can be found:

(6) Walman (van, Torricelli, New Guinea, Brown & Dryer 2008: 534)

W-ru	chuto	rounu	alpa	**w-aro-l**	nyanam	nngkal
GEN-3SG.F	female	old	one[F]	**3SG.F**-and-**3SG.DIMIN**	child	small

ngo-l	pa	y-an	nakol.
one-DIMIN	DEM	3PL-be.at	village

'(Only) one old woman and a small child were left behind in the village.'

Observe how *-aro-* 'and' agrees with both conjuncts: it has the third singular prefix *w-* to agree with the first conjunct and the diminutive suffix *-l* to agree with the second (just as Walman verbs can mark subject and object). It is morphologically a verb (and can be used as a verb with the sense 'be with'), though syntactically it can behave like a conjunction. This shows that the typological picture is not clear; we know in principle that inflection is more frequent in certain parts of speech than in others, but we lack a full typological survey.

3.2 Determining which items (fail to) inflect

Naturally we ask why some items fail to inflect (or indeed why some items inflect). We may frame the question in terms of "two ancient dichotomies": we ask what is unpredictable as opposed to predictable/rule-governed, and what is arbitrary as opposed to functionally motivated/natural (Bye 2015: 107). For the first dichotomy, unpredictable as opposed to rule governed, there are several factors adduced as determining whether or not a particular item will inflect (given that it belongs to a part of speech where inflection is found); Fedden (2019) points to these: phonotactic, phonological, morphological and semantic (and compare Doleschal 2000: 17–19 for systematic vs unsystematic uninflectability). With the part of speech *noun* in Slavonic languages, Doleschal (2000: 48–151) considers the specifics of personal names, geographical names,[11] the names of organizations and products, foreign words and abbreviations. And specifically in a study

[11] Russian personal names are well described in Timberlake (2004: 153–158); see also the discussion in Spencer (this volume). The suggestion that in World War II, Russian official sources did not inflect place names to avoid confusion between similar names goes back to Mirtov (1953: 105).

of Archi (see Table 4 above), Chumakina & Corbett (2015) show the need for reference to phonology, derivation, and morphological type (e.g. dynamic vs stative verbs) to determine which items inflect.

The second dichotomy contrasts arbitrary and functionally motivated (compare Spencer 2020: 147). A recurring and convincing type of example here involves items where the expected inflection would bring the item into the wrong class. Take borrowed female names like *Dolôres* in Serbo-Croat (Browne 1993a: 378). This name has the form of a bare stem and so "ought" to inflect in the same way as nouns like *Jòvan*, a man's name. But this would make it like a masculine noun, and so it remains uninflectable.[12] A related type is that where the motivation is better alignment of the item with a part of speech or subtype within it. In German, for instance, proper names of different types have increasingly formed a subclass of nouns, reducing their degree of inflection when compared with the part of speech as a whole (see Nübling 2017: 344–349). For Italian, D'Achille & Thornton (2003) and Thornton & D'Achille (this volume) track the rise of uninflectability over several centuries, including some nouns which do not fit the main correspondences between inflection and gender. For Polish see Gaszewski (this volume) and for Japanese see Köhlich (this volume).

It is here that Slavonic languages are particularly apposite. Recall from Section 1.3 Worth's (1966: 11–12) point that the languages form a 'great semicircle', running from Serbo-Croat, where there are few uninflectable items, through Czech, Polish and Ukrainian, to Russian, where there are more uninflectable items.[13] The situation is more nuanced in that, for instance, Russian has many uninflectable nouns, yet retains inflection for numerals, while Serbo-Croat is strict on inflecting nouns but has developed uninflectable numerals (as I noted above, and to be discussed in Section 6.2.2). We might expect that borrowings would fail to inflect, if their phonological shape did not fit with the inflectional system, and that they might be accommodated over time; hence that the underlying trend would be for uninflectables to become inflectable. But there is clear evidence of instances of borrowings initially being inflected, and only later becoming uninflectable. In Russian in particular, nouns which were borrowed (in many instances from French) were in some instances inflected, but later were established

[12]There is another theoretical alternative, the third inflection class, where nouns like *svâst* 'sister-in-law' would be a potential model. This class includes few animates and does not attract borrowed names.

[13]Naylor (1982) suggests a phonological explanation for why Serbo-Croat has so few uninflectable nouns, namely the presence of vowel clusters, which offers a route for assimilating loans with a shape that would be problematic for Russian. However, many of the uninflectable nouns of Russian have a phonological shape which could fit readily into the inflectional system.

as uninflectable. Mučnik (1964: 161–164) shows how in 18th century Russian, borrowed nouns tended to be inflected, and that this tendency was reversed during the 19th century, with the uninflectable status of many nouns becoming firmer in the early 20th century. This is the result of the pressure of educated speakers of the standard language. A convenient summary with sources is given by Comrie et al. (1996: 117–123); see Panov (1968: 45–55) for speakers' negative reactions when asked about inflecting such nouns; there is useful discussion in Dunn (1988) and a wealth of data in Doleschal (2000).[14] This discussion shows that, more generally, to demonstrate what the determining factors are, diachronic evidence can prove valuable; for helpful use of diachronic data see further **chapters/08** (this volume), **chapters/07** (this volume), **chapters/09** (this volume) and **chapters/13** (this volume).

4 Extent of uninflectability within items

I now turn to uninflectability within items. Here we might anticipate that uninflectability would be a black and white affair: lexemes inflect or they do not. But the situation is more subtle. Paradigms may be split into an inflectable and an uninflectable portion (or segment); in such an instance we need to define the split (Section 4.1).[15] There are also situations where the system of feature specification may need elaboration: we need to check whether a simple specification is adequate or a multiple feature specification is justified (Section 4.2). These two criteria allow us to build a full typology (Section 4.3).

4.1 Definition of inflectability splits

Consider the Polish (pol) data in Table 5.

Polish has straightforward, inflecting nouns, like *wino* 'wine' in Table 5, and there are different patterns (inflection classes); it also has uninflectable nouns

[14]It is not only Russian which shows an increase in uninflectable items. Auty (1968) gives evidence to suggest a partly similar development in Czech.

[15]There are instances where subclasses of lexemes lack full paradigms. Examples included pluralia tantum nouns, location nominals and so on (Baerman et al. 2017: 71–74, 79). Such items have a paradigm structure different from that of the main class (they lack certain cells) and are not our concern here. We are concerned with instances where, given the paradigm, some cells are filled but lack inflection.

[16]The genitive plural ending is not what we would expect (namely the bare stem, as with *win*, the genitive plural of *wino* 'wine'). The *-ów* inflection is found with the main inflection class of masculine nouns and with some pluralia tantum nouns. Note also that according to Swan (2002: 120) Polish uninflectable nouns do sometimes inflect.

Table 5: Three Polish nouns (Swan 2002: 116, 118, 121)[16]

		wino 'wine'	muzeum 'museum'	hobby 'hobby'
SINGULAR	NOM	wino	muzeum	hobby
	ACC	wino	muzeum	hobby
	GEN	wina	muzeum	hobby
	DAT	winu	muzeum	hobby
	LOC	winie	muzeum	hobby
	INS	winem	muzeum	hobby
PLURAL	NOM	wina	muzea	hobby
	ACC	wina	muzea	hobby
	GEN	win	muzeów	hobby
	DAT	winom	muzeom	hobby
	LOC	winach	muzeach	hobby
	INS	winami	muzeami	hobby

like *hobby* 'hobby'. But Table 5 shows an instance which falls between, namely *muzeum* 'museum'. The paradigm of this noun is split by uninflectability. In the plural we have full inflection (for number and case), while in the singular there is a different stem, and uninflectability. We could in principle measure from either extreme: that is, the noun inflects (apart from the singular) or it is uninflectable (apart from the plural). For consistency, we have been measuring from full inflection. *Muzeum* 'museum' is an unusual instance of heteroclisis, in that nouns like this take their forms from two different inflection classes (one being uninflectable), as indicated by the differences in shading. We see that the uninflectable portion consists of a slab of the paradigm, that is, in a motivated portion (hence we have featural uninflectability).

The majority of Polish nouns inflect (whether like *wino* 'wine' or according to some other pattern); a minority are uninflectable (like *hobby* 'hobby'), and a few behave like *muzeum* 'museum', as listed by Swan (2002: 118). However, for the purposes of the current section that is interesting but secondary information. Here we abstract away from the number of items which behave in a particular way (see Section 3.1.1). The key point is that there is more going on than a simple difference between fully inflecting and not inflecting. Nouns like *muzeum* 'museum' suggest that a single dimension (inflecting – uninflectable) will not be adequate. As Brown & Chumakina point out more generally (2013: 6):

> … the space defined by the criteria may contain dimensions which differ in their nature (i.e. in terms of the number of points in the dimension, and whether the dimension is discrete or gradient).

We therefore need to investigate further. I drew attention to the fact that the uninflectable portion of nouns like *muzeum* 'museum' consists of a slab of the paradigm, in this instance all the singular values. We ask whether partial uninflectability has to be this way, or whether the logical possibility of uninflectedness according to particular cells, can be found. This distinction is valid elsewhere. Thus pluralia tantum nouns are defective for a slab of the paradigm; such motivated portions are sometimes called "sub-paradigms". And this is a dimension that can contribute to our typology (as I show later in Table 7).

There is a suggested instance, the Russian noun *èxo* 'echo'. Unbegaun (1947: 133–134) states that this noun is more likely to inflect in the instrumental singular (*èxom*) than elsewhere. Thomas (1983: 182–183) goes further, saying that this noun inflects *only* in the instrumental singular (and gives references).[17] We should reflect on this claim: a given lexeme can inflect or not inflect according to the specific cell in the paradigm. If correct, this claim implies a dramatic expansion of the notion 'possible lexeme'.[18]

To test this, I searched the Russian National Corpus (available at http://www.ruscorpora.ru), taking the automatically disambiguated texts only, which yielded over 5000 potential examples. There were extremely few plurals, and just the singular examples were retained.[19] The parser allows us to establish, for each cell, how many times the noun is inflected, and how often it remains uninflected. Compare the following examples, both involving the instrumental:

[17] The original claim was made many years ago, and of course change is likely (as with other items cited from earlier periods); see Comrie et al. (1996: 120) for sources of the 1960s which suggested that *èxo* 'echo' was less frequently inflected than it had been previously. Our concern is with what is possible, even if a previous situation no longer holds. See Esher (2025) for a complex series of incremental changes in French, correlated with inflectional class, phonological shape and gender; the loss of case contrasts involves regular sound change, analogy and syntactic change, while the loss of number contrasts is principally due to sound change.

[18] Items can be defective cell by cell, as illustrated for Latin by Rhodes (1987: 229); thus *vīs* 'force' lacks the genitive and dative singular. The suggestion I am discussing is "worse" in the sense that a given item is claimed to be non-defective, yet not to inflect in all cells of the paradigm.

[19] Examples from two authors were omitted: one had the noun in a different inflection class, with nominative *èxa*, the other had *Èxe* as a character's name.

(7) Ee golos razmeta-l-sja po perexod-u
her voice(m)[sg.nom] disperse-pst[sg.m]-refl round passage-sg.dat
slab-ym **èxo**.
weak-sg.ins.n echo
'Her voice scattered around the passage as a weak echo.'
(Evgenij Proškin, *Mexanika večnosti* 2001)

In (7) the syntax and the agreeing adjective make clear that the case is instrumental, and the noun *èxo* 'echo' remains uninflected. Now compare (8):

(8) Moj golos gulk-im **èx-om**
my[sg.nom.m] voice(m)[sg.nom] resonant-sg.ins.n echo-sg.ins
raznës-sja po zal-e.
spread.pst[sg.m]-refl round hall-sg.dat
'My voice spread as a resonant echo through the hall.'
(Evgenija Pankratova, *Angel čerdačnogo okna* 2015)

Example (8) is similar to (7), but with an inflected form of *èxo* 'echo'. As I will show, the proportions do indeed vary from cell to cell, with the instrumental showing the highest proportion of inflected forms. Here we hit a problem. The number of instances of the use of *èxo* 'echo' in the instrumental case is suspiciously high, but we need a standard of comparison. We need to know what the distribution of a typical Russian noun looks like. Progress towards this goal is recorded in Kopotev (2008); he gives data on case but does not distinguish according to number. As a step towards this, I offer for comparison the frequency ranking of the singular case values of nouns in three corpora: first a corpus of 20th century Russian fiction (Sharoff 2006). Second, a web corpus (~100 million words) from 2013 (many thanks to Dunstan Brown for his help here). Third, counts from Slioussar & Samoilova (2015: Table 6), also using the Russian National Corpus (the grammatically disambiguated subcorpus SynTagRus yielding ~1.2 million singular forms). These three corpora allow us to derive comparative data on individual case values, to compare with those of *èxo* 'echo', as in Table 6.

Table 6 first gives the number of examples of *èxo* 'echo' found in the Russian National Corpus and then the portion of those which are inflected. For instances which appear in syntactic positions requiring nominative or accusative, we cannot tell whether the noun is inflected or not, given that the form *èxo* 'echo' is a fine nominative or accusative singular, hence the indicator +/-. For the remaining case values, we see that the noun is more often inflected than not. This is not as expected, but it is still directly relevant: this noun is neither simply inflecting

Table 6: Optional inflection: Russian *èxo* 'echo' (rank of singular only) vs typical noun

Russian *èxo* 'echo' SG	examples (RNC)	% inflected	rank by total	rank for typical noun		
				Corpus 1	Corpus 2	Corpus 3
NOM	2996	+/-	1	1	2	1
ACC	444	+/-	3	2	3	3
GEN	274	86.9	4	3	1	2
DAT	98	80.6	5	6	6	6
LOC	63	66.7	6	4	4	4
INS	1138	98.5	2	5	5	5

RNC: Russian National Corpus
Corpus 1: 20th century fiction: rank order from Brown et al. (2007: 522)
Corpus 2: web corpus: Araneum Russicum III Minus
Corpus 3: RNC, Slioussar & Samoilova (2015: Table 6)

nor uninflectable but rather it is partially inflecting. And indeed the instrumental is the case value where we find the highest degree of inflection (98.5%). There is an interesting pattern: as the instances of a particular case value increase, so does the likelihood of inflection. And this leads to the question of what the expected distribution of frequencies is. For this, I turn to other corpora, where a partial answer can be obtained. For each I give the rank order of the singular case values (Table 6). In all three corpora we find the instrumental as less frequent than all but the locative (for singular nouns as a whole). There is indeed something unusual about the noun *èxo* 'echo': the frequency distribution of its case values does not fit the general pattern of nouns,[20] since the instrumental is its most frequent case value, apart from the nominative. It does inflect, but not fully (Unbegaun 1947 was partially right), and we are interested in partial inflection, whatever the distribution of inflection vs lack of inflection. Moreover, the likelihood of inflection for *èxo* 'echo' depends on the case value involved – that is the key point for our typology.

[20]There is clearly more to be done to understand the typical pattern. For instance, Slioussar & Samoilova (2015: Table 6) compare animate and inanimate nouns. Here the relative frequencies of the different cases vary considerably; however, the instrumental singular for inanimate nouns like *èxo* 'echo' again ranks fifth.

I draw three points from the discussion of Russian *èxo* 'echo'. First, since it (frequently) inflects, but may also not inflect, this means that we find instances of overabundance (Thornton 2019b): there are two forms available to realize the same cell in the paradigm. Other instances of this kind have been noted, in Czech (Auty 1968: 12), Polish (Swan 2002: 120; Krzyżanowski 2020: 53), and Belarusian (Mayo 1993: 905 on *maci* 'mother').

Second, Russian *èxo* 'echo' shows how idiosyncratic a noun can be in inflectional terms, specifically in terms of the impact of case values on inflectability. The interaction of different types of inflectional irregularity was highlighted in Corbett (2011). Further, we know that there are nouns which have special distributions with respect to number. We find number dominance (for instance, English *lips* occurs much more frequently in the plural than in the singular, while the usual distribution is for the singular to be more common). There is also number orientation, whereby nouns have singular and plural forms available, but do not use them in an unconstrained way (for instance, Russian *ogurcy* 'cucumbers' has a regular singular, but there are contextual limitations on the use of this singular). For examples and sources on number dominance and number orientation see Corbett (2019: 67–71). And there can be very specific phonetic properties of given items (see Schlechtweg & Corbett 2023: 69–70 for references). All this shows that there is much more to be understood about the profile of individual lexical items, including their propensity to inflect.

Third, and most important for current purposes, inflectability can vary according to individual paradigm cells. I have shown this for Russian *èxo* 'echo'; Doleschal (2002b) suggests that the case value has an effect on inflectability in Czech, and further evidence of optional inflection, from Bulgarian and Serbian dialects, is provided by Krasovitsky et al. (this volume). And more generally, there is more to paradigm cells than their featural specification (see Bonami et al. 2023: 317).

For current purposes, by contrasting examples from Polish and Russian we have demonstrated an opposition according to the inflectable portion of the paradigm (see Table 7).

As Table 7 indicates, lexemes may be split for inflectability into portions, which may be defined as a slab of the paradigm or by particular cells.

4.2 System of feature specification

I now turn to the second, cross-cutting criterion; this concerns the feature system. Consider this example from Mian (Mountain Ok family, within Trans New Guinea):

Table 7: Partial inflectability: examples of the first dimension of comparison

definition of inflectability split	
slab (featural)	cell(s) (morphomic)
Polish *muzeum* 'museum'	Russian *èxo* 'echo'

(9) Mian (mpt, Fedden 2022: 284)
Máam=e **a**-nâ'-n-**o**=be.
mosquito=SG.M **3SG.M.OBJ**-hit-REAL-**3SG.F.SBJ**=DECL
'She hit the mosquito.'

The verb has two featural specifications for person, number and gender. We have a verb that marks agreement with the object by means of a prefix, agreement is in person (3), number (SG) and gender (M). It also agrees with its subject (3 SG F) through the *-o* suffix. In many languages we can specify items (verbs in this instance) simply; that is, for each feature there will be one value. But this will not work in Mian; the verb in (9) is both masculine and feminine. And just specifying both gender values is insufficient, we need to distinguish the two; it is masculine for the subject and feminine for the object. As pointed out by Zwicky (1986), accounting for inflection may require multiple feature specifications; in Mian we need separate feature specifications for subject agreement and for object agreement.[21] (Mian does not require an overt subject in examples like (9).) We may find multiple specifications for inherent and contextual features (as in Upper Sorbian, Corbett 1987) and for agreement with different controllers (as in (9)).[22] Typically these distinct featural specifications arise from different syntactic requirements.

We shall see that one set of forms can occur without the other, which helps confirm that these features are structured. While Mian verbs all have obligatory subject agreement, they may be uninflectable for object agreement, as in (10):

[21]This does not imply that other featural specifications are unstructured. We may find, for instance, that gender values are distinguished only in the singular (as in Russian); and the Polish paradigm of *muzeum* 'museum' in Table 5 shows case values being distinguished only in the plural. But these are conditions imposed by feature *values* (plural, in these instances); multiple feature values involve complete features. Thus in Mian, object agreement (of the type in (9)) requires person, number and gender (just as subject agreement does).

[22]As a related phenomenon, potentially involving multiple feature specifications, in Nuer just a few nouns retain a construct state form, while all nouns (including those with the construct state form) inflect for case (Matthew Baerman, personal communication 15 June 2023).

(10) Mian (Fedden 2022: 284)
Máam=e bou-n-o=be.
mosquito=SG.M swat-REAL-3SG.F.SBJ=DECL
'She swatted the mosquito.'

More generally, the subject agreement forms (various specifications) appear on all verbs; the object agreement forms (various specifications) appear on some verbs (as in (9)) and not on others (as in (10)). Thus there is a slab of the paradigm which does not appear with certain verbs.

These data are sufficient to make the main point, but Mian is intriguingly more complex in having a second system of nominal classification. This has been called a system of verbal classifiers; however, the difference between gender and classifier is not clear-cut (Corbett et al. 2017), and one may want to call this system 'GENDER 2'. An example is given in (11):

(11) Mian (Fedden 2022: 288)
Máam=e **dob**-ò-n-o=a…
mosquito=SG.M **3SG.GENDER2.OBJ**-take.PFV-SEQ-3SG.F.SBJ=MED
'She picked up the mosquito and then…'

This second gender system is found mainly with verbs of object handling and movement: 'give', 'take', 'put', 'throw', 'lift', 'turn', 'fall' (some 50 verbs in all). For the verbs involved, gender 2 is obligatory. This system works on an absolutive basis, but only one verb is intransitive, and the remaining verbs show agreement with the object.[23] Like the object agreement discussed above, agreement is found prefixally. The form in (11) is the masculine ('mosquito' is masculine in this system). In brief, all verbs take subject agreement, and some additionally take object agreement. This object agreement is of two types: (i) some verbs take object agreement including gender as in (9), while for some this slab of the paradigm is missing; (ii) some take object agreement including gender 2 as in (11), and some do not. Finally, some verbs are uninflectable for both types of object agreement (see Corbett et al. 2017 for the detail). The canonically complete verb, with subject agreement and both types of object agreement, does not exist in Mian. There are no external conditions (Section 5) on whether a verb inflects for the object, and this has no effect on syntax (Section 6).

Let us turn to the number of verbs involved (as shown in Table 8):

[23]There are also seven verbs which arguably show verbal number, alternating according to the number of the object (Fedden 2019: 311–314).

Table 8: Object agreement in Mian verbs: types (Fedden 2022: 293)

Transitive verb type (object agreement)	n	%
never agree, e.g. *bou* 'swat', as in (10)	248	84%
agree in person, number and gender, e.g. *nâ'* 'hit, kill', as in (9)	7	2%
agree in gender 2, e.g. *ò* 'take', as in (11)	40	14%

We see from Table 8 that the number of verbs inflecting for object agreement is low. The key point, however, is that all verbs inflect for subject agreement, and a minority inflect for object agreement (in different ways). I return to Mian in Section 6.1.2.

Before leaving multiple feature specifications in a slab of the paradigm, I note three further examples. The first is object agreement in Nankina (nnk), a Trans New Guinea language. Nine verbs have a full paradigm of object agreement, some others distinguish number only, and the rest are uninflectable for object agreement (Spaulding & Spaulding 1994: 40–42; Corbett 2012: 164n15; Fedden & Paciaroni 2022). The second is possessed nouns in Dano (aso), also a Trans New Guinea language. Some possessed nouns inflect for possessor person and number, and for case, while others inflect for possessor person and number, but not case (Strange 1972; Baerman et al. 2017: 10–12, 28). Additional examples can be found in Windschuttel (2019). The third is number and gender agreement in Archi; Table 4 showed that in different parts of speech, some items show agreement while others do not. This cross-cuts other inflectional possibilities, so that some items are fully uninflectable and some are only partially uninflectable; for present purposes, the number and gender agreement realizes one feature specification, and there are others which may or may not be realized.

I have established two dimensions of partial inflectability. First splits in the paradigm, where I contrasted slab and cells. Second the distinction in the system of feature specification, between a simple featural specification and multiple feature specification. We have seen that Mian shows multiple feature specification involving a slab of the paradigm. The remaining possibility would be multiple feature specification involving the cell(s) in the paradigm. This configuration seems highly unlikely, and yet it does occur. For this I return to Archi. Consider example (12):

(12) Archi (Kibrik 1994: 349, repeated from (5))

Buwa-mu	**b**-ez	dit:a<b>u	χ:ʷalli
mother(II)-SG.ERG	**SG.III**-1SG.DAT	early<SG.III>	bread(III)[SG.ABS]

a<b>u.
make<SG.III>PFV
'Mother made bread for me early.'

The pronoun *bez* is first person singular, and it stands in the dative. Besides this specification for person, number and case, it is further specified for the number and gender of the absolutive argument, with which it agrees. The forms are given in Table 9, where the prefix marks agreement with the absolutive argument in the clause.

Table 9: The first singular dative pronoun in Archi

		NUMBER	
		SINGULAR	PLURAL
GENDER	I	w-ez	b-ez
	II	d-ez	
	III	**b-ez**	ez
	IV	ez	

It is significant that the agreeing forms are illustrated from the first singular dative pronoun. The phenomenon is not general across the personal pronouns; rather it is limited to a few cells of the paradigm (see Tables 9a and 9b).

The paradigms in Tables 9a and 9b are partial since some thirty case values are omitted. We see that certain cells in these paradigms inflect for agreement, while the majority do not.[24] To be clear what is going on, each of the cells shaded in grey in the main paradigm houses a sub-paradigm; for instance, the forms of the first person singular dative are structured as in Table 9, and the particular form in example (12) above is indicated in bold in Table 9 and Table 9a.

Each of the shaded cells in Tables 9a and 9b includes an additional paradigm comparable to Table 9, with the same pattern of syncretism. (For comparison with other languages of the family see Kibrik & Kodzasov 1990: 220–223.) This is precisely the situation which seemed unlikely. Its existence means that we have a full typology, as we shall see in Section 4.3.

[24]For the remarkable first person plural inclusive absolutive case form in Table 9b, which agrees with itself, see Corbett (2006: 68–69).

Table 9a: Archi singular personal pronouns (partial paradigms), based on Kibrik (1977: 257–260) and Chumakina & Corbett (2015: 102)

	singular	
	1 person	2 person
absolutive	zon	un
ergative	zari	
genitive	w-is d-is } b-is b-is is } is	wit
dative	w-ez d-ez } b-ez **b-ez** ez } ez	wa-s
comitative	za-ɬ:u	wa-ɬ:u
similative	za-qˤdi	wa-qˤdi
comparative	za-χur	wa-χur
substitutive	za-kɬ'ena	wa-kɬ'ena
superessive	za-t	wa-t
superelative	za-t:i-š	wa-t:i-š
superlative	za-t:i-k	wa-t:i-k
superterminative	za-t:i-kəna	wa-t:i-kəna
contelative	za-ra-š	wa-ra-š
contlative	za-ra-k	wa-ra-k
contallative	za-r-ši	wa-r-ši
contterminative	za-ra-kəna	wa-ra-kəna

Note:
'cont' in the names of spatial cases indicates contact location.

Table 9b: Archi plural personal pronouns (partial paradigms), based on Kibrik (1977: 257–260) and Chumakina & Corbett (2015: 102)

	PLURAL		
	1 PERSON		2 PERSON
	EXCLUSIVE	INCLUSIVE	
ABSOLUTIVE		nen<t’>u	
ERGATIVE	nen	nena<w>(u) nena<r>u nena<b>u nen<t’>u etc.	ž^{w}en
GENITIVE	ulu d-olo b-olo olo etc.	la<w>u la<r>u la<b>u la<t’>u etc.	wiš
DATIVE	w-el d-el b-el el etc.	w-ela<w>(u) d-ela<r>u b-ela<b>u el<t’>u etc.	wež
COMITATIVE	la-ɬ:u		ž^{w}a-ɬ:u
SIMILATIVE	la-qˤdi		ž^{w}a-qˤdi
COMPARATIVE	la-χur		ž^{w}a-χur
SUBSTITUTIVE	la-kɬ’ena		ž^{w}a-kɬ’ena
SUPERESSIVE	la-t		ž^{w}a-t
SUPERELATIVE	la-t:i-š		ž^{w}a-t:i-š
SUPERLATIVE	la-t:i-k		ž^{w}a-t:i-k
SUPERTERMINATIVE	la-t:i-kəna		ž^{w}a-t:i-kəna
CONTELATIVE	la-ra-š		ž^{w}a-ra-š
CONTLATIVE	la-ra-k		ž^{w}a-ra-k
CONTALLATIVE	la-r-ši		ž^{w}a-r-ši
CONTTERMINATIVE	la-ra-kəna		ž^{w}a-ra-kəna

Note: ‘etc.’ in a cell indicates the pattern of syncretism where the plural of GENDERS I and II matches GENDER III SINGULAR, and the plural of GENDERS III and IV matches GENDER IV SINGULAR.
‘CONT’ in the names of spatial cases indicates contact location.

4.3 The typology of partial inflectability

Putting together the analyses in Section 4.1 and Section 4.2 we arrive at the typology shown in Table 11.

Table 11: Partial inflectability: Two dimensions of comparison: summary

		definition of segments	
		slab (featural)	cell(s) (morphomic)
system	simple specification	Polish *muzeum* 'museum'	Russian *èxo* 'echo'
	multiple feature specification	Mian object agreement	Archi pronouns

We see that our typology of partial inflectability is complete (Table 11). This opens the way for future investigation of the distribution of examples over the four types. The degrees of inflectability found in each type also deserve further work: the examples found to date typically have fewer lexical items inflecting for the full featural possibilities than those which are partially uninflectable (see, for instance, Table 8 on Mian verb types).

5 Conditions (*incoming* factors)

In the canonical world, inflecting items realize their feature specifications fully, irrespective of external conditions. Thus a verb may have feature values for tense and aspect, and agreement specifications, and this is sufficient to determine its form. That is, external conditions such as word order are irrelevant: a verb does not change its tense value when in clause-initial position, for instance. However, in the real world we do find significant instances where items inflect or do not inflect, subject to external conditions (in addition to the expected featural requirements imposed, for instance, by government and agreement). Such conditions are non-canonical, since the syntax would be simpler without them. The result of such conditions has been termed "constructional non-inflectedness" by Spencer (2020: 144), and "syntagmatic/contextual uninflectedness" by Doleschal (2000: 14). I first present the different types of conditions which lead to uninflectedness (full or partial), then in Section 5.5 I review the forms which result.

5.1 Syntactic position: German

A classic example is provided by German adjectives. In attributive position these inflect for case, number and gender, with two levels of reduced agreement (Corbett 2006: 95–96) to which I return in Section 5.5. Elsewhere, notably in predicate position, adjectives are uninflected:

(13) Dies-er Artikel ist sehr interessant.
this-SG.M.NOM article(M)[SG.NOM] be.PRS.3SG very interesting
'This article is very interesting.'

In (13) the adjective *interessant* 'interesting' lacks any inflection.[25] Hence the condition determining whether the adjective inflects is a syntactic one.

5.2 Position in the phrase: Tsova-Tush

In several languages we find inflection restricted to the right edge of a nominal phrase (this may involve conjuncts or an attributive modifier and a noun). We might anticipate a simple external condition on inflection: the item on the right edge inflects, others do not. For instance, an attributive occurring phrase-finally would inflect, while if standing before a noun (hence not at the right edge), it would not. With this in mind, consider the data in Table 12, from the Nakh language Tsova-Tush (also known as Batsbi), as analysed by Sims-Williams & Baerman (2021).

Table 12: Tsova-Tush (bbl, Holisky & Gagua 1994: 171–172, Sims-Williams & Baerman 2021: 28)

	Substantivized adjective in final position 'bad one'	Attributive adjective in phrase 'bad child'
ABSOLUTIVE	mos:i^{n}	mos:i^{n} bader
ERGATIVE	muis:čo-v	muis:čO badre-v
GENITIVE	muis:čo-^{n}	muis:čO badre-^{n}
DATIVE	muis:čo-n	muis:čO badre-n

Note: ^{n} indicates nasalization, O indicates a reduced vowel (word-final position).

[25] For the complex issue of the conditions on case marking and lack of inflection for case in German, see Spencer (2009) and references there.

In Table 12 we see that the substantivized adjective in final position is indeed inflected; but when there is attributive adjective, followed by a noun, it is the noun which takes the inflection. Sims-William & Baerman stress the word-order effect, given that group inflection is an area feature shared by various languages of the Caucasus. From this perspective, we have an external condition on the paradigm: inflection is found at the right edge of the nominal phrase (and note that the absolutive singular has the bare stem, a significative absence, Section 2.3). However, this adjective is one that also has a stem alternation, contrasting absolutive and oblique (the remaining case values). Even when it does not bear the main inflection, it retains some inflection, that of the stem alternation.

5.3 Featural condition: Sanskrit

I now turn to a different type of condition, found in the Sanskrit (san) future, as presented by Stump (2013). The tense in question, termed the periphrastic future,[26] consists of a masculine agent noun and, in the first and second persons, a form of the copula; this is set out in Table 13.

Table 13: Periphrastic future-tense formation in Sanskrit (Stump 2013: 111)

<table>
<tr><th></th><th></th><th colspan="2">components</th></tr>
<tr><th></th><th></th><th>verb derivative</th><th>copula</th></tr>
<tr><td rowspan="2">PERSON</td><td>1 and 2</td><td>agent-noun derivative
MASCULINE
NOMINATIVE
SINGULAR</td><td>PRESENT INDICATIVE ACTIVE (very rarely MIDDLE) forms of as 'be'</td></tr>
<tr><td>3</td><td>agent-noun derivative
MASCULINE
NOMINATIVE
SINGULAR / DUAL / PLURAL</td><td>[none]</td></tr>
</table>

The forms of *as* 'be' are given in Table 14. This verb has all forms available, though the third person forms do not appear in the periphrastic tense in question.

More importantly, for our purposes, the agent noun also has a full paradigm, distinguishing number, gender and case. With that in mind, consider the paradigm in Table 15.

[26] See further Lowe (2017) for critical discussion of the periphrastic status of this future tense at different periods of Sanskrit and for additional examples.

Table 14: Present indicative active of *as* 'be' in Sanskrit (Stump 2013: 111)

	SINGULAR	DUAL	PLURAL
1	ásmi	svás	smás
2	ási	sthás	sthá
3	ásti	stás	sánti

Table 15: Periphrastic future-tense forms of *dā* 'give', without sandhi (Stump 2013: 111)

	SINGULAR	DUAL	PLURAL
1	dātā́ asmi	dātā́ svas	dātā́ smas
2	dātā́ asi	dātā́ sthas	dātā́ stha
3	dātā́		

Presenting the paradigm in this form is helpful, since it shows the partial inflectability which interests us. Of the full paradigm of the agent nominal, we find just the number distinction, and that only for third person forms. This can be seen, though less obviously, in the "surface" forms, those which arise as a result of automatic sandhi processes, where the two parts are pronounced as a single phonological word (see Table 16). In the examples, sandhi is indicated by .

Table 16: Periphrastic future-tense forms of *dā* 'give', with sandhi (Stump 2013: 111)

	SINGULAR	DUAL	PLURAL
1	dātā́smi	dātā́svas	dātā́smas
2	dātā́si	dātā́sthas	dātā́stha
3	dātā́		

The key point for our analysis is that the agent nominal has a full paradigm, but when used in this future tense it inflects partially: it does not inflect when the periphrastic verb is first or second person; for the third person the agent nominal inflects for number but not for gender (only the masculine is found).[27] Let us see how this works in examples:

[27]We see distributed exponence here: where the agent nominal is uninflectable for number, the auxiliary is present, and inflects for number; when the auxiliary is absent (in the third person) the agent nominal marks number.

(14) Sanskrit (Stump 2013: 112)
Tatas tvā pārayitāasmi.
from.that 2SG.ACC rescue.NMLZ.SG.Mbe.PRS.1SG
'I will rescue (lit: I am rescuer) you from that.'
(*Śatapatha Brāhmaṇa* 1.8.1.2)

In (14) we have a first person singular subject, and the agent nominal is masculine singular. There is also a direct object (*tvā* 'you'), in the accusative, showing that this tense is not an ordinary predicate nominal construction. Compare the plural in (15):

(15) Sanskrit (part of example in Stump 2013: 113)
Mahac choka-bhayam prāptāsmaḥ.[28]
great.SG.ACC pain-fear.SG.ACC meet.NMLZ.SG.Mbe.PRS.1PL
'We are going to meet with great pain and dread.'
(*Gopatha Brāhmaṇa* 1.1.28)

Note that although we have a plural subject, the agent nominal remains singular here (since the verb is first person). I now move to the third person:

(16) Sanskrit (Stump 2013: 113)
Ekā jananayitā putram.
one.SG.NOM.F produce.NMLZ.SG.M son.SG.ACC
'One (feminine) will bear a son.' (*Rāmāyaṇa* 1.38.8)

Example (16) is significant: the subject is third person (so no copula); it is clearly feminine, but the agent nominal maintains masculine gender. However, the agent nominal can inflect (for number) in future tense use, as example (17) shows:

(17) Sanskrit (Stump 2013: 114)
Na jāne yātāras tava ripavaḥ kena ca
not know.1SG go.NMLZ.PL.M thy enemy.PL.NOM which.SG.INS and
pathā.
path.SG.INS
'I don't know by which path your enemies will go.' (*Bhojaprabandha* 55)

[28] *smaḥ* is the pre-pausal sandhi variant of the first person plural *smas* (Greg Stump, personal communication, 5 August 2013).

Here we see a third plural subject and the agent nominal is plural. This fits with the pattern in Table 13. The agent nominal has a full paradigm, but when used for the future it does not inflect at all when the subject is first or second person; when the subject is third person it inflects for number (17) but not gender (16). The agent nominal's pattern of partial inflection is conditioned by the person of the subject, that is, by a featural specification.

5.4 Further instances

Other examples of external conditions include the following varied selection. Proper names may be accompanied by classifier-like nouns (sortals) and the proper name may then remain uninflected. See Matushansky (2022, 2023) for careful discussion of these constructions in Russian. An example is given in (18):

(18) Russian (Matushansky 2023: 235)
na ulic-e / *ulic-a
on street-SG.LOC / *street-SG.NOM
Jakimank-e / Jakimank-a
Yakimanka-SG.LOC / Yakimanka-SG.NOM
'on Yakimanka Street'

In this instance (see Matushansky 2023 for the other types), while the sortal noun must take the case value required by the syntactic context, the proper name may be in that case or may stand in the nominative. For appositives in Polish see Cetnarowska (2021).

In Kayardild, the external condition is that the set of words on which a morphosyntactic feature is realized is determined by concentric domains of features (Round 2013: 78–84). In Lithuanian Arkadiev (2013, 2020), participles agree or lack agreement morphology depending on the construction (and there is a complex web of interrelated constructions). Giger (2016) describes the loss of inflection of participles as a participial perfect emerges in Slovak. And for the interesting complexities of Hungarian inflecting infinitives and Latvian genitive compounds see Spencer (this volume).

5.5 Forms used and distinctions retained in constructional non-inflectedness

I have so far examined the types of construction which condition (often partial) non-inflectedness. I now switch my focus to the forms used in examples of constructional non-inflectedness, and the distinctions retained. The clearest type is that found in the German predicate adjective (example (13)); this takes a form lacking all inflection, and retaining none of the featural distinctions of inflected forms. This implies two distinctions. First, there are uniquely uninflected forms, as in our German example, opposed to normal inflected forms, used for constructional non-inflectedness, such as the Russian nominative (Section 5.4). Second, there are forms where no distinctions are retained, as in the German example, opposed to those where at least one distinction is retained, as in Tsova-Tush. Significant examples are given in Table 17, and then key points are highlighted.

Somali deserves mention since it has reduced agreement forms which are unique to the reduced paradigm; furthermore, the distinctions made in the reduced paradigm are not a simple collapsing on those in the full paradigm (as in Tsova-Tush), but rather the patterns of syncretism are different. Inari Saami is included because the reduced paradigm has no unique forms; it retains the third person singular and the third plural from the full paradigm, and these fill out the reduced paradigm. Reduced agreement (Fedden & Paciaroni 2022, and Loporcaro, this volume), a subset of partial inflection determined by external conditions, is a fruitful area to work on, because agreement allows us to manipulate the feature values involved, which can be useful for establishing the nature of the partially inflected forms. Continuing with the data in Table 17, the Sanskrit periphrastic future (Section 5.3) is illuminating because we find possible inflected forms, but used beyond their normal range. Here we can ask whether we have a default form (an inflected form used when there is no specification). This is a plausible analysis for the Sanskrit future, where we find masculine as the gender default, and singular as the number default. (The nominative is expected as the case value in the predicate.) However, in the third person, distinctions of number are retained: we have a reduced paradigm rather than a single default form. Thus the features in Sanskrit have sufficient values for it to be clear what is going on; we need to beware of small systems, which can be misleading.[29] Finally, we need to note the complexities of German adjectives in attributive position. There are two stages of reduction: the paradigms are traditionally labelled strong, mixed and weak. There is a notional paradigm with 16 cells (four case values, plural vs singular and – in the singular – three gender values). However, there are extensive syncretisms so that even the strong paradigm has only five distinct forms. The mixed paradigm has four of those forms, and the weak has only two; however, we do not find a straightforward collapsing of distinctions, but rather we see different patterns of syncretism. The mixed and weak paradigms retain some distinction within each of the features (case, number and person).

[29] There is careful discussion of default agreement vs lack of agreement inflection in Lithuanian in Arkadiev (2020: 383–387, 403-404, 408-412). And one needs to establish the default instance by instance; thus in Serbo-Croat (Browne 1993a: 329, 374) and Polish the accusative is an attractor for length-of-time and quantity constructions.

Table 17: Forms and distinctions in constructional non-inflectedness

example	unique vs normal forms	featural distinctions retained	source
German predicative adjective	unique	none	see Section 5.1
Tsova-Tush attributive adjective (not phrase-final)	unique	ABSOLUTIVE VS OBLIQUE	see Section 5.2
Somali verb (subject focus)	unique	3SG.F / 1PL / rest	Corbett (2006: 93, 203–204) and references there
Inari Saami reduced agreement	normal	SG VS DU-PL	Corbett (2006: 94–95), Toivonen (2007)
Russian sortal noun	normal: NOMINATIVE	SG VS PL	see Section 5.4
Sanskrit periphrastic future	normal: MASCULINE (and SINGULAR in person 1 and 2)	SG VS DU VS PL (in person 3)	see Section 5.3, Stump (2013)
German attributive adjective	normal	CASE and NUMBER (limited degree)	Corbett (2006: 95–96)

We have observed that there are lexemes which do not inflect, even though this might be expected (Section 3), and this uninflectability is a matter of degree (Section 4). In this section (Section 5), we have seen that there are external conditions which determine whether otherwise inflecting items inflect. And here too it is a matter of degree. I now turn to the final logical possibility, the possible outward effect of uninflectability.

6 External consequences (*outgoing*)

Since uninflectability is a matter of inflection, one would expect no syntactic consequences. We do find instances of this canonical situation (Section 6.1), but there are also claims in the literature that uninflectability has consequences, and I discuss some of these (Section 6.2).

6.1 No external consequences

Often this situation is simply assumed: researchers report the inflectional situation and imply that there is no more to be said. However, there are also studies which look carefully for external consequences, and do not find any. I consider in turn Nichols (2018) on Ingush, and Fedden (2022) on Mian.

6.1.1 Agreement in Ingush (Nakh-Dagestanian)

Ingush (inh), which has some 300,000 speakers living mainly in Ingushetia in the North Caucasus, is a Nakh language (hence a relative of Tsova-Tush, discussed in Section 5.2 and a more distant relative of Archi, introduced in Section 1.3 and Section 3.1.2). It has “an arbitrary and strictly lexical bifurcation of verbs” (Nichols 2018: 845), in that some have gender and number agreement and some do not. There are 400 simple verbs at most, of which around 30% show gender and number agreement (change of the initial consonant); however, there are other possibilities, including light verbs and tense auxiliaries, so that inflected verb forms have some gender and number agreement in probably well over 50% of the instances (Nichols 2011: 141n63, 235).[30] Given this situation, we can use variation in one phenomenon (inflection for agreement or lack of it) to test proposed explanations for another. Thus it is sometimes suggested that the function of agreement is to facilitate reference tracking. By comparing agreeing with non-agreeing verbs, one can ask whether we can see differences in other means of reference tracking, in particular the presence of an overt argument (which might be introduced to facilitate reference tracking when the verb does not agree). Nichols (2018) has measured carefully, using a corpus of narrative texts (mainly spoken). The results are summarized in Table 18.

We see that there is no difference between the examples with an agreeing verb and those with a non-agreeing verb. The morphological difference is not

[30]That is 90/294 (30.6%) of the simple verb roots; if those with number distinctions (verbal number) are included, the figure is 109/393 (27.7%). Of the simple adjectives, fewer than 10% take agreement (Nichols 2011: 141). See Nichols (2011: 141–145) for details on the relation of gender and number in agreement.

Table 18: Ingush: verb agreement and overt arguments (tokens) (from Nichols 2018: 858)

		S	A	O/T	total
verb root agrees	n	100	41	37	178
	% overt argument	88	59	92	82
verb root does not agree	n	41	38	24	103
	% overt argument	85	76	83	82

Note:
S: single argument of intransitive verb; A: agent-like argument of a monotransitive; O: direct object of a monotransitive; T: theme-like or patient-like object of a ditransitive

reflected in the syntax. For comparable results from the Dagestanian language Lak, see Forker (2018).

6.1.2 Agreement in Mian

Mian was introduced in Section 4.2. Based on a fully annotated Mian test corpus of 14 texts, some 4,000 words, with around 1,700 clauses, Fedden (2022) undertook a study similar to that of Nichols (2018), discussed in Section 6.1.1. Here we are concerned only with object agreement, since all verbs show subject agreement. There are three types of transitive verb; recall from Table 8 that only 16% of Mian verbs show object agreement, 2% showing agreement in person, number and gender and 14% taking gender 2. Thus object agreement is relevant for a lower proportion of the transitive verbs than that found in Ingush. Due to the high type and token frequency of non-agreeing verbs, the syntax is constantly encountering gaps. What then can we learn about reference tracking? The data on the presence of overt arguments are given in Table 19.

The difference between the highlighted cells is not statistically significant. And so, although the situation in Mian is rather different from that in Ingush, we find again that inflectability has no syntactic effect.

6.2 External (syntactic) consequences

There are various places in the literature where it is claimed that uninflectability does have syntactic consequences. I will report the data on some of the suggested

Table 19: Mian object agreement: verb agrees or not (Fedden 2022: 300)

		gender	gender 2	combined
verb agrees	n	61	218	279
	% overt	33	39	38
verb does not agree	n			397
	% overt			43

examples (I have carried out only limited fieldwork and cannot vouch for them). I will note the others briefly. Typically there is speaker variation.

6.2.1 Serbo-Croat *doba* 'time, season, age, period'

This noun is unusual in several ways, including the fact that it is the only neuter noun in the language with the nominative singular in *-a*. In a language where uninflectability is rare (Browne 1993a: 378; Browne 1993b), this noun is often cited in grammars and dictionaries as uninflectable, with *doba* as its unique form. However, the situation is more complex, as was pointed out as early as Kovačević (1951: 246), see also Stevanović (1975: 212–214) and Nikolić (1995–1996: 16–17). The potential paradigm is given in Table 20, though the use of the form *doba* is found in other cells of the paradigm too.

Table 20: Inflection of Serbo-Croat *doba* 'time' (Benson 1971: 79)

	SINGULAR	PLURAL
NOMINATIVE	doba	doba
ACCUSATIVE	doba	doba
GENITIVE	doba	dobā
DATIVE	dobu	dobima
LOCATIVE	dobu	dobima
INSTRUMENTAL	dobom	dobima

Our concern here is those instances when the noun remains uninflectable, since attributive modifiers can be affected by the uninflectability of the noun. There are examples in both Kovačević (1951) and Stevanović (1975) of phrases with *doba* 'time', governed by a preposition, where the attributive modifier appears to follow the form of *doba* rather than standing in the appropriate case. Consider these alternatives:

(19) Serbo-Croat (Kovačević 1951 and web sources)

a. u t-om doba
in that-SG.LOC.N/M time
'at that time'

b. u t-o doba
in that-SG.ACC.N time
'at that time'

The use of the locative modifier with the uninflected *doba* is rare (19a); much more common is the use of the *to* form of the demonstrative (19b). We can treat this as the accusative, and note two factors here. First the accusative tends to be used, rather than the locative, more generally in such time expressions: this is true both for other attributive modifiers of *doba* 'time' (Browne 2015: 11–13) and for other nouns denoting time which do not have the unusual characteristics of *doba*. And second, *to doba* 'that time' could be analysed as moving towards being an adverbial expression, a fixed phrase unaffected by external government. We therefore take a preposition governing the genitive case: then the first factor no longer applies, while the second does. Browne (2015) reports a web count of three relevant examples:

Browne (2015: 12): web count 31.12.2014

(20) od t-og doba — 285 instances: 43.5%
from that-SG.GEN.N/M time
'from that time'

(21) od t-oga doba — 242 instances: 36.9%
from that-long.SG.GEN.N/M time
'from that time'

(22) od t-o doba — 128 instances: 19.5%
from that-SG.NOM/ACC.N time
'from that time'

Since *od* 'from' takes the genitive, (20) and (21) are as expected. But a minority of instances as in (22) show that *doba* 'time' does not always behave as we would expect from an uninflectable noun. Rather we find a syncretic nominative-accusative form, which we take as accusative for the reasons given above. But this would imply that we have different behaviours, depending on the particular cell of the paradigm (Section 4.1). Given that, the suggestion that *to doba* 'that time' is becoming a fixed adverbial expression of time, impervious to case government, seems a reasonable one.

6.2.2 Serbo-Croat numeral phrases

Serbo-Croat numerals have largely lost inflection. The higher numerals, *pet* 'five' and above, typically do not inflect (and they take the genitive plural of the quantified noun).[31] The numerals *dva* 'two' (masculine and neuter, with *dve* feminine), *tri* 'three' and *četiri* 'four' are losing inflection for case, but have not done so completely, and this has engendered a substantial literature.[32] These lower numerals require the *adnumerative* of the quantified noun, and this is realized as the genitive singular or the nominative plural, depending on the inflection class of the noun (Corbett 2009: 159–160). For our purposes, the typical situation is the interesting one, that where numeral phrases increasingly function as non-inflecting units; that is, the numeral does not inflect, and it determines the inflectional form of the quantified noun, giving a constituent which is impervious to external syntax. In other words, a numeral + noun unit (given in square brackets in

[31]They lost inflection in the 16th to 17th centuries, according to Rogić (1954: 138). However, the higher, more noun-like numerals (*stotina* 'hundred', *tisuća* and *hiljad* 'thousand', *milion* and *milijun* 'million') can inflect, particularly in complex numerals, or stand in the accusative (Popović 1979: 11–12).

[32]Marković (1951) pointed out that *dva* 'two', and even more so, *tri* 'three' and *četiri* 'four' were becoming uninflectable for case; this change was well under way in the 19th century, and he discusses instances where inflection is preserved in relevant texts. He noted already the tendency for numeral phrases which would be in the instrumental to be replaced with *s* 'with' and the uninflectable form of the numeral. Rogić (1954) gives examples illustrating the loss of inflection of these numerals. Hamm (1956) discusses grammarians' statements on the topic and usage in the press. M. Popović (1967) talks of "blocks" which inflect, meaning that in constructions consisting of preposition + numeral + noun, the numeral and noun form a 'block' whose case is registered in the preposition. The journal *Naš jezik* had three papers on the topic in 1976: Kraljević (1976) gives a wealth of data, including dialect data, showing that lack of inflection is more prevalent in the east; Nikolić (1976) gives summary information on 28 dialects, showing that inflection of the numeral is more likely if the noun is feminine, more likely in the dative, locative and instrumental cases than in the genitive, more likely with *dva* 'two' than with *tri* 'three', and more likely with *tri* 'three' than with *četiri* 'four'; Radović-Tešić (1976) concentrates on the range of forms of the numerals. Lj. Popović (1979) discusses uninflectability within the general morphosyntactic system of the numerals; his papers on uninflectability and the dative are discussed in the main text. Ďurovič (1985) also gives a general account, including uninflectedness, as does Browne (2022), while Grubišić (1994) discusses the grammatical tradition on the topic. Kuzmić (2007) examines Čakavian legal texts (13th–18th centuries) while Kuzmić & Kuzmić (2007) analyse Kajkavian legal texts (16th–18th centuries); both papers provide a wealth of textual examples from the respective varieties, including evidence on the rise of uninflectability. Stefanović (2019: 148–193) provides many examples, of both inflectedness and uninflectedness.

our examples) behaves somewhat like an uninflectable noun. Such phrases are fine when they occur in syntactic positions requiring the nominative, accusative or genitive case; the latter is illustrated in (23):

(23) Serbo-Croat (Wechsler & Zlatić 2003: 150)
Seć-am se [ova četiri glumca].[33] [GENITIVE]
remember-PRS.1SG REFL DEM four actor
'I remember these four actors.'

They are also fully acceptable with prepositions taking various cases, as in (24):

(24) Serbo-Croat (Wechsler & Zlatić 2003: 150)
Razgovar-am sa [ova četiri glumca]. [*sa* + INSTRUMENTAL]
talk-PRS.1SG with DEM four actor
'I am talking to these four actors.'

However, these numeral phrases do not fit everywhere. As pointed out by Lj. Popović (1981–1982, see also 1982), the case frame matters: they do not work as a dative:[34]

(25) Serbo-Croat (Wechsler & Zlatić 2003: 150)
*dati knjig-e [ova četiri studenta] [*DATIVE]
give.INF book-PL.ACC DEM four student
'to give books to these four students'

This is indeed surprising; in some case frames the lack of morphological realization of the case value is unproblematic, while for the dative there is an issue (25). The non-inflecting unit behaves as though it is defective, lacking a dative case form. There is further discussion in Wechsler & Zlatić (2003: 148–152), Franks (2024) and for higher numerals in Bošković (2006).

I have concentrated on numeral phrases; according to (Wechsler & Zlatić 2003: 133–140) similar issues of inflectability depending on the case value arise with uninflectable *nouns* in Serbo-Croat. (There are relatively few of these, but female

[33]Here *glumca* 'actor' is in the adnumerative, realized as the genitive singular; *ova* 'these' is also in the adnumerative, realized as the nominative plural neuter (Corbett 2023: 17–18).

[34]I give examples from the extended discussion by Wechsler & Zlatić, since they are simpler than those in the original. See also Browne (1993a: 374–375). As for the remaining two case values: (i) the locative is found only with prepositions, and so it causes no problem; (ii) the instrumental, interestingly, can be replaced in these constructions by *sa* 'with' plus the uninflected unit, to give an acceptable form comparable to (24). There is no comparable escape for the dative (Lj. Popović 1981–1982: 606).

names such as *Dolôres*, which do not match the appropriate Serbo-Croat inflectional pattern, are uninflectable, as noted in Section 3.2.) Ivić (1972: 107n3) pointed out that their use in an oblique case is facilitated by the presence of a modifier or a preposition. See also Bošković (2006, 2009); Horvath (2014) and Adamson (2019: 103–111) on this; there is speaker variation here. External effects have also been suggested for uninflectable *adjectives*; Serbo-Croat also has some instances, many of them orientalisms, as documented in Nikolić (1996). Bošković (2013: 22) claims that these have syntactic effects, notably that an uninflectable adjective must be adjacent to its head noun. His account is discussed in Adamson (2019: 115–121). There is speaker variation here too.[35] Adamson (2019: 160–184) further analyses German prenominal uninflectable adjectives and argues that here there is a syntactic effect in that they do not allow noun phrase ellipsis.

6.2.3 Definiteness in Bulgarian

The relevance of definiteness in Bulgarian was first pointed out by Halpern (1995: 150–153); see also Spencer & Luís (2012: 128–130); it is of ongoing interest, as shown by Adamson (2019: 83–103; 2022) and Georgieva (2023). In Bulgarian, definiteness is inflectional, and a nominal phrase can be marked definite on the first markable item:[36]

Bulgarian (bul, Spencer & Luís 2012: 128–130)

(26) knig-a-ta
book(f)-sg-def.sg.f
'the book'

(27) xubav-a-ta knig-a
nice-sg.f-def.sg.f book(f)-sg
'the nice book'

Example (26) shows definiteness marking on the noun since it is the only item in the phrase, while in (27) it is on the adjective, as the first markable item. The issue arises where we have an item like *serbez* 'quarrelsome', which the cited researchers analyse as an uninflectable adjective. When the phrase is indefinite, all is well, as in (28):

[35] Adamson (2019: 121–131) discusses Italian uninflectable adjectives, and concludes that their inability to appear in prenominal position, when unmodified, is not a direct result of their lack of inflection. Hence this is not a case for inclusion here.

[36] I thank Vladimir Zhobov for his help with the Bulgarian data.

(28) Bulgarian (Halpern 1995: 150–153, Spencer & Luís 2012: 128–130)
serbez žen-a
quarrelsome woman(F)-SG
'a quarrelsome woman'

The issue arises when the phrase is definite. If *serbez* 'quarrelsome' counts as the first element (as we would expect given (27)), ungrammaticality results (the alternatives show the ways that definiteness might, theoretically, be shown on it):

(29) *serbez-ăt/ta žen-a
quarrelsome-DEF.SG woman(F)-SG
'the quarrelsome woman'

Clearly, then, *serbez* 'quarrelsome' does not count as the first element for definiteness. But if it is not counted, the result is also ungrammatical:

(30) *serbez žen-a-ta[37]
quarrelsome woman(F)-SG-DEF.SG.F
'the quarrelsome woman'

Thus neither way works: uninflectable *serbez* 'quarrelsome' neither accepts definiteness marking nor allows it on the noun; it neither counts as the first element nor allows the noun to be the first element. Hence we do appear to have an instance where uninflectability has an external effect on the syntax.

6.3 External (syntactic) consequences: summing up

I have presented three possible instances and noted others; South Slavonic is well represented. Each case is challenging, and there is speaker variation to be reckoned with. In this section, the key issue is not whether the item inflects, but how similar it is syntactically to regular inflecting items. The Serbo-Croat numeral phrases (Section 6.2.2) are significant, since they show defectiveness, in that they fail to fit into syntactic slots where a dative would be required.[38]

[37]Halpern states (1995: 165n22) that this example is possible as an ''ironic or humorous quotation'' of a previous description.

[38]Testelets (2010) offers interesting discussion of case deficiency more generally, based on Russian data.

7 Conclusions

I have explored the logical space of uninflectability. First, I calibrated uninflectability across items, determining which items have the property, and how they can be specified. Second, I determined the extent of uninflectability within items. Third, I noted the possible *incoming* conditions on uninflectability. And fourth, I evaluated the *outgoing* requirements imposed by uninflectable items.

What then can we learn when the dog doesn't bark, when an item doesn't inflect? This can challenge our assumptions, and take us forward, in four related areas. First, there is much to be gained for *morphological typology*. I have shown that in similar morphological systems, uninflectability can vary substantially; recall the way in which Russian has favoured uninflectable items, particularly nouns, while the related Serbo-Croat accepts uninflectable items with reluctance. There are also considerable differences across systems: in more familiar languages we commonly find a minority of items being uninflectable, but we have now seen several languages in which a majority can be uninflectable (whether completely, or in respect of specific features). Given the reasonable assumption that syntactic and morphological classes will align, mismatches between the two are of interest; this is equally true whether the majority of the members of a class are inflectable or uninflectable. And individual items can be very different, both in being fully uninflectable or allowing inflection as an alternative, and in showing various splits in inflectability. In all this, *diachrony* is a valuable window into how uninflectability is determined; for instance, loss of inflection can reflect a change of part of speech. And we should recall that change is not always from uninflectable items (often the result of borrowing) gaining inflection as they are integrated into the morphological system; I have also given instances of borrowings which gained inflection subsequently losing it. Uninflectability is significant for *morphological theory*. Spencer (2020: 143) argues that uninflectability itself makes a nonsense of some variants of Distributed Morphology. More generally, it is key to a better understanding of lexical insertion (Spencer, this volume). Further, as we have seen, uninflectability raises the question of how inflectional morphology can have external consequences. Thus uninflecting items (and constructions such as Serbo-Croat numeral phrases) may be partially defective, giving unexpected syntactic effects. And finally, uninflectability leads us to question the *role of inflection*. We saw instances (Russian, Mian, Ingush) where uninflectability appears to make no difference. Similarly, when we examine the complex agreement rules of Archi, we found that many agreement targets fail to inflect, this is perplexing. It appears that speakers invest a good deal, in syntactic terms, to ensure the appropriate feature specification for governees and agreement targets, and yet there may be no morphological realization. Why keep a dog when it does not bark? The lack of a bark told Holmes a good deal. Perhaps we need to revisit our assumptions about inflection; its function cannot be the straightforward conveying of grammatical information since it may fail to realize that information, apparently without consequence.

Abbreviations

Abbreviations are the Leipzig ones with additions:

abs	absolutive	med	medial verb
acc	accusative	n	neuter
dat	dative	nmlz	agent nominal
decl	declarative	nom	nominative
def	definite	obj	object
dem	demonstrative	pfv	perfective
dimin	diminutive	pl	plural
erg	ergative	prs	present
f	feminine	real	realis
gen	genitive	refl	reflexive
ins	instrumental	sbj	subject
loc	locative	seq	sequential
m	masculine	sg	singular

Furthermore, [ ] indicates information that can be inferred from the use of the bare stem; < > marks infixation; ( ) is for inherent feature values; bold is used as a flag to draw attention to relevant characteristics of examples; Roman numbers are used for different gender values in Archi.

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Chapter 2

Uninflectability: Concepts and consequences

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Uninflectable lexemes are to be distinguished from simply uninflecting lexemes (e.g. English prepositions) or defective lexemes. Uninflectability can be an inherent lexical property or a contextually/constructionally defined phenomenon exhibited by otherwise normally inflecting lexemes. Uninflectability can also be total (Russian nouns such as *kenguru* 'kangaroo') or partial (Polish *muzeum*-nouns, Nakh-Dagestanian "sporadic agreement"). This implies a four-way typology, of which three types are attested in the literature. I argue that the missing type, contextual partial uninflectability, is instantiated by infinitive phrases in Hungarian and genitive plural based compounds in Latvian. I explore which types of inflectional processes are subject to uninflectability, providing evidence that it applies solely to the "core" synthetic inflection of a language, and not, say, to affix-clitics, periphrasis, evaluative morphology. I sketch a formal account of uninflectability within the framework of Paradigm Function Morphology, Version 2 (Stump 2016): an uninflectable lexeme is associated with the expected complete content paradigm for lexemes of its class, but lacks the form/realized paradigms, in whole or in part. I define lexical insertion as the insertion of the most highly specified value of the (multivalued) FORM attribute in the lexical representation (following Spencer's 2013 model of lexical relatedness). Uninflecting lexemes such as English prepositions lack both content and form/realized paradigms and so lexical insertion is defined over the (sole) value of FORM, the lexeme's root. In effect, an uninflectable lexeme likewise defaults to the root form, since this is the only form available in the absence of a form/realized paradigm. This account raises open questions about the architecture of morphology, for instance, the nature of 'core' synthetic inflection as opposed to other types of lexical relatedness, and the way that defectiveness should be treated in a formal model.

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1 Introduction

In inflecting languages, some (classes of) lexemes are *uninflecting*, i.e. are not associated with any inflectional paradigm: *of, the, almost, …*, but some members of otherwise inflecting classes may (unexpectedly) fail to inflect, i.e. they show *(lexical) uninflectability*. Spencer (2020) outlines three types of uninflectability, distinguishing lexical from constructional/contextual uninflectability, and total uninflectability from partial uninflectability. In this paper I argue that the two parameters of uninflectability define a four-way typology. Cases of total and partial lexical uninflectability are well-known, and it is not difficult to identify cases of total constructional/contextual uninflectability. In Section 2 I suggest that Hungarian and Latvian may provide instances of the fourth type, thus establishing the phenomenon of uninflectability as typologically robust.

An account of uninflectability owes an account of what it means for a lexeme to be inflected in the first place. In Section 3 I cite data suggesting that otherwise uninflectable lexemes nonetheless permit affix-like clitics, periphrastic exponence and evaluative morphology (as well as straightforward derivational morphology). I propose that what the uninflectable lexeme is unable to inflect for is some kind of "core" synthetic inflectional morphology, closer to the canonical ideal (see Corbett, this volume, on Canonical Typology).

In Section 4 I propose a formal account based on the distinction drawn in Paradigm Function Morphology, Version 2 (PFM2) (Stump 2016) between the content paradigm on the one hand, and the form/realized paradigms on the other. The leading idea is very simple. An uninflecting lexeme, such as an English preposition, has just a (root) form and neither content nor form paradigm. An uninflectable lexeme, on the other hand, has a content paradigm and therefore behaves as though it were an inflecting lexeme, but it lacks a form/realized paradigm. Uninflectability is not the same as defectivity: it is not the case that an uninflectable lexeme cannot be used in sentences. But this raises the question of what form of an uninflectable lexeme it is that undergoes lexical insertion. I propose a solution to this problem which appeals to an enriched notion of lexical representation, based on the model presented in Spencer (2013).

The discussion illustrates an important point made by one of the first linguists to investigate uninflectability (in Slavic languages) on a large scale: "Unflektierbarkeit zeigt die Grenzen und damit die innere Struktur des morphologischen Systems einer Sprache" (Doleschal 2002: 60; see also Doleschal 2001): "Uninflectability points to the boundaries and hence the inner structure of the morphological system of a language". I would argue, though, that the importance of uninflectability goes beyond the implications for any given language, but has repercussions for the way we view morphological systems in general, and provides further evidence in support of an autonomous morphology (cf Aronoff 1994), and in support of the idea that inflectional systems are structured in paradigms.

2 A typology of uninflectability

In Spencer (2020) I argue for two axes or parameters of uninflectability. The familiar instances are those where a single lexeme fails to inflect in all morphosyntactic contexts, so that uninflectability is an idiosyncratic lexical property of that particular lexeme. This is *lexical uninflectability* (corresponding to Doleschal's (2001) 'paradigmatische Uninflektierbarkeit'). However, there are also cases in which an otherwise inflecting class of lexemes fails to inflect in a specific morphosyntactic context or construction. This is *contextual* or *constructional uninflectability* (Doleschal's 2001 "systematische Uninflektierbarkeit"). For instance, the non-head noun of an English noun-noun compound typically fails to take (plural) inflection. The second parameter is that of total vs. partial uninflectability. In the most familiar cases an uninflectable lexeme has no inflected forms at all – *total uninflectability*. However, there are well-documented instances in which a lexeme inflects for one set of features, say, for nominal case in singular number forms, but fails to show inflection for other feature combinations, e.g. nominal case in the plural. This is *partial uninflectability*. The hypothetical example just give in fact corresponds to the Polish *muzeum*-nouns discussed below, p. 61.

This classification of uninflectability in terms of total vs. partial and lexical vs. constructional/contextual uninflectability implies a four-way typology, shown schematically in Table 1: Type 1 total/lexical, Type 2 partial/lexical, Type 3 total/contextual, Type 4 partial/contextual. All but Type 4 have been discussed in the literature, but before I come to Type 4 I very briefly exemplify the first three types.

Table 1: Four theoretical types of uninflectability

	lexical	contextual
total	Type 1	Type 3
partial	Type 2	Type 4

A clear case of Type 1, total lexical uninflectability, is found in the Russian noun lexicon. Russian nouns distinguish 6 x 2 = 12 case/number forms but about

3000 are indeclinable (uninflectable): all the forms of their paradigm are identical to the root, e.g. *kenguru* 'kangaroo'. We distinguish such uninflectable lexemes from uninflecting lexemes on the one hand and defective lexemes (e.g. Russian *mečta* 'dream', lacking genitive plural) on the other. Uninflectable lexemes can occur in all the contexts open to inflecting lexemes (unlike defective lexemes), but the form is invariable, as shown in the Russian examples in (1) (where *uninfl* indicates an uninflecting or uninflectable form):

(1) a. odin vombat/kenguru
one.NOM.M.SG wombat(M)NOM.SG/kangaroo(UNINFL)
'one wombat/kangaroo'

b. s èt-imi vombat-**ami**/kenguru
with these-INS.PL wombat-**INS.PL**/kangaroo(UNINFL)
'with these wombats/kangaroos'

Crosslinguistically, any part of speech, not just nouns, can show uninflectability.

In a number of languages, certain classes of adjective borrowed from languages with rather different inflectional systems fail to inflect: German *prima*, 'great', *rosa*, 'pink', Greek *ble* 'blue', *gri* 'grey'. In Bulgarian and Macedonian a number of adjectives borrowed from Turkish are uninflectable, such as Macedonian *taze* 'fresh' (Friedman 1993. See also Thornton & D'Achille, this volume, for discussion of uninflectable adjectives in Italian.)

Some languages have inflecting adpositions. These are often derived historically from nouns, and they typically take possessive agreement morphology. For instance, Welsh prepositions are said to fall into three inflectional classes: AR 'on', *arnaf* '1SG', *arno* '3SG.M', *arni* '3SG.F' etc, TROS 'over', *trosof* '1SG', *trosto* '3SG.M', *trosti* '3SG.F' etc, GAN 'with', *gennyf* '1SG', *ganddo* '3SG.M', *ganddi* '3SG.F' etc. However, many prepositions fail to inflect, and instead just take ordinary pronominal complements (Williams 1980: 47, 127–130): GYDA 'with' *gyda mi* 'with me'.

While less commonly found, even verbs can be uninflecting, as in the case of French loans from Romani, verbs in the Verlan criminal argot, and more recently loans from English such as *follow* 'follow (e.g. on the social media platform X, formerly known as "Twitter")' (Boyé 2019). Examples of the latter can be found in social media and in the lyrics of French rap songs, as in (2). (These examples are of interest for the way we will later come to characterize the notion of *inflection*.)

(2) a. Baby girl, si je te follow.
baby girl, yes I you follow
'Yes, babe, I'm following you.' (*Sirène* – Luidji)
https://music.youtube.com/watch?v=0EeheHnqQio

b. Elle m'-a follow.
she me-has follow
'She (has) followed me.'
https://www.youtube.com/watch?v=GbzBfr2aBus

c. Elle m'-a pas followback quand je l'-ai follow.
she me-has NEG followback when I her-have follow
'She didn't follow me back when I followed her.'
https://www.azlyrics.com/lyrics/niska/rseaux.html

Lexemes can also be partially (un)inflectable, Type 2 in Table 1 (Baerman et al. 2015: 26–28; see Corbett, this volume, for a carefully documented typology of partial uninflectability): verbs of the English *hit* class have only *-s* and *-ing* inflections; Macedonian adjectives such as *kasmetlija* 'lucky, SG', *kasmetlii* 'lucky, PL' fail to show the expected three-way gender contrast in the singular (Friedman 1993). A particularly interesting case is that of a small group of neuter gender Latin loanwords into Polish with the endings *-um* or *-o*, *muzeum* 'museum', *studio* 'studio' (Swan 2002: 118; see Corbett, this volume, for detailed discussion). In addition to having the 'wrong' genitive plural form (*muzeów, studiów* for the expected zero ending form) these words are uninflectable in all the five otherwise distinct case forms of the singular 'slab' of their paradigm. Thus, while a native noun such as PISMO 'document' has singular case marked forms *pismo, pisma, pismu, pismem, pismie* (for nominative/accusative/vocative, genitive, dative, instrumental, locative), MUZEUM, STUDIO have just *muzeum, studio.*[1]

It is rarer for verbs to show partial uninflectability, but a form of this is seen in the Russian copular/verb of existence BYT′. In the past tense this verb is regular, formed on the root *by-* with the *-l* suffix of the 'l-participle' form, glossed L in examples (3).

(3) a. U menja by-l-a kniga.
at me[GEN] COP.PST-L-F.SG book(F).SG
'I had a book.' [literally 'with me there.was a.book']

b. Kniga by-l-a interesnaja.
book(F).SG COP.PST-L-F.SG interesting.F.SG
'The book was interesting.'

[1]Loanwords borrowed as masculine gender nouns inflect normally, though they abstract out the *-um* component as the NOM.SG ending: *album, albu, albem, albie* 'album (nominative/accusative/vocative, genitive/dative, instrumental, locative singular)'.

In the present tense the copular/verb of existence is usually omitted altogether, (4a, b), but under certain circumstances it is possible to add the invariable present tense form *jest′*. In (4c) the overt copula adds rather subtle nuances of meaning so that (4c) tends to imply an individual-level or temporary predication, while the form without copula, (4a), tends to be interpreted as a stage-level or permanent state.

(4) a. U menja kniga.
at me[GEN] book(F).SG
'I have a book.'

b. Kniga interesnaja.
book(F).SG interesting.F.SG
'The book is interesting.'

c. U menja jest′ kniga.
at me[GEN] COP.PRS book(F).SG
'I have a book.'

Unlike all other Russian (non-impersonal) verbs, which agree with the subject in person/number, the present tense copular form is invariant for all persons/numbers, whether it is used purely as a copular, or whether it has an existential meaning. Example (5) is the title of a Russian pop song (https://www.youtube.com/watch?v=_oM4CQVXats), while example (6) is the title of a novel by V. Tokoreva.

(5) U tebja jest′ ja.
at you[GEN] COP.PRS I[NOM.SG]
'You have me.'

(6) Ja jest′. Ty jest′. On jest′.
I[NOM.SG] COP.PRS you[NOM.SG] COP.PRS he[NOM.SG] COP.PRS
'I am. You are. He is.'

The Russian copular, then, is an instance of a verb which inflects normally except in the present tense, where it is uninflectable for the expected person/number agreements.[2]

[2]In languages like Russian, Arabic and others in which the present tense copular is generally omitted altogether one might say that we have a maximally non-canonical instance of partial uninflectability, in which uninflectability manifests itself as total absence.

Russian *kenguru* is uninflectable in all its occurrences, a case of *lexical uninflectability*. However, English *kangaroo* is inflectable except in compounds: *kangaroo(*s) tails*, (arguably) an instance of *contextual uninflectability*, in which an otherwise inflecting lexeme has to be realized as a non-inflecting word form, or even as an uninflected bound stem, as in many cases of noun incorporation. Where the uninflectability seems to be an integral part of a grammatical construction, as in the case of English compounding, we can talk of *constructional uninflectability*, a subset of the contextual uninflectability type. When a lexeme becomes completely uninflectable because of the construction or context it is in we have an instance of Type 3, total constructional/contextual, uninflectablity.

One type of constructional uninflectability closely related to English compounding is the Light Verb Construction (LVC) in certain languages (Spencer 2020). In most languages LVCs don't show (contextual) uninflectability, but straightforward uninflectedness. However, in some languages we find the same kind of contrast between inflectability and uninflectability in LVCs that we see in compounds in many languages. For instance, the well-known Japanese LVC formed with the empty verb SURU 'do' often alternates between a form with completely inert noun complement, as in *benkyoo suru* 'study' with bare noun BENKYOO 'studying', and a variant based on a grammatically marked form of the noun, *benkyoo o suru*, in which the noun complement of the light verb bears the object marker O (Miyamoto & Kishimoto 2016).

A particularly interesting case of contextual uninflectability is the predicative adjective in German, illustrated in (7).

(7) a. Ich bin ein **kleines** Känguru.
I am a little.N.SG.NOM/ACC kangaroo
'I am a little kangaroo.'

b. Das Känguru ist **klein**(*-e/*-es/…).
the kangaroo is little(UNINFL)
'The kangaroo is little.'

This type of uninflectability is found not just with predicative adjectives after the copula SEIN. It is found with secondary predication, postnominal attributive adjectives, and generally in any position other than canonical attributive modification, including adjective-noun compounds such as *Kleingeld* 'small change' (literally 'small_money', cf **Kleinegeld*, **Kleinesgeld*), and we could also extend this to adverbial use (Trost 2006, Dürscheid 2002). For these reasons it is difficult to associate this contextual uninflectability with any specific construction type.

The German predicative adjective construction leaves an interesting, largely unexplored, question for paradigm-/lexeme-based models of inflection: precisely what grammatical features are borne by the predicative form of an otherwise inflectable German adjective? For instance, what number/gender/case feature values does the predicate adjective *klein* in (7b) actually bear, if any? And if it fails to bear any of the standard adjectival feature values, how is its form specified in the grammar, given that the bare root form *klein* itself never occurs in the position of an attributive modifier, but always bears some kind of inflectional suffix?

More generally, we can ask this question of any instance of contextual uninflectability, even those cases where it appears as though the uninflected lexeme has the form of an inflected word form in a cell of the lexeme's paradigm. Cetnarowska (2021) reports an interesting case in point. In Polish composite lexical units (a type of compound construction), an otherwise inflecting modifier word may appear as an uninflected element. She discusses cases such as *statek widmo* 'ghost ship', (8), formed by modifying the noun *statek* 'ship' with the (normally inflectable) noun *widmo* 'ghost'.

(8) a. o statk-ach widm-o
about ship-LOC.PL ghost-NOM.SG

b. o statk-ach widm-ach
about ship-LOC.PL ghost-LOC.PL
'about ghost ships'

The examples (8a) and (8b) are identical in meaning and are equally acceptable. In example (8b) both components of the compound inflect for locative plural, but in (8a) the modifier noun *widmo* 'ghost' appears in the citation, nominative singular form. The phenomenon illustrated in (8a) is different from either the lexical uninflectability shown by Russian *kenguru*-words, or the contextual uninflectability of German predicative adjectives. The form *widmo* is, in fact, an inflected form, the nominative singular, part of the inflectional paradigm, and not an uninflected stem or root form.

Cetnarowska (2023) proposes a Construction Morphology (Booij 2010) solution to this paradoxical situation: take the N-*widmo* construction to be a kind of compound in which *widmo* is (optionally) specified to be in the nominative singular form only (taken to be the default form by Cetnarowska). In effect, this redefines one of the morphosyntactic/morphosemantic functions of the property set *nominative singular*. Another solution is to assume that when uninflecting, *widmo* is a distinct (though related) lexeme whose root form is specified as identical to the nominative singular form of the standard lexeme. This would have the advantage of allowing us to code the metaphorical meaning change as a separate lexical property. However, there is little difference between the two analyses.

In some cases it is not immediately apparent whether we are dealing with lexical uninflectability or contextual uninflectability. Spencer (2020) discusses the case of indeclinable foreign names in Russian. Many borrowed or cited foreign words, and especially names, have a phonological shape which is difficult to accommodate to the Russian morphological system, like *kenguru* (no native nouns have a stem ending in a vowel) or *Si (Czinʹ pin)* 'Xí (Jìnpíng)'. Other foreign names, however, can easily be inflected as though they were Russian: *Bil* 'Bill', genitive singular *Bil-a* (cf *Kiril*, *Kiril-a*), or *Klinton* 'Clinton', genitive singular *Klinton-a* (cf *Anton* 'Anton', genitive singular *Anton-a*). However, native female referent names do not have phonological forms such as *Klinton*, and native given names do not end in *-i* for either sex, so that *Xilari Klinton* 'Hillary Clinton' is indeclinable, like *Si* 'Xí', (9).

(9) rečʹ 'the speech of …'

a. Bill-a Klinton-a
Bill-GEN.SG Clinton-GEN.SG
'…Bill Clinton'

b. Xillari Klinton
Hillary(UNINFL) Clinton(UNINFL)
'…Hillary Clinton'

c. Xilari Benn-a
Hilary(UNINFL) Benn-GEN.SG
'…Hilary Benn'

(Hilary Benn is a (male) British politician.) Not all foreign female referent names are indeclinable. Where the form of a name resembles that of a declinable Russian name it will be inflected, as in the case of the Spanish name in (10).

(10) knigi Teres-y Solan-y
books Teresa-GEN.SG Solana-GEN.SG
'the books of Teresa Solana'

A different instance is that of the Russian form of the name of the Roman orator, Marcus Tullius Cicero, *Mark Tullij Ciceron*. All three components of this name are inflecting. In each case, Russian takes the bound stem of the Latin name and uses this as the unbound stem/nominative singular form, cf the original Latin genitive form *Marc-i Tulli-i Ciceron-is*.

Generally, there is no reason in principle why a female name like *Klinton* couldn't be inflected just like the male name but taking feminine gender agreements, in much the same way that thousands of male-referent names and kin terms (especially diminutives, such as *Andrjuša* from *Andrej*) inflect like feminine gender nouns but take masculine agreements: *Ja znaju èt-ogo mužčin-u* 'I know this-M.ACC/GEN.SG man-ACC.SG. Alternatively, Russian could have supplied female-referent names with a special feminine gender ending, like the Czech *-ová*. What this means is that we have to regard *Klinton*(M) and *Klinton*(F) as distinct, though related(!), lexemes, and we should analyse the female-referent names as an instance of total lexical uninflectability.

The partial lexical uninflectability seen with Polish *muzeum*-nouns is found more systematically in cases of "sporadic agreement" (Fedden 2019). In a number of Nakh-Dagestanian languages many verbs agree with (principally absolutive case) arguments. However, this agreement is often only shown by a minority of verbs. Other lexemes fail to show corresponding agreement, but otherwise show regular inflection for other verbal (TMA) categories. In this respect the verb system of these languages is similar to the prepositional lexicon of Celtic languages such as Welsh. Sporadic argument agreement is to be distinguished from differential object marking/agreement, where the presence of absence of agreement is semantically or pragmatically significant, for instance, indicating focus, definiteness, specificity and so on (see Corbett, this volume, for a detailed typology.)

The extant literature which is devoted explicitly to uninflectability has only clearly identified three distinct types: (i) lexically conditioned total uninflectability (Russian *kenguru*-nouns); (ii) lexically conditioned partial uninflectability (Polish *muzeum*-nouns); (iii) sporadic agreement. This raises the question of whether there might be clear cases of a fourth type, *partial contextual/constructional uninflectability*, in which all lexemes of a given class are expected to inflect in a given construction but in which some lexemes fail to inflect for a subset of features in those constructions. In other words can we find exemplars of the missing type in the simple typology shown in Table 2?

Table 2: Incomplete typology of uninflectability

	lexical	contextual
total	Type 1: *kenguru*-noun	Type 3: German predicative adjective
partial	Type 2: sporadic agreement	Type 4: ???

Polish *widmo*-compounds: The first candidate to consider is the Polish construction illustrated in (8a) above. The uninflecting element is, in fact, the nominative singular form, as we have seen, but that form is fixed by the construction itself. Though the *widmo*-compounds broadly fit the characterization of type (iv) uninflectability they are not perfect exemplars of that type, because only a relatively small number of lexemes seem to enter into the construction. It is therefore worth examining whether there are more general, lexically less restricted, cases of contextual partial uninflectability.

Latvian genitive compound: One candidate which is similar to the *widmo*-compounds but not lexically restricted is illustrated by a class of noun-noun (N N) compounds in Latvian. In this language, a very common way for a noun to serve as attributive modifier to a head noun is through a noun-noun headed compound construction type in which the modifier noun obligatorily appears in the genitive case, either singular (11a) or plural (11b) (Mathiassen 1997: 56).

(11) a. kafij-as tasīte
coffee-GEN.SG cup
'coffee cup'
b. latvieš-u valoda
Latvian-GEN.PL language
'Latvian (language)' [literally 'language of the Latvians']

Although Latvian has N N compounds in which the modifier is in the root form, as well as relational adjectives, the $\mathrm{N}_{\mathrm{genitive}}$ + N construction has a high token frequency. It is much more common than, say, the corresponding construction in the closely related Lithuanian. Thus, while 'stone house' is normally rendered with the $\mathrm{N}_{\mathrm{genitive}}$ + N construction in Latvian, in Lithuanian a relational adjective would be found, agreeing like any other adjective in number, gender, case (Kalnača & Lokmane 2016: 174).

(12) a. Latvian
akmen-s nams
stone-GEN.SG house(M).NOM.SG
'stone house'
b. Lithuanian
akmen-in-is namas
stone-REL.ADJ-M.GEN.SG house(M).NOM.SG
'stone house' [literally 'petrous house']

In Latvian, many of the $N_{genitive}$ + N expressions, the genitive attributive noun is itself modified, by nouns, adjectives, numerals, prepositions and other elements (Kalnača & Lokmane 2016: 171–172).

(13) a. liel-zied-u ceriņi
large-flower-GEN.PL lilac
'large-flowered lilac'

b. deviņ-stāv-u māj-a
nine-storey-GEN.PL house
'nine-storeyed house'

(14) a. vien-zilb-es vārds
one-syllable-GEN.SG word
'monosyllabic word'

b. bez-maks-as pakalpojums
without-charge-GEN.SG service
'free service'

In a number of such cases the compound $N_{genitive}$ modifier is modified by a preposition which would normally govern a case other than the genitive, comparable to 'paratactic' adjectives in other European languages, such as Russian *za-rubež-nyj* 'overseas', from *za* 'beyond', *rubež* 'border', *-nyj* 'adjective-forming suffix'. The difference is that in Latvian the noun remains a noun and does not have to be transposed into a (relational) adjective. The structure is thus akin to that of the English *over-seas*. In the Latvian construction the noun remains in the genitive case, irrespective of what case is governed by the compounded preposition (Kalnača & Lokmane 2016: 181). This shows that the genitive has been frozen as an exponent of the modifier function of the noun. For instance, when governing a noun phrase in the plural, all Latvian prepositions take dative case (whatever case they govern in the singular), but in the corresponding compound genitives the noun remains in the genitive plural, (15–16). The $N_{genitive}$ + N constructions, including the compound genitives can also function as nominal predicates, (17).

(15) a. N + P + DAT
tablet-es pret sāp-ēm ⇒
pill-NOM.PL against pain-DAT.PL
'pills against pain'

b. compound genitive
pret-sāpj-u tablet-es
against-pain-GEN.PL pill-NOM.PL
'painkillers '

(16) a. N + P + DAT
autobus-s starp pilsēt-ām ⇒
bus-NOM.SG between city-DAT.PL
'bus between cities'

b. compound genitive
starp-pilsēt-u autobus-s
between-city-GEN.PL bus-NOM.SG
'intercity bus'

(17) Šis pakalpojums būs bez-maksas.
this.NOM.SG.M service(M)NOM.SG COP.FUT.3 without-charge.GEN.SG
'This service will be free.' (Kalnača & Lokmane 2016: 178)

The attributive genitive marked noun can also be used in elliptical contexts which represent a non-canonical form of attributive modification. Thus, a set of teaching materials by Nau (2008) entitled *Gramatikas modulis* ('grammar-GEN.SG module'), has a Latvian-Polish glossary with the title (18).

(18) student-u vārdnīca (latvieš-u – poļ-u)
student-GEN.PL vocabulary (Latvian-GEN.PL Pole-GEN.PL)
'students Latvian-Polish vocabulary'

Here, the genitive plural forms *latvieš-u* 'Latvian' and *poļ-u* 'Polish' function as attributive modifiers to an elided head noun *vārdnīca* 'dictionary'.

The conclusions from the Latvian genitive attributive construction are that genitive case functions as a kind of 'frozen' inflection, much like the nominative singular form of Polish *widmo*-nouns. However, the attributive nouns retain the number contrast, so they are not completely uninflectable. In this respect they are akin to Polish *muzeum*-nouns, except that they are constructionally uninflectable, not lexically uninflectable. The Latvian genitive compounds therefore not only present a plausible example of the missing type (iv) of uninflectability, but also highlight the point made by Corbett (this volume), that uninflectability is not a binary or absolute category, but admits of degrees of variation.

Hungarian inflecting infinitive: Another possible instance of type (iv) uninflectability is the Hungarian inflecting infinitive. The infinitival complement to certain auxiliary(-like) verbs, *kell* 'need', *lehet, szabad* 'is possible/allowed', inflects like a possessed noun, shown in boldface in (19), agreeing with the (understood) subject in person/number ((19a), Kenesei et al. 1998: 314, (19b), Kiss 2002: 210, (19c), Kenesei et al. 1998: 318).

(19) a. Nekem most men-n-**em** kell.
me.DAT now go-INF-**1SG.PX** need
'I have to go now.' [present tense]

b. Korábban kell-ett volna neked haza men-n-**ed**.
earlier need-PST CONDITIONAL you.DAT home go-INF-**2SG.PX**
'You needed to have gone home earlier.' [past conditional]

c. Nem fontos magyarul beszél-ni-**e**.
not important Hungarian speak-INF-**3SG.PX**
'It isn't important that s/he speak Hungarian.' [complement to *fontos*]

With other (auxiliary-like) verbs, there is no inflection, (20), (Kenesei et al. 1998: 314).

(20) Nem tud-ok/bír-ok fel-áll-ni.
not able-1SG.PRS up-stand-INF
'I am not able to stand up.'

No inflection is found with unspecified subjects, presumably because as far as the verb is concerned, there is no subject, (21), (Kenesei et al. 1998: 314).

(21) Mit kell a kerttel csinál-ni?
what need the garden.to do-INF
'What needs to be done to the garden?'

As in the case of German predicative adjective uninflectability, it is not clear what counts as a 'construction' here, so it is safer to label the phenomenon as 'contextual uninflectability'.

The Hungarian inflecting infinitive is the constructional counterpart to the sporadic agreement phenomenon found in Nakh-Dagestanian languages and elsewhere. The difference is that it is not lexically conditioned, it is determined by the wider syntactic context and applies to all appropriate lexemes. Together with the Latvian genitive compounds, the Hungarian (partially) inflecting infinitive serves to fill in the missing cell in the partial typology of uninflectability given in Table 2, allowing us to propose the typology in Table 3.

3 What counts as inflection?

We have so far been tacitly assuming that we know what it means for a lexeme to inflect. However, any serious study of the nature and typology of uninflectability must first establish what exactly is meant by *inflection*. There are

Table 3: Four-way typology of uninflectability

	lexical	contextual
total	*kenguru*-noun	German predicative adjective
partial	*muzeum*-noun	Latvian genitive compounds
		(Polish *widmo*-compounds)
	sporadic agreement	Hungarian inflected infinitive

several types of morphological or morphosyntactic process that are closely related to inflection, and which in some ways could be said to count as inflection, so it is worthwhile asking whether an uninflectable lexeme can undergo those processes. These include clitic-like affixation (or conversely, affix-like clisis), periphrasis, and various types of evaluative morphology, for instance, diminutive formation. (Admittedly superficial) examination of extant descriptions of uninflectable lexemes suggests that none of these processes count as inflection for the purposes of defining uninflectability. In other words, uninflectability only seems to relate to what we might call 'core' inflection: uncontroversial affixation or stem modification serving to realize a central repertoire of functional categories (hence, excluding evaluative morphology).

According to some authors, pronominal and other elements that are traditionally referred to as clitics in a variety of languages should really be thought of as a particular type of affix, which effectively means that the properties those 'clitics' realize are inflectional features (e.g. Monachesi 1999 for Italian, Joseph 1988 for Greek, Lyons 1991 for Macedonian, Miller & Sag 1997, Bonami & Boyé 2007 for French amongst others; see also Spencer & Luís 2012). The uninflectable French verb FOLLOW, however, seems to cooccur freely with these bound pronominals in (2a), suggesting that such clitic-affixes do not count as inflection for the purposes of uninflectability.

Similarly, it has been claimed that the postpositive definite articles in Bulgarian and Macedonian are better thought of as phrasal affixes or edge inflections, rather than straightforward clitics (Sadock 1991, Halpern 1995; see also Spencer & Luís 2012: 129–30). But these, too, can attach to uninflectable adjectives in those languages, e.g. Macedonian *taze-to* 'fresh-DEF' 'the fresh (one)' (multiple attestations on the internet).

In (2b) we see an instance of an uninflectable French verb licensing the periphrastic past tense ('passé composé') construction, suggesting that periphrasis, too, fails to count as core inflection (Spencer & Popova 2015: 208).

The status of evaluative morphology with respect to traditional notions of inflection and derivation is notoriously unclear (for discussion see Spencer 2013: 113–122; Grandi 2015). I shall adopt the view that certain types of evaluative morphology have to be distinguished from derivation or word-formation proper, that is, from lexeme formation. We can encapsulate this stance as the *Novel Lexeme Criterion*: If a morphological process applies to some form of a lexeme and can only sensibly be said to deliver a form belonging to the same lexeme then that process is not one of derivation/word-formation. A case in point would be diminutive/augmentative forms of proper names. In Russian, a name such as *Andrej* can give rise to a number of expressive diminutive forms such as *Andrjuša*, *Andrjušenʹka*. However, these diminutive forms will all denote the same individual as that denoted by the more formal variant *Andrej*. In this sense, the diminutives are just forms of the basic name. (This does not necessarily mean that it makes sense to call diminutive formation an inflectional process, of course.)

The first point to establish is that evaluative morphology can often be applied to uninflecting words, that is, words which belong to classes which are not expected to take any kind of inflection in any case. The most obvious such instances are seen in languages with no or virtually no inflectional morphology in the first place. Chinese, for instance, lacks (uncontroversial) inflections, but has non-affixal evaluative morphological processes: reduplication, ablaut, tone shift (Arcodia 2015). In inflecting languages, we find that evaluative morphology can be applied to uninflecting word classes such as adverbs or quantifiers. This is true of the Basque diminutive suffix *-txo* (Artiagoitia 2015: 197–198), and the Jingulu evaluative suffix *-kaji* (Pensalfini 2015: 416).

The propensity for evaluative morphology to apply to uninflecting words, even if with strong lexical limitations, seems rather widespread. Even in Russian, for instance, which has adjective-based evaluative morphology in addition to very productive noun-based evaluation, but no productive evaluative morphology for other parts of speech, we find isolated instances such as the diminutive form of the word *spasibo* 'thank you' ⇒ *spasib-k-i* (diminutive suffix *-k-* with an apparent plural ending *-i*). Given this background we can ask what potential an uninflecting name in Russian might have for diminutive formation. In my extended family there is a dog whose English name is Harry, but who also responds to the Russian form of the name, *Xari*. The Russian form is uninflectable. However, it readily takes the diminutive form *Xari-k* (which is how he is usually addressed by Russian-speaking family members). If the set of situations just described are typical then it indicates that evaluative morphology, too, does not count as *inflection* for the purposes of uninflectability.

Uninflectable lexemes can sometimes undergo derivation-like inflectional processes. For instance, comparative/superlative formation in Indo-European adjectives and adverbs does not seem to create a new lexeme, yet it adds an important meaning component to the original base word and changes its syntactic potential considerably. In this respect it is far from canonical as inflection. It is therefore not too surprising that the German contextually uninflectable predicate adjectives have comparative forms, (22).

(22) German

a. Das Känguru ist klein.
the kangaroo is little(UNINFL)
'The kangaroo is little.'

b. Aber dieses ist noch klein-er.
but this-one is even little-COMPARATIVE
'But this one is even smaller.'

It should equally come as no surprise to learn that uninflectable lexemes can undergo uncontroversially derivational morphology. More significantly, such lexemes can undergo morphology which sits on the boundary between canonical derivation and canonical inflection. In Table 4 we see examples of standard derivation (*konoščnik*, *kengurjonok*), and the formation of what Spencer (2013) refers to as a transpositional lexeme, in this case a relational adjective that adds no semantic predicate to the base noun but which nevertheless behaves like an autonomous lexeme, *kofejnyj* (for a justification of this claim and an extensive crosslinguistic typology of relational adjectives see Nikolaeva & Spencer 2019).

Table 4: Derivation applied to uninflectable Russian nouns

kino	'cinema'	⇒	*konoščnik*	'cinema worker'
kenguru	'kangaroo'	⇒	*kengurjonok*	'joey (baby kangaroo)'
kofe	'coffee'	⇒	*kofejnyj*	'pertaining to coffee' (relational adjective)

Finally, uninflectable lexemes may undergo compounding, including the more morphological type of compounding which involves an intermorph, as in the Russian examples shown in (23), where *klin* 'wedge' is an ordinary inflecting noun while *kino* 'cinema' is uninflectable.

(23) Russian compound formation

kino ‘cinema’; *operator* ‘operator’ ⇒ *kin-o-operator* ‘cameraman’

cf

klin ‘wedge’; *obraznyj* ‘having form X’ ⇒ *klin-o-obraznyj* ‘wedge-shaped’

4 A paradigm-based solution

We turn now to the problem of how we can best represent the various types of uninflectability in a formal model of morphology. This is a non-trivial problem for at least two reasons. First, as we have seen, and as Corbett (this volume, Section 4) demonstrates at length, uninflectablity turns out to be a multi-dimensional phenomenon which is best thought of in terms of degrees of uninflectability rather than an absolute property. Second, there are a number of conceivable ways of stating the facts of uninflectability in various frameworks. However, we should aim to go beyond simple description and strive towards the more interesting goal of stating those facts in a way that reveals some kind of insight into the underlying nature of (the various types of) uninflectability, and which might thus provide the beginnings of an explanation for the observed patterns.

Total uninflectability is a relatively neglected problem, especially in the theoretical literature. I have already mentioned the important monograph-length descriptive study by Doleschal (2001), but uninflectability is not mentioned in Stump (2016), nor can I find reference to it in recent handbooks such as Baerman (2015), including psycholinguistic studies (Stoll 2015, Walenski 2015). The discussion in Brown & Hippisley (2012: 65, 156–159, 162), in which uninflectedness is treated as a kind of syncretism with the root form, is an honourable exception (see also Baerman et al. 2017: 26–28.).

In this section I will develop a formal account based on a realizational model of inflection combined with the approach to lexical representation advocated in Spencer (2013). It is important to be able to propose a formal account that encompasses all four types in the typology of Table 2 in essentially the same way. Otherwise, we will have no guarantee that we are not dealing with apparently similar but in fact disparate phenomena. What this means is that there has to be some core formal property or collection of properties that is shared uniquely by the four types of uninflectability but not by other phenomena.

Traditional morpheme-based models (including classical LFG) would presumably be obliged to postulate up to twelve zero suffixes for *kenguru*-nouns and for the Russian (Hillary) *Klinton*. Distributed Morphology (Embick 2015) might be able to deploy the device of Impoverishment to neutralize formal distinctions between otherwise inflected forms by retreating to some maximally unmarked form identical to the lexical root. But it is hard to see how this would work for languages such as Greek, in which all inflected words are affixed forms of a bound stem (cf the Latin noun declension illustrated below in Table 9). For such cases it would presumably be necessary to posit a special zero suffix for some arbitrarily chosen cell/feature set and then Impoverish the rest of the paradigm to that cell. Uninflecting lexemes thus present a particularly severe challenge to such models. Inferential-realizational models such as PFM1 Stump (2001) would have to posit massively neutralizing rules of referral (always to the lexical base form), not much of an improvement.

Spencer (2020) argues for an analysis which makes crucial appeal to the distinction between two types of paradigm proposed in Stump (2016), in a model that has come to be known as PFM2. PFM2 distinguishes crucially between content and form paradigms (Π_C, Π_F). The leading idea is very similar to that which motivates the distinction between s(yntactic)-features and m(orphological)-features in Sadler & Spencer's (2001) analysis of the Latin perfect passive periphrasis. The s-features are those features that govern syntactic constructions and which may receive some special non-lexical semantic interpretation. For example, in English definiteness is a property of noun phrases which gives rise to (very subtle and poorly understood) semantic and pragmatic contrasts. It is also an obligatory property, much like a purely inflectional property such as verb tense. However, it is never expressed by a purely morphological form, rather it is expressed by specific function words (in/definite articles, demonstratives) and other constructional properties (e.g. possessive phrases). This means that it is a purely syntactic feature. In Scandinavian languages, for instance, Danish (Lundskær-Nielsen & Holmes 2015) definiteness is expressed both by function words and by special inflected forms of the noun, while in Latvian (Kalnača & Lokmane 2021) definiteness is expressed by special morphological forms of attributive adjectives. For those languages, then, definiteness is a feature which regulates not only syntactic structure, but also morphological form.

In PFM2 Π_C defines all the syntactically accessible inflectional contrasts a lexeme is obliged to make. It serves as the interface between morphological forms and syntactic-semantic structures, in the sense that the cells of the Π_C are defined by features which represent all the morphosyntactic contrasts visible to the syntax, as well as all the functional semantic contrasts that are interpreted by the semantics. This is Stump's (2016: 105) *interface hypothesis*: "Paradigms are the interfaces of inflectional morphology with syntax and semantics". Given this characterization, the features, σ, defining the Π_C are a subset of the s-features, the total set of features required to specify syntactic structures and the interpretation of functional features (as opposed to lexical meanings).

By contrast, the form paradigm, Π_F, serves as part of the definition of the set of morphophonological forms expressing the contrasts defined at the level of the Π_C (Stump 2016: 106–115). The Π_F for a lexeme $\mathcal{L}$ is a set of pairings of complete property sets[3] with a set of appropriately selected stems for $\mathcal{L}$. The cells in Π_F are therefore ordered pairs of the form $\langle \mathrm{Z}, \tau \rangle$, where Z is a member of the lexeme $\mathcal{L}$'s stem set and τ is a set of (morphomic) morphosyntactic properties possibly distinct from the set σ. It is Π_F that serves as the point of departure for the application of the rules of morphology, the Paradigm Function, which maps the cells of Π_F to a realized paradigm (Π_R), consisting of pairs of the form $\langle \omega, \tau \rangle$, where ω is the inflected word form (or a set of word forms in the case of overabundance, Thornton 2012). In this respect the Π_F is a theory internal construct whose role is to serve as the input to the battery of realization rules that make up the PFM2 morphological engine.

The paradigms Π_C and Π_F are related through a principle of Paradigm Linkage, mediated by *Corr*, a Correspondence function which specifies the mapping $\Pi_C \mapsto \Pi_F$, in part defined by the function *pm* ('property mapping') defined over the feature sets, Σ, T, of Π_C, Π_F. The Π_C of a lexeme is thus a matrix of cells consisting of ordered pairs of the form $\langle \mathcal{L}, \sigma \rangle$, where σ is some subset of Σ. The component $\mathcal{L}$ is a lexical index which identifies an equivalence class of lexemes by their inflectional paradigm, ignoring irrelevant homophony/homonymy where appropriate. In this respect, a word such as the English verb DRAW belongs to a single lexemic equivalence class, even though it has a wide range of meanings, some of them totally unrelated to each other (*draw a tooth, draw a conclusion, draw a picture, …*). In other words, the lexemic equivalence class with respect to morphology is defined by the shared inflectional paradigm. This is not the only way that lexemes need to be individuated, of course. As Fradin & Kerleroux (2003) point out, derivational morphology often targets very specific meanings of homonymous or polysemous lexemes, even if those lexemes share all their inflectional features (see also Bonami & Crysmann 2017 for further discussion). For the cases I shall be discussing, however, only very coarse grained distinctions between lexemes are required. The component σ is a complete and coherent morphosyntactic property set, which the syntax-semantics needs to have access to, so as to express purely syntactic dependencies such as agreement and govern-

[3] Stump refers to 'properties' rather than features. For our purposes we can treat these terms as synonymous, but I use 'property' when discussing Stump's own proposals.

ment, and for realizing or specifying the values of obligatory meaningful categories such as nominal number, verb tense and so on. The set σ crucially does not include purely morphological (morphomic) properties such as inflectional class indices.

In a canonical inflectional system the Π_C and Π_F are isomorphic: each lexeme $\mathcal{L}$ is associated with a unique root, Z, its sole inflectional stem, and $\tau = \sigma$, and *Corr* and *pm* are the identity function. This is canonical paradigm linkage as defined in Stump (2016: 113, example (2)). Note that the properties which define Π_C and Π_F are taken from the same vocabulary of properties and in the canonical case the two sets are identical, not *merely* isomorphic, but actually identical (Stump 2016: 113; see below). However, Stump (2016) catalogues a variety of non-canonical situations, in which $\tau \neq \sigma$. The deviations from the canonical isomorphic mapping between Π_C and Π_F include syncretism, deponency, heteroclisis, overabundance, defectiveness. In a number of cases the deviation from canonicity is purely morphomic. Heteroclisis, for instance, is found when a lexeme inflects according to more than one inflectional class in different parts of its paradigm (say, singular vs plural forms). In the case of deponency, e.g. Latin verbs which are 'passive in form but active in meaning', we have an unexpected mapping from the content property 'active voice' to a set of forms which would normally realize passive voice. In the case of syncretism we have a pair of contrasting Π_C properties which map to identical Π_F properties, neutralizing the Π_C property contrast. For instance, Stump (2016: 174–75) handles a set of syncretisms in the Bhojpuri verb by 'deleting' certain property values from the Π_F paradigm in the *pm* and hence halving the number of cells.

In a few of the case studies presented in Stump (2016), notably the Nepali 'long present negative' conjugation (following the analysis proposed in Bonami & Boyé 2008) the set τ includes properties which Stump (2016: 139–145) describes as 'morphomic'. These properties, which Stump, following Bonami and Boyé labels 'row' and 'column', have no correspondent at all in the Π_C. However, in the great majority of cases the properties in the set τ are formally exactly the same objects as the properties in σ. This majority situation reflects the property-set preservation principle as described by Stump (2016: 113), (24).

(24) Property-set preservation: Content cells and their form correspondents have the same property set; that is, for any property set σ associated with a content cell, $pm(\sigma) = \sigma$.

Property-set preservation conceals a mild conceptual inconsistency in PFM. That model is based on the principle of the autonomy of morphology or *morphology-by-itself* (Aronoff 1994), as discussed extensively in Stump (2016: Chapter 8):

"… morphomic properties can all be seen as an effect of properties that are restricted to the level of form paradigms" (Stump 2016: 120). But the reverse should hold too, given the architectural assumptions of PFM: all the properties of Π_F (and Π_R) should themselves be morphomic, for the simple reason that neither syntax nor semantics have access to those levels of representation (cf Boyé & Schalchli 2019; Spencer 2021). Exploring these issues goes beyond our remit here.

I will illustrate the virtue of the Π_C/Π_F distinction with a brief survey of German noun declension. This case illustrates the extent to which inflection can be morphomic but more importantly for our present purposes it also illustrates the kind of case which might be mistaken for a case of (partial) uninflectability when in fact it is something else.

In standard descriptions, German nouns inflect for four cases and two numbers (Durrell 2002), so that the German noun phrase is defined by an 8-celled paradigm, illustrated in Table 5 by the declension of the definite article in the masculine singular form. In Table 6 we see the declension of two typical nouns, Vater 'father', and Hand 'hand', and in Table 7 we see the declension of the'weak' noun Student 'student'. Nouns of the *Student*-class, traditionally called weak nouns, differ from so-called 'strong' nouns in that all their forms end in *-en* except for their nominative singular form. We might be tempted to assume that weak nouns are therefore indeclinable for 7/8ths of their paradigm and only inflect (subtractively) in the nominative singular. However, this would be to completely misanalyse German noun inflection.

Table 5: German definite article masculine singular

	singular	plural
nominative	der	die
accusative	den	die
genitive	des	der
dative	dem	den

Let us call the features defining the Π_C 'σ-features' (a subset of the s-features) and those defining the Π_F 'μ-features' (for 'morphological/morphomic' features, broadly equivalent to m-features). In German there is a considerable mismatch between the σ-feature set required for the Π_C and the μ-feature set (Boyé & Schalchli 2019; Spencer 2009). What is unusual about the weak nouns is that they show that the morphomic feature inventory depends on lexical class. In the strong nouns the inflectional split is between singular and plural subparadigms

Table 6: Standard description of German 'strong' noun declension: Vater, Hand

	Vater 'father'		Hand 'hand'	
	singular	plural	singular	plural
nominative	Vater	Väter	Hand	Hände
accusative	Vater	Väter	Hand	Hände
genitive	Vaters	Väter	Hand	Hände
dative	Vater	Vätern	Hand	Händen

Table 7: Standard description of German 'weak' noun declension: Student

	Student 'student'	
	singular	plural
nominative	Student	Studenten
accusative	Studenten	Studenten
genitive	Studenten	Studenten
dative	Studenten	Studenten

(mainly), rather than nominative singular and the rest of the paradigm. Notice that the Student class is not the only one whose paradigm only has two distinct forms. The same is true of Mädchen 'girl', Wohnung 'flat, apartment', Park 'park', Table 8.

Table 8: Standard description of German noun declension: Mädchen, Wohnung, Park

	Mädchen 'girl'		Wohnung 'apartment'		Park 'park'	
	sg	pl	sg	pl	sg	pl
nom	Mädchen	Mädchen	Wohnung	Wohnungen	Park	Parks
acc	Mädchen	Mädchen	Wohnung	Wohnungen	Park	Parks
gen	Mädchens	Mädchen	Wohnung	Wohnungen	Parks	Parks
dat	Mädchen	Mädchen	Wohnung	Wohnungen	Park	Parks

In fact, no German noun distinguishes more than four of the eight theoretically possible forms (the noun Beamte 'civil servant' and nouns such as Angestellte(r) 'employee' inflect like adjectives, not nouns). The 'weak' nouns are the only nouns which ever distinguish nominative from accusative case and then only in the singular (a small set of masculine irregular nouns such as Name 'name' also distinguish nominative singular from accusative singular, but they only have a total of three distinct forms each in their complete paradigm: *Name*, nominative singular, *Namens*, genitive singular, *Namen* elsewhere). It is therefore more in keeping with our morphomic assumptions to characterize the German noun Π_F by just giving arbitrary labels to the various form classes, as illustrated in (25), where I conflate in part the (abstract) Π_F with the concrete realized paradigm.

(25) a. Form1=Vater; Form2 = Form1⊗s; Form3=Umlaut(Form1); Form4=Form3⊗n
b. Form1=Hand; Form2 =Umlaut(Form1)⊗e; Form3=Form2⊗n
c. Form1=Mädchen; Form2 = Form1⊗s
d. Form1=Wohnung; Form2 = Form1⊗en
e. Form1=Park; Form2 = Form1⊗s
f. Form1=Student; Form2 = Form1⊗en

Notice that these specifications give the superficial impression that Wohnung and Student belong to exactly the same lexical class, but that impression is dispelled when we look at the *Corr* function. This specifies Form2 for the *Wohnung*-class as the Π_F correspondent of Π_C [Number:plural], while for the *Student*-class, Form2 is the default correspondent and Form1 is specified as the Π_F correspondent of [Number:singular, Case:nominative]. Sample rules are provided in (26).

(26) (i) Class:Vater
[number:plural, case:dative] ⇒ Form4
[number:plural] ⇒ Form3
[number:singular, case:genitive] ⇒ Form2
Elsewhere ⇒ Form1
(ii) Class:Student
[number:singular, case:nominative] ⇒ Form1
Elsewhere ⇒ Form2

(iii) Class:Wohnung
[number:plural] ⇒ Form2
Elsewhere ⇒ Form1[4]

What we learn from systems like the German noun paradigm is that there is no necessary simple correspondence between inflectional form and inflectional function. But more than that, the divergence between form and function can become so great that a lexeme hardly appears to inflect at all. The German weak nouns are just one step away from becoming uninflectable. This illustrates yet again the point made by Corbett (this volume, especially Section 4) that uninflectability is not an all-or-nothing affair. It is then the task of the formal grammarian to find adequate ways to represent this fact.

We are now in a position to propose a formal model of uninflectability within a paradigm-based morphology. The leading intuition will be that an uninflecting lexeme has a full Π_C but a depleted or even non-existent Π_F. Recall that the original motivation for the Π_C~Π_F distinction is the idea that the Π_C represents the interface with syntax/semantics, while the Π_F represents the interface with (morpho)phonology. One way of thinking of this is to say that PFM2 is a model of *lexical insertion*: the Π_C defines a set of properties that a lexeme is obliged to realize in a given language, while the Π_F specifies what kinds of morphophonological forms will be available to realize those properties. In other words, the *Corr* function regulates the syntax/semantics~morphology interface by specifying which Π_F cell occupant corresponds to a given Π_C feature set. In this way, *Corr* regulates lexical insertion for inflecting lexemes (cf Stump 2016: 27–30, 115).

Central to the argument, then, is the claim that the problem of uninflectability is intimately related to the question of lexical insertion. We can reframe the problem in this way: how can a lexical model of inflection license the insertion of lexical forms which apparently fail to bear inflection? We must therefore address the question of how lexical insertion is conceived in a paradigm-/lexeme-based model of morphosyntax. We begin with the most innocuous looking of lexical elements, the uninflecting lexeme.

The uninflecting lexemes of a language, such as English prepositions, can easily be defined as members of a lexical class which is not associated with any Π_C. In the architecture of PFM2 the cells of the Π_F are defined by projection from Π_C by means of the *Corr* mapping. Thus, in the absence of a Π_C the Π_F will be undefined. How, then, can we determine the form such a lexeme takes when it is selected to be part of a sentence (lexical insertion)?

[4]Choice of Form1 as the default is arbitrary, of course, mandated only by the fact that singular tends to be the default number in languages like German.

One possibility is that we treat uninflecting lexemes as having a trivial inflectional paradigm, consisting of one Π_C cell, $\langle \mathcal{L}, \sigma \rangle$ mapped to a Π_F cell ⟨Z,∅⟩, where ∅ is a null form property of some kind. The feature ∅ is not associated with any realization rule, so we still have no way of specifying the final form of the inserted lexeme. The obvious step would be to invoke Stump's Identity Function Default (IFD). However, it is not clear how this could be done given the standard PFM architecture.

The motivation for the IFD is to provide an output for a maximally general, underspecified input. However, this application is paradigm-driven, in the following sense. The inflectional paradigm of any lexical class is defined in terms of some ensemble of features. Consider the simplest case of a paradigm with just two cells, say, a noun inflecting for NUMBER:{singular, plural}. The IFD guarantees that the cell, say, ⟨DOG, [NUMBER:singular]⟩ will be mapped to a Π_F correspondent, even in the absence of a realization rule defined over the feature specification [NUMBER:singular], namely, the form given by the stem selection rule for English nouns, which itself by default delivers the lexical root form /dog/.

Implicit here is the assumption that inflectional defectivity is to be avoided, ceteris paribus. The grammar demands that an English count noun have an inflectional paradigm with two forms, one for singular and one for plural (even if those forms are morphophonologically identical, as in the case of SHEEP). The IFD guarantees that DOG has a singular form, by applying the totally underspecified realization rule (27) in the first (sole) inflectional rule block, where X = /dog/ (cf. Bonami & Stump 2016: 465).

(27) $X_U, \{\} \rightarrow X$

Rule (27) states that a form 'X' (whether basic or complex), realizes a completely underspecified morphosyntactic property set {U}, by being mapped to itself.

Table 9: Paradigm of Latin DOMINUS 'lord'

	SINGULAR	PLURAL
NOMINATIVE	domin-u-s	domin-iː
VOCATIVE	domin-e	domin-oː-s
GENITIVE	domin-iː	domin-oː-rum
DATIVE	domin-oː	domin-iː-s
ABLATIVE	domin-oː	domin-iː-s

In some cases it will appear that the IFD fails to apply. Latin nouns inflect for five cases and two numbers. A 2nd declension noun such as DOMINUS 'lord' has a root form /domin/ which is bound and thus never appears in an unsuffixed form, as can be seen from Table 9. Even if we assume that its inflected forms are based in part on a derived stem form (say, *domin-u* or *domin-o*), there will be no cell in the noun's paradigm that is not formed with an inflectional suffix, so that the whole paradigm has to be defined by the application of non-trivial, specified realization rules, including the nominative singular form. When applied to a lexeme such as DOMINUS the IFD would define a root form *domin* (and possibly also an uninflected bound derived stem *domin-u*, *domin-o* or whatever, depending on the details of the analysis). However, such a bound form would never be realized as a fully inflected word form, and would thus never be eligible for lexical insertion into a syntactic phrase, because all the cells in the paradigm of DOMINUS are filled by forms which are defined by inflectional rules which are more specific, and which thus pre-empt the IFD.[5]

Given the architecture of PFM it is clear that the IFD is irrelevant to the question of how to specify the form of an uninflecting lexeme for lexical insertion, for the simple reason that the IFD specifies forms in inflection paradigms, but an uninflecting lexeme, by definition, has no inflectional paradigm, no paradigm function, and a fortiori, is not subject to any realization rule, not even the maximally underspecified rule (27) which instantiates the IFD.[6]

The only way we could deploy the IFD for uninflecting lexemes, it seems, is to provide such lexemes with a paradigm. This would entail creating a special inflected form, realizing some property, say, [LEXICALINSERTION:yes], whose inflectional (one-celled) paradigm is not specified by any stipulated realization rule. This would mean that its form is invariably defined by the IFD. In other words, each uninflecting lexeme would have a Π_C whose cell realized the [LEXICALINSERTION:yes] feature, and whose form correspondent was ι(Z), where ι is the identity function and Z is the lexical root. In fact, given the assumptions of PFM2, such a treatment is the only one possible. The Π_C, recall, represents the interface between morphological form (the contents of the Π_F cells) and syntax-semantics. Therefore, either we have to assume that uninflecting lexemes do, after all, have a Π_C, and are hence "covertly" inflecting, or we have to conclude

[5]The inflectional rules of Latin can be expected to take nominative (and singular) as default values of the CASE and NUMBER features, in which case there would be an important sense in which the IFD *does* apply even to DOMINUS. It is just that its effects are hidden.

[6]This conclusion is equally valid for the morph-based interpretation of PFM proposed by Crysmann & Bonami (2016) and Crysmann (2020) under the rubric of Information-based Morphology (IbM).

that uninflecting lexemes fall outside the PFM architecture in its current form, and that they interface with the rest of the grammar in some other way.

The ploy of creating a 'dummy' inflectional paradigm for uninflecting lexemes is, in essence, is the solution proposed by Sag (2012: 119). He assumes a "trivial" *Zero Inflection Construction*, shown in (28), "which pumps noninflecting lexemes to word status. This construction takes a lexeme as daughter, licensing mothers that are of type *word*, but otherwise identical to their daughter."

(28) *Zero Inflection Construction* ($\uparrow$*infl-cxt*)

$$\textit{zero-infl-cxt} \Rightarrow \begin{bmatrix} \text{MTR} & X\,!\,\textit{word} \\ \text{DTRS} & \langle X : \textit{invariant-lxm} \rangle \end{bmatrix}$$

The type feature *invariant-lxm* defines the class of uninflecting lexemes, and the symbol ! in MTR X ! Y means that the mother X is identical to the daughters except for the specification Y, in other words that the only difference between the daughter and the mother is that the daughter is of the supertype *lexeme*, while the mother is of the type *word*. The annotation ($\uparrow$*infl-cxt*) means that *zero-infl-cxt* is a subtype of the *infl-cxt* construction, (29), (Sag's (57) Sag 2012: 108), which states "The mother of an inflectional construct is of type *word*; the daughters must be lexemes."[Footnote omitted]

(29) Inflection construction in SBCG

$$\textit{infl-cxt:} \begin{bmatrix} \text{MTR} & \textit{word} \\ \text{DTRS} & \textit{list} : (\textit{lexeme}) \end{bmatrix}$$

The problems with the type shifting account, whether in Sign-Based Construction Grammar or in PFM, are the following. The principal problem is that it is clearly no more than an arbitrary device deployed solely to meet the parochial architectural needs of a particular formalization of the model. Second, it fails to capture the very clear intuition that an uninflecting lexeme is a lexeme that fails to inflect, not a lexeme with some strange 'zero' inflection. In particular, what feature specification is an uninflecting lexeme associated with? Do all uninflecting lexemes 'realize' the same 'zero feature'? The counter-intuitiveness of the solution becomes obvious when we consider languages such as Vietnamese, which have no inflectional morphology at all. The type-shifting procedure would have to create a 'trivially' inflected form for every lexeme in the language, even though the language lacks any inflection. Finally, we have a question which is not addressed in Sag's account, but which is central to this paper: what is the relationship between lexemes which belong to a non-inflecting class such as English prepositions (members of Sag's *invariant-lxm* type, our uninflecting lexemes), and uninflectable lexemes, whether lexically uninflectable, like KENGURU or contextually/constructionally uninflectable, like German predicative adjectives?

An alternative to a type-shifting account, sketched in Spencer (2019a,b) says that an uninflecting lexeme has literally no inflectional paradigm, whether Π_C or Π_F. It is not associated with any Π_C in any form, because it is not, by definition, associated with any morphologically expressed morphosyntactic property. A fortiori it is not associated with a Π_F. This leaves the question of how to specify the form of such a lexeme when it occurs in a syntactic construction, that is, how to specify lexical insertion for such a lexeme. Ideally, we should do this without appeal to the PFM equivalent of the type shifting operation (28).

In Spencer (2019a, 2019b; see also Spencer 2020) I proposed a Default Exponence Principle (DEP), which states informally that, by default, the form of any word undergoing lexical insertion (however this notion is characterized) is the lexical root, even in the case of a standardly inflecting lexeme. To see how that account was supposed to work let Σ be the set of features defining a content paradigm, and let T be the set of features defining the corresponding form paradigm. As things stand the Paradigm Linkage mapping will be undefined for cases where $\Sigma = \mathrm{T} = \emptyset$. Spencer (2020) therefore assumes mapping (30), the Default Exponence Principle, as the default morphosyntactic expression of all lexemes defined over null feature sets.

(30) Default Exponence Principle (DEP) [to be rejected!]
$\mathrm{PF}(\langle \mathrm{Z}, \emptyset \rangle) = \mathrm{Z} = \mathrm{STEM}_0(\boldsymbol{\mathcal{L}})$

The intended upshot is that the default realization of all lexemes is STEM_0 (≡ Z, the lexical root). For inflecting lexemes the DEP (30) is overridden by the more specific *Corr* function. However, for an uninflecting lexeme DEP has the same effect as Sag's Zero Inflection Construction. This would mean that we would not require uninflectable lexemes to have a non-null inflectional paradigm. The uninflected lexemes whose form is defined by (30) have just a root form but no associated paradigm, content paradigm or form paradigm.

The idea behind this formulation is that the DEP is overridden by any genuinely inflected word form, in the sense that any lexeme which has a Π_F will have a definition of lexical form which is more specific that the DEP, and this will pre-empt the default. However, there is an important flaw in this way of viewing default root exponence. The difficulty relates to the apparently trivial and obvious aspect of the model mentioned above: what is the relationship between the lexical insertion of an inflected form and lexical insertion of an uninflected form? In the present case, the technical problem with our informal DEP is that it is not possible to treat uninflected default root forms and inflected word forms as being in (Paninian) competition with each other. Therefore, it is not possible for a specific inflected form to override the root insertion default. Such an override would only be possible if the root insertion were formalized as a kind of zero inflection, of the sort we have just rejected.[7]

I therefore abandon the proposal based on the DEP. In order to account for uninflectability, however, we need to make completely explicit what we mean by lexical insertion. I shall sketch such an account, what I take to be essentially the notion of lexical insertion that is invariably presupposed in the literature, and then show how it serves as the basis of an account of uninflectability. My account will appeal crucially to the model of lexical representation and lexical relatedness offered in Spencer (2013).

The intuition we wish to capture is very simple: by default, lexemes are uninflecting and the phonological form which realizes a syntactic node, v, is the sole phonological form associated with that lexeme, the lexical root. Where we have an inflecting lexeme the default root insertion operation has to be pre-empted by a more specific operation which inserts a more narrowly specified phonological form. That form is the Π_F correspondent of the most narrowly specified cell in the lexeme's Π_C corresponding to the featural content of syntactic node v. I shall code the notion of lexical insertion in relatively informal terms, borrowing and adapting the notational conventions introduced in the Parallel Architecture model of Jackendoff & Audring (2020).

First, we assume that lexical insertion is coded as a co-indexing of a syntactic node, v, with a morphophonological representation of a word form, ω. Thus, to say that the forms /ðə/, /blak/, /dog/ are inserted into the NP frame $[_{NP}[_{D}$ the$]$ $[_{A}$ black$]$ $[_{N}$ dog$]]$ is simply to provide the appropriate indices: $[_{D}$ the$]_1$ $[_{A}$ black$]_2$ $[_{N}$ dog$]_3$$\Leftrightarrow$/ðə/$_1$ /blak/$_2$ /dog/$_3$.

Second, we should observe that such co-indexation is always feature-driven in the sense that the node v is necessarily specified by at least a lexical category feature (or equivalent), and that the phonological form it is coindexed with has to be associated with a compatible feature specification. The leading idea is this. Let us denote the set of all the morphosyntactic, morpholexical interface features (morphosyntactic features) as $\mathcal{I}$. The set $\mathcal{I}$ will contain all the possible

[7] The appeal to the DEP also makes it more difficult to understand the relationship between uninflectedness and defective lexemes, since one of the entailments of DEP is that all licensed collections of σ-features for a lexeme correspond to some form, and this entailment is incompatible with the existence of defective lexemes. However, the question of defectiveness and uninflectability has to remain the subject of work in progress.

features which could govern morphosyntactic structure, including complementation/selection features and others, perhaps including sociolinguistic properties where these interact with morphosyntactic well-formedness. Lexical insertion is then a relation which takes a node in a syntactic structure and matches it with a word form or set of word forms which offer the best fit with the specification on that node. Where the syntactic construction is underspecified we may find that this partial specification or description is compatible with more than one form. For instance, in the syntactic context *The* ___ *sat on the mat* the unspecified position is compatible with a noun in either singular or plural form. However, for simplicity of exposition let us assume that the syntactic node is fully specified and that the semantics specifies a lexeme with a unique meaning, so that lexical insertion gives a unique output up to perfect synonymy, including semantically neutral overabundance.

(31) Lexical Insertion (LexIn)
Given a syntactic node, v, defined by some feature set $\mathcal{F} \subset \mathcal{I}$ and given a lexeme, lexemic index $\mathcal{L}$. Then $\mathcal{L}$ and v can be coindexed iff there is some ω, a phonological word form of $\mathcal{L}$, which expresses/realizes the largest set $\mathcal{G} \subset \mathcal{I}$, of features such that $\mathcal{F} \subseteq \mathcal{G}$.

As formulated in (31) LexIn will also license the lexical insertion of an uninflecting form such as *on* in *The cat sat* ___ *the mat*. In the case of a singular or plural noun the coindexation takes place between the syntactic node and the form correspondent of the appropriate cell in the lexeme's Π_C, which by Paradigm Linkage maps to the appropriate cell in the Π_F. In the case of an uninflecting preposition, I have argued that there is no Π_C, hence, no Π_F, so that the only form available for insertion will be the lexical root, realizing the purely syntactic features of the preposition, such as its lexico-syntactic category, P.[8]

The important point about the formulation of LexIn is that it makes absolutely no (direct) reference to inflection. However, the set $\mathcal{I}$ clearly includes the set of (non-morphomic) inflectional features which define the content paradigms of lexemes, and which therefore serve as (part of) the interface between morphology and syntax. It is thus reasonable to say that the lexical insertion of an inflected word form is a special case compared to the insertion of an uninflecting word form. The featural characterization of an inflected word form will include as a

[8]In a complete formalization care will have to be taken to ensure that the morphophonological characterization of the inflected word forms in the realized paradigm, and the morphophonological characterization of the bare root form of an uninflecting lexeme are couched in the same morphophonological vocabulary, so that both types of representation will be "pronounceable" at the appropriate level of description.

bare minimum the categorial information associated with that lexeme (however such categories are defined) together with its own inflectional content.

We now have nearly all the machinery need to account for uninflectability. I will crucially appeal to the Π_C~Π_F distinction, though, as far as I can tell, all that is required for the analysis to work is any equivalent distinction between syntactically relevant and syntactically invisible (morphomic) features, such as the morphological vs. syntactic feature distinction of Sadler & Spencer (2001) or the HEAD vs. INFL features distinction of HPSG-oriented treatments of morphology such as IbM (Crysmann & Bonami 2016, Crysmann 2020). In this respect, the problem of lexical and contextual uninflectability provides additional strong support for some such distinction.

It remains for us to ensure that the mechanisms by which the inflected forms of an inflecting lexeme are defined are essentially the same as the mechanism by which the root form of an uninflecting lexeme is defined. The model of lexical organization proposed in Spencer (2013) provides the necessary machinery. We assume that all lexemes are individuated by means of a unique lexemic index, LI, $\mathcal{L}$. For simplicity, in the case of concrete lexemes I will use the name of the lexeme in SMALL CAPITALS as its lexemic index: SHEEP, HIT etc. We further assume that a lexeme has to be furnished with at least three more representations or attributes: FORM, specifying as a minimum the unpredictable aspects of the lexeme's phonology and morphology, such as the shape of the root, any idiosyncratic stem forms or inflected forms, and so on; SYN, a representation of crucial aspects of the syntax of the lexeme, such as syntactico-lexical category membership, argument structure or complementation features and so on; SEM, a semantic/pragmatic representation. These representations can be thought of as the output of three functions over the LI, e.g. FORM(HIT), SYN(HIT), SEM(HIT). Consider an uninflecting lexeme such as the preposition UNDER. The morphophonological root form, /ʌndə/ (or whatever) is defined by FORM(UNDER). For an inflecting lexeme such as DOG we have a (maximally simple) Π_C, say, {⟨DOG, SG⟩,⟨DOG, PL⟩}. English morphology defines the *Corr* function in such a way that the Π_R of DOG is {⟨*dog*, SG⟩,⟨*dogz*, PL⟩}.

Our aim now is to provide a mechanism for the formal representation of an uninflecting lexeme such as UNDER and the formal representation of an inflecting lexeme such as DOG which allows both type of lexeme to be treated in the same way by the principles of lexical insertion. In effect, this amounts to uniting the FORM attribute for lexemes such as UNDER, and the *Corr* function as applied to inflected words, but without introducing a (covert) disjunction. We can achieve this by defining lexical insertion over an expanded lexical representation of inflected lexemes in which the content of the FORM attribute is generalized so as to cover all possible form correspondents of a lexeme, including those defined initially in terms of a (non-trivial) Π_C. In other words, the expanded lexical entry of a lexeme incorporates that lexeme's realized paradigm, effectively coding the common notion of "full-listing" of inflected forms in lexical representations (Butterworth 1983).

This expanded lexical representation effectively formalizes the traditional notion that one of the purposes of the lexeme concept is to unify all the inflected forms of a lexeme as members of a single unit. In fact, such a generalization seems to be required even for uninflecting lexemes. For instance, some lexemes exhibit allomorphy which is determined in purely morphophonological terms independently of morphosyntactic context. The English indefinite article, *a*~*an* is a case in point. The FORM attribute applied to A, the LI of the indefinite article, therefore yields two outputs, {/a/, /an/}, perhaps together with some morphophonological featural annotation specifying which contexts each form occurs in.[9] Other cases of this sort might be found with uninflecting lexemes that have full form vs clitic alternants, high vs low register variants, and so on, or more exotic alternations of the kind illustrated by English *in my stead* ~ *instead of me*. Such alternations are not inflectional in character, but they do show that defining the root form of an uninflecting lexeme can be non-trivial. Against this background we now consider (non-defective) inflecting lexemes.

Consider the 2nd declension Latin noun DOMINUS, illustrated earlier in Table 9. Let's assume that its root form is /domin/. We now appeal to the architecture of lexical representation proposed in Spencer (2013: 184–89), in which the FORM attribute includes the lexeme's morpholexical signature, MORSIG, a subattribute of FORM, which specifies the morphological class (MORCLASS) to which the lexeme belongs for the purposes of the inflectional morphology. For instance, the lexical representation for the Latin DOMINUS will have a FORM attribute roughly as in (32).

(32) $$\begin{bmatrix} \textsc{form} \begin{bmatrix} \textsc{morsig} & \begin{bmatrix} \textsc{morclass} & \textit{declension2} \end{bmatrix} \\ \textsc{stem}_0 & \textit{domin} \end{bmatrix} \end{bmatrix}$$

The attribute FORM for DOMINUS will specify this stem (directly) and also all the inflected forms of Table 9 (indirectly, via the rules of Latin morphology). This gives us the schematized lexical representation shown in (33), where ⟨…⟩ stands for the representations of the cells of the rest of the paradigm.

[9] This may mean that FORM is a relation rather than a function, sensu stricto. However, this depends on exactly how such allomorphy is implemented, and so I will continue to speak of functions here.

$$(33)\quad \left[\text{FORM}\begin{bmatrix}\text{MORSIG} & [\text{MORCLASS}\ \ \textit{declension2}] \\ \text{STEM}_0 & \textit{domin} \\ \text{NOM,SG} & \textit{dominus} \\ \text{NOM,PL} & \textit{domini:} \\ \langle\ldots\rangle & \end{bmatrix}\right]$$

Similarly, the expanded FORM attribute for the German adjective KLEIN will be roughly that of (34).

$$(34)\quad \left[\text{FORM}\begin{bmatrix}\text{MORSIG}\ [\text{MORCLASS}\ \ \textit{adjective}] & \\ \text{STEM}_0 & \textit{klein} \\ \text{MASCULINE NOM.SG} & \textit{kleiner} \\ \text{NEUTER NOM.SG} & \textit{kleines} \\ \langle\ldots\rangle & \end{bmatrix}\right]$$

In both the Latin and the German examples the STEM_0 form is the basis for the inflectional rules of morphology. The crucial difference is that in the case of KLEIN, the German morphosyntax includes rules which make direct appeal to the STEM_0 form, whereas this is not true for Latin, even if that STEM_0 form happens to have the morphophonology of a pronounceable word.[10]

Now consider an uninflectable lexeme such as Russian KENGURU. I assume that as a lexical property its *Corr* function fails to define a mapping to Π_F. This means that none of the cells in the lexeme's Π_C has a form correspondent. We code this fact by stipulating that KENGURU belongs to an inflectional class [MORCLASS:uninflectable]. Lexemes of this class are associated with a property mapping which states that the feature set for the form/realized paradigm of the lexeme is undefined: $pm(\sigma)$ = undefined. Effectively, this means that there will be no expanded lexical entry for KENGURU, so that its lexical representation will remain that shown in (35).

$$(35)\quad \left[\text{FORM}\begin{bmatrix}\text{MORSIG} & [\text{MORCLASS}\ \ \textit{uninflectable}] \\ \text{STEM}_0 & \textit{kenguru}\end{bmatrix}\right]$$

However, our definition of lexical insertion makes no direct mention of a form correspondent to an inflectional cell, or indeed to any kind of paradigm, whether

[10]This account opens up the possibility that a lexeme might have two STEM forms, one the basis for morphology, whether free or bound, the other specifically applicable only to certain syntactic constructions. The special combining forms found in some of the compounds of German (Nübling & Szczepaniak 2013) and other languages might be a candidate.

Π_C, Π_F, or Π_R. LexIn just asks for a phonological word associated with the features on the syntactic node. In the absence of the (generally very specific) form-feature pairings defined by the Π_F, Π_R, all we have is the lexical root (or what I have labelled 'STEM$_0$'). It is therefore this root that undergoes the coindexation which instantiates lexical insertion in our informal characterization. It is in this sense that the root of KENGURU is a default form for inflection. The effect is to define a lexeme which behaves morphologically like an uninflecting lexeme, but which can undergo lexical insertion just as though it were an inflecting lexeme. The case of Russian female referent foreign names such as (*Hillary*) *Clinton* can be dealt with in the same way that we handle the *kenguru*-nouns, provided the grammatical representation of proper names has access to the referent's sex/gender, and provided we allow the female referent form of the name to constitute a distinct lexeme from the male referent form. Note that this only applies at the level of individual name lexemes; it does not apply to the class of non-native female names as a whole, or to the entire NP containing that name. A woman with a Russian (-sounding) given name, such as *Melania Trump* or *Marina Warner* will have that given name inflected but not the surname: *kniga Melani-i Tramp/Marin-y Vorner* 'Melania Trump's/Marina Warner's book'.

In the case of contextual uninflectability the main descriptive burden generally has to be placed on the syntax. Thus, the obvious way of handling the contextually uninflecting German predicative adjective is to stipulate that in predicative use (however defined) LexIn targets that value of the adjective's FORM attribute which has just categorial features and no others, namely, the STEM$_0$. This will mean that none of the features in the adjective's Π_C will be applicable, so that in predication position the adjective will behave exactly like an uninflecting (NB *not* uninflectable) lexeme. Again, since LexIn is effectively defaulting here to the lexical root, the grammar will not have to pretend that *klein* in (7b) (repeated here in adapted form) bears any (spurious) inflectional features.

(7b) Das Känguru ist **klein**.
the kangaroo is little(UNINFL)
'The kangaroo is little.'

This makes the case of predicative adjectives identical to that of other complement predicates to copular verbs in German. For instance, in (36) we see an instance of a NP functioning as the complement to the copular.

(36) a. Das Känguru ist mein Freund.
the kangaroo.SG is my.M.NOM.SG friend(M).NOM.SG
'The kangaroo is my friend.'

b. Die Kängurus sind unsere Freunde.
the kangaroo.PL are our.NOM/ACC.PL friend.NON-DAT.PL
'The kangaroos are our friends.'

The complement NPs *mein Freund* 'my friend' and *unsere Freunde* 'our friends' are in the nominative singular/plural form (unambiguously so in the case of the singular). Here we have to say that the syntax demands a nominative case marked predicative NP complement. This is not a given. In Polish, for instance, the NP complement to any copular verb will normally be in the instrumental case, while in Russian the choice of case depends on lexical and semantic factors. Adverbial phrases, including prepositional phrases, can also serve as complements to the copular verb, as seen in example (37) from German.

(37) Das Känguru ist hier/hinter dem Baum.
the kangaroo is here/behind the tree
'The kangaroo is here/behind the tree.'

In such constructions the adverbials *hier, hinter dem Baum*, do not realize any morphosyntactic property at all.

We have now largely achieved one of our principal aims. In both Type 1 and Type 3 total uninflectability LexIn targets an uninflected stem form, either because the syntax demands it (contextual/constructional uninflectability) or because the lexeme lacks any other forms (lexical uninflectability).

Turning now to partial uninflectability, consider first the Polish *muzeum*-words. Recall that lexemes of this class, MUZEUM, STUDIO and so on, typically retain, it seems, their latinate morphological structure: *muze-um, studi-o*. This can be seen from the plural inflected forms: *muze-a, muze-ów, …, studi-a, studi-ów, …*. These are based on a bound stem, *muze-, studi-,* which also appears in derived forms: *muze-al-n-y* 'pertaining to museums'. In effect, such lexemes exhibit heteroclisis. The plural part of the paradigm inflects according to a relatively standard inflectional class (though with an unusual genitive plural form), based on a bound stem. The singular part of the paradigm is uninflectable, and is based on what historically is generally the nominative singular ("citation form") of the original Latin word, i.e. that part of the paradigm is governed by the feature MORCLASS:uninflectable. Thus, the singular and plural paradigms belong to distinct inflectional classes. More specifically, following Stump (2016: 189–91) such a lexeme is associated with two distinct (though in this case, phonologically related) basic stems, each of which belongs to a distinct inflectional class. The singular subparadigm is defined over the invariant stem form *muzeum*, and this stem is marked as being uninflectable. This means that it is not associated with any Π_F. In the plural, however, the stem, *muze-* inflects according to the specific inflectional class it belongs to.

Consider again the phenomenon of 'sporadic agreement' (Fedden 2019). There are two important differences between sporadic agreement as seen in, say, the Nakh-Dagestanian languages and the partial uninflectability seen in cases such as the Polish *muzeum*-class. The *muzeum*-class words are very much the exception in the Polish lexicon, and for half their paradigm they are completely uninflectable. However, in the case of sporadic agreement it is generally the verbs showing agreement morphology that are in the minority, and those verbs that fail to show agreement nonetheless inflect normally for other inflectional categories. Thus, there is a sense in which sporadic agreement is a species of overdifferentiation, though on such a scale that it has to be coded explicitly in the grammar and not just ignored as an aberration. But this simply means that the agreeing lexemes are given a richer *morpholexical signature* than other lexemes, that is, they are associated with a richer set of inflectional features for which they are obliged to inflect (here, agreement features). As far as I can tell, the sporadic agreement cases do not therefore pose any special problems compared with other types of partial uninflectability: only a part of the verb lexicon bears specification for the agreement features.

Now consider again the Polish *widmo*-construction. The nominative singular form *widmo* (and likewise, the form *muzeum*), is, of course, traditionally regarded as the citation form of the lexeme WIDMO. It is presumably for this reason that such forms are chosen to serve as uninflecting variants in languages such as Polish or Latin, whose inflectional stems are generally bound morphemes. In such languages, citation forms seem the natural choice of default form with respect to lexical insertion. However, from a formal point of view this seems to entail that lexical insertion can in part be defined over the lexeme's morpholexical signature, i.e. the set of inflectional features for which it must inflect. Specifically, it appears to require that default application of LexIn can appeal somehow to those feature sets which function as defaults. It is not obvious how such an effect can be achieved formally, however.

A somewhat similar situation is found when we see the use of a citation form in titles of literary works and similar.[11] As a consequence, in certain contexts such titles and similar types of name (e.g. names of journals or newspapers), appear to be uninflecting in languages where we could otherwise expect them to inflect. In Russian, for instance, the title *Vojna i mir*, 'War and peace', inflects as a normal

[11]I am grateful to Grev Corbett for urging me to consider such constructions.

conjoined noun phrase, as in (38), with both nouns appearing in the expected genitive case.

(38) poslednee izdanie "Vojn-y i mir-a"
latest edition War-GEN.SG and peace-GEN.SG
'the latest edition of "War and Peace"'

However, somewhat more natural, perhaps, is a construction in which the title is in apposition to a generic noun such as ROMAN 'novel', as in (39).

(39) poslednee izdanie roman-a "Vojn-a i mir"
latest edition novel-GEN.SG War-NOM.SG and peace.NOM.SG
'the latest edition of the novel "War and Peace"'

In (39) it is the noun 'novel' which receives genitive case, while both nouns in the title itself remain in the nominative singular citation form.

The appositional construction illustrated in (39) is reminiscent of compound nouns in a number of languages with a phrasal non-head, such as *his why-does-it-always-have-to-happen-to-me attitude*.[12] One class of such phrasal compounds is generally taken to be quotative: the phrasal non-head expression is not used as such, but rather quoted. If we take the title *Vojna i mir* in (39) to be the citation of the title then it is less mysterious to see it appearing in the citation form. It might even be possible that citation accounts for the use of the default nominative singular in the Polish *widmo*-constructions, though further speculation would take us too far afield.

Finally, we turn to the cases of partial constructional/contextual uninflectability, as exemplified by the Hungarian agreeing infinitive and the Latvian genitive compounds. The Latvian genitive compounds exhibit a structure which is in many ways comparable to the compounds in which the non-head of the compound has to appear in a special "combining form", as found in Sanskrit, Greek, German and other languages. For instance, German has a large number of compounds of the form *Liebeslied* 'love song', formed by combining the nouns *Liebe* 'love' and *Lied* 'song' with a meaningless intermorph *-s-*: *Liebe-s-lied* (Nübling & Szczepaniak 2013), and many Russian compounds take the form of the example (23) *klin-o-obraznyj* 'wedge-shaped' cited above. The crucial difference is that the Latvian 'combining form' is always identical to a genitive case form of the non-head noun (singular or plural). This means that the grammar has to select

[12] See the contributions to Trips & Kornfilt (2017) for discussion of phrasal compounding and specifically Pafel (2017) for interesting typological observations.

a specific cell in the lexeme's paradigm, as opposed to a particular stem form or the root form.

The Hungarian infinitive construction is essentially homologous to the spontaneous agreement described for Nakh-Dagestanian languages and others, with the difference that the construction applies to all verbs and not just to a (sizeable) minority. The solutions proposed for spontaneous agreement will therefore presumably carry over to the Hungarian case, but without the lexical restrictions needed for lexically-governed spontaneous agreement.

5 Conclusions and queries

This paper has had two principal goals.

First, I have extended the typology of uninflectability, arguing for the four-way typology shown in Table 2. This establishes two parameters of uninflectability. A word can be either totally or partially uninflectable. A word can be uninflectable either as an inherent lexical property or the uninflectability can be induced by the syntactico-semantic context (contextual uninflectability) or actually introduced as part of a grammatical construction (constructional uninflectability). The literature attests only three of the four possibilities, but additional evidence from Hungarian and Latvian suggest that the full typology is viable.

As part of this conceptual endeavour I have asked just what inflectional features it is that a lexeme can be uninflectable for. The answer seems to be just the core inflectional features that are associated in that language with (more or less canonical) synthetic morphology.

Second, I have asked how uninflectability can be accommodated in a formal grammar. Taking as my point of departure the PFM2 model of Stump (2016) I have argued that we can treat uninflectability as a total or partial lack of a form/realized paradigm in a lexeme that nonetheless has a fully-fledged content paradigm. The fact that the uninflectable lexeme is associated with a content paradigm distinguishes the uninflectable lexemes from the common-or-garden uninflecting classes of lexemes, such as English prepositions. By defining lexical representations in the right way, however, we can make the principles of lexical insertion (the syntactic licensing of a concrete form of a lexeme) apply in the same way to uninflecting lexemes and uninflectable lexemes. By defining the FORM attribute of a lexeme as the output of inflectional morphology it is then also possible to account for the lexical insertion of ordinarily inflected word forms using the same machinery.

A number of questions arise from this study.

I have suggested, on the basis of admittedly limited evidence, that uninflectability only applies in the case of thorough-going synthetic inflection. To what extent is this actually true? And to the extent that it is true, does it provide a test bed for identifying the notion of *core synthetic inflection* in a given language? If so, this might have implications for the notion of paradigmatic organization and for the putative distinction between inflection and derivation.

Uninflectability is not an all-or-nothing, binary phenomenon. It remains to be seen, therefore, whether the account sketched here can accommodate all gradient cases of uninflectability in a convincing fashion.

The PFM2 model presupposes a very specific view of inflectional morphology, and the level of form paradigm, in a sense, is an artefact of that model. Other closely related models do not appeal to such a level, for instance, Network Morphology (Brown & Hippisley 2012) or the morph-based HPSG-oriented implementation of PFM2 in Information-based Morphology (Crysmann & Bonami 2016). Moreover, the notion of form paradigm itself raises conceptual questions: if the feature ensemble defining the content paradigm has to be interpreted by syntax/semantics then those features cannot, by definition, be morphomic (in the sense of Aronoff 1994 and many subsequent references, including Stump 2016). But if the form paradigm represents a level that is 'informationally encapsulated' from syntax/semantics and just defines formal, morphological relations, then it is odd to see that the features which define it in the canonical cases of linkage are precisely the non-morphomic content paradigm features (cf Boyé & Schalchli 2019, Spencer 2021). Future research should therefore be devoted to generalizing the account offered here so as to make it compatible with any model of inferential-realizational inflectional morphology.

Finally, we have seen that it is important to distinguish uninflectability from defectivity, but does the current proposed analysis of uninflectability have any (possibly unwelcome) implications for defectivity? A defective lexeme is one which lacks one or other forms from its paradigm, so that there is no way of expressing that particular cell's feature content for that lexeme. For instance, the Russian noun MEČTA 'dream' cannot be used in the genitive plural form, even though all Russian speakers know what that form 'ought' to be, namely, $^{*!}$*mečt* (where the annotation $^{*!}$ is intended to mean 'form which is expected, but which is unacceptable'). Stump (2010), examining the interaction of defectiveness with syncretism, suggests that defectiveness might be a property either of content paradigms, or of form paradigms. My approach to uninflectability suggests that it is better to treat defectivity as a property of the lexeme's content paradigm, as the safest way of ensuring that LEXIN has no member of the form paradigm/FORM attribute to fall back on. In other words, a defective lexeme is one with a defective morpholexical signature, which simply denies the existence of certain cells in the content paradigm, hence, a priori in the form/realized paradigm.

Abbreviations

ACC	accusative
Corr	Correspondence (function)
DAT	dative
DEP	Default Exponence Principle
F	feminine
GEN	genitive
HPSG	Head-driven Phrase Structure Grammar
IbM	Information-based Morphology
IFD	Identity Function Default
INF	infinitive
INS	instrumental
LFG	Lexical Functional Grammar
LI	Lexemic Index
LOC	locative
LVC	light verb construction
M	masculine
MWE	multiword expression
N	neuter
NOM	nominative
PF	Paradigm Function
PFM	Paradigm Function Morphology
PL	plural
pm	property mapping
PX	possessor
SBCG	Sign-Based Construction Grammar
SG	singular
TAM	tense-aspect-mood
UNINFL	uninflectable
VOC	vocative

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Chapter 3

The conditions of uninflectability in nouns in the Slavic languages

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In Slavic languages, major word classes like nouns, adjectives, and verbs typically inflect for various grammatical categories. However, some lexemes within these parts of speech are uninflected, meaning they do not display any grammatical markers. I first address how to define uninflectedness – and uninflectability, which is straightforward for nouns but more complex for adjectives and verbs. Next, I explore which lexeme classes are most commonly uninflected in languages such as Czech, Polish, Russian, Serbian, Slovak, and Slovene, focusing primarily on proper names, loanwords, and abbreviations, which are mostly nouns. I highlight that uninflectedness exists on a continuum, with different degrees of uninflectedness within each class, and this continuum is inversely related to other factors like numeral inflection. I argue that the lack of integration of certain nouns into the inflectional system is linked to specific morphological properties like stem ending, base form, and grammatical gender. These properties vary in restrictiveness across languages, with some (e.g., Russian) showing higher uninflectedness sc. uninflectability in nouns, while others (e.g., Slovene) are more permissive. Additionally, noun classes with feminine gender tend to be more restrictive in terms of stem structure and base form than those with masculine gender.

1 Introduction

In this article I will present some findings of my habilitation thesis (Doleschal 2000) where I investigated the phenomenon of uninflectedness and uninflectability in six Slavic languages – Russian, Polish, Slovak, Czech, Slovene and Serbian. The aim of this investigation was to find out how and why in highly inflecting

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languages, both single words such as the Russian noun *metro* as in *stancija metro* 'metro station', the Czech adjective *blond* in *blond vlasy* 'blond hair'or the numeral *dve* in Serbian as in *s dve devojke* 'with two girls' and entire classes of words as Russian and Polish nouns ending in the vowel /u/, e.g., *s kenguru* 'with (a) kangaroo' remain partly or even fully uninflected, and if eventually, interjections as *pryg* 'hop' in Russian might qualify for uninflectable verbs: *my pryg* 'we hopped'.

In the course of this research, other phenomena also related to uninflectedness emerged, most notably constructions with company names in compound-like structures, such as *Duracell* in *Duracell nagradna igra* (Slovene) 'Duracell prize draw', *Milka* in *50 Milka plišanih kravica* (Serbian) '50 Milka fluffy toy cows' or a historical growth of inflectedness as wittnessed by the (relatively) new inflectional class *kuli* in Czech, Slovak and Polish. These issues have been addressed in Doleschal (1999, 2008, 2015, 2017).

The structure of the present article is the following: Firstly, the Slavic languages will be characterised as inflecting languages (Section 2). Secondly, the different forms of the phenomenon which have been identified in the material will be described (Section 3). Finally, the ways in which uninflectability manifests itself in the six languages under investigation, and how this uninflectability is related to the inflectional classes in Slavic, will be examined, as will the research question about the conditions of uninflectedness (Section 4).

2 Slavic languages as inflecting languages

The Slavic languages are traditionally classified as highly inflecting and belonging to the fusional type (cf. e.g., Comrie & Corbett 1993: 6). This means that they signal grammatical meanings by endings which are attached to lexical morphemes, usually stems. Endings are characterised by cumulative exponence, i.e., one ending expresses more than one grammatical meaning (feature value). Inflection is different and characteristic for the different parts of speech.

The inflected parts of speech are: noun, pronoun, adjective, numeral, determiner, verb (including participle; Sussex & Cubberley 2006: 222), and – depending on one's theoretical premises – adverb (cf. Nagórko 2009: 743). Within the inflected parts of speech, word forms are organised in paradigms forming inflectional classes so that any lexeme of these parts of speech necessarily belongs to (usually) one such class (cf. Table 1). The grammatical (morphosyntactic and morphosemantic) categories involved in inflection are the following:[1] case, number,

[1]Cf. Spencer & Wiemer (2020) for a discussion of inflectional categories in Slavic languages.

gender, animacy, (gradation) in nouns and adjectives; person, number, gender, tense, mood, voice, (aspect), gerund, participle in verbs, determining the content paradigm of a lexeme that belongs to a certain part of speech. For example, the Polish nouns *pan* 'gentleman, mister' and *żona* 'woman' take different sets of inflectional endings for the same values of the categories case and number (Table 1). Furthermore, these two nouns are classified differently with regard to the category of gender – while *pan* has the feature masculine, *żona* displays feminine gender. Whether the gender feature is also expressed by the sets of endings along with the case and number features is controversial (cf. Doleschal 2009: 148).

Table 1: Inflectional classes (Polish): *pan* (M) 'gentleman, mister', *żona* (F) 'woman'

	SG	PL	SG	PL
N(OM)	pan-Ø	pan-owie	żon-a	żon-y
G(EN)	pan-a	pan-ów	żon-y	żon
D(AT)	pan-u	pan-om	żon-ie	żon-om
A(CC)	pan-a	pan-ów	żon-ę	żon-y
I(NSTR)	pan-em	pan-ami	żon-ą	żon-ami
L(OC)	pan-u	pan-ach	żon-ie	żon-ach
V(OC)	pan-ie	pan-owie	żon-o	żon-y

3 Types and manifestations of uninflectedness

3.1 General

In the general area of uninflectedness in inflected parts of speech, different manifestations of the phenomenon can be distinguished:

1. classes of words that never take any endings (Section 3.2)
2. single words that never take any endings (Section 3.2)
3. words that may or may not take endings and vary by syntactic context, register, sociolect, speaker (or freely) (Section 3.3).

3.2 Uninflectability (paradigmatic uninflectedness)

In the first place, there are words that never take any inflectional endings and are therefore paradigmatically uninflected, as the borrowings *miss* F. 'miss', e.g., in *miss Polonia* or *gnu* N., F., M. (in Russian, Polish, Slovak) 'gnu' for a type of antelope. Paradigmatic uninflectedness pertains to the lexical level. It is an inherent property of the lexeme. This manifestation of uninflectedness will be called "uninflectability".

Uninflectable lexemes can be subdivided into two groups:

1. In the first group uninflectability is regular, i.e., it can be described by rules based on narrowly linguistic (e.g., phonological or lexical) properties: in Russian, e.g., nouns ending in the vowel /u/ cannot be inflected: *tabu* 'taboo', *ragu* 'ragout', *kenguru* 'kangaroo', *Lulu* 'Loulou', *Subaru* 'Subaru'. This case shall be called *systematic uninflectability*.

2. The second group contains words whose uninflectability cannot be explained by properties of the language system, as in the case of some toponyms in Slovak ending in the vowel /e/ as *Skopje* or in a so-called soft consonant[2] as *Providence* (/s/) while *Opole*, *Bystrice* (both ending in /e/), *Provence* (ending phonologically in /s/) with the same type of phonological ending are inflectable. This case shall be called *unsystematic uninflectability*.

On this basis we can now define what exactly will be understood by *uninflectability*:

A word is called uninflectable,[3] if

1. it belongs to an inflectable part of speech, and
2. does not take any endings normally found with words of that kind, but
3. may nevertheless appear in any syntactic context typical for the part of speech in question.

[2]In the Slavic languages, consonants underwent several palatalisations, leading to correlations between soft (palatal or palatalised) and hard (non-palatal) consonants. This difference is widely morphologised, so that "hard" stems and "soft" stems select different sets of endings which are not conditioned by phonology. Moreover, there are "neutral" sc. "unpaired" consonants which do not take part in the correlation (cf. Comrie & Corbett 1993: 6, 9–10).

[3]Cf. also Krzyżanowski (2013: 24–35) for a thorough terminological discussion.

Condition 1 is necessary to distinguish the words in question from uninflected parts of speech, such as conjunctions and prepositions, in the Slavic languages. Condition 2 distinguishes uninflectable words from semi-inflectable ones as, e.g., Polish *muzeum* 'museum' (Table 2), which is uninflected in the singular but regularly inflected in the plural, and from contextual uninflectedness (see below Section 3.3).

Table 2: Semi-inflectable nouns in Polish

	SG	PL
N	muzeum	muze-a
G	muzeum	muze-ów
D	muzeum	muze-om
A	muzeum	muze-a
I	muzeum	muze-ami
L	muzeum	muze-ach
V	muzeum	muze-a

Condition 3 is necessary to distinguish between uninflectability proper and defectivity (cf. Doleschal 2000, 2002, Spencer 2020): for example, the Russian indefinite numeral *malo* 'a little, few' has only one form and appears solely in Nominative and Accusative Singular expressions (1):

(1) a. Tam malo detej.
there few.NOM children.GEN
'There are few children there.'

b. Ja znaju malo detej.
I know few.ACC children.GEN
'I know few children.'

c. *Ona interesuetsja malo detej.
she is-interested-in few.INSTR children.GEN
'She is interested in few children.'

It is thus also uninflectable, but not in the same way as are nouns as *kenguru* 'kangaroo' which as such can express all syntactic cases and both singular and plural, e.g. (2):

(2) a. Tam molod-oj kenguru.
there young-M.NOM.SG kangaroo.NOM.SG
'There is a young kangaroo there.'

b. Ja tam vižu kenguru.
I there see kangaroo.ACC.SG/PL
'I see a kangaroo there.' / 'I see kangaroos there.'

c. Ona interesuetsja kenguru.
she is-interested-in kangaroo.INSTR.SG/PL
'She is interested in a kangaroo / kangaroos.'

This definition can be exemplified with two female names taken from Czech – *Jana* and *Mimi* (cf. Table 3). As proper names, both are clearly nouns and thus fulfil condition 1. Secondly, it can be seen from Table 3 that *Mimi* does not take any case endings but occurs with the same case meanings as the inflected lexeme *Jana*, i.e., it has the same content paradigm as *Jana*, and thus fulfills condition 2. It can therefore appear in all necessary syntactic contexts as is illustrated by the use of prepositions in the second column and so also fulfils condition 3.

Table 3: Inflectable and uninflectable name in Czech

N		Jan-a	Mimi
G	(bez)	Jan-y	Mimi
D	(k)	Jan-ě	Mimi
A	(pro)	Jan-u	Mimi
L	(při)	Jan-ě	Mimi
I	(s)	Jan-ou	Mimi
V		Jan-o	Mimi

Condition 3 also implies that an uninflectable word possesses and represents all morphological categories and their grammatical meanings (feature values) of the part of speech in question. But how large can such a paradigm be? Would it also have to cover otherwise periphrastic morphology? Does it have to cover word-class changing morphology?

The Russian verb paradigm has up to 124 forms (e.g., the verb *ljubit'* 'to love'), including the four participles with their paradigms and the two gerunds (regarding aspect as a lexical feature, cf. Genis 2021). Does an uninflectable item have to substitute for all of these forms in order to be called uninflectable and not defective? For my material, I have come to the following conclusion both on an empirical and a theoretical basis: an uninflectable lexeme must be able to independently represent all synthetic forms of the paradigm of a part of speech in question. For the nominal parts of speech this means all the case-number forms in the noun and case-number-gender forms and adverb in the adjective, and case forms in the numeral as well as case-gender forms for some numerals, but not the periphrastic forms, such as periphrastic gradation.

This is mainly due to the fact that words that are usually perceived as uninflectable can combine with periphrastic markers such as Czech *více* in *více blond* 'more blond' for the comparative or as Russian *pryg by* 'would hop' for the conditional. From the theoretical point of view, it has been argued (cf. e.g., Haspelmath 2000) that obligatory preposition-noun combinations, such as in the locative in the Slavic languages, are usually seen as phrases although they could just as well be analysed as periphrastic case forms. However, we do not expect an uninflectable noun in the (syntactic) locative to co-signal a preposition. Another viewpoint could be that periphrasis is on the borderline between syntax and inflection (Brown et al. 2012), hence not a prototypical inflectional marker and could therefore be neglected. As to the word-class changing forms (adverb, participle), Spencer (2020) has argued for excluding them from the paradigm, and I will follow him here. Under these premises, in Slovene an uninflectable adjective (as *super*) would substitute for a paradigm as presented in Table 4.

Since gradation can be periphrastic in all Slavic languages, this category need not be expressed by an uninflectable adjective by itself. In Slovene, adjectives can form synthetic forms of comparative and superlative as *nov-ejš-i* 'new-GRADATION-M.SG' 'newer' and *naj-nov-ejš-i* 'SUP-new-GRADATION-M.SG' 'newest' along with a periphrastic comparative and superlative, which is then also applied to uninflectable adjectives (Table 5).

3.3 Contextual (syntagmatic) uninflectedness

Having defined uninflectability, let us now turn to a closely related phenomenon which is often treated together with uninflectability under the same heading ("nesklonjaemost", "neohebnost") but is distinct from the first: contextual uninflectedness.[4] Contextual uninflectedness means that a word, which is fully inflectable on the lexical level, remains uninflected for contextual reasons, such as text type, style, sociolect and last but not least – syntactic context, e.g., apposition or use of a classifier as in *v naselj-u Njivice* (in village-LOC.SG Njivice 'in the village

[4] In Doleschal (2000) "syntagmatic" was used throughout. However, as one of the reviewers pointed out, this term is too narrow. Therefore, the phenomenon is called *contextual uninflectedness* here.

Table 4: Inflectable and uninflectable adjective in Slovene

	‘new’			‘super’		
	M	F	N	M	F	N
			SG			
N	nov-Ø/nov-i*	nov-a	nov-o	super	super	super
G	nov-ega	nov-e	nov-ega	super	super	super
D	nov-emu	nov-i	nov-emu	super	super	super
A	nov-Ø/nov-ega†	nov-o	nov-o	super	super	super
L	nov-em	nov-i	nov-em	super	super	super
I	nov-im	nov-o	nov-im	super	super	super
			DU			
N	nov-a	nov-i	nov-i	super	super	super
G	nov-ih	nov-ih	nov-ih	super	super	super
D	nov-im	nov-im	nov-im	super	super	super
A	nov-a	nov-i	nov-i	super	super	super
L	nov-ih	nov-ih	nov-ih	super	super	super
I	nov-imi	nov-imi	nov-imi	super	super	super
			PL			
N	nov-i	nov-e	nov-a	super	super	super
G	nov-ih	nov-ih	nov-ih	super	super	super
D	nov-im	nov-im	nov-im	super	super	super
A	nov-e	nov-e	nov-a	super	super	super
L	nov-ih	nov-ih	nov-ih	super	super	super
I	nov-imi	nov-imi	nov-imi	super	super	super

*Indefinite/definite form (cf. Greenberg 2008: 36).
†Inanimate/animate gender.

Njivice’) in Slovene or *w województw-ie Białystok* (in province-LOC.SG Białystok ‘in the province Białystok’). However, such uninflectedness is usually not obligatory, e.g., in Polish, masculine surnames ending in /e/ or /i/ are inflectable, but especially in collocation with a first name or a title often remain uninflected, e.g., *Maksimilian-a Kolbe* Maksimilian-GEN.SG Kolbe [GEN.SG] instead of *Maksimilian-a Kolb-ego* Maksimilian-GEN.SG Kolbe-GEN.SG (cf., e.g., Krzyżanowski 2013: 39–40). In Slovene this happens with feminine surnames in /a/, e.g., *z Ad-o Muha*

Table 5: Gradation of adjectives in Slovene

	Positive	Comparative	Superlative
synthetic	nov-Ø, -a, -o	nov-ejš-i, -a, -e	naj-nov-ejš-i, -a, -e
periphrastic	nov-Ø, -a, -o	bolj nov-i, -a, -o	najbolj nov-i, -a, -o
	super	bolj super	najbolj super

'with Ada-INSTR.SG Muha [INSTR.SG]' 'with Ada Muha' instead of *z Ad-o Muh-o* 'with Ada-INSTR.SG Muha-INSTR.SG'. In Serbian, male surnames can remain uninflected in collocation with first names, if they precede the first name: *Petrović Ivan-a* Petrović Ivan-GEN.SG 'Petrović Ivan's'.

On the other hand, in Czech abbreviations as *NATO* or *FIFA* remain uninflected in official language, but are inflected in casual speech (*Internetová jazyková příručka* 2008-2025). Another case in point are Russian place names formed with the suffixes *-ovo* and *-ino* such as *Perovo*, *Puškino* which are in principle inflectable and should be inflected according to the norms of Contemporary Standard Russian (cf. Paxomov 2024, Redakcija portala «Gramota.ru» 2024; Zaliznjak 2008: 739; Ageenko 2010: 35). Nevertheless, they often remain uninflected both in spoken and written language and seem to show a tendency towards uninflectability (Graudina 1980: 156–159; Panov 1968: 56–59; Zaliznjak 2008: 739, but cf. Tejkin 2021 for an opposite view).

One of the interesting issues is if and how contextual and paradigmatic uninflectedness relate to each other, i.e., is contextual uninflectedness a preliminary stage of uninflectability? This may be the case as with the aforementioned Russian place names, but it is certainly not the case, when an uninflected noun occurs together with an appositive classifier in Slovene or in Serbian. This problem needs an investigation of the relationship between the effects of register and style and language change caused by both sociolinguistic and inherently morphologocial factors, such as productivity of inflectional classes, and has to be left for further exploration.

3.4 Geographical distribution of uninflectability in six Slavic languages

As a result of the study by Doleschal (2000) the manifestation of uninflectability in the six Slavic languages under investigation can be characterised as follows: Uninflectability occurs regularly, but in different ways, with the parts of speech *noun*, *adjective*, *(cardinal) numeral*. The degree of uninflectability manifests itself as a kind of continuum with two trajectories for different parts of speech: one from west to east and the other from south to north (cf. Table 6). This means that for nouns the highest degree of uninflectability is found in Russian and the lowest in Slovene, whereas for both adjectives and numerals the highest degree is found in Serbian and the lowest (if any) in Russian.

Table 6: Characterisation of uninflectability in six Slavic languages

	Noun				
N	Polish			Russian	N
↑	Czech			Slovak	↑
S	Slovene			Serbian	S
	W	→	→	E	

	Numeral	(only cardinal numeral)			
N	Polish			Russian	N
↓	Czech			Slovak	↓
S	Slovene			Serbian	S
	W	→	→	E	

	Adjective				
N	Polish			Russian	N
↓	Czech			Slovak	↓
S	Slovene			Serbian	S
	W	→	→	E	

In Table 6, dark slots indicate a high degree of uninflectability, while light slots mean a low degree. A white slot indicates total lack of uninflectability for a certain part of speech, as is, e.g., the case for cardinal numerals in Russian, Polish and Czech where all members of this class display overt inflectional forms. In Serbian, on the other hand, numerals from 'five' to '999' are all uninflectable. A black slot thus means that a considerable number of lexical items is affected by uninflectability.

The shading does not correspond to absolute numbers, but rather to a qualitative criterion: how many lexical classes are affected? How many types of words are affected by non-assignment and how regular is this? In Russian, all classes of names, foreign words and abbreviations are affected by uninflectability. In addition, nouns with any phonological ending and any gender are affected, but systematically those ending in /e/, /i/ and /u/. Most varieties of abbreviations are also uninflectable. Thus, uninflectability of nouns also shows a high degree of regularity or systematicity. This is why the cell for the noun in Table 6 is shaded black for Russian. In Slovene, on the other hand, names or abbreviations are not affected by uninflectability at all. Moreover, there is no phonological ending that would preclude nouns beforehand from being inflected. There are some uninflectable nouns in Slovene, but they cannot be described by rules. Therefore, the corresponding cell is shaded in light grey.

The conditions for this uninflectability are different for the different parts of speech: In numerals we find a diachronic reduction of inflection over the past 100-150 years (Suprun 1969; Doleschal 2000: 297–313). In this case we are dealing with contextual uninflectedness as a step towards uninflectability. Uninflectable adjectives, on the other hand, are almost always borrowings (cf. Doleschal 2000: 277–297). Therefore, the reason for uninflectability in adjectives can be found in their lack of integration and their foreignness. In the case of nouns, the situation is more complicated and will therefore be analysed in detail in the following section.

4 Uninflectability in nouns

4.1 Introductory considerations

Uninflectable nouns can be divided into three lexically and morphologically defined classes, which overlap logically to some extent: *proper nouns*, *abbreviations* and *borrowings*. These groups may intersect in the following way: many of the uninflectable proper nouns are at the same time borrowings or foreign names, e.g., the (Italian) male name *Mario*, or the (German) female name *Sigrid* (in Russian, Polish, Czech, Slovak), the toponyms *Oslo*, *Cetinje* (in Russian), and chrematonyms[5] as *Subaru* (in Russian, Polish, Czech, Slovak). It is very seldom the case that genuinely native names are uninflectable, as, e.g., the Russian male first name *Sadko*[6] or Russian surnames as *Dolgix*, *Živago* in Russian and the Czech type of surnames ending in /u:/ (graphic <ů>, such as *Janů*, *Petrů*) in Slovak.[7] Abbreviations may of course also be borrowings and/or names (e.g., of organisations, such as *IBM*), but they may also be native, such as *SNG* 'CIS' (Russian),

[5]Also called pragmatonyms, see Zgusta (1996).

[6]A mythical hero of the bylinas, cf. the opera by Rimsky-Korsakov.

[7]Both the Russian and the Czech surnames of these types are frozen Genitive forms. In Czech such surnames may be inflected.

ORMO (Ochotnicza Rezerwa Milicji Obywatelskiej 'Volunteer Reserve of the Citizens' Militia') (Polish) or compounds as *kompolka* 'commander of a regiment' (Russian). The class of borrowings includes (besides names) all borrowed common nouns that are not morphologically integrated for various reasons, e.g. *alibi*, *tabu* (Russian, Polish, Czech, Slovak), *miss* 'miss', *ledi* 'lady' (all investigated languages).

Apart from these, there are small subgroups of native uninflectable common nouns in some languages, e.g., female designations of persons (*mami* 'mummy' in Slovak and Slovene, *professor* 'woman-professor' etc. in Polish),[8] extragrammatical[9] formations, such as *cucu* (Slovak) or *ciuciu* (Polish) 'candy', *psipsi* (Polish) 'wee-wee' or the designations of letters and musical notes (*be*, *ce*, *de* 'b, c, d'; *do* "C", *re* "D", *mi* "E" in all investigated languages besides Slovene). Other native common nouns are very rarely uninflectable (cf. e.g., Krzyżanowski 2013: 18).

In the literature on uninflectability various explanations have been brought forward for motivating the uninflectability of nouns. The main reason is that they do not fit the requirements of inflection, for phonological as well as morphological reasons (cf., e.g., Krzyżanowski 2013: 37). Thus, Isačenko (1954: 225) justifies the uninflectability of borrowings on /o/ in Russian, which in the majority of cases bear final stress, with the lack of a morphonological model among Russian nouns (the accent type *okno* /oˈkno/ 'window' comprises only a few Russian native words and is not productive), and Worth (1966) attributes uninflectability in Russian to a vocalic stem ending, as would be the case in *bo-a* 'boa'. Krzyżanowski (2013: 30) mentions a word-final stressed vowel as one cause of uninflectability of abbreviations in Polish, such as *EKG* [ɛkaˈgɛ] 'electrocardiogram'.

A notorious reason is also the mismatch between gender and phonological shape of a noun, as with *miss* 'miss', *madam* 'madam' in Russian (Timberlake 2004: 148). Another reason given is the recognisability of names, e.g., surnames that are homonymous with common nouns, such as Polish *Lato*, *Musik* (Krzyżanowski 2013) which is especially important in official style. This reason is also mentioned as a motivation for not inflecting Russian place names as (neuter) *Puškino*, which could be confused with the paronymous (masculine) place name *Puškin* due to identical endings in the oblique cases (cf. Tejkin 2021 for a discussion of this argument). Finally, foreignness is presented as a key factor for uninflectability, cf. the discussion by Krzyżanowski (2013: 35–42), also Tejkin (2021: 1075–1076).

[8] The Polish cases are, however, defective, in that they cannot be used in the plural **panie professor* 'Ms:N.PL professor'.

[9] See Dressler (2000) for a discussion of this term.

It is also put forward that some nouns are assigned more easily or regularly to inflection than others. The latter either remain uninflectable for a span of time or vary between inflection and uninflectedness. This is, e.g., the case with borrowed common nouns and names ending in /o/ as well as with male names ending in /e/ or /i/ in Polish (cf. Krzyżanowski 2013: 20–21).

But what does the fact that certain nouns do not (easily) fit into inflection imply for linguistic explanation? In this regard, two points can be made which will be explored in the next section:

1. Systematic uninflectability demonstrates the limits and thereby the inner structure of the inflectional system of a language. This is to say that in the case of systematic uninflectability there is a mismatch between some property of the noun in question and the rules conditioning inflection in a specific language.

2. Unsystematic uninflectability is most often a phenomenon of prescriptive norms, but it may also reveal tendencies within inflection.

4.2 Uninflectability and inflectional classes

We shall commence with an examination of the phenomenon of unsystematic uninflectability. In Russian, for instance, French names ending in stressed /a/ are uninflectable (as *Nikol'a* [nʲɪkʌˈla] "Nicolas", *Dega* [Dɛˈga] "Degas". This is not true for names in stressed /a/ from other languages (e.g., *Abdulla* [ʌbdulˈla] "Abdullah"). It is typical for unsystematic uninflectability that such a norm may change, e.g., *Marseille* (ending in the soft consonant /j/) was uninflectable in Slovak according to the norm in 1998 and is now codified as both inflectable and uninflectable (cf. Považaj 1998, 2013). Unsystematic uninflectability may therefore be indicative of a transitory state of a (set of) noun(s) towards inflection.

In other cases, unsystematic uninflectability may be an indicator of a tendency, as with nouns ending in /o/ in Polish, where uninflectability is gaining ground both following and counter to prescriptive norms (cf. Doleschal 2000: 56, 68–70, 98–100, 113, 126, 138 for discussion and references). E.g., male surnames in /o/ should be generally inflectable, but in dictionaries many foreign names are marked as uninflectable, such as *Eco*, *Franco*. Other names in /o/, too, often remain uninflected in usage (cf. Doleschal 2000: 68–70; Krzyżanowski 2013: 21, 41). Borrowed common nouns in /o/ with neuter gender, on the other hand, are uninflectable at first, but usually come to be inflected over time, e.g., *agio* 'agio' already in the 19th century, *auto* 'car', *radio* 'radio', *kino* 'cinema' in the period before the Second World War, and more recently *molo* 'pier', *kakao* 'coacoa', which are classified as variably inflected and uninflected (cf. Doleschal 2000: 126; Żmigrodzki 2004-2021; Woliński et al. 2020). The integration of neuters in /o/ into a corresponding inflectional class is therefore not automatic, but slower and not always complete (see Dressler et al. 1997: 102; Orzechowska 1974: 112). This points to a development towards systematic uninflectability for nouns ending in /o/ in Polish.

Let us now discuss systematic uninflectability as an indicator of the limits of the inflectional system. Systematic uninflectability means that a certain type of noun cannot be assigned to any inflectional class of a given language and thus shows what the requirements of inflectional classes are.

The fundamental question is: what characteristics must a word possess, in order to be integrated into the inflectional system of a language, i.e. to be assigned to an inflectional class? One important factor in the Slavic languages is the phonological ending. As previously stated, nouns ending in the vowel /u/ cannot be assigned to any inflectional class in Russian or Polish. This implies that the Russian and Polish inflectional classes do not permit a base form in /u/. Consequently, inflectional classes can presuppose a certain base form type, which is a conditio sine qua non for the assignment of a word to them. This type of property of an inflectional class will be referred to as a *constitutive property* (cf. Doleschal 1997, 2006).

Usually native words correspond to the requirements of the inflectional system, because lexicon and regular word formation of a given language provide words with a permissible base form. Therefore, it is often foreign words that are affected by uninflectability, although this is not the only case: relict forms created with synchronously unproductive means of word formation can also be affected, e.g., the (archaic) Russian first names *Mixajlo*, *Danilo* (cf. Zaliznjak 2008: 739). Therefore, the inflectional system may also change in a way that leaves native words outside. Therefore, uninflectability enables us to identify, firstly, the constitutive properties associated with inflectional classes and, secondly, the degree of productivity of the latter (cf. Wurzel 1984, Dressler 1997, 2003).

In the literature on Slavic languages, the integration of a noun into an inflectional class is usually related to gender and to the phonological ending of the base form as well as to the ending of the stem (Hentschel & Menzel 2009). In the case of the present Slavic languages, four properties have turned out to characterise inflectional classes of nouns: *base form type*, *stem type*, *gender* and *semantic animacy*.

As an example, let us consider first names. First names fall into two groups: male and female, which behave differently as to uninflectability. In the Slavic languages first names are characterised by the following semantic and morpho(no)logical properties: they are inherently animate and belong to either the masculine or the feminine gender, where a canonical male first name belongs to the masculine gender and a canonical female name to the feminine gender. Depending on the respective language, their base forms may either end in a vowel, which in Czech, Serbian, Slovak and Slovene may be phonologically short or long (while in Russian and Polish vowel length is not distinctive), or they may end in a consonant. Their stems may end in a vowel, a hard consonant, a soft consonant or a neutral (Czech, Slovak) sc. unpaired (Russian) consonant (cf. footnote 2).

It has now to be made clear, how a first name is assigned to an inflectional class, e.g., a borrowed name, such as the male name *Pedro*. We can think of this assignment as a process of "feature checking": First animacy and gender are established – in the case of *Pedro*, masculine animate gender. Then the phonological ending of the name is matched with base forms of existing inflectional classes that contain masculine nouns. If one or several such classes exist, it has to be checked if the final vowel /o/ should be treated as part of the stem or as the Nominative ending for this particular class. If the latter is the case, the word form is analysed as *Pedr-o* 'STEM-ending' and it has to be checked in a final step if the resulting stem conforms to the stem-structure of the chosen inflectional class. If it does, the word becomes a member of this class. If it does not, it remains uninflectable, unless it is adapted phonologically or morphologically, e.g., by changing the word ending and/or the stem ending.

E.g., if the name *Pedro* enters the Russian language, it is assigned to masculine-animate gender (which means that it selects a specific class of agreement targets). Afterwards it is checked, if there is an inflectional class with a base form ending in /o/ that allows for nouns with masculine-animate gender. It turns out that the only class with nouns ending in /o/ is reserved for neuter gender (*okno* N. 'window'), whereas the classes allowing for masculine-animate nouns require a base form ending in a consonant (*Vladimir*) or in /a/ (*Nikita*). Therefore, *Pedro* cannot be assigned to any inflectional class and remains uninflectable in Russian.

If we take Slovene on the opposite end of the noun continuum in Table 6, the same process yields a different result: In Slovene, there is one class specialized for masculine-animate nouns which at the same time allows for a base form in any vowel or consonant. In order to assign *Pedro* to this class, the status of its stem has to be checked. The declension in question allows both for stems ending in a consonant and in a vowel. Since the final /o/ in *Pedro* is unstressed, it is analysed as a NOM.SG ending and hence the noun is inflected, like the native name *Marko*, as *Pedr-o* (NOM), *Pedr-a* (GEN), *Pedr-u* (DAT), *Pedr-a* (ACC), *(o) Pedr-u* (LOC), *(s) Pedr-om* (INSTR).

Tables 7 and 8 show the degree of uninflectability in first names in the six languages under investigation. This degree is again marked by different colourings of the cells – a darker colouring indicates a higher degree of uninflectability. A plus sign means that there are cases of systematic uninflectability for the base form ending in question, while a plus in brackets indicates cases of unsystematic uninflectability for this cell. White colouring combined with a minus sign means total lack of uninflectability. Shaded cells with question marks indicate that such cases could not be verified and probably do not exist.

As can be deduced from Tables 7 and 8, uninflectability is much more widespread with female first names in all Slavic languages under investigation.[10] As to male first names, only Russian and Polish display cases of uninflectability, while in Slovak, Czech, Slovene and Serbian all male first names can be assigned to an inflectional class, regardless of their phonological make-up.

Table 7: Uninflectability of male first names

	base form ending in	C#	a	a:	o	o:	e	e:	i	i:	u	u:
male first names	Russian	-	+		+		+		+		+	
	Polish	-	+		(+)		(+)		(+)		+	
	Slovak	-	-	-	-	-	-	-	-	-	-	-
	Czech	-	-	-	-	-	-	-	-	-	-	-
	Slovene	-	-	-	-	-	-	-	-	-	-	-
	Serbian	-	-	-	-	-	-	-	-	-	-	-

We will now concentrate on Russian first names and compare them to Slovene first names.

Russian inflectional classes are traditionally divided into three macroclasses[11] according to common sets of endings (cf. for Russian, e.g., Timberlake 2004: 130).

[10] In Polish and Russian vowel length is not distinctive, therefore the cells for the respective vowels are coalesced.

[11] According to Dressler (1997, 2003) a class is a set of similar paradigms that can be subdivided into subclasses and ultimately into "microclasses which share exactly the same morphological generalizations. (…) An inflectional macroclass is the highest, most general type of class, which comprises several (sub)classes (…) Prototypically, its nucleus is a productive microclass and it has at least two microclasses. The interior coherence of a macroclass, in terms of shared properties, must be higher than affinities between microclasses of different macroclasses" (Dressler 2003: 35–36).

Table 8: Uninflectability of female first names

	base form ending in	C#	a	a:	o	o:	e	e:	i	i:	u	u:
female first names	Russian	+	+		+		+		+		+	
	Polish	+	-		+		+		+		+	
	Slovak	+	-	?	+	+	+	+	+	+	+	+
	Czech	+	-	?	(+)	(+)	(+)	(+)	+	-	+	+
	Slovene	(+)	-	?	(+)	(+)	(+)	(+)	+	+	+	+
	Serbian	(+)	-	?	(+)	(+)	(+)	(+)	+	+	+	+

The 1st macroclass displays (among others) a class containing nouns with masculine animate gender and a base form ending in a consonant, which is divided further into subclasses according to the stem ending (identical to the base form ending) and animacy. Typical Russian male names belonging to this class are, e.g., *Boris* (ending in a hard consonant), *Igor'* (ending in a soft consonant), *Jurij* (ending in an unpaired consonant). The 2nd macroclass, on the other hand, requires a base form ending in /a/ and a stem ending in a consonant and can again be divided into subclasses depending on animacy. Animate nouns in this class may have feminine or masculine gender. Typical Russian female names here are, e.g., *Anna* (stem ending in a hard consonant), *Marija* (stem ending in an unpaired consonant), (hypocoristic) *Anja* (stem ending in a soft consonant), typical male names are *Nikita* (stem ending in a hard consonant) and (hypocoristic) *Kolja* (stem ending in a soft consonant). The 3rd macroclass has a subclass with feminine animate gender and a base form (as well as stem) ending in a soft consonant which contains also traditional Russian female names, e.g., *Ljubov'*, *Raxil'* "Rachel".

Let us first look at male names. As can be seen from Table 7, in Russian male names ending in a vowel are affected by systematic uninflectability. But only in case of the vowels /e/, /i/ and /u/ is this correlation 100%, i.e., it is the base form ending alone that conditions uninflectability. In other words, masculine-animate proper nouns with a base form ending in /e/, /i/ or /u/, do not conform to the conditions of any inflectional class in Russian. Therefore, foreign names as *Arne, Teddi, Lu* are uninflectable:

(3) Ja vstretila Arne/Teddi/Lu.
I met:F Arne.ACC/Teddy.ACC/Lou.ACC
'I met Arne/Teddy/Lou.'

With names ending in /o/ and /a/, other factors come into play, as can be seen in Table 9. Names ending in /o/ are uninflectable, if the /o/ is stressed (*Sadko*):

(4) Ja vstretila Sadko.
I met:F Sadko.ACC
'I met Sadko.'

Nevertheless, some traditional Slavic names ending in unstressed /o/ (*Mixajlo*)[12] may be inflected according to the 2nd inflectional macroclass:

(5) Ja vstretila Mixajl-u.
I met:F Mixajlo-ACC
'I met Mixajlo.'

This fact is reflected in Table 9 by putting this base form in brackets. (Unstressed /o/ is pronounced like unstressed /a/ as [ə], which may partly motivate this class assignment).

In male names ending in /a/, uninflectability is conditioned by the stem ending. The corresponding nouns are assigned to the 2nd declension, if /a/ is analysed as the Nominative suffix -a, as in *Nikit-a*):

(6) Ja vstretila Nikit-u.
I met:F Nikita-ACC
'I met Nikita.'

But membership in this class is allowed only if the remaining stem ends in a consonant. Therefore, a name such as (French) *Fransua* "François" whose stem would end in /u/ (*Fransu-a) cannot be inflected:

(7) Ja vstretila Fransua.
I met:F François.ACC
'I met François.'

But (Japanese) *Akira* is inflectable, since its stem ends in the consonant /r/ (Akir-a):

[12] This form of the name "Michael" (nowadays *Mixail*) was the original given name of the great Russian scholar Mixail Lomonosov. Such forms (as also *Danilo*, *Ivanko*) are obsolete as given names today. But they are still presented in dictionaries (Petrovskij 2024) and characterised as inflectable according to the 2nd declension which normally requires a base form ending in /a/, see Zaliznjak (2008: 739, also on *Sadko*).

(8) Ja vstretila Akir-u.
I met:F Akira-ACC
'I met Akira.'

Hence in Russian only male names ending in a consonant can be integrated into inflectional classes unconditionally, because in a concrete noun, all four constitutive properties – base form type, stem type, gender, subgender – have to have the fitting value. This implies that the constitutive properties of the Russian nominal inflectional classes are applied in a restrictive way and thus motivates the high degree of systematic uninflectability in male names.

Table 9: Constitutive properties of Russian masculine inflectional classes[13]

Class	Base form	Gender	Stem type	Inflectable example	Uninflectable example
1. hard	C-Ø	M-ANIM.	hard C	Boris-Ø, Bob-Ø	Pedro, Arne, Lu "Lou"
1. soft	C-Ø	M-ANIM.	soft C	Igor'-Ø	Mikele "Michele", Teddi "Teddy"
1. unpaired	C-Ø	M-ANIM.	unpaired C	Jurij-Ø	Mat'je "Mathieu", Luidži "Luigi"
2. hard	-a	M-ANIM.	hard C	Nikit-a	Fransua "François"
	(-o)			(Mixajl-o)	Sadko (stressed /o/)
2. soft	-a	M-ANIM.	soft C	Kolj-a	Nikolja "Nicolas" (stressed /a/ and French origin)

The inflectional classes of Slovene are essentially analogous to those of Russian. Consequently, we will adhere to the numerical classification previously outlined.[14] The 1st macroclass displays again a class for nouns with masculine-animate gender. The 2nd macroclass is characterised by a base form in /a/ and includes one class comprising feminine and masculine-animate nouns. (The 3rd

[13]In Russian /j/ belongs to the "unpaired consonants".

[14]Note that Slovene grammaticography defines macroclasses on the basis of gender, cf. Herrity (2000: 40–61); Greenberg (2008: 28–31).

macroclass is reserved for feminine gender nouns, and is not productive in integrating first names).

With regard to male first names, these can belong to either the 1^{st} or the 2^{nd} macroclass (Table 10). In the 1^{st} class, the base form of native nouns typically ends in a consonant (e.g., *Borut, Matjaž* “Matthew”), in the 2^{nd}, it typically ends in /a/ (as in *Miha*[15] “Michael”). Stems typically end in a consonant, as in Russian. However, in contrast to Russian, these inflectional classes permit a wide range of base forms and also allow for vocalic stems, as evidenced by the Latin name *Leo*, where the final, unstressed /o/ is analysed as an ending and the noun inflected accordingly as *Le-o* (NOM), *Le-a* (GEN), *Le-u* (DAT), *Le-a* (ACC), *(o) Le-u* (LOC), *(s) Le-om* (INSTR). If the name ends in a stressed vowel or in /e/, /i/, /u/, the stem is automatically augmented by /j/[16] (or sometimes /t/) in the oblique cases, thereby creating a consonantal stem ending. This can be observed in the examples of *Teddy-Ø, Teddyj-a* (GEN), *Lou-Ø, Louj-a* (GEN), *Tone-Ø, Tonet-a* (GEN) “Tony”. Consequently, all male names are automatically assigned to an inflectional class. Therefore there are no uninflectable male names in Slovene.

We will now consider female first names. As previously noted, female names are significantly affected by uninflectability, as illustrated in Table 8. In Russian, all female names ending in a vowel other than /a/ are systematically uninflectable, e.g., *Renate, Mimi, Ildiko, Lulu.* Furthermore, names with a base form in a consonant are also characterised by a high degree of uninflectability, e.g., *Èdit* (German) “Edith”, *Kler* “Claire”, and even certain names with a base form in /a/ may be uninflectable, such as *Klaudia* “Claudia”, *Rea* “Rhea” (with two coda vowels).

In Slovene, uninflectability is also found in female first names, but systematically only with names ending in /i/ and /u/, such as *Mimi*, *Lulu.* Similarly, names with a base form ending in the vowels /e/ and /o/ as well as in a consonant are also affected by uninflectability, although not in a systematic manner: a corpus search for *Renate Götschl*[17] in the Slovene corpus *Gigafida* yields 675 tokens for the wordform *Renate*, and one token for the inflected form *Renat-i* (DAT) (excluding items that may be related to a base form *Renata*). The names *Isabel* or *Madeleine* usually appear in an uninflected form: A total of 3,123 tokens of the form *Madeleine* are observed, along with one instance of the form *Madelein-o* (ACC) and two instances of the form *Madelein-i* (DAT). The same is true, mutatis mutandis, for *Isabel*: there are 1,926 tokens of *Isabel* and two of *Isabel-i* (DAT) (where the relation to this base form, rather than an adapted *Isabela* is undisputable). Names ending

[15] Male names in unstressed /a/ can inflect either according to the 2^{nd} or the 1^{st} macroclass.

[16] In Slovene /j/ belongs to the “soft consonants”.

[17] A well-known Austrian athlete (1990’s-2000’s).

Table 10: Constitutive properties of Slovene masculine inflectional classes

Class	Base form	Gender	Stem type	Inflectable	Uninfl.
1. hard	C-Ø, -o, -a, -e (unstressed)	M-ANIM.	hard C or V	Borut-Ø, Mark-o, Mih-a Bob-Ø, Pedr-o, Le-o, Akir-a	-
1. soft	C-Ø V-Ø	M-ANIM.	soft C (stem-augmentation after vowel)	Matjaž-Ø, Franz-Ø Nicolas-Ø (stressed /a/), François-Ø (stressed /a/) Arne-Ø, Teddy-Ø, Lou-Ø	-
2.	-a	M-ANIM.	C	Mih-a, Akir-a	-

in consonant or /e/ are therefore classified as unsystematically uninflectable. In contrast, names ending in /a/ are unconditionally inflectable and go into the 2nd (macro)class.

Let us again turn our attention to the inflectional classes of Russian and Slovene. In Russian, the conditions for female first names are similar to those for male first names: in order to be assigned to an inflectional class, a female name with feminine-animate gender has to provide a stem ending in a consonant and a base form ending in either /a/ or a soft consonant (the 3rd macroclass). In reality, however, this latter case is obsolete, since the third declension is no longer productive (in the sense of Dressler 1997, 2003). This means that it is not open for borrowings or extragrammatical formations, even if they fulfil the conditions of the constitutive properties, e.g., the Chinese name *Junin'* "Yuning". Nevertheless, older borrowings, such as biblical names (e.g., *Raxil'* "Rachel"), or the Soviet extragrammatical formation *Ninel'* (an anagram of the name *Lenin*) are inflectable (where *Ninel'* often remains uninflected, as a search of the "Nacional'nyj korpus russkogo jazyka 2.0: Novye vozmožnosti i perspektivy razvitija"[18] confirms). Therefore, this type is not classified as systematically uninflectable. New femi-

[18]The research was conducted using the "Nacional'nyj korpus russkogo jazyka 2.0: Novye vozmožnosti i perspektivy razvitija" (ruscorpora.ru), Savčuk et al. (2024).

nine[19] female names can only be integrated into inflection if they have a base form that ends in /a/ and a stem that ends in a consonant. Therefore, names as *Rea* (*Re-a), *Klaudia* (*Klaudi-a) remain uninflectable.

Female names can of course be adapted by changing an unfitting base form ending into /a/ (e.g., *Renate* > *Renata*) or by inserting the glide /j/ between two coda vowels (e.g., *Klaudia* > *Klaudija*).[20] (This occurs frequently in casual speech and in the dialects). However, this is not a common practice in Contemporary Standard Russian, where usually unfitting names remain as they are and are therefore uninflectable.

Table 11: Constitutive properties of Russian feminine inflectional classes

Macroclass	Base form	Gender	Stem type	Inflectable	Uninfl.
2. hard	-a	f-anim.	hard C	Anna	Rea “Rhea”
2. soft	-a	f-anim.	soft C	Anja	
3.	C-Ø	f-anim.	soft C, unpaired /š/, /ž/	Raxil’	Solanž “Solange”, Junin’ “Yuning”

In Slovene the situation is more fluid, although the tendency is essentially the same: only female names ending in /a/ are fully inflectable and are assigned to one of the subclasses of the 2nd macroclass. In contrast to Russian, this class displays no restrictions as to the properties of animacy and stem type. Therefore, both animate and inanimate nouns with a consonantal or vocalic stem can be assigned to it. Furthermore, names with a base form ending in /o/, /e/ or a consonant can be assigned to this class. The part before such a vowel is analysed as the stem to which inflectional suffixes are attached. However, such cases are very rare and often historical, as with names from Greek mythology or from the Bible (*Kli-o* (nom), *Kli-e* (gen)). The 3rd macroclass of Slovene, finally, does not contain or integrate female names at all.

[19] This gender feature has to be emphasized, because in Russian there are hypocoristic female names with masculine gender, which are derived by a native suffix and are inflected according to the 1st macroclass, e.g.: *Liza* f. > *Liz-oček* m.

[20] Ageenko (2010) lists *Klaudija Kardinale* (with glide-insertion) and *Klaudia Schiffer* (without), Klyšinskij (2010) recommends the form with /j/ for the transcription of Italian names in <ia> but without /j/ for German names with this ending.

Table 12: Constitutive properties of Slovene feminine inflectional classes

Class	Base form	Gender	Stem type	Inflectable	Uninflectable
2.	-a, (o, e, C-Ø)	f	any	Alina, Lea, (Klio, Melpomene, Ruth, Madeleine, Isabel)	Lulu, Mimi, Ildiko
3.	C-Ø	f	C	no names	no names

This discussion has shown that the nominal inflectional classes of Russian and Slovene are characterised by the constitutive properties *gender* (including the subgender of *animacy*), *base form type* and *stem type*. In these cases, the semantic category of *animacy* is grammaticalised both as a subgender and a subclass of the corresponding inflectional classes.

However, animacy may also be a semantic constitutive property, as evidenced by Czech and Slovak. In Czech, there is one inflectional class for feminine nouns ending in a soft consonant, analogous to the respective 3rd class in Russian and Slovene. This class is productive, but it does not contain animate nouns. Because of this semantic constitutive property, it integrates common nouns such as *kartridž-Ø* 'cartridge', toponyms as *Marseille* (ending in /j/) and *Provence* (ending in /s/), but a female name such as *Florence* (ending in /s/) cannot be assigned to this class and remains uninflectable.

5 Summary

The objective of this article is to demonstrate the following points: Uninflectedness in inflecting languages is not a uniform phenomenon. In the first place, two fundamentally distinct manifestations can be identified: contextual and paradigmatic uninflectedness. The former represents a temporary absence of inflection in an otherwise inflectable lexeme, whereas the latter is a permanent feature of either a single lexeme or a class of lexemes with identical properties. The first type has been termed *contextual uninflectedness*, while the two varieties of the second type have been termed *systematic* and *unsystematic uninflectability*. These manifestations can be attributed to different causes which have been spelled out for the noun.

Contextual uninflectedness is caused by contextual factors, such as syntactic context or register, but may also be sociolinguistically conditioned. Systematic uninflectability of nouns is caused either by a mismatch between the constitutive properties of inflectional classes and the properties of the nouns in question, or by the fact that an otherwise matching class is unproductive. Finally, unsystematic uninflectability can on the one hand occur for reasons of style or register. On the other hand, it can be an indicator of ongoing changes, showing either that an inflectional class is changing its constitutive properties or that it is becoming unproductive.

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Chapter 4

Uninflectedness as a factor in agreement loss

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In some weakly inflecting-fusional Romance languages, there has been a significant shift towards an isolating word structure, where uninflectedness has become more common, especially in certain areas of grammar. French nominal morphology is a well-known example, but many Italo-Romance varieties, both north and south of Tuscany, have similarly reduced inflection distinctions. This has led to a less systematic expression of agreement compared to Latin or standard Italian. In this paper, I expand on previous research to examine how uninflectedness emerges in these Romance varieties and how it interacts with agreement. The primary focus is on verb agreement with direct objects, particularly through participles. Most Romance languages exhibit four-cell paradigms in this area, which can show syncretisms or uninflectability due to regular sound change. The notion of uninflectability can also be extended to cases where one feature, usually expressed cumulatively in inflected words, fails to be expressed, or where there is asymmetrical marking of features, as seen in some Italo-Romance dialects with uninflecting roots and inflecting endings in the same participle. This paper explores the implications of these phenomena for agreement.

1 Introduction

Among the (weakly) inflecting-fusional Romance languages, some have gone a long way towards generalizing an isolating word structure, making uninflectedness the rule rather than the exception, at least in some areas of grammar. The best-known case is French nominal morphology, but a large subset of the Italo-Romance varieties – both north and south of Tuscany, whose dialects provided the basis for the standard language – parallels French in having reduced

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distinctions in inflection in general, in such a way that agreement has come to be signalled much less systematically than it used to be in Latin and still is in standard Italian.

In the present paper, elaborating on previous work (see e.g. Loporcaro 2010, 2015, Loporcaro & Paciaroni 2021), I show the ways in which uninflectedness enters these Romance varieties and the various ways in which it interacts with agreement. Among the evidence considered, special attention will be paid to verb agreement with the direct object, whose target is a participle within a periphrastic verb form. In such participles, most Romance languages display four-cell paradigms which may show syncretisms (along patterns classified in Loporcaro 2011) or uninflectability, usually as a product of regular sound change (the case exemplified above with French). Spencer (2020: 143) stretches the notion uninflectability a little bit, to also cover cases in which just one feature among those otherwise expressed cumulatively in the inflection of words from a given part of speech (its *inflectional signature*) fails to be expressed. Another way in which the notion can be extended is to cover asymmetrical marking of distinct features in multiple exponence: in Italo-Romance dialects, thus, one most often encounters uninflecting roots combined with inflecting endings in the same participle, but one can also observe inflecting roots co-occurring with either uninflecting or inflecting endings and, in case both the root and the suffix inflect, they need not encode exactly the same feature-value combinations. I will explore the consequences of these different distributions for agreement.

To pave the way for this, after a quick prologue (Section 2), I will first sketch (in Section 3) an inventory of the different kinds of uninflectability, elaborating on Spencer (2020) and focusing in particular on the types of partial uninflectability. Among these types, I will finally concentrate on one, which I will call *sublexemic partial uninflectability*, showing (in Section 4) how this is involved in the cases of agreement loss mentioned above. As a by-product, the present discussion also reads as a plea for the investigation of dialect variation in Romance, which provides language typologists with exciting and at times unparalleled data.

2 Prologue: inflecting the uninflecting in Italo-Romance

Given what has been said in Section 1, it is clear that, adopting Spencer's (2020: 143) distinction between *uninflecting* and *uninflectable* lexemes, the present paper is about the latter, which belong to parts of speech which do otherwise inflect in the relevant language. However, for a contribution focusing on Romance in a volume on uninflectedness, it is appropriate to at least quickly mention the Indo-European variety which probably goes the farthest in inflecting the uninflecting

(or what is usually uninflecting cross-linguistically), that is Ripano. In this Italo-Romance dialect, spoken in Ripatransone (in the Marches, central-eastern Italy), through the vagaries of sound change even adverbs (of manner, as in (1), but others too) or wh-words (as in (2)) have come to agree with the clause subject inflecting for gender and number (examples from Paciaroni & Loporcaro 2018):
Ripano

(1) a. ˈva a l-a ˈʃkɔːle **ˈntsjeːm-i**
go.PRS.3 to DEF-F.SG school(F).SG together-M.PL
'They go to school together.' (male referents)

b. ˈva a l-a ˈʃkɔːle **ˈntsjeːm-a**
go.PRS.3 to DEF-F.SG school(F).SG together-F.PL
'They go to school together.' (female referents)

(2) **ˈkɔmː-u** / **ˈkɔmː-e** ˈʃtje // **ˈkɔmː-i** ˈʃteːni // **ˈkɔmː-a** ˈʃteːma
how-M.SG / how-F.SG be.2SG // how-M.PL be.3PL // how-F.PL be.1PL
'How are you (SG.F/M)?' 'How are they (M)?' 'How are we (F)?'

The reader is referred to Paciaroni (2023) for a discussion of unusual agreement targets in Ripano (including parts of speech that are generally uninflecting elsewhere) and for earlier references on this fascinating variety, which, however, will not detain us any further here.

3 Types of partial uninflectability

Well-known examples such as Russian *pal'to* 'coat' or Polish *muzeum* instance total (or whole) vs. partial uninflectability (Baerman et al. 2005: 30; Baerman et al. 2017: 28; Spencer 2020: 143).[1] With a variation on the theme of Anna Karenina's principle, while wholly uninflectable lexemes are all alike, once one admits of partial uninflectability, it follows that each partially uninflectable lexeme (or class thereof) can be such in its own way. A tentative taxonomy of these different ways, elaborating on Spencer (2020), is provided in (3):

[1] While Spencer (this volume) uses "total uninflectability", some native English speakers I consulted (who are also distinguished morphologists) opine that the adjective *total* sounds slightly awkward when used to modify a noun with the negative prefix *un-*. However, as an anonymous reviewer kindly points out to me, data from the enTenTen21 corpus show that *totally un*ble* occurs more often than *wholly un*ble.*

(3) An inventory of the types of partial uninflectability:
  a. constructional uninflectability/non-inflectedness (Spencer 2020: 144)
  b. featural
    i. context-free; categorical
    ii. context-free; optional
    iii. contextual
  c. sublexemic (paradigmatic: subset of the paradigm cells)
  d. sublexemic (syntagmatic: morpheme-related)

In the rest of this section, I will briefly comment on constructional and featural uninflectability (3a–3b), to then move on to sublexemic uninflectability in Section 4, which is the type involved in the cases of agreement loss to be discussed in Section 4. Before proceding further, a preliminary remark is in order here. Many of the examples of partial uninflectability to be reviewed in what follows will turn out to involve syncretism, as defined in (4a), from Baerman et al. (2005), where it is contrasted with neutralization and uninflectedness (4b):

(4) a. Baerman et al. (2005: 2): "*syncretism* is the failure to make a morphosyntactically relevant distinction […] under particular (morphological) conditions".
  b. Baerman et al. (2005: 32): "*neutralization* is about syntactical irrelevance as reflected in morphology, *uninflectedness* is about morphology being unresponsive to a feature that is syntactically relevant". [emphasis added]

Thus, one might wonder whether positing a notion "partial uninflectability" runs counter to Occam's razor, multiplying entities beyond necessity. A tentative answer could be that, while there is indeed some overlap, a) framing (what are in part) the same data in terms of partial uninflectability helps us to see them in a new perspective; and, b) at least some instances of partial uninflectability cannot fall under the scope of the notion syncretism.

The first subtype distinguished by Spencer is constructional uninflectability (3a), as observed e.g. in English nouns used as modifiers (*cat food*) or in the German predicative adjective (5a), contrasted with agreeing attributive adjectives in (5b–5d; in the glosses, case values are omitted for simplicity):

(5) a. D-er Mann / D-ie Wohnung / D-as Haus
DEF-M.SG man(M)[SG] / DEF-F.SG flat(F)[SG] / DEF-N.SG house(N)[SG]
ist schön.
be.PRS.3SG beautiful
'The man is handsome. // The flat / The house is beautiful.'

b. ein schön-er Mann
INDF[M.SG] nice-M.SG man(M)[SG]

c. ein-e schön-e Wohnung
INDF-F.SG nice-F.SG flat(F)[SG]

d. ein schön-es Haus
INDF[N.SG] nice-N.SG house(N)[SG]
'a handsome man // a nice flat / house'

The *muzeum*-type of uninflectability can be termed featural (3b), whose first subtype (3b-i), context-free or categorical, is illustrated by Italian or Spanish Class 2 adjectives such as *verde* 'green' (6b)/(7b) which depart from what Spencer (2020: 146) calls the *inflectional expectation* or *inflectional signature* for that part of speech in the relevant language, in that they only signal number, not gender, contrary to Class 1 adjectives like *alto* in (6a)/(7a) where both feature values are expressed cumulatively.

Two different analyses are discussed in Bond (2019) and Thornton (2019), that regard this lack of inflectional distinction as either syncretism (7b) (given an exhaustive paradigm, some of whose cells host forms that are ambiguous), or featural inconsistency, if the paradigm is taken to be non-exhaustive (6b):[2]

(6) a. A paradigm with forms distinguished by two intersecting features (Class 1)

Italian/Spanish *alto* 'tall'

	SG	PL
M	alto	alti/altos
F	alta	alte/altas

b. A paradigm with forms distinguished by a single feature (Class 2)

Italian/Spanish *verde* 'green'

SG	PL
verde	verdi/verdes

[2]The relevant definitions and criteria for (6–7) are the following (from Thornton 2019: 91, elaborating on Corbett 2005: 33; Corbett 2007: 9; Corbett 2015: 157; Spencer 2003: 252; Stump 2012: 255, and Bond 2019: 413–414):

> "exhaustiveness (there is a cell for every possible feature combination, within a given part of speech), completeness (all cells are filled by a form) and non-ambiguity (each cell contains a form that is different from all the forms contained in the other cells of the same paradigm)."

(7) a. A paradigm with forms distinguished by two intersecting features

Italian/Spanish *alto* 'tall'

	SG	PL
M	alto	alti/altos
F	alta	alte/altas

b. A paradigm with same forms for one of two intersecting features (Class 2)

Italian/Spanish *verde* 'green'

	SG	PL
M, F	verde	verdi/verdes

Whatever analysis one chooses for Spanish/Italian Class 2 adjectives such as *verde*, the partial uninflectability (for one feature) is categorical. However, featural uninflectability can be also optional (3b-ii), as illustrated by noun inflection in Wolof (Atlantic, Niger-Congo; Babou & Loporcaro 2016: 10):[3]

Table 1

	SG	PL		Wolof
		= SG	≠ SG	gloss
a.	këf ki	*këf yi	yëf yi	'the thing/-s'
b.	bët bi	bët yi	gët yi	'the eye/-s'
c.	mbagg mi	mbagg yi	%wagg yi	'the shoulder/-s'
d.	bant bi	bant yi	†want yi	'the bit/-s of wood'
e.	góor gi	góor ñi	–	'the man/men'

Most Wolof nouns are of type (e) from Table 1, as a distinct plural form is never attested in the documented history of the language. For some others, a distinct plural form is attested in the past but has now become ungrammatical (Table 1d) or is not (anymore) acceptable for all speakers (Table 1c).[4] At the other extreme, just a handful of nouns (exemplified with *këf* 'thing' in Table 1a) still have a plural form that must be employed categorically when the syntax requires it. Yet

[3] In each example, the noun is followed by the form of the proximal definite article, which is added in Table 1 because it marks number (in addition to noun class/gender).

[4] As an anonymous reviewer aptly remarked, Table 1d and Table 1e, are synchronically identical, yet information about the documentation of a distinct plural form in past stages of the language for nouns such as the one in Table 1d is not irrelevant in describing the drift towards generalized uninflectability of Wolof nouns, some of which have been uninflectable since the beginning of documentation (Table 1e), while others had a distinct plural form which has now fallen into disuse for all speakers (Table 1d), and still others have a distinct plural form that is acceptable to at least some speakers, as indicated by the % sign in Table 1c.

another group of nouns (Table 1b) still displays a plural form but this is not selected categorically in plural contexts and, as a matter of fact, those nouns are more likely to occur in such contexts, too, in the same form as used in the singular. Obviously, this is a way in which in a language an entire part of speech can become uninflecting, a process now approaching its final stage in the case of Wolof nouns. As long as the change, whose stages are mirrored in Table 1a-e, is not completed, nouns such as those exemplified in Table 1b-c display overabundance (see Thornton 2011) – that is, variation between two cell-mates (in the sense of Loporcaro & Paciaroni 2011: 420) – in the plural cell and thus provide an example of optional featural uninflectability. This optionality is an instance of free variation and thus differs from the next subtype of partial featural uninflectability (3b-iii), which is context-dependent. Consider the forms of the Italian definite article in (8):

(8) Italian definite article

a. __#C-

	SG	PL
M	il	i
F	la	le

b. __#V-

<table>
<tr><th></th><th>SG</th><th>PL</th></tr>
<tr><td>M</td><td rowspan="2">l</td><td>ʎ</td></tr>
<tr><td>F</td><td>le</td></tr>
</table>

l'ultimo/-a 'the last.M/F.SG'

Before consonant-initial nouns (8a), the article has a complete paradigm (see footnote 2), with four cells filled with four distinct forms.[5] Before vowel-initial nouns, however (8b), the singular masculine/feminine forms are syncretic or, once partial uninflectability is admitted, uninflectable for gender, whereas the number contrast is preserved. Compare Romanesco in (9), where the prevocalic allomorph (9b) has become unresponsive to both gender and number and reduced to just one form:

(9) Romanesco definite article

a. __#C-

	SG	PL
M	er	(l)i
F	(l)a	(l)e

b. __#V-

<table>
<tr><th></th><th>SG</th><th>PL</th></tr>
<tr><td>M</td><td rowspan="2" colspan="2">l</td></tr>
<tr><td>F</td></tr>
</table>

l'urtimo/-a/-i/-e
'the last.M/F.SG//M/F.PL'

[5] For completeness, it may be mentioned (as an anonymous reviewer reminds me) that the M.SG cell shows an allomorph *lo*, selected (as its plural counterpart *ʎi*) before consonants/consonant clusters that are not exhaustively syllabified in the onset. However, this allomorph is omitted for simplicity in (8a), as it is not relevant to the present argument.

4 Sublexemic uninflectability and loss of object agreement in Romance

Having paved the ground, we now finally come to the last two in the tentative inventory of types of partial uninflectability in (3), which I propose could be called *sublexemic* (3c–3d) and which are those involved in the cases of agreement loss to be discussed in this section.

The first type (3c) is sublexemic in a paradigmatic sense , since uninflectability concerns a subset of the paradigm cells of the given lexeme. This the usual case of participles in languages such as French, where affixal inflection (and hence agreement) has been heavily impacted by regular sound change: this resulted in a complete loss of suffixal nominal inflection (apart from *liaison* contexts) and a strong reduction of verb inflection as well. From now on, we will focus on past participles since these are the target of direct object agreement across Romance. As seen in (10a), erosion of all affixal inflection through sound change resulted in uninflectability for all regular participles, exemplified with Class 1 *chanté* 'sang' in (10b):

(10) Past participle inflection in French, affixal

a.

	M = F
SG = PL	-Ø

b. weak
ʃɑ̃t-e 'to sing'

	M = F
SG = PL	ʃɑ̃t-e

c. strong
pʁɑ̃d-ʁ 'to take'

	M	F
SG = PL	pʁi	pʁiz

This is an instance of partial uninflectability, since the verb itself still inflects, if only weakly so, while it is just a subset of the paradigm cells, those of the participle, which have become unresponsive to the features gender and number. Since participles are the target of object agreement, the latter cannot be signalled anymore with regular participles. There are just two groups of root-stressed participles, exemplified with *pris/prise* in (10c) and *fait/faite* in (11), which still inflect, just for gender, never for number, resulting in the mirror image of Italian and Spanish Class 2 adjectives, which inflect for number, not for gender:

(11) a. Proto-Romance > b. Old French > c. Modern French[6]

a. Proto-Romance	b. Old French	c. Modern French	
FAC-T-U	ˈfai-t	fɛ	M.SG
FAC-T-OS	ˈfai-t-s		M.PL
FAC-**T**-A	ˈfai-**t**-ə	fɛ**t**	F.SG
FAC-**T**-AS	ˈfai-**t**-əs		F.PL

‘done’

Even though with regular participles (10b) agreement is trivially gone, the object agreement rule is still functioning in French, as can be ascertained whenever the participle has feminine forms ending in /z/ or /t/ (see footnote 6), as exemplified in (12b), a context where the conditions for participle agreement are met, though here, too, some variation is reported for the spoken language so that the default masculine form of the participle, which occurs categorically in syntactic contexts where agreement is ungrammatical (12a), can be selected optionally:

(12) French (Guasti & Rizzi 2002)

a. Il a mis/*mise la voiture dans le garage.
he have.3SG put.M/put.F DEF.F.SG car(F) into DEF.M.SG garage(M)
‘He put the car in the garage.’

b. La voiture, il l’=a mise/%mis dans le garage.
DEF.F.SG car(F) he DO.3F.SG=have.3SG put.F/put.M into DEF.M.SG garage(M)
‘The car, he put it in the garage.’

Crucially, the syntactic agreement rule is itself independent from the morphology, as shown by cases such as the Walloon dialect of Liège (North-Eastern Gallo-Romance; see Remacle 1956: 148), where the gender contrast is still marked on all participles, including weak ones such as those in (13), yet object agreement in perfective periphrastics is completely gone (just as in Spanish or Portuguese), even in the last syntactic context which resists the gradual demise of participle agreement in other Romance varieties, viz. initially transitive clauses with direct object clitics, as shown in (14):

[6] See Kilani-Schoch & Dressler (2005: 145–146): “Dans une minorité de paradigmes et micro-classes (14 sur 57), le genre féminin est marqué sur le participe passé par les consonnes /z/ et /t/, lorsque la base n’est pas longue.” As seen in (11), the consonant /t/, which has become the exponent of feminine agreement in the Modern French paradigm exemplified in (11c), stems from the original (inherent) inflection of the participle, which, by contrast, was deleted in the masculine forms after the following final vowel had already been deleted by the Old French stage (11b). Conversely, final -/ə/ < Lat. -A in the feminine endings prevented final consonant deletion from applying, thereby preserving the contrast with masculine.

(13) Liégeois

M	F		M	F
trompé	trompêye		vèyou	vèyowe
‘deceived’			‘seen’	

(14) (Èle) dji l’=a vèyou / *vèyouwe.
(her) I DO.3F.SG=have.1SG seen.M / seen.F
‘I’ve seen her.’

Of course, if phonetic erosion is complete and no single inflecting participle is left, then the language acquirer loses all evidence for a syntactic rule of object agreement. This has not happened so far in French (see (12b)) but has in Gardiol/Guardiolo, the dialect spoken in the Occitan enclave of Guardia Piemontese, Northern Calabria (Loporcaro 1998: 183f.). On the model of the neighbouring Italo-Romance dialects, Gardiol has merged all final unstressed vowels into schwa, which now occurs only variably. Consequently, for all participles there is now just one form in all syntactic contexts and no inflection for gender or number, even in initially transitive clauses with direct object clitics (as exemplified for French in (12b) and Liégeois in (14)):

(15) Gardiol/Guardiolo (Occitan)
a. tə l a saˈrːæ̈/dəˈvɛrtə (la ˈpːɔrtə/ʊ kːanˈtʃɛɖːʐ^{ə})
‘You closed/opened it (the door(F)/the gate(M)).’
b. ˈiʃta ˈkːoːz/ɪˈkːe ˈdːʒiə̯k ɔ β^{w} ɛ pa ˈmːai̯ ˈfɑi̯t
‘This thing(F) I never did. / That game(M) I never played.’
c. ˈiʃtə ˈdːʊtːərə z ɛ pa ˈʃkrɪtːə ˈmːɪ
‘These letters(F) I did not write.’
d. kːe ˈfɵɟːə l ɛ pa ˈʃkrɪtːə ˈmːɪ
‘I didn’t write that leaflet(M).’
e. la ˈpːɑʃt^{ə}/ʊ tːʃaˈbːrɛ tə l a ˈcːeu̯t
‘The pasta(F), did you cook it? / The kid(M), did you cook it?’
f. ɟi mːakːaˈrːʊn iɟ ɛ ˈcːeu̯t
‘The macaroni(M), I cooked them.’

Many dialects of a large area of Central-Southern Italy in which final unstressed vowels merge into schwa show a situation comparable with standard French, in the sense that all regular non-root-stressed participles have become uninflectable, as exemplified with Altamurano (an Apulian dialect) in (16b) (Loporcaro 1998: 66):

(16) Past participle inflection in Altamurano, affixal

a.

	M	F
SG	-ə	
PL		

b. weak laˈvɛ ‘to wash’

	M	F
SG	laˈvɛtə	
PL		

c. strong ˈsːœlvə ‘to untie’

	M	F
SG	ˈsːeltə	ˈsːœltə
PL		

Yet, here the few strong (i.e., root-stressed) participles that still inflect, such as (16c), show agreement even in the very first context from which direct object agreement has disappeared elsewhere, namely transitive clauses with lexical (i.e., non-clitic) direct object, demonstrating that it is still possible to detect crucial syntactic differences (in our case, between this Apulian dialect and French) despite most of the morphological evidence having dissolved because of sound change:

(17) Altamurano

a. ˈaɟːə ˈsːœltə / *ˈsːeltə la ʃʊˈmːwɛnd
have.1SG unfastened.F / unfastened.M DEF.F.SG mare(F)
‘I have unfastened the mare.’

b. ˈaɟːə laˈvɛtə / ˈnːʏtːə la ʃʊˈmːwɛnd
have.1SG washed / brought DEF.F.SG mare(F)
‘I have washed/brought the mare.’

As the glosses in (17) show – which are morphological, rather than morphosyntactic – only gender agreement is signalled, as in French, albeit by different means, namely stem vowel alternations that arose via metaphony. As seen in (18a), the original masculine singular and plural endings, consisting of final high vowels, caused metaphony, resulting in either raising, as in Maceratese (18b), or diphthongization, as in Leccese (18b), Altamurano and Neapolitan (18c):

(18) a.

	ProtoRom.
M.SG	COC-T-U
M.PL	COC-T-I
F.PL	*COC-T-E
F.SG	COC-T-A

‘cooked’

b.

Macer.	=	Lecc.
ˈkotː-u	=	ˈkwɛtː-u
ˈkotː-i	=	ˈkwɛtː-i
ˈkɔtː-e	=	ˈkɔtː-e
ˈkɔtː-a	=	ˈkɔtː-a

c.

Altam.	=	Neap.
ˈkwetːə	=	ˈkwotːə
ˈkœtːə	=	ˈkɔtːə

A more articulate, synchronic display of the Maceratese data is in (19) (from Paciaroni 2017: 54):

(19) Participle inflection in Maceratese (Paciaroni 2017: 54)

a.

		SG	PL
N	laˈat-	-o	-
M		-u	-i
F		-a	-e

‘washed’

b.

		SG	PL
N	ˈkotː-	-o	-
M		-u	-i
F	ˈkɔtː-	-a	-e

‘cooked’

c.

		SG	PL
N	ˈrutː-	-o	-
M		-u	-i
F	ˈrotː	-a	-e

‘broken’

Regular (i.e. non-root-stressed) participles in Maceratese inflect only affixally (19a) for number (a two-valued feature like elsewhere in Romance) and gender, which has a third value, the neuter, occurring only in the singular alongside masculine and feminine. The paradigms of strong metaphonic participles, exemplified in (19b–19c), are more complex, since alongside affixal inflection, root allomorphy also co-signals morphosyntactic feature values.[7] Here, the neuter form shares the root alternant with the masculine, contrasting with the feminine. In other words, number is signalled just on the ending, while there is double exponence of gender. Thus, it is fair to say that the participle stem in (19b–19c) inflects for gender, in a way that synchronically does not just replicate the contrasts marked affixally.[8] Contrary to the affixal contrasts, that are always there whatever the class of the participle, inflectional marking on the root may occur or not, depending on the class of the participle.

A possible way of framing the inflection class difference between weak (19a) and strong participles (19b–19c) consists in saying that the stem is uninflectable in the former while inflecting for gender in the latter. Thus, weak participles are an

[7]The data in (19–26) are slightly simplified, in that in these dialects there are also root-stressed strong participles which do not show any root alternation, depending on the quality of the (etymological) stressed vowel: all over the area considered here, only stressed mid vowels underwent metaphony, which resulted in the morphonological alternations discussed.

[8]Both an anonymous reviewer and Anna M. Thornton (p.c., May 1, 2024) objected to the terminology adopted here, since they “would not say that roots / stems directly inflect or do not inflect, but rather that roots / stems can be, or contain, exponents of inflectional feature values” (Reviewer 2’s report). I fully realize that this criticism is in line with an orthodox application of a word-and-paradigm approach to morphology. However, I believe that the wording adopted here is descriptively adequate, once a) it is admitted that morphemes can be segmented and have a status in morphological theory and b) one observes that the root allomorphy in (18b–18c) and (19b–19c) – as well as in (20b–20c) below – covaries with contrasts in feature-value specifications (here, gender). This wording is also in line with the terminology used in Italian dialectology, where alternations such as those just mentioned are defined as “flessione interna metafonetica” (cf. e.g. D’Elia 1975–1976; Paciaroni et al. 2013: 98; Maturi 2023: 60–61; Mileti 2024: 55).

instance of partial sublexemic uninflectability, not in the sense that uninflectability concerns a subset of the paradigm cells, as in (3c), but rather because it is seen just on the stem, independently of inflectional endings: in other words, this could be termed syntagmatic (i.e. morpheme-related) sublexemic uninflectability (3d).

It would probably not be sensible to claim that the stem of weak participles in (19a) is syncretic for gender: after all, we usually talk of syncretism for affixal morphology or, in inferential-realizational models, for whole word forms occupying paradigm cells, not just for stems which, consequently, have no "inflectional expectation/signature" (Spencer 2020: 146) by themselves.

In previous work of mine (see Loporcaro 2015: 119–123), I argued that among dialects showing multiple exponence of gender along the lines in (19b–19c), a few, spoken in an area straddling Northern Calabria and Southern Basilicata, display a puzzling situation, which results in a violation of the morphology-free syntax principle (see Zwicky & Pullum 1983), according to which

> "syntax can be sensitive to abstract properties realized in morphology, but not to specific inflectional marks for these properties (to dative case, say, but not to a particular dative case marking, or to a declension class for nouns)"
>
> (Zwicky 1996: 301)

The violation consists in the fact that in this dialect area, the syntax of direct object agreement is sensitive to the class of the participle. This is illustrated in the following, where the data from one such dialect, Castrovillarese, spoken in the province of Cosenza, Northern Calabria (see Pace 1993-1994: 128–147; Loporcaro 2010: 167–171; Loporcaro 2015: 119–123),[9] are compared with parallel data from Maceratese, a dialect in which the morphology of participles is similar but that does not show this syntactic peculiarity. The Castrovillarese past participle paradigms, displayed in (20), appear less complex than in Maceratese (19):

(20) Participle inflection in Castrovillarese (Loporcaro 2010: 167)

a.

<table>
<tr><td></td><td></td><td>SG</td><td>PL</td></tr>
<tr><td>M</td><td rowspan="2">laˈvat-</td><td>-ʊ</td><td rowspan="2">-ɪ</td></tr>
<tr><td>F</td><td>-a</td></tr>
</table>

'washed'

b.

<table>
<tr><td></td><td>SG</td><td>PL</td></tr>
<tr><td>ˈkutː-</td><td>-ʊ</td><td rowspan="2">-ɪ</td></tr>
<tr><td>ˈkɔtː-</td><td>-a</td></tr>
</table>

'cooked'

c.

<table>
<tr><td></td><td>SG</td><td>PL</td></tr>
<tr><td>aˈpɛrt-</td><td>-ʊ</td><td rowspan="2">-ɪ</td></tr>
<tr><td>aˈpɪrt-</td><td>-a</td></tr>
</table>

'opened'

[9] In addition to the dialect of Castrovillari, a similar situation has been described (Loporcaro & Silvestri 2011: 350–353) for those of Verbicaro, San Biase (also in the province of Cosenza), and Viggianello (in the province of Potenza, Basilicata).

First, there are just two gender values and, second, all inflectional suffixes converge into -/ɪ/ in the plural as a product of regular sound change. If we compare weak and strong participles, we can say that the former (20a) have a syncretic plural. Another way of putting it, once partial sublexemic uninflectability is assumed, is that weak past participle stems are uninflectable whereas those of strong metaphonic participles inflect for gender, regardless of number. By contrast, on endings gender is neutralized (rather than syncretized), according to the definition in (4b), since affixal endings just mark number, never gender.
Maceratese

(21) a. rɔsa ɛ rvinut-a / *-o jeri
Rose be.3sg come-f.sg / -n.sg yesterday
'Rose has come yesterday.' (unaccusative, weak PtP)

b. rɔsa ɛ mmɔrt-a / *mmort-o
Rose be.3sg died\f-f.sg / died\n-n.sg
'Rose has died.' (unaccusative, strong PtP)

(22) a. (l ú-a) rɔsa l=a rlaat-a/*-o
def.f.sg grapes(f).sg Rose do3f.sg=have.3sg washed-f.sg/-n.sg
'(The grapes) Rose has washed them.' (DO clitic, weak PtP)

b. (l ú-a) rɔsa l=a rkɔrd-a / *rkord-o
def.f.sg grapes(f).sg Rose do3f.sg=have.3sg picked\f-f.sg / picked\n-n.sg
'(The grapes) Rose has picked them.' (DO clitic, strong PtP)

(23) a. rɔsa a rlaat-o/*-a l ú-a
Rosa have.3sg washed-n.sg/-f.sg def.f.sg grapes(f).sg
'Rose has washed the grapes.' (lexical DO, weak PtP)

b. rɔsa a rkord-o / *rkɔrd-a l ú-a
Rosa have.3sg picked\n-n.sg / picked\f-f.sg def.f.sg grapes(f).sg
'Rose has picked the grapes.' (lexical DO, strong PtP)

The (a) vs. (b) examples contain weak vs. strong participles, a morphological difference which, as expected, does not impact on the agreement rule, which prescribes agreement in (21–22) but excludes it in (23), much as in Standard Italian.

Compare now the Castrovillarese counterparts in (24–26):
Castrovillarese

(24) a. rɔsa jɛ vvinʊt-a/*-ʊ
Rose be.3SG come-F.SG/-M.SG
'Rose has come.' (unaccusative, weak PtP)

b. rɔsa jɛ mmɔrt-a / *mmurt-ʊ
Rose be.3SG died\F-F.SG / died\M-M.SG
'Rose has died.' (unaccusative, strong PtP)

(25) a. (l átʃin-a) rɔsa a llavat-a/*-ʊ
DEF.F.SG grapes(F)-SG Rose DO3F.SG=have.3SG washed-F.SG/-M.SG
'(The grapes) Rose has washed them.' (DO clitic, weak PtP)

b. (l átʃin-a) rɔsa a kkɔt-a / *kkut-ʊ
DEF.F.SG grapes(F)-SG Rose DO3F.SG=have.3SG picked\F-F.SG / picked\M-M.SG
'(The grapes) Rose has picked them.' (DO clitic, strong PtP)

(26) a. rɔsa a llavat-ʊ/*-a l átʃin-a
Rose have.3SG washed-M.SG/-F.SG DEF.F.SG grapes(F)-SG
'Rose has washed the grapes.' (lexical DO, weak PtP)

b. rɔsa a kkɔt-a / *kkut-ʊ l átʃin-a
Rose have.3SG picked\F-F.SG / picked\M-M.SG DEF.F.SG grapes(F)-SG´
'Rose has picked the grapes.' (lexical DO, strong PtP)

In Castrovillarese, the expected irrelevance of morphology for the operation of the agreement rules is observed in all syntactic constructions – exemplified in (24) with unaccusative clauses and in (25) with transitive clauses with a clitic direct object – except for the first one from which agreement started to retreat across Romance, viz. transitive clauses with lexical direct objects (26). Here, the occurrence of a participle stem that inflects for gender (26b) correlates with the application of a less restrictive rule (27a) which provides for direct object agreement, while the occurrence of an uninflectable participle stem (26a) correlates with the application of a more restrictive rule (27b) (cf. Loporcaro 2010: 171 for a formal statement of the syntactic rule(s)):

(27) Participle agreement sensitive to morphology in Castrovillarese, Verbicarese, ecc.:

a. participles with double exponence of gender (on both the root and the suffix)
⇒ agreement rule 1 (including agreement with lexical direct objects)

b. participles with an uninflectable root, only marking gender on the ending
⟹ agreement rule 2 (more restrictive, excluding agreement with lexical direct objects)

Within the framework of the present discussion, we can conclude that, in this highly unexpected case of interaction of morphology with syntax, the stem uninflectability for gender correlates with the selective demise of agreement and can thus be conceived – as the title goes – as a factor in agreement loss.

5 Conclusion

To conclude, looking at the Romance data from the point of view of uninflectability leads in part to alternative labelling of some phenomena that are usually defined otherwise, mostly as syncretism. However, the discussion in Section 4 of the interplay between partial uninflectability and agreement loss has revealed some phenomena for which this definitional optionality does not seem to hold. In several Romance dialect areas, just a minority of irregular participle stems – those of strong metaphonic participles – inflect for gender while the overwhelming majority of participles (all regular and several irregular too) do not, as was the case in Latin and still is in the major Romance languages (with French a special case, as seen in (10–11)). In this case, following Spencer (2020: 146), it seems hard to argue that gender marking belongs in the "inflectional expectation" or the "inflectional signature" of just the participle stem, which in turn implies that it would be inappropriate to say that non-metaphonic participle stems show gender syncretism.

Finally, the relevance of this kind of sublexemic partial uninflectability has been shown to play a role in the case of the highly surprising morphosyntactic change in the working of direct object agreement that, in a small area of Southern Italy, has resulted in a violation of the morphology-free syntax principle.

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Chapter 5

Diachronic paths to uninflectedness in South Slavic

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We argue that the ongoing loss of case in Serbian and Bulgarian dialects has led to a split system with competing co-grammars: one where nouns inflect and another where they do not. This division between *forerunners* (without inflection) and *underachievers* (with inflection) results in partial rules of different kinds. There are three types of splits depending on the dialect. First, a split between morphological classes, where some retain case distinctions while others do not. Second, classes that retain inflection may still exhibit uninflectedness in specific constructions, such as partitive constructions. Third, an otherwise inflecting morphological class may split on a semantic basis, such as animate vs. inanimate. These asymmetries in morphological change lead to the rise of partial synchronic rules, whether morphological, syntactic, or semantic. These rules persist over time and evolve as historical processes move into new phases. By studying these rules in contemporary dialects and arranging them chronologically, we can uncover the detailed stages of this historical process.

1 Introduction

The loss of inflectional case marking is an ongoing process in the dialect continuum that stretches across Serbian and Bulgarian linguistic territory. This has led to substantial variation in the dialect case systems of the region. We observe the first signs of this historical change in systems with rich case distinctions, where

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case marking is obligatory on all classes of nominals, as in the western part of the continuum. Further east, we find dramatically reduced case paradigms, with just a few distinct case forms, attested with only part of the lexicon, while the rest of the lexicon has lost the ability to inflect for case. At the same time, in these dialects we find competing rules of case marking, which appear to reflect different historical periods in the breakdown of the case system. The result is either competition between more or less distinct inflection (towards the west, in Serbian dialects), or competition between inflection and uninflectedness (towards the east, in Bulgarian dialects). We investigate factors which underlie this competition and show how they vary at different stages of case decline. This provides us with an insight into the historical mechanisms underlying case loss, and to the conditions that may foster this change.

While our analysis has confirmed the relevance of the factors previously proposed to explain variable case marking synchronically and diachronically (e. g. Moravcsik 1978, Bossong 1991, König 2008, Dalrymple & Nikolaeva 2011, Cristofaro 2013, Kurumada & Jaeger 2015, Witzlack-Makarevich & Seržant 2018), the data obtained from Serbian and Bulgarian dialects shed new light on these factors and show that they do not in fact operate in the way that has previously been suggested.

First, extensive typological data show that explicit case marking is used in environments where it is essential to avoid ambiguity (e.g. between subjects and objects) and is omitted where syntactic roles are transparent (Comrie 1977). For example, it has been shown that the omission of morphological case marking goes hand-in-hand with the use of adpositions, which to a certain extent duplicate the information conveyed by case inflection. This has been shown to play a role in diachronic processes affecting case systems: the information redundancy of case inflection in adpositional constructions may make these constructions more prone to the loss of morphological case than those without adpositions, where case inflection provides critical information about syntactic roles (Hewson & Bubenik 2006). In the dialects of East Serbia, while the use of adpositions (prepositions) correlates with a decline in case distinctions, the dialects display a pattern which is the opposite of what is predicted by the above principle: the original case marking is better preserved precisely when governed by a preposition, and more prone to loss elsewhere.

Second, pragmatic requirements, in particular discourse salience, have been proven to be another important factor in variable case marking cross-linguistically. To mention just two examples, McGregor (2018) and Montaut (2018) present somewhat similar situations in unrelated languages (Southern

African Khoisan and Dravidian respectively) where the rise of object case marking led to pragmatically motivated choices between overt case marking (accusative) and zero marking: prominent objects receive overt case marking while backgrounded objects do not. In Bulgarian border-region dialects, where just two cases have been preserved (nominative and accusative), explicit case marking on non-subject nouns (accusative) is also conditioned by pragmatic factors. However, it is not associated with pragmatic salience, but rather the opposite: explicit (accusative) marking on non-subjects is more frequent in pragmatically less prominent parts of an utterance (e.g. containing background information). Pragmatic prominence instead tends to trigger nominative marking on non-subject nouns, which makes them morphologically indistinguishable from subjects.

We present two case studies that show how different factors shape the transition from a larger to a smaller case system, and further to uninflectedness. After the essential background information in Section 2, we consider the role of structural factors, which are dominant in Serbian dialects (Section 3), and analyse the impact of pragmatic conditions, which are dominant in the Bulgarian dialects under study (Section 4). We suggest that this shift in prominence is due to the diachronic stage each variety has reached: structural factors are prominent at earlier stages of case decline and pragmatic factors at an advanced stage.

2 Background

South Slavic varieties inherited from Proto-Slavic a case system with six morphological cases (plus the vocative whose status as case is debatable[1]) and six major inflection classes (Mirčev 1958: 147–160; Slavova 2017: 127–178; Popović 1955: 115). The further history of case largely correlates with the division of the South Slavic territory into eastern and western zones. In the western zone (contemporary Serbian, Croatian, Bosnian, Montenegrin, Slovenian) the original Proto-Slavic case system has been largely preserved. In the eastern zone (contemporary Bulgarian and Macedonian), most varieties have seen the reduction and loss of nominal case marking and the rise of analytical constructions (e.g. prepositional constructions), as can be seen by contrasting the data from standard Bulgarian and standard Serbian in Table 1.

[1]Vocative forms continue to be widely used in contemporary Serbian and Bulgarian, but do not express grammatical relationships (Blake 2004: 8) and do not interact with the types of configurations we have investigated.

Table 1: Nominal inflection in Serbian and corresponding forms in Bulgarian[2]

Standard Serbian inflected forms	**Standard Bulgarian** prepositional phrases with an uninflected form	**Translation**
Ovo je **Kipar**. (NOM)	Tova e **Kipâr**.	'This is Cyprus.'
Idu na **Kipar**. (ACC)	Otivat na **Kipâr**.	'They go to Cyprus.'
Pomažu **Kipru**. (DAT)	Pomagat na **Kipâr**.	'They help Cyprus.'
Stanovništvo **Kipra**. (GEN)	Naselenieto na **Kipâr**.	'The population of Cyprus.'
Žive na **Kipru**. (LOC)	Živejat na **Kipâr**.	'They live in Cyprus.'
Upravljaju **Kiprom**. (INS)	Upravljavat **Kipâr**.	'They govern Cyprus.'

In between these two types there is a substantial transitional dialect zone. A characteristic feature of these transitional dialects is the competition between these two strategies, as seen both in Serbian (1) and Bulgarian dialects (2).

(1) Zove ga otac telefon-om / na telefon.
call[PRS.3SG] 3SG.M.ACC father[NOM] phone-INS.SG / on phone[ACC](=NOM)
'The father calls him by phone.' (Miloradović 2003: 183, 202)

(2) Petr-a / Petâr go njama.
Peter-SG.ACC / Peter[SG.NOM] 3SG.M.ACC EXIST.NEG
'Peter is absent.' (Stojkov 1981: 163)

The dialects in this transitional area correspond to different postulated chronological stages of the decline of the Proto-Slavic case system. A group of dialects in the west of the transitional area preserve all six original Proto-Slavic cases (plus vocative), but speakers regularly replace three of them with the accusative form, which serves as a general oblique case (Figure 1, Transitional System 1). Thus, the

[2]Bulgarian demonstrates an asymmetry between nominal and pronominal systems with respect to case marking. Nouns and adjectives do not inflect for case (in standard varieties and in a significant number of dialects). Personal pronouns distinguish full and clitic forms (the choice being syntactically determined). Full personal pronouns retain a two-case distinction, nominative vs. accusative, the latter being used in all non-subject syntactic roles: ***tja*** (3SG.F.NOM) *običa* ('she loves') vs. *običam* ***neja*** (3SG.F.ACC) ('I love her') vs. *kazax na* ***neja*** (3SG.F.ACC) ('I told her', literally 'to her'). Clitic personal pronouns are used in non-subject roles and distinguish between accusative and dative: *običam* ***ja*** (3SG.F.ACC) vs. *kazax* ***ì*** (3SG.F.DAT) ('I told her').

original six-case system competes in these dialects with an innovative three-case system, consisting of nominative, dative, and accusative only (Miloradović 2003, Simić 1972). The further east one goes, the greater are the reductions. The dative case is the next to be replaced (Transitional System 2). Here, the more conservative three-case system competes with a two-case system, consisting of nominative and accusative (Belić 1905, Ivić 1956). Finally, in the east of the transitional area, the accusative itself starts to yield to a single uninflected form (Transitional System 3). Here, the two-case system coexists with a system without case (Stojkov 1975, 1981). Thus, three structural dialect types can be distinguished within the transitional area, according to the size of the case paradigm: six-, three- and two-case systems, as summarised in Figure 1.[3]

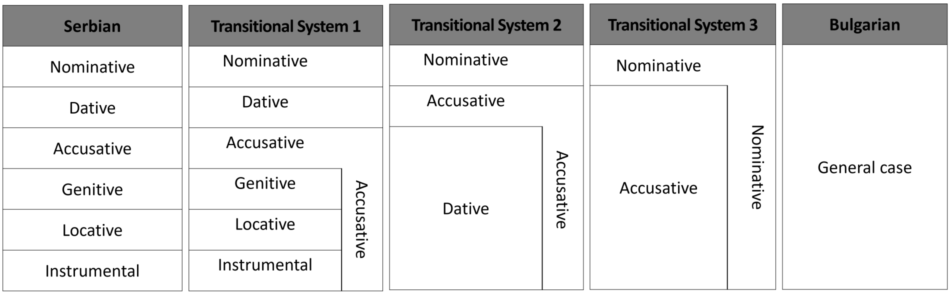


Figure 1: Case systems in the Serbian-Bulgarian dialect continuum

The chronology that this transition reflects has been reconstructed primarily on the basis of liturgical texts produced within various regional written traditions in the South Slavic territory from the 10^{th} century onwards (Tsonev 1984, Duridanov 1956, 1958, Rusek 1964, Češko 1970, Belić 1905) and, for later periods, on the basis of clerical and household documents (Bernštejn 1948).

It is believed that the decay of the case system originated in the South Slavic eastern zone (Bulgarian and Macedonian) around the 10^{th} century (Duridanov

[3]In distinguishing between the largely syncretic dative and locative cases in Serbian, we follow the Slavic tradition of case description (Piper & Klajn 2013, Miloradović 2003). The non-syncretic locative endings were used at least until the end of the 15^{th} century (Belić 1999), while the loss of case started sometime before the 12^{th} century (Belić 1905). It follows, then, that when the process of the case loss started, the dative and locative were still morphologically distinct cases. An additional argument in favour of differentiating between the two cases comes from the fact that it facilitates the description of the process of case loss. The original locative case is frequently replaced by the general oblique in the whole transitional area, including the western region where it is the only case to be replaced. The original dative case, on the contrary, is preserved in most Serbian dialects with the exception of the eastern zone where the only remaining cases are nominative and accusative (Sobolev 1991).

1958: 25). By the end of the 14th century morphological case marking on nominals was completely lost in at least some of the dialects spoken in the eastern zone (Bulgarian border-region dialects) (Češko 1970: 11). Even earlier, by the end of the 13th century, case loss was at an advanced stage in a number of Macedonian dialects (Rusek 1964). In Serbian, the process of case decline started sometime before the 12th century and affected the South-East dialects (Belić 1905). It did not, however, lead to the complete loss of case distinctions: rather it resulted in different types of reduced systems.

Historical evidence points to the following processes involved in this change: the spread of prepositional constructions replacing bare noun phrases; the redistribution of functions between indirect cases and displacement of less productive indirect cases; the spread of accusative forms in the function of an oblique case generalising across original indirect cases;[4] and the spread of uninflected forms replacing this oblique case.

3 Constructional variation: the loss of the genitive in Serbian dialects

The loss of case in European languages has been reported to go hand in hand with the increased use of prepositions (Blake 2004, Hewson & Bubenik 2006, Kulikov 2006). Blake (2004: 178) illustrates this with an Old English version of Luke 15:15 (3a) and Wyclif's fourteenth-century translation (3b) (glossing and translation are as in Blake 2004):

(3) a. He folgode an-um burg-sitt-end-um menn thaes rices.
he followed one-DAT town-dwell-ing-DAT man.DAT that.GEN land.GEN

b. He clevede [=attached himself] to oon of the citizens of that contré.
'He hired himself out to one of the citizens of that country.'

In (3a), the directional ('to the citizen') and the possessive ('of that country') relationships are expressed via inflection, while in (3b), they are expressed by

[4] The chronology of the displacement of individual indirect cases varies. The most stable, as dialectological data of the Serbian language show (Belić 1905, Miloradović 2003) is the dative case. The historical data from Bulgarian also testify to this: the dative forms of nouns were found in Bulgarian texts of the 17th, 18th and even 19th centuries, while the genitive, instrumental and local cases had already fallen out of use in Bulgarian dialects in the 13th-14th centuries (Mirčev 1958: 255–260; Češko 1970: 301–310).

prepositional constructions. The relationship between the two processes, however, is not clear. Did the spread of prepositional constructions cause case to disappear as it became semantically redundant (as claimed in Hjelmslev 1935, Hewson & Bubenik 2006)? Or was it, by contrast, a repair strategy necessary after the loss of explicit case marking, as proposed, for example, in Skautrup (1944)?

The variation found in Serbian dialects allows us to investigate the interplay of inflectional case marking and prepositional constructions using synchronic data. We focus here on the genitive, which is in the process of being replaced by the accusative. Specifically, we ask: do prepositional constructions particularly favour this replacement? This is what an informativeness principle would predict: since the preposition provides semantic context, genitive case marking is redundant and the more general accusative case can be used without any loss of information (cf. Comrie 1977, 1989). Our data, however, show the opposite tendency: prepositional constructions favour the *retention* of the genitive, while its replacement by the accusative is more advanced in non-prepositional contexts. That means that in Serbian dialects, some principle other than informativeness is at play. We suggest below that this principle could be understood as a speaker-oriented strategy of grammatical change.

In Section 3.1, we introduce the data used in this case study. Section 3.2 outlines the use of the genitive forms in Serbian and discusses the results of previous research on the variation between genitive and accusative in South-East Serbian dialects. Sections 3.3 and 3.4 present the patterns of variation found in our data; Section 3.5 discusses the theoretical implications of our findings.

3.1 Data

The data for this study come from sociolinguistic interviews with speakers of Serbian dialects located in or near Kosovo. These come from two sources: the archive of the Institute for Balkan Studies and our own fieldwork. The archive provided a subset of recordings collected in the project "Research of Slavic Vernaculars in Kosovo and Metohija" (2002 – 2003) led by the Institute for the Serbian Language of SASA and financed by UNESCO. Our data come from fieldwork in the municipality of Brus, Central Serbia, conducted by Maria Kyuseva within the project "Declining case: inflectional loss in progress" in 2022.

These data were collected in a uniform fashion that conforms to the Serbian dialectological tradition. The participants were native speakers of local dialects, aged between 70 and 81 years, who had been long-term residents of the region and had engaged in traditional activities throughout their lives, such as cattle breeding or agriculture. This sampling ensured the minimal interference of the high-prestige standardised variety of Serbian. The topics of the interviews included history, tradition, culture, crafts, cuisine, everyday life, and biographical stories.

The total corpus compiled from these two sources contains over 274,800 tokens, 30 hours and 36 minutes of recordings. Overall, twenty-three speakers from eleven villages were recorded. Figure 2 shows the locations of the villages, which are colour-coded according to dialect. The Zeta-South Sandžak dialect is marked in green, the Kosovo-Resava dialect is marked in pink, and the Prizren-South Morava dialect is marked in purple.

Figure 2: Locations of the interviews in Serbia

● Zeta–South Sandžak ● Kosovo–Resava ● Prizren–Timok

With respect to case marking, these three dialects have the same system (Transitional System 1 in Figure 1): all six original cases are preserved to various degrees, but three of them (genitive, locative, and instrumental) can be replaced by accusative forms. The dialects differ in the frequency of such replacements. Zeta-South Sandžak is the most conservative: it has a stable six-case system, and the three peripheral cases are only occasionally replaced by the accusative. Kosovo-Resava represents a more advanced stage of case loss: the genitive, locative, and instrumental cases frequently give way to the accusative. Finally, Prizren-Timok shows the most advanced stage: the locative and instrumental are virtually nonexistent (occurring with only a handful of lexemes), and the use of the genitive is highly restricted. We compare the usage of genitive forms in these three dialects and project the synchronic tendencies onto the diachronic plane. This allows us to draw conclusions regarding the historical dynamics involved in the decline of this case.

3.2 Genitive case in Serbian

The Serbian genitive is highly polysemous. Its typical function is to mark a relationship between two entities, such as possessor-possessee, part-whole, and source-goal. Additionally, it can denote time, reason, goal, and manner. From the point of view of syntax, it has a wide distribution. It can be used in adnominal, adjectival, and verbal constructions, as well as in prepositional constructions and constructions with a quantifier. Out of all the cases, the genitive occurs with the widest inventory of prepositions (Feleško 1955).

Various studies of individual dialects of South-East Serbia have already established the heterogeneous behaviour of the genitive in different constructions. For example, Miloradović (2003) analyses the competition between the genitive and accusative in the Kosovo-Resava dialect of the Paraćin region in Central Serbia. She observes that while the genitive is dominant in prepositional constructions, its use is highly limited elsewhere. In verbal constructions, the genitive is replaced by the accusative (4a), in adnominal constructions – by a prepositional phrase (4b), and in adjectival constructions, it can be replaced either by the accusative or by a prepositional phrase (4c).

(4) a. Nije bi-l-o **vod-e**. → Nije bi-l-o **vod-u**.
AUX.NEG.3SG be-PTCP-N.SG water-GEN.SG AUX.NEG.3SG be-PTCP-N.SG water-ACC.SG
'There was no water.'

b. kraj **mačug-e** → kraj **od mačug-e** / **od mačugu**
end[NOM.SG] cane-GEN.SG end[NOM.SG] of cane-GEN.SG / of cane-ACC.SG
'cane end'

c. Pun **vod-e**. → Pun **vod-u**. / Pun **s vod-u**.
full[M.NOM.SG] water-GEN.SG full[M.NOM.SG] water-ACC.SG / full[M.NOM.SG] with water-ACC.SG
'[It is] full of water.'

Stanković (2022) analyses the speech of the younger population in the town of Vranje in South Serbia, where the Prizren-South Morava dialect is spoken. This dialect has a three-case system consisting of the nominative, accusative, and dative. The speech of the younger generation, however, is influenced by the standard dialect. Therefore, it is susceptible to the penetration of other cases found in the standard, especially the genitive. Stanković observes that the possibility of using the genitive depends on the construction. Focusing on the non-prepositional uses, she finds that adnominal constructions and constructions with a numeral permit the occasional appearance of the genitive case (5a-5b), while constructions where the complement depends on a verb or a predicative adjective do not (5c-5d):

(5) a. udic-a za **pecanj-e** **rib-a**
rod-NOM.SG for catching-ACC.SG fish-GEN.PL
'fishing rod'

b. Ima-m **pet** **različit-ih** **igar-a**.
have-PRS.1SG five different-GEN.PL game-GEN.PL
'I have five different games.'

c. Nije **je-o** **čokolad-u** / *čokolad-e.
AUX.NEG.3SG eat-PTCP.M.SG chocolate-ACC.SG / *chocolate-GEN.SG
'He didn't eat the chocolate.'

d. Dvorišt-e **pun-o** **s** **igračk-e** / *pun-o igračak-a.
yard-NOM.SG full-N.NOM.SG **with** toy-ACC.PL / *full-N.NOM.SG toy-GEN.PL
'The yard is full of toys.'

In our study, we bring these individual observations together by comparing the use of the genitive case forms in different constructions across the three South-East Serbian dialects: Zeta-South Sandžak, Kosovo-Resava, and Prizren-South Morava.

3.3 Constructions with the genitive in South-East Serbian

We focus on three constructions with the genitive case: prepositional constructions, constructions with a quantifier, and adnominal constructions. In South-East Serbian dialects, these constructions allow both the original genitive and the innovative accusative forms:

(6) Prepositional construction

a. Napravljen-i **od** **drv-a**.
made-M.NOM.PL from wood-GEN.SG
'[They are] made from wood.'

b. Ima i **od** **drv-o**.
EXIST.PRS and from wood-ACC.SG
'There is [some of it] from wood as well.'

(7) Construction with a quantifier

a. Malo **trav-e** gi stavi-m.
a.little grass-GEN.SG 3PL.DAT put-PRS.1SG
'I put out a bit of grass for them.'

b. I ondak turi malo **trav-u**.
and then put[IMP] a.little grass-ACC.SG
'And then put out a little bit of grass.'

(8) Adnominal construction

a. Tamo odnese-š čaš-u **vin-a**.
there take.away-PRS.2SG cup-ACC.SG wine-GEN.SG
'You take there a cup of wine.'

b. I uzme-š čaš-u **vin-o**.
and take-PRS.2SG cup-ACC.SG wine-ACC.SG
'And you take a cup of wine.'

The overall number of examples in the dataset is 960. The prepositional construction is represented by 676 examples, the adnominal construction – by 169 examples, and the construction with a quantifier – by 115 examples. The ratio represents the occurrence of these constructions in spontaneous speech. For the prepositional construction, we include only phrases with the preposition *od* 'from, of', which is the most frequent preposition that governs the genitive. Constructions with a quantifier include the following quantifiers: *mnogo* 'a lot', *malo* 'a little', *lek* 'very little', *pomalo* 'a little bit', *puno* 'much', *više* 'more', and *manje* 'less'. Finally, adnominal constructions include a variety of governing nouns, the most frequent meanings in the sample are: quantity (a cup of tea, a bowl of rice), and part-whole (top of the table, edge of the village). True possession (a son of my brother), otherwise typical for the construction, is rare in our examples. It is usually expressed in the dialects by alternative means of expression, such as possessive adjectives and constructions with the dative case.

The bar chart (Figure 3) shows the variation between the original genitive and the innovative accusative in the three dialects.

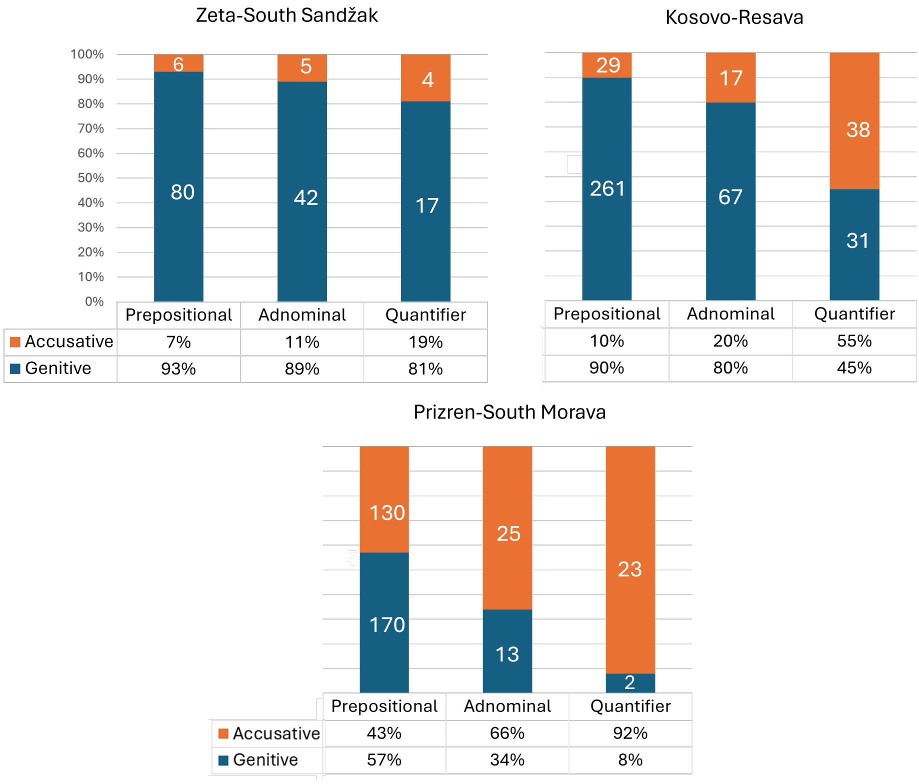


Figure 3: Competing strategies of case marking in South-East Serbian dialects

As the figure illustrates, in Zeta-South Sandžak (the top left section of the bar chart), the most conservative dialect in the sample, the genitive prevails in all three constructions. It appears in around 90% of the examples in the prepositional and adnominal constructions and is used somewhat less in the construction with a quantifier. As an ANOVA[5] test shows, there is no significant effect of the construction on the case in this dialect, with p=0.2429 and F=1.4284. The difference between the three constructions is more prominent in the Kosovo-Resava dialect, which has a more advanced stage of case decline. Here, the genitive still dominates with 90% of occurrences in the prepositional construction, it is slightly

[5] The ANOVA analysis of the corpus data here and below was carried out by Dr Alexander Stewart (University of St Andrews, UK).

less frequent in the adnominal construction, and its use falls dramatically in the construction with a quantifier, where it is no longer a dominant strategy. The ANOVA test shows a significant effect of the construction on the case in this dialect with p=3.3092e-18 and F=44.1388. Finally, the most innovative Prizren-South Morava dialect (the lower section of the bar chart) shows yet another picture. Here, the use of accusative is the dominant strategy in both the adnominal construction (66% of examples) and the construction with a quantifier (92% of examples), and the only construction where the genitive appears in a slight majority of examples (57%) is the prepositional construction. Again, ANOVA shows a significant effect of the construction with p=1.1018e-06 and F=14.2548. (The null hypothesis for each of the three dialects assumed no effect of the construction on case selection.)

These data show that the use of the genitive versus the accusative in the Serbian dialects under study is at least partly conditioned by the type of construction. The exact conditions depend on the dialect, or if we accept the diachronic interpretation of the data, on the stage of case decline. The general tendency is the following: constructions with a quantifier are the most susceptible to the penetration of the innovative accusative, adnominal constructions come second, and prepositional constructions preserve the original genitive the longest. This effect is clear when we look at the distribution of cases in individual dialects, but it can also be seen if we trace the linguistic behaviour of the constructions across dialects. The use of the genitive case consistently declines in each construction as we move across the transitional zone from west to east. For each construction, we find a significant effect of the dialect: with p=2.1082e-25 and F=61.8976 (prepositional construction), p=9.1127e-10 and F=23.6043 (adnominal construction), and p=1.1889e-06 and F=15.4478 (construction with a quantifier).

3.4 Morphologically conditioned variation in Serbian dialects

Another factor that conditions variation between the original genitive and innovative accusative case is inflection class. This plays a role in the whole dialectal continuum, but in our data it is especially pronounced in the Prizren-South Morava dialect.

In standard Serbian, nouns cluster into four inflection classes (Table 2). The paradigms exhibit a high degree of syncretism (see the comment in footnote 3 concerning dative-locative syncretic forms).

In South-East Serbian dialects, there is a tendency for nouns of inflection class III to adopt the forms (and the gender) of inflection class I. For this reason, we do not have enough examples to be able to draw reliable generalizations about this

Table 2: Inflection classes for selected nouns

	Inflection class I	Inflection class II	Inflection class III	Inflection class IV
	prostor 'space' (MASC)	kuća 'house' (FEM / some MASC)	stvar 'thing' (FEM)	polje 'field' (NEUT)
SG.NOM	prostor	kuć-a	stvar	polj-e
PL.NOM	prostor-i	kuć-e	stvar-i	polj-a
SG.ACC	prostor	kuć-u	stvar	polj-e
PL.ACC	prostor-e	kuć-e	stvar-i	polj-a
SG.GEN	prostor-a	kuć-e	stvar-i	polj-a
PL.GEN	prostor-ā	kuć-ā	stvar-ī	polj-ā
SG.DAT	prostor-u	kuć-i	stvar-i	polj-u
PL.DAT	prostor-ima	kuć-ama	stvar-ima	polj-ima
SG.LOC	prostor-u	kuć-i	stvar-i	polj-u
PL.LOC	prostor-ima	kuć-ama	stvar-ima	polj-ima
SG.INS	prostor-om	kuć-om	stvar-ju	polj-em
PL.INS	prostor-ima	kuć-ama	stvar-ima	polj-ima

class, and we exclude it from further analysis. The bar chart in Figure 4 shows the variation between the genitive and the accusative for the remaining three inflection classes in the Prizren-South Morava dialect.

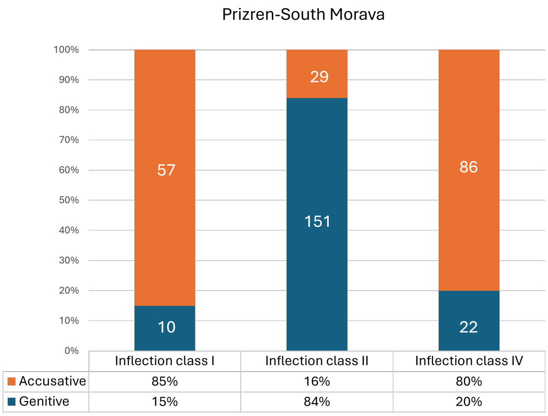


Figure 4: Prizren-South Morava dialect: original genitive vs. innovative accusative in different inflection classes

The choice of case is clearly conditioned by inflection class: class II favours the genitive form, while class I and class IV favour the accusative form. This pattern can be illustrated with examples of the same meaning of the genitive (material) being expressed differently depending on the inflection class, see (9a-9b).

(9) a. Se ispred-e **od** **vun-e**.
REFL spin-PRS.3SG from wool(II)-GEN.SG
'It is spun from wool.'

b. Pravi-l-i **od** **drv-o**.
make-PTCP-M.PL from wood(IV)-ACC.SG
'They used to make it from wood.'

In (9a), the noun belongs to class II (*vuna* 'wool') and takes the genitive; in (9b), the noun belongs to class IV (*drvo* 'wood') and takes the accusative. This tendency to preserve the original case form in class II is even more pronounced in Bulgarian border-region dialects, where case marking with class II occurs most frequently and is semantically unrestricted (see Section 4).

3.5 Discussion: structural factors in case decline

These data illustrate both syntactic and morphological conditions on variation between the genitive and accusative. In terms of syntax, certain constructions favour the use of the accusative over the genitive, in particular prepositional phrases. In terms of morphology, certain inflection classes are more likely to adopt the innovative accusative, in particular class II.

The tendency of prepositional phrases to preserve the original case marking contradicts commonly held assumptions about the role played by prepositions in case loss (Blake 2004, Hewson & Bubenik 2006). Since genitive case marking in these prepositional constructions is semantically redundant, it could be omitted without incurring any ambiguity. One might assume then that prepositional constructions would be the first to lose the genitive. The fact that our data show the opposite means that the preservation of the genitive is not driven by the need to disambiguate. We propose here a possible alternative motivation.

Following Norde (2002), we distinguish two types of less-effort strategies for grammatical change: a speaker-oriented and a hearer-oriented strategy. These can be thought of as two opposing forces that shape any language change. The speaker "wants" to make less effort in encoding the message, and the hearer "wants" to make less effort in decoding the message. The informativeness principle aligns with a hearer-oriented strategy. In this instance, the genitive should first be replaced in prepositional constructions, as it is not needed for the hearer to be able to correctly interpret the message, while it has a function in other constructions. A speaker-oriented strategy would be one that favours ease of production. The genitive case after the preposition *od* 'from' is syntactically determined: *od* governs the genitive, creating a high degree of predictability: every time the preposition *od* 'from' is used, it can be expected to be followed by the genitive. In adnominal constructions and constructions with a quantifier, there is no such syntactic determination, in the sense that the nouns, such as 'cup' or 'edge', do not obligatorily govern a noun phrase, and quantifiers can similarly be used on their own. In this way, the syntactic information supports the case morphology and contributes to the preservation of the original case forms. Crucially, there is no gain for the hearer in preserving the case forms in prepositional constructions while omitting them in non-prepositional constructions. Rather, it is in the speaker's interest to keep an inflected form in a context that always requires it.

With these results in mind, we turn to the contrasting patterns of Bulgarian border-region dialects.

4 Competing case marking in Bulgarian border-region dialects: pragmatic factors

In Bulgarian border-region dialects, case marking is lost in all inflection classes of nouns except class II and, to some extent, class I (restricted to animate nouns). In these classes, singular nouns retain two case forms, nominative and accusative; plural forms are not inflected for case. The nominative marks nouns in the subject position and the accusative marks nouns in non-subject positions. Nouns in classes III and IV, and inanimate nouns in class I have retained only one (historically nominative) morphological form for each number value (singular and plural) which is used both in the subject and any non-subject position. This is illustrated in examples (10), (11), (12) and (13) from our corpus (data represent various locations in the municipalities of Belogradchik and Trân).

Inflection class I, masculine. Nominative and accusative (animate nouns only)

(10) a. Dojde **doktor**.
come[PST.3SG] doctor[SG]
'A doctor came.' subject NP
Filipovtsi, Trân

b. Nije natovari-m pak **doktor-a**.
we task-PRS.1PL again doctor-ACC.SG
'We task the doctor again.'
Vrabcha, Trân

c. Bašta mi me kara u Lom **pri doktor-a**.
father 1SG.DAT 1SG.ACC took[PRS.3SG] to Lom to doctor-ACC.SG
'My father took me to the doctor in Lom.'
Chuprene, Belogradchik

Inflection class II, feminine and masculine. Nominative and accusative

(11) a. **Vod-a** gi e kara-l-a.
water-NOM.SG 3PL.DAT AUX.3SG transport-PTCP-SG.F
'Water moved them.' subject NP
Goren Chiflik, Belogradchik

b. Sipe-mo **vod-u**.
pour-PRS.1PL water-ACC.SG
'We pour water.' non-subject NP
Repljana, Belogradchik

c. Ide i **za vod-u**.
go[PRS.3SG] and for water-ACC.SG
'And s/he is going to get water.' non-subject PP
Gorni Lom, Belogradchik

Inflection class III, feminine. Caseless (historically nominative)

(12) a. … **sol** se posolva.
… salt[SG] REFL salt[PRS.3SG]
'… salt is being added.' subject NP
Salash, Belogradchik

b. **Morsk-a sol** turi-m tri ložits-e.
sea-SG.F salt[SG] pour.in-PRS.1SG three spoon-PL
'I pour in three spoons of sea salt.' non-subject NP
Ezdimirtsi, Trân

c. Posolva-š sâs **sol**.
salt-PRS.2SG with salt[SG]
'You add salt / one adds salt.' non-subject PP
Salash, Belogradchik

Inflection class IV, neuter and masculine. Caseless (historically nominative)

(13) a. … **mlek-o** se podsirva.
… milk-SG REFL ferment[PRS.3SG]
'… milk is fermented.' subject NP
Chuprene, Belogradchik

b. Ako ima-š **mlek-o**.
if have-PRS.2SG milk-SG
'If you have milk.' non-subject NP
Ezdimirtsi, Trân

c. Dve kofišk-i **ot mlek-o**.
two pot-PL of milk-SG
'Two pots of milk.' non-subject PP
Ezdimirtsi, Trân

The clear-cut distribution of the nominative and accusative across subject and non-subject positions within inflection classes I and II as presented in (10) and (11), however, does not hold in the border-region dialects. These dialects show a tendency to generalise the nominative for nouns of inflection classes I and II across non-subject positions (making them like Standard Bulgarian where this process has been completed in all historical inflection classes, Mirčev 1958: 146–147). This results in competing choices for case marking on non-subject NPs and PPs as illustrated in (14) and (15) for inflection class II:

(14) a. **Vod-u** gi dava-mo.
water-ACC.SG 3PL.DAT give-PRS.1PL
'We give them water.' non-subject NP (accusative)
Repljana, Belogradchik

b. **Vod-a** smo kara-l-i.
water-NOM.SG AUX.PRS.1PL carry-PTCP-PL
'We used to carry/haul water.' non-subject NP (nominative)
Ezdimirtsi, Trân

(15) a. Ne može **bez vod-u** da peče-š.
NEG possible without water-ACC.SG COMP roast-PRS.2SG
'One can't roast [meat] without water.' non-subject PP (accusative)
Filipovtsi, Trân

b. Da ne ostane **bez** **vod-a**.
COMP NEG remain[PRS.3SG] without water-NOM.SG
'So that s/he won't be left without water.' non-subject PP (nominative)
Salash, Belogradchik

From a diachronic perspective, the situation in Bulgarian border-region dialects reflects one of the final stages in the transition to uninflectedness: only two of four historical inflection classes retain case distinctions, classes I and II, and in only one part of the paradigm (singular); in addition, all inanimate nouns of class I have already lost the ability to inflect for case. This is different from what we see at the other end of the dialect continuum, the three groups of Serbian dialects in Section 3, which illustrate the initial stages in the decomposition of the Proto-Slavic case system. This raises the question as to whether the factors that condition the loss of case distinctions remain stable over time, or whether they change from one stage of the process to the next. We address this question by looking into the synchronic competition between accusative as a specific non-subject case and nominative as a single form for all syntactic functions, and identify what might trigger speakers' choices where both alternatives are available. Given the limited number of examples containing animate nouns of class I, further analysis focuses on the nouns belonging to the more populous class II.

4.1 Data

The study is based on annotated transcripts of sociolinguistic interviews recorded in West Bulgaria (the dialects of Belogradchik and Trân, part of the border-region dialects). The fieldwork was conducted within the project "Declining case: inflectional loss in progress" by Vladimir Zhobov (University of Sofia, project consultant for Bulgarian) and Alexander Krasovitsky in 2021 and 2022 (Figure 5). Additional recordings provided by Vladimir Zhobov stem from his previous fieldwork in North-West Bulgaria.

Forty-six residents born between 1926 and 1950 were recorded (one interview per language consultant; the interviews range in length from forty minutes to two hours). Thirteen of the recorded consultants retain two cases, nominative and accusative, in inflection class II. These thirteen interviews make up a corpus of approximately 168,000 words containing 2292 phrases with inflection class II nouns in non-subject positions. All thirteen interviews demonstrate a competition of accusative and nominative in non-subject positions, however, the probabilities vary considerably between speakers (Figure 6).

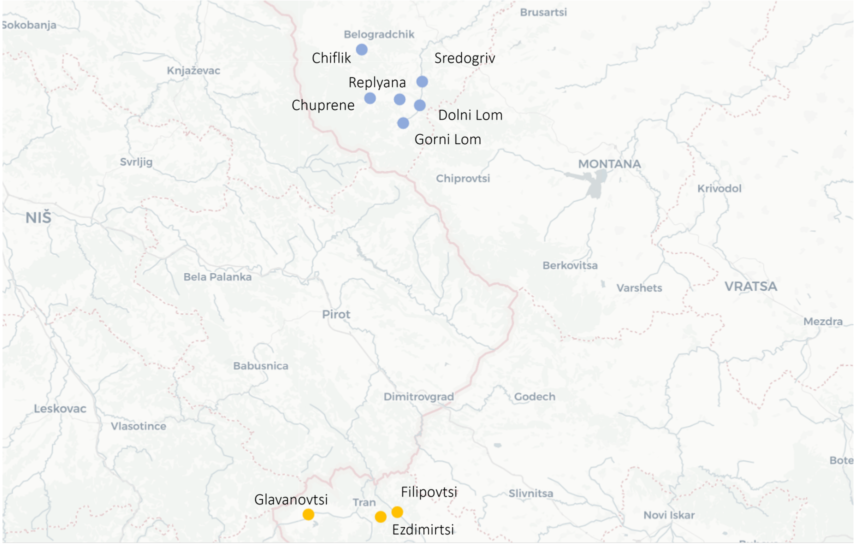


Figure 5: Locations of the interviews in Bulgaria. Border-region dialects

● Belogradchik sub-group ● Trân sub-group

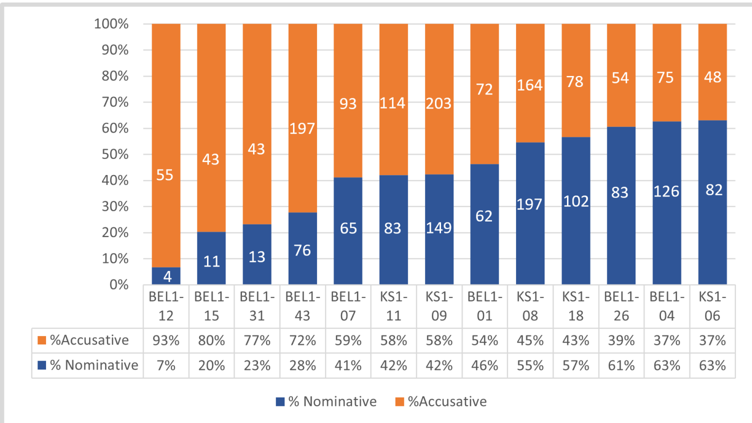


Figure 6: Relative frequency of nominative and accusative forms in inflection class II nouns in non-subject positions (by speaker). Corpus size: around 168, 000 tokens. 13 sociolinguistic interviews, 2292 occurrences of inflection class II nouns in non-subject positions. Codes below the columns identify individual speakers.

4.2 Hypothesis: information structure conditions case selection

Immediate observations made during the sociolinguistic interviews and further elicitation work led us to the following hypothesis. Variation between the nominative and accusative on non-subjects is conditioned by information structure and pragmatic salience: nouns associated with new and pragmatically important information tend to lose specific non-subject case marking and attract nominative; nouns associated with background information and pragmatically less salient parts of an utterance tend to retain specific non-subject marking and show stronger preference for the accusative.

In what follows, we briefly consider the role of pragmatic factors in case selection within a general linguistic context (Section 4.3) and then proceed to the analysis of our data (Section 4.4) and a discussion (Section 4.5). We argue that pragmatic conditions play a crucial role in speakers' choices between the two competing cases which mark non-subjects and in the decline of the accusative as distinct morphological marking for non-subjects. At the same time, we will show that the relationship between case and information structure in Bulgarian border-region dialects differs significantly from that observed cross-linguistically, and offer an explanation for this.

4.3 Information structure as a condition on case marking

Information structure is claimed as a cause for variable case marking cross-linguistically (e.g. Aikhenvald 2010, Iemmolo 2010, Valle 2011, Dalrymple & Nikolaeva 2011). First, a large body of typological research links differential object marking (DOM) to disambiguation: typical objects which cannot be confused for subjects are less likely to receive specific grammatical marking than atypical ones, where such marking is essential for disambiguation (Silverstein 1976, Comrie 1977 *inter alia*). With respect to information structure, this would mean that topical objects need to be marked to avoid ambiguity with subjects, while objects occupying their typical slot as part of the pragmatic focus may remain unmarked. In some languages specific non-subject markers are used with highly topical objects dislocated from their default position (Iemmolo 2010, Aikhenvald 2003). Second, the degree of pragmatic prominence is another factor which may affect case marking: it has been shown for a number of DOM languages that more prominent objects with a higher pragmatic status are more likely to be marked for case than pragmatically less salient objects (Aissen 2003, Malchukov 2008).

It is useful to assess the South Slavic situation against this broad cross-linguistic picture. The data from Bulgarian border-region dialects provide no evidence for case marking being used as a tool in contexts where atypical objects (e.g. topicalised and dislocated to non-default positions or used with omitted governors) may be confused with subjects (see the analysis of the relationship between word order, pragmatic structure and case marking in Section 4.4). Thus, the situation in the border-region dialects is different, in broad terms, to a situation in some of the DOM languages (such as those considered in Aissen 2003, Malchukov 2008) where pragmatically more salient nouns attract case marking while less pragmatically salient nouns do not. In the Bulgarian dialects in question, case marking on non-subject NPs is conditioned, to a significant extent, by whether they are associated with background or new information in an utterance, or with the topic–focus distinction. However, a significant peculiar feature of these dialects is that specific non-subject case marking (accusative) is more frequent on nouns which are backgrounded. Nominative forms (which may be used in subject and non-subject position, as examples in (14) and (15) demonstrate) are more frequent in the pragmatically salient part of an utterance. We will consider this in greater detail in the next section. It should be noted that in qualifying parts of an utterance as, topic or focus we follow the tradition of defining topic as existing, background knowledge, or "common ground", and focus as part of an utterance which brings in new information and thus modifies the "common ground" (Hoop & Swart 2000, Krifka 2007). Since originally our data collection was not intended to test the impact of information structure on case marking, and such analysis was applied later, we used broad definitions based on this opposition, and excluded from the analysis utterances where a non-ambiguous definition of information structure was not possible. Under this view, three types of utterances required special decisions: utterances with contrastive topic as in (16), those with repeated focus, as in (17) and those with contrastive focus, as in (18).

(16) **Interviewer**:
Kak platiš na toj kojto gledal ovci celoto leto?
'How do you pay someone who looked after sheep the whole summer?'

Consultant:
Ne ima-l-o takâv. **U naš-a-ta**
NEG EXIST-PTCP-SG.N such at 1PL-NOM.SG-DEF.NOM.SG
bačij-a ne ima-l-o ovčar-e.
village.sheep-NOM.SG NEG EXIST-PTCP-SG.N shepherd-PL
'There was no such person. For our village sheep, there were no shepherds.' (contrastive topic)

(17) A bab-a mi ode-še… pase-še
and grandmother-nom.sg 1sg.dat go-impf.3sg graze-impf.3sg
krav-a… Ja njakolko pât otiš-l-a sâs nju*, ama
cow-nom.sg 1sg.nom several time go-ptcp-sg.f with 3sg.acc.f but
poveče ona si e pas-l-a **krav-u**.
more 3sg.nom.f part aux.prs.3sg graze-ptcp-sg.f cow-acc.sg
'And my grandmother used to go… grazed the cow. I went with her a few times but mostly she grazed the cow.' (repeated focus)

*Dialectal form used in the border-region and some other dialects corresponding with the Standard Bulgarian *neja* 'she [3sg.acc.f]' (cf. Todorov 2002: 82).

(18) **Interviewer:**
A kâde go slagate kopâra?
'Where (i.e. in which dishes) do you add dill?'

Consultant:
U tarator tura-m mnogo. Daže i **u salat-a** si tura-m.
in tarator[sg] put-1sg much even and in salad-nom.sg part put-1sg
'I put a lot of dill in tarator. I even put it in salad.' (contrastive focus)

Following the approach taken in this study, we grouped together utterances where non-subject nouns appear in the repeated focus, as in (17), and in the topic on the one hand, and those where non-subject nouns are part of the contrastive topic, as in (16), and of the focus. The rationale behind both decisions is based on information novelty. In the former case, information in the repeated focus is already brought into the discourse and cannot be defined as changing the 'common ground'. In the latter case, contrastive topic turns out to be very close to focus, in that the contrast itself contributes to existing knowledge and thus modifies the 'common ground'. The similarity of focus and contrastive topic has been demonstrated in a number of studies within Alternative Semantics, where both concepts are understood as parts of an utterance which provide new information by singling out existing alternatives (Rooth 1992: 75–77; Büring 2016: 64–68). A small number of examples were annotated as contrastive focus, as in (18). Contrastive focus is not different from focus in that it, like focus, provides new information and modifies the 'common ground', under some specific conditions such as polarity of unexpectedness (e. g. Zimmermann 2008: 348; Goodhue 2022: 117–126). For this reason, in the analysis presented in Section 4.4 we did not distinguish between the two types of focus. However, a possible effect of contrastive focus will be discussed separately in the account presented in Section 4.5.

We will consider the effect of information structure on non-subject case marking with respect to two other factors: word order and phrase type (NP vs. PP).

Previous research has suggested that the diachronic loss of case marking is accompanied by compensatory freezing of word order as a means of distinguishing grammatical roles (Blake 2004). Among other things, this implies that if an object noun occurs outside of its expected position, it will be more likely to preserve its original case marking. This straightforward complementarity between fixed word order and case marking does not necessarily hold (cf. Allen 2006 for Old English and Detges 2009 for Old French); in fact, the opposite correlation may occur. Data from Japanese show that word order alternations may loosen grammatical relationships within a sentence so that dislocated objects may be *less* likely to receive expected case marking than their counterparts in neutral positions within a basic constituent structure (cf. Japanese scrambling of accusative-marked objects to a position where only the nominative may be assigned, Kasai 2018).

The effect of phrase type (noun phrases vs. prepositional phrases) on case selection and decline of one of the competing cases was surveyed in Section 3. Here we will only reiterate the observation made above that the correlation between the use of prepositions and the tendency to maintain an original case form has been shown to be strong for the dialect types under study. Therefore, we found it important to disentangle different conditions on case marking in order to assess their effect individually without interference from other conditions. To do so, we will analyse the effect of topic-focus distinction separately for different types of word order and different phrase types (NP and PP).

4.4 The effect of topic and focus on case marking in Bulgarian border-region dialects

To investigate the effect of information structure on competing choices in case marking, a sub-corpus of approximately 90,000 words (six sociolinguistic interviews) was annotated with respect to information structure. The percentage of nominative on non-subjects in the six included interviews is as follows: 7% (BEL1-12), 20% (BEL1-15), 23% (BEL1-31), 42% (KS1-09), 55% (KS1-08), 57% (KS1-18). A dataset of 956 utterances containing inflection class II nouns in non-subject positions and allowing unambiguous assignment of the five values (topic, contrastive topic, focus, contrastive focus, repeated focus) was extracted. The data were combined for (i) topic and repeated focus, and for (ii) focus, contrastive focus and contrastive topic, as discussed in Section 4.3. The analysis presented in this section is based on these two subsets.

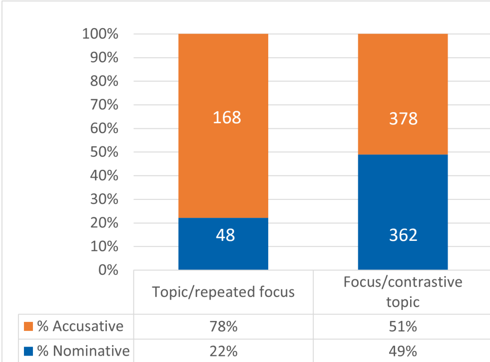


Figure 7: Nominative and accusative on non-subject nouns

As Figure 7 demonstrates, there is a significant difference in the frequency of the nominative and accusative across the two subsets. The accusative dominates in topic (19a) or repeated focus position (19b): 78% accusative forms. Frequency of the accusative falls significantly in focus (19c) and contrastive topic positions (19d): 51%. In other words, non-subject nouns are more likely to take the accusative in pragmatically less salient positions; pragmatically more salient parts of a sentence show a stronger preference for the nominative on non-subjects.

(19) a. I slam-a-ta otzad. I otvârlj-u
and straw-NOM.SG-DEF.NOM.SG behind and toss-PRS.3PL
slam-u-tu nastranu.
straw-ACC.SG-DEF.ACC.SG aside
'And the straw is behind. And they toss the straw aside.' (topic)

b. Čorapete ot **dvojk-a** si plete-mo.
socks from double.thread-NOM.SG PART weave-PRS.1PL
'We weave socks with double threads.' (focus)

I ot **dvojk-u** i ot trojk-u
and from double.thread-ACC.SG and from triple.thread-ACC.SG
sam ple-l-a.
be.PRS.1SG weave-PTCP-SG.F
'I have weaved both with double and triple threads.' (repeated focus)

c. Ako nema rek-a nosi-mo **vod-a**.
if EXIST.NEG river-NOM.SG carry-PRS.1PL water-NOM.SG
'If there is no river, we carry water.' (focus)

d. **Interviewer:**
S râka li pravete?
'Do you do it manually ("with a hand")?'

Consultant:
S râk-u, a s kvo? **S lev-a-ta**
with hand-ACC.SG and with what with left-NOM.F.SG-DEF.NOM.SG
tegli-š otgore, **s desn-a-ta**
pull-PRS.2SG from.above with right-NOM.F.SG-DEF.NOM.SG
vârti-š vreten-o-to.
rotate-PRS.2SG spindle-SG-DEF.SG.N
'Manually, how else? With your left hand you pull it from above, with your right hand you rotate the spindle.' (contrastive topic)

We assessed the effect of information structure on case marking in connection with two potentially relevant factors, phrase type (presence or absence of a preposition) and word order. As shown above, phrase type proved to have an effect on case marking in the Serbian dialects at an earlier stage of case decline (Section 3). Word order, although not independent of information structure, may still vary within the same topic–focus structure, therefore this factor was also disentangled from the other two factors under study. In Bulgarian, direct and indirect objects under basic default word order occur postverbally. Non-default ordering (e.g. preverbal objects) is motivated by a variety of conditions, which may or may not be related to information structure (Georgieva 1974, 1983, Dyer 1992). Thus, topicalised objects are normally fronted, but this is not a universal rule, and under appropriate contextual conditions topicalised objects may occur postverbally (20a).

(20) a. **Consultant 1:**
Kâde utiva-š?
Where go-PRS.2SG?

Consultant 2:
Po rabot-a, majk-o.
for work-NOM.SG mother-VOC

Consultant 1:
Ah, mrazj-a az **tazi rabot-a**!
ah hate-PRS.1SG 1SG.NOM this.NOM.SG.F work-NOM.SG
'C1: Where are you going? C2: On business, mother. C1: Oh I hate this business!'

On the other hand, as expected in SVO languages, objects in focus normally occur postverbally, but strong logical stress may result in emphatic word order, under which an object occurs in preverbal position (20b).

b. **Cjal-a-ta** **planin-a** vdišva-m.
whole-NOM.SG.F-DEF.NOM.SG.F mountain-NOM.SG inhale-PRS.1SG
'I inhale all this mountain range.' (adapted from Georgieva 1983: 284).

In addition to postverbal and preverbal ordering, we found a significant number of elliptic constructions, for example, objects with a missing governing verb, or interruptions where a governor and a governee stand far apart from each other in different syntagms. To see whether the absence of an overt governor has an effect on case marking, such phrases (tagged as "isolated") were considered separately.

Therefore, in order to disentangle the contribution of potentially conditioning factors, we assessed the effect of information structure separately for six different combinations of word order and phrase type:

(21) a. NP, postverbal
Ne smo slaga-l-i **vod-u**.
NEG AUX.PRS.1PL put-PTCP-PL water-ACC.SG
'We added no water.'

b. NP, preverbal
Krav-a smo ima-l-i.
cow-NOM.SG AUX.PRS.1PL have-PTCP-PL
'We had a cow.'

c. NP, isolated
Ne sâm ima-l-a bab-u, samo **majk-a**.
NEG AUX.1SG have-PTCP-SG.F grandmother-ACC.SG only mother-NOM.SG
'I didn't have a grandmother, just a mother.'

d. PP, postverbal
I ojde-mo večerom **na večerj-u**.
and go-PRS.1PL in.the.evening to dinner-ACC.SG
'And in the evening we will go to have dinner.'

e. PP, preverbal

Na	**bašt-a**	*mi*	pomaga-x.
to	father-NOM.SG	1SG.DAT	help-AOR.1SG

'I helped my father.'

f. PP, isolated

Interviewer:

Ima li češma?

'Is there a tap?'

Consultant:

Otzad	češm-a-ta.	**Na**	**kuxnićk-u**.
in.the.rear	tap-NOM.SG-DEF.NOM.SG	in	kitchen-ACC.SG

'The tap is in the rear. In the kitchen.'

Relative frequencies of the two alternative cases, nominative and accusative, on non-subject nouns with respect to the six combinations of conditions are presented in Figure 8 (for NPs) and Figure 9 (for PPs).

A key outcome of this analysis is the uniform pattern which we found across all six data subsets (factor combinations): while pragmatically more important parts of an utterance (focus or contrastive topic) demonstrate a strong tendency to generalize the nominative on non-subject nouns, the accusative as a distinct non-subject case is better preserved in pragmatically less salient positions (topic and repeated focus). In the light of the findings reported above for Serbian dialects, it is particularly striking that structural conditions, that is, the absence or presence of a preposition, seem to have no impact on speakers' choices, and neither enhance the use of a distinct non-subject case form (accusative), nor contribute to its loss and the generalisation of an undifferentiated morphological form (nominative). In other words, non-subject NPs and PPs in terms of their case marking show similar sensitivity to information structure. With respect to word order, a strong effect of information structure is found for all three linear structures: postverbal, preverbal and isolated positions. In all three data subsets, the accusative is better retained in topic and repeated focus position, and is less frequent in focus and contrastive topic position where it is replaced by the nominative.

The statistical significance of the results presented in Figure 8 and Figure 9 was assessed by ANOVA analysis. The effect of each of the three factors, phrase type, information structure and word order, was considered as a sole factor, and in connection with the other two factors (the null hypothesis assumed that there was no effect of either of these factors on case selection). The outcome of this analysis is presented in Table 3.

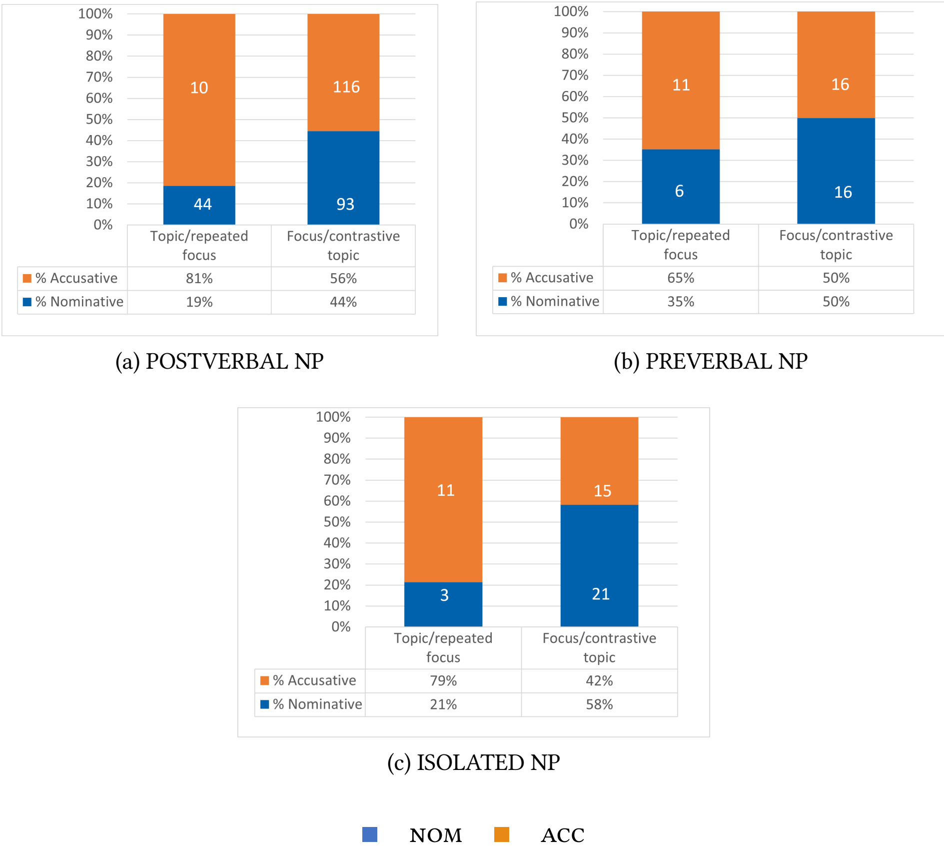
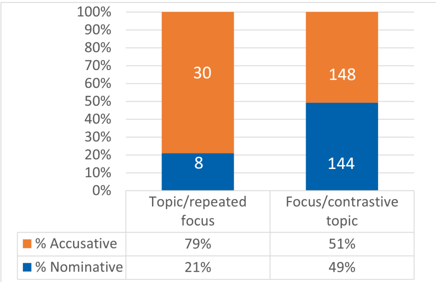
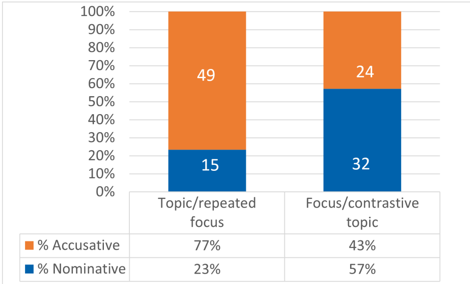


Figure 8: Preverbal, postverbal and isolated NPs in topic and focus positions

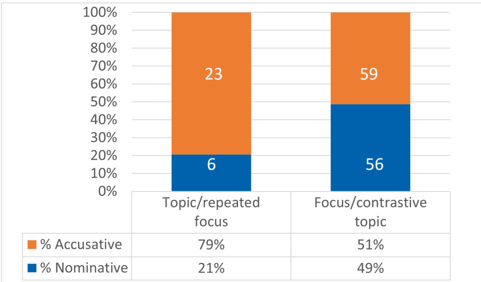
The results of the statistical analysis presented in Table 3 confirm the observation that phrase type (NP or PP), whether taken separately or in connection with word order and information structure, has no effect on case marking on non-subjects (NP: p=0.29 and PP: p=0.35, respectively). On the other hand, we see a strong effect of information structure, either considered on its own (as in Figure 7), or in connection with phrase type and word order, for the sub-sets presented in Figures 8 and 9 ($p < 0.001$ either taken separately or in connection with other conditions). We cannot, however, arrive at such an unequivocal conclusion on the role of word order. Within each of the six sub-sets presented in Figures 8 and 9, word order does not reach clear statistical significance (p=0.18). Taken as a sole factor without considering other factors (structure of the phrase and its affiliation with different pragmatic roles) postverbal positions seem to enhance the



(a) POSTVERBAL PP



(b) PREVERBAL PP



(c) ISOLATED PP

NOM ACC

Figure 9: Preverbal, postverbal and isolated NPs in topic and focus positions

use of the accusative, and preverbal positions favour the nominative ($p < 0.001$). However, it should be noted that the latter outcome may reflect a natural bias in the data. Non-subject postverbal focus NPs are more frequent than preverbal ones. Given that focus tends to attract nominative marking according to our data, this may significantly bias the analysis of the role of word order as a sole factor in case marking. Therefore, we think it important that word order, when taken as a factor in combination with information structure, fails to reach unambiguous statistically significant values. Firm conclusions, however, could only be reached from data normalised with respect to different types of word order and different pragmatic roles, which would require a larger corpus than we currently have.

4.5 Discussion: pragmatic factors in case decline

The reported findings leave little doubt that information structure has a crucial effect on variation in case marking in Bulgarian border-region dialects. What

Table 3: The effect of phrase type, word order and information structure on case marking in Bulgarian border-region dialects

Phrase type (NP vs. PP)	Word order	Information structure
NO EFFECT	WEAK EFFECT (?)	STRONG EFFECT
$p = 0.35$ in connection with WO and IS	$p = 0.18$ in connection with WO and phrase type	$p < 0.001$ under any condition(s) or taken separately
$p = 0.29$ taken separately	$p < 0.001$ taken separately	

might this synchronic variation tell us about the diachrony of case decline? We hypothesise that the historical competition of case marking here is motivated by pragmatic salience. What we observe, however, is not a tight coupling between a pragmatic role and grammatical form, but rather a tendency for caseless forms (historically the nominative) to be generalised first across the pragmatically most salient positions, and only then to spread further to other parts of the utterance. Key evidence supporting this view comes from the comparison of different parts of the subcorpus that have been annotated for information structure. Preferences for case marking on non-subject nouns in this subcorpus vary enormously across speakers, ranging from a total of 7% to 57% of nominative forms. We divided the subcorpus into two parts depending on the by-speaker frequency of nominative forms. The first part includes the three most conservative speakers (from 7% to 23% of nominative forms in non-subject positions). The second part includes interviews recorded from more innovative speakers with an overall frequency of nominative forms between 42% and 57% (T-test confirms the significance of the difference between the two groups with respect to case variation: $p<0.0001$, $t=8.4$).

The analysis presented below shows which parts of the information structure are most susceptible to the loss of case distinctions at a very early stage of this process. We looked at the data from the three most conservative speakers in order to identify conditions which make these speakers deviate from their dominant pattern, accusative marking of non-subjects. We assessed the impact of each pragmatic condition for which our data are annotated. Unlike the findings from the whole subcorpus analysed in the previous section (six interviews), our results here show that in the three most conservative individual dialects, the accusative is strongly associated not only with the background or repeated information (topic and repeated focus), but also with new information (focus). If, however, we look separately at the pragmatically most salient NPs and PPs, contrastive topic and contrastive focus, we find that these pragmatic conditions favour the nominative (Table 4, illustrated by examples in (22)).

Table 4: Distribution of nominative and accusative forms with non-subject nouns for the three most conservative individual dialects. Raw numbers and % of nominative

	accusative	nominative	% nominative
Topic	17	0	0%
Repeated focus	13	4	24%
Focus	104	18	15%
Contrastive topic	0	3	100%
Contrastive focus	0	2	100%
Total	134	27	17%

(22) a. **Interviewer**:
Kak platiš na toj kojto gledal ovci celoto leto?
'How do you pay someone who looked after sheep the whole summer?'

Consultant:
Ne ima-l-o takâv. U **naš-a-ta**
neg exist-ptcp-sg.n such at 1pl-nom.sg.f-def.nom.sg.f
bačij-a ne ima-l-o ovčar-e.
village.sheep-nom.sg neg exist-ptcp-sg.n shepherd-pl
'There was no such person. For our village sheep, there were no shepherds.' (contrastive topic)

b. **Interviewer**:
Da mu platiš posle?
'Do you pay him afterwards?'

Consultant:
Ne, da ne mu platiš.
'No, you do not pay him.'
Odi-l-i smo **na zared-a**.
go-ptcp-pl aux.prs.1pl on work.for.smb-nom.sg
'We went to work for him (in exchange for his favour).'
(contrastive focus)

Under non-contrastive focus we can expect the accusative (22c):

c. I posle ide-mo **za nevest-u-tu**.
and after go-PRS.1PL for bride-ACC.SG-DEF.ACC.SG
'And then we go to pick up the bride.'
(non-contrastive focus, listing consecutive actions)

The data from the three most innovative dialects reveal the spread of nominative from the most salient pragmatic conditions, such as contrastive topic and contrastive focus, to the parts of the utterance containing new information in general (as demonstrated in Table 5).

Table 5: Distribution of nominative and accusative forms with non-subject nouns for the three most innovative individual dialects. Raw numbers and % of nominative

	ACCUSATIVE	NOMINATIVE	% NOMINATIVE
Topic	70	33	32%
Repeated focus	69	10	13%
Focus	272	326	55%
Contrastive topic	1	4	80%
Contrastive focus	1	9	90%
Total	413	382	48%

The data for a subset of the most conservative speakers in the corpus, however small, may be taken as an indication of a diachronic tendency when compared to the whole corpus. While accusative dominates in the speech of the three conservative speakers overall, the pragmatically most salient parts of an utterance favour the nominative on non-subject nouns. This process has further developed with more innovative speakers with whom nominative has spread further into the focus position, where it dominates, and into topic position, though here it is still in the minority.

The fact that pragmatics play a crucial role in speakers' choices in Bulgarian border-region dialects may be due to both language-internal and external factors. As we noted earlier in Section 4, nouns of inflection classes I and II are the only domain where case distinctions are still preserved, while nouns of the other two classes (and adjectival forms which agree with them) are uninflected. This contributes to the marginalisation of case and restricts accusative to less salient positions. This process is supported by external factors, namely exposure to language varieties without case distinctions on a daily basis (younger speakers, TV, written language).This can marginalise accusative forms, restricting them to less highlighted parts of an utterance.

5 Conclusion

The ongoing loss of case in Serbian and Bulgarian dialects results in the competition between two forms, one of them more specific and one of them more general. How exactly this competition is manifested depends on the particular case system it occurs in. In more conservative varieties that retain in principle a six-case system (Serbian), competition is between a more specific case (e.g. the genitive) and a general oblique (accusative). In more innovative varieties (Bulgarian), where case loss is more advanced, the competition is between the general oblique form (accusative) and an uninflected stem. A particularly interesting result of our study has been to uncover key factors underlying these systems of competing case marking. Broadly construed, they match what has been observed cross-linguistically, but the principles that govern them are different. Elsewhere, prior work has shown that in case systems which allow for more explicit and less explicit marking for a given syntactic role, more explicit case marking has two major functions: (i) indicating the syntactic role of nominals in contexts where this is not otherwise obvious, and (ii) highlighting nominals that are semantically or pragmatically prominent. In the Serbian and Bulgarian dialects investigated here, the factors conditioning the competition of cases appear to be the exact opposite.

First, contexts where the syntactic role of a nominal is unambiguously signalled, as in prepositional phrases, in fact provide a more favourable environment for the preservation of the original case forms. We have illustrated this with the data from Serbian dialects where the original genitive case forms alternate with the innovative accusative forms. We analysed three construction types: prepositional constructions, adnominal constructions, and constructions with a quantifier. In all the dialects of the sample, prepositional constructions preserved the original case forms better than the other two. The three constructions vary with respect to the strength of connection between their elements. While the use of a noun (such as 'cup' or 'top') or a quantifier does not necessarily entail the subsequent use of a noun, a preposition obligatorily governs a noun phrase, marked with the genitive in our case. Thus, the established association of a form with its apparent context turns out to be a key factor which determines speakers' choices and contributes to the preservation of the original case forms.

Second, higher pragmatic salience, rather than triggering case distinctions, as has been widely observed cross-linguistically, in fact leads to the opposite result in the Bulgarian border-region dialects. These dialects retain at most two cases, nominative and accusative. The accusative is normally used for non-subjects, but this is not always so: they may also take the nominative, making them morphologically indistinguishable from subjects and thus effectively uninflected. This variation is conditioned by information structure: a subject (nominative) vs. non-subject (accusative) distinction is better maintained where background (given) information is being presented, while pragmatically salient parts of an utterance show a strong tendency to eliminate this distinction. Additional factors that play a role in limiting the use of the accusative are the fact that only two classes of nouns are affected (inflection classes I and II), and the overall marginalisation of case distinctions on nouns in the context of the prestige standard language, which lacks nominal case.

The fine-grained analysis of the competing patterns of case marking provided here contributes to our understanding of factors that condition the loss of inflectional case. In particular, different factors appear to be activated at different stages of the change. A synchronic cross-dialect analysis therefore allows us to uncover the intricate detail of this historical process.

Abbreviations

For interlinear glosses, we follow the conventions of the Leipzig Glossing Rules (https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf) with the following additions:

AOR	aorist
EXIST	existential
IMPF	imperfect

Acknowledgements

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Chapter 6

French nominal inflection and diachronic pathways to uninflectability

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This study addresses the question of whether uninflectability can arise in diachrony by mechanisms other than language contact. The study identifies loss of inflection as a logically possible source of *internal* uninflectability, and examines the well–documented loss of NUMBER and CASE inflection for French nouns as potential examples. While French nouns are shown to develop stable uninflectability for NUMBER as the majority inflectional pattern, the feature CASE is neutralised and then entirely lost, without uninflectability. The contrast between these examples indicates probable conditions on the emergence of uninflectability via loss of inflection.

1 Introduction

In synchrony, UNINFLECTABILITY is characterised as a lack of expected inflectional contrast for lexemes within a given word class: "[u]ninflectable lexemes can occur in all the morphosyntactic contexts open to inflecting lexemes (unlike defective lexemes), but their form is invariable" (Spencer 2020: 143). This paper adopts a diachronic perspective, exploring pathways through which uninflectability may be expected to emerge and conditions which promote or prevent its development.

I first review what is known about the diachronic development of more familiar inflectional phenomena which existing typological studies relate to uninflectability. Baerman et al. (2005) treat uninflectability (termed 'uninflectedness' in their monograph)[1] as a specific subtype of SYNCRETISM, while Spencer (2020)

[1]For consistency I will use the term UNINFLECTABILITY throughout.

Louise Esher. 2026. French nominal inflection and diachronic pathways to uninflectability. In Enrique L. Palancar & Sebastian Fedden (eds.), *Voids in morphology: Exploring "uninflectedness"*, 191–216. Berlin: Language Science Press. DOI: ??

contrasts uninflectability with DEFECTIVENESS (for which see Sims 2015). These comparators suggest initial expectations for the diachrony of uninflectability, particularly with regard to its emergence (Section 2).

I then examine two case studies illustrating reduction of nominal inflection in the history of French. Focusing on NUMBER (Section 3), I show that the majority of noun lexemes in modern standard French are uninflectable, and date this phenomenon to a seventeenth-century sound change (Pope 1934: 222–223) which creates syncretism between singular and plural forms. The conclusion to be drawn here is that uninflectability can readily arise within a small paradigm where there is minimal formal contrast between inflected forms.

Turning to CASE (Section 4), I show that these conditions nevertheless do not guarantee stable uninflectability. In early mediaeval French, a set of sound changes and analogical changes promote emergence of a declension class which is uninflectable for case, but these same changes also render the feature CASE syntactically irrelevant in the context of feminine gender. As the uninflectable declension class is entirely populated by feminine nouns, the result is the loss of the feature CASE from the inflectional paradigm of these lexemes, rather than stable uninflectability. The conclusion to be drawn here is that stable uninflectability is more likely where inflectional expression of a given feature differs robustly across multiple word classes.

Existing discussion of uninflectability focuses on examples involving language contact and the borrowing of lexical items (Baerman et al. 2005: 33; Spencer 2020). The case studies described here establish that alternative, internal pathways to uninflectability exist, subject to specific inflectional conditions (Section 5).

2 Insights from the diachrony of syncretism and defectiveness

2.1 Synchronic approaches to syncretism

The standard definition of syncretism in contemporary morphological theory, set out by Baerman et al. (2005), comprises three key properties:

> A morphological distinction which is syntactically relevant (i.e. it is an inflectional distinction)
>
> A failure to make this distinction under particular (morphological) conditions
>
> A resulting mismatch between syntax and morphology (Baerman et al. 2005: 2)

Illustrative classical Latin examples, which will be relevant to the French data discussed later in this paper, are shown in Table 1. A morphosyntactic distinction between dative and ablative values of the feature CASE is justified because the associated forms occur in contexts which are syntactically distinguishable and not interchangeable: dative forms are typically used to encode indirect objects (recipient, goal) and also appear as the complement of certain verbs, such as PLACET 'be pleasing to'; while ablative forms occur as the complement of many prepositions, such as CUM 'with', and to encode the agent in passive structures. However, the featural distinction does not always map to a distinction between phonological forms: in specific morphological contexts, namely in the presence of the value 'plural' for the morphosyntactic feature NUMBER across declensions, and in the presence of the value 'second declension' for the purely morphological (morphomic) feature INFLECTIONAL CLASS, the dative and ablative forms are of identical phonological shape. These forms thus instantiate a mismatch between syntax and morphology, or syncretism.

Table 1: Singular and plural forms of first–declension *rosa* 'rose', second–declension *murus* 'wall' and third–declension *pater* 'father' in classical Latin.

	rosa 'rose' I		murus 'wall' II		pater 'father' III	
	SG	PL	SG	PL	SG	PL
NOM	rosa	rosae	murus	murī	pater	patrēs
ACC	rosam	rosās	murum	murōs	patrem	patrēs
GEN	rosae	rosārum	murī	murōrum	patris	patrum
DAT	rosae	rosīs	murō	murīs	patrī	patribus
ABL	rosā	rosīs	murō	murīs	patre	patribus

The dative–ablative syncretisms qualify as examples of CANONICAL SYNCRETISM (Baerman et al. 2005: 34), which involves 'in certain contexts, a loss of distinctions between some but not all values of a particular feature F' where those values and F itself are still distinguished by other syntactic objects and thus remain syntactically relevant (Baerman et al. 2005: 34). Instances where distinction between all values of F is lost, and the feature is no longer syntactically relevant (i.e. visible on other syntactic objects) are classed instead as NEUTRALISATION (Baerman et al. 2005: 28–30); this would be the equivalent of having no morphosyntactic case distinctions in Latin. UNINFLECTABILITY is defined specifically as involving:

> in certain lexemes only, a loss of all values of a particular feature F found elsewhere in the language. [...] Other syntactic objects distinguish values of feature F, either generally or in the given context, and feature F is therefore syntactically relevant. (Baerman et al. 2005: 33)

In the Latin example, uninflectability would amount to having some nouns in which a single inflected form is used for all twelve combinations of case and number, while other nouns continue to maintain formal distinctions as shown in Table 1, and noun modifiers such as determiners and adjectives also express formal distinctions, whether the noun which they modify is uninflectable or not.

Under Baerman et al.'s (2005) definitions, uninflectability can be understood as massive, generalised syncretism: for a given inflectional feature, multiple feature values are visible in morphosyntax, but all feature values are expressed via an identical inflected form (as opposed to canonical syncretism, where only a subset of feature values are expressed via an identical form, other values being associated with distinct forms).

2.2 Sources of syncretism in diachrony

If uninflectability is understood as a subtype of syncretism, it is reasonable to expect that it may arise via the same routes as more familiar instances of syncretism.

The source of syncretism most commonly discussed in scholarly literature is sound change (see e.g. Meiser 1992: 190; Baerman et al. 2005: 5; Baerman 2008: 229; Esher forthcoming). Where inflectional forms were initially distinguished by a contrast between segments, a sound change which involves either neutralisation of the contrast, or deletion of the contrasting segments, is liable to cause syncretism. Both types can be simply illustrated from Latin first-declension nouns (Baerman et al. 2005: 5; Pinkster 1990, Sornicola 2011, Esher forthcoming). In classical Latin, for most first-declension nouns, the accusative singular was uniquely identifiable by the presence of final -m, and the ablative singular by final -ā (e.g. rosa 'rose(f).nom.sg', rosam 'rose(f).acc.sg', rosā 'rose(f).abl.sg' in Table 1). The subsequent deletion of final -m resulted in syncretism between the nominative singular and accusative singular for such nouns, while the loss of phonemic vowel length additionally caused syncretism of both these forms with the ablative singular (e.g. *rosa* 'rose(f).nom/acc/abl.sg').

Because the source of such changes is morphology-external, and their results depend upon the existing phonological shape and paradigmatic distribution of inflectional forms, they produce highly diverse and language-specific patterns of syncretism which may or may not align with natural classes of feature values (see Esher forthcoming). Once introduced, these patterns may nevertheless be reanalysed as a systematic property of the inflectional system, and replicated by analogy (Baerman et al. 2005: 10, 166–169; Hansson 2007; Baerman 2008: 230; Hinzelin 2011, 2012, 2022, Esher 2020).

More rarely, a syncretism pattern may arise via morphological analogy. Syncretism of present and imperfect subjunctive forms in the first and second person of some Occitan and Catalan varieties (e.g. *cantèssem* 'sing.PRS/IPF.SBJV.1PL' supplanting the inherited distinction between *cantem* 'sing.PRS.SBJV.1PL' and *cantèssem* 'sing.IPF.SBJV.1PL') is due to analogical changes which align multiple overlapping morphomic distribution patterns within the inflectional paradigm (Esher 2022). Syncretism may also occur as an incidental by-product of sporadic analogical changes (Esher 2024, 2025). Both these types are distinct in origin from morphological analogies which replicate existing syncretism patterns, such as the northern Occitan replacement of etymological *anem* /anã/ 'go.PRS.IND.1PL' by *vam* /vã/ 'go.PRS.IND.1PL' introduced from the third person plural form *van* /vã/ 'go.PRS.IND.3PL' on the model of an existing syncretism between first and third person plural present indicative forms, originating in sound change (Esher 2020).

It is a logical expectation that, like syncretism, uninflectability may arise either via sound change or via analogy; but that, as with syncretism, the sound change pathway is likely to be more commonly attested. The examples of uninflectability in French nominal inflection discussed in this paper illustrate the route via sound change.

2.3 Synchronic approaches to defectiveness

Theoretical analyses of inflectional defectiveness can be found in Baerman & Corbett (2010) and Sims (2015). Baerman & Corbett (2010: 1) adopt Matthews' (1997: 89) definition of defective lexemes as having a paradigm which "is incomplete in comparison with others of the major class that it belongs to", while Sims (2015: 26) develops a formal definition framed according to the conventions of Paradigm Function Morphology (Stump 2001, 2016):

> IF there exists a set of morphosyntactic and/or morphosemantic feature values F that is well-defined and morphologically encoded for at least one lexeme belonging to part of speech C;
>
> AND IF there exists a well–formed syntactic structure S that requires F in combination with some lexeme L belonging to C;

BUT any form of LC that is inserted into S produces an ungrammatical construction;

THEN the paradigm cell defined by <LC, [F]> is defective. (Sims 2015: 26)

In synchrony, typological approaches classify instances of defectiveness according to whether gaps occur at the level of form or of function (Baerman & Corbett 2010, Sims 2015). Gaps in the inventory of feature value sets may be arbitrarily specified (as in the lack of feminine indefinite patient forms for some Mohawk kinship terms, Sims 2015: 78–79), or reflect a lack of semantic need (as in the lack of first and second person forms for weather verbs and impersonal verbs, Baerman & Corbett 2010). Gaps in the inventory of inflected forms (the REALISED PARADIGM, in the terms of Stump 2016) are frequently associated with phonological infelicity of typical inflectional patterns when applied to a given lexeme (e.g. the lack of indefinite genitive singular forms for sibilant–final nouns in Swedish, Sims 2015: 60). Gaps in the inventory of forms may also match established morphomic patterns of paradigmatic distribution (e.g. the lack of all cells expected to share the 'infinitive stem' in some Finnish verbs, Baerman & Corbett 2010; see also Maiden & O'Neill 2010).

2.4 Sources of defectiveness in diachrony

Baerman & Corbett (2010) propose two separate pathways for the emergence of defectiveness: ARRESTED DEVELOPMENT, and DECAY.

In arrested development, a lexeme arises with an incomplete set of inflected forms, but 'the expected expansion of forms to the whole paradigm stops short' (2010: 11). Arrested development typically originates in borrowing or derivation (Baerman & Corbett 2010: 12). Expansion through the paradigm is constrained by phonological shape and existing paradigmatic distribution patterns: for example, in French, the conversion of *dénué* 'devoid', *dévolu* 'allotted' and *contrit* 'contrite' from adjectives to past participles generates all periphrastic forms comprising auxiliary and past participle, but only *dénué* develops an infinitive form (and ultimately a complete paradigm) since it matches the productive first-conjugation pattern of syncretism between the past participle and the infinitive (Bach & Esher 2015: 145–146). Arrested development may also affect previously complete lexemes if syntactic or semantic changes expand the content paradigm of the lexeme or of the word class to which it belongs (Baerman & Corbett 2010: 12–13; see also Sims 2015: 63–73).

The phenomenon of arrested development offers an interesting comparison with the examples of uninflectability given by Spencer (2020), which involve loanwords with a phonological shape that precludes their integration into existing inflectional patterns. Both phenomena can be understood as adaptive strategies for dealing with a novel lexeme of awkward phonological shape: in defectiveness, speakers' response is to fill only those cells for which a phonologically acceptable realised form is available and satisfies existing inflectional patterns; in uninflectability, speakers' response is instead to fill all cells with an invariant form, creating a novel inflectional class.[2]

The intersection between uninflectability and defectiveness in the case of arrested development intimates that it is also worth entertaining decay as a possible source of uninflectability.

In decay, a lexeme originates with a full array of inflected forms and feature value sets, some of which progressively cease to be used (Baerman & Corbett 2010: 14). Many of the defective French verb lexemes listed by Grévisse & Goosse (1991: 1273–1287) result from decay: for example, in modern French, *ardre* 'burn' and *choir* 'fall' retain few forms other than the infinitive, but their Latin etyma *ardere, CADERE, and the mediaeval French forms of these verbs are known to have been complete, i.e. non-defective (Buridant 2019: 392). Baerman & Corbett (2010: 14–15) suggest multiple routes to decay, including SPONTANEOUS LOSS, in which specific inflected forms are replaced by an alternative construction rather than novel synthetic forms for the same lexeme, and FUNCTIONAL REANALYSIS, in which feature-value sets are lost from the paradigm altogether.

Exploring the possible parallel between uninflectability and decay, Sections 3 and 4 investigate case studies involving loss of inflection (loss of feature-value sets from the content paradigm; loss of distinct forms from the realised paradigm; or both) and assess how such loss may contribute to the development of uninflectability.

3 Uninflectability and the feature NUMBER in the history of French

3.1 The two–cell content paradigm of modern French nouns

In modern standard French, the only morphosyntactic feature for which nouns inflect is NUMBER (Bach 2024; for morphosyntactic features including NUMBER,

[2] I diverge from Spencer (2020) in assuming that uninflectable lexemes have a form paradigm which stipulates a set of cells to be filled with an identical form, rather than lacking a form paradigm and defaulting to a stipulated "root" form.

see Corbett 2012: 49–50, 73–74, 121).[3] This feature has two values, singular and plural, and thus French nouns canonically have a content paradigm comprising only two cells, as shown in Table 2 following the formalism developed by Stump (2016: 103–115).

Table 2: Content paradigm of Fr. *livre* 'book'.

<LIVRE, sg>
<LIVRE, pl>

The contrast between singular and plural values of number on nouns is discernable via agreement patterns (for which see Corbett 2006: 4–5, 35, 40–43). In example (1),[4] the forms of agreement targets such as the demonstrative determiner *ce* 'this', the copula *être* 'be', and the adjective *original* 'original' (though not *fragile* 'fragile', to which I return below in Section 4.4), all vary according to the number value of the agreement controller, the noun *journal* 'newspaper'. Contextual realisations of plural nouns preceding a vowel–initial adjective (i.e. VARIABLE LIAISON) will be discussed in Section 3.5.

(1) a.

Ce	journal	**original**	**est**	fragile.
sə	ʒuʁnal	oʁiʒinal	ɛ	fʁaʒil
this.M.SG	newspaper(M).SG	original.M.SG	is	fragile.M.SG

'This original newspaper is fragile.'

b.

Ces	journaux	**originaux**	**sont**	fragiles.
se	ʒuʁno	oʁiʒino	sɔ̃	fʁaʒil
these.M.PL	newspaper(M).PL	original.M.PL	are	fragile.M.PL

'These original newspapers are fragile.'

The same morphosyntactic contrast remains even when the noun itself does not show any contrast in inflectional form correlated with number value (see

[3] Nouns in French do not inflect for grammatical GENDER (Bach 2024); instead, gender on nouns is a lexically stipulated property. Each noun in French is assigned to one or other gender class, either masculine or feminine, and acts as a controller for gender on agreement targets, principally adjectives and determiners (see also Corbett 2012, Bach & Esher 2023).

[4] Constructed examples based on the sentence *Tous ces documents originaux sont fragiles et leur préservation constitue un enjeu fondamental du métier d'archiviste* 'All these original documents are fragile and conserving them is a fundamental role for professional archivists', taken from an advert for a European Heritage Days event held in Rennes, France, on 17 September 2023, https://www.infolocale.fr/evenements/evenement-rennes-exposition-musee-prendre-soin-des-documents-707121907, accessed 17 November 2023.

also Loporcaro 2011: 329), as in example (2) for *document* 'document'. It is this contrast which motivates the assumption of a content paradigm with two cells.

(2) a.

Ce	document	**original**	**est**	fragile.
sə	dokymɑ̃	oʁiʒinal	ɛ	fʁaʒil
this.M.SG	document(M).SG	original.M.SG	is	fragile.M.SG

'This original document is fragile.'

b.

Ces	documents	**originaux**	**sont**	fragiles.
se	dokymɑ̃	oʁiʒino	sɔ̃	fʁaʒil
these.M.PL	document(M).PL	original.M.PL	are	fragile.M.PL

'These original documents are fragile.'

For nouns such as *document*, there is systematic syncretism between the singular and plural forms, which is to say between all cells of the realised paradigm. As the paradigm thus has only a single, invariant form, such nouns qualify as UNINFLECTABLE in the sense of Spencer (2020).

3.2 Uninflectable nouns constitute the majority inflectional class in modern French

The contrast between inflecting nouns, such as *journal*, and uninflectable nouns such as *document*, is lexically specified: it is an inflectional class contrast which may be formalised as a morphological feature in the terms of Corbett (2012: 50–61).

In modern standard French, uninflectable nouns constitute the majority inflectional class. The extent of this inflectional class can be estimated from the figures given by the *Grande Grammaire du Français* (Abeillé & Godard 2021), based on the database *Lexique.org* (New et al. 2004). Excluding compound nouns, and phenomena such as LEXICAL PLURALS[5] (Acquaviva 2008), leaves 17,751 simplex nouns the paradigms of which are expected to contain forms realising the values 'singular' and 'plural' for the morphosyntactic feature NUMBER. Of these, only 229 lexemes have distinct singular and plural realised forms: the remainder, 98.7%,

[5]A familiar French example of a lexical plural, also termed PLURALIA TANTUM, is the lexeme *ciseaux* /sizo/ 'scissors'. The inflectional paradigm of *ciseaux* 'scissors' does not contain a morphosyntactically singular form. Its only available form is morphosyntactically plural (i.e. controls plural on agreement targets) and this form is used whether the intended referent is semantically singular or plural: *des ciseaux* 'a pair of scissors, several pairs of scissors'. By contrast, both values of morphosyntactic number are available in the paradigm of the separate lexeme *ciseau* /sizo/ 'chisel': *un ciseau* 'a chisel', *des ciseaux* 'chisels'.

with identical singular and plural realised forms, are uninflectable (Abeillé et al. 2021).

Lexemes with distinct singular and plural forms instantiate diverse inflectional alternation patterns, including contrasts between affixal material, as in *journal* /ʒuʁnal/ 'newspaper', *journaux* /ʒuʁno/ 'newspapers' and *vitrail* /vitʁaj/ 'stained glass window', *vitraux* /vitʁo/ 'stained glass windows'; contrasts between presence and absence of material, as in *bœuf* /bœf/ 'ox', *bœufs* /bø/ 'oxen' and *os* /ɔs/ 'bone', *os* /o/ 'bones'; and contrasts between suppletive forms, as in *œil* /œj/ 'eye', *yeux* /jø/ 'eyes' (see Abeillé et al. 2021 for an overview of patterns and their lexical incidence). For some lexemes, there is evidence for variation between singular/plural contrast and singular–plural syncretism; speakers may prefer one or other pattern in general, or associate the distinctive plural with specific contexts or registers, often fossilised expressions (see Abeillé et al. 2021 for examples). Such variation is indicative of a progressive trend favouring the uninflectable pattern.

Figures for compound nouns are not available, but it can be assumed that uninflectable patterns remain an overwhelming majority, since the only compounds for which distinct singular and plural forms are available are those involving one of the few base lexemes which have distinct forms.

3.3 Exemplifying the distinction between the canonical and the typical

Throughout the development of the Canonical Typology approach, scholars have consistently stressed that canonical inflectional phenomena represent a theoretical absolute which may not necessarily be encountered in real–life inflectional systems (Corbett 2007, 2009, 2023, Brown & Chumakina 2013, Corbett & Fedden 2016, Audring 2019, Bond 2019):

> The canon is where we calibrate from, just as we measure length from zero metres, and temperature from zero kelvins. The canon is not what is frequent, functional, unmarked, or prototypical. No value–judgement is attached: zero metres is not a good length for a typical object, and zero kelvins is not a desirable temperature, but each is a good point to measure from.
>
> (Corbett 2023: 188)

The inflectional behaviour of French noun lexemes serves as an excellent illustration of this principle. Among the expectations of a canonical lexeme is that "every cell in its paradigm will realize the morphosyntactic specification in a way distinct from that of every other cell" (Corbett 2009: 2); uninflectable lexemes behave non–canonically in having an identical inflectional form for all paradigm cells. Uninflectability occurs in the vast majority of French nouns, including many of the highest token frequency nouns and those referring to basic concepts such as kinship terms or parts of the body. Indeed, even if all noun lexemes in French were uninflectable without exception, the phenomenon would remain non–canonical since the essential facts about the paradigm structure of French nouns remain unchanged: number is a syntactically relevant feature with two values which are visible in agreement phenomena, and which would remain so visible even if no lexical noun exhibited an overt distinction between the inflectional forms realising singular and plural.

3.4 Uninflectability on French adjectives

A clarification is necessary with respect to examples (1) and (2), which illustrate a contrast between inflecting and uninflectable adjectives. As with nouns, uninflectability is a lexically specified property of adjectives, and is not related to syntactic structure. Reversing the position of *original* and *fragile* (3) shows clearly that, whether these adjectives appear in attributive or predicative position, *original* always presents distinct singular and plural forms, while *fragile* is consistently uninflectable for number. Thus, French adjectives do not instantiate the phenomenon of constructional uninflectability described by Spencer (2020: 148–150) for German.

(3) a.

Ce	document	fragile	**est**	**original**.
sə	dokymɑ̃	fʁaʒil	ɛ	oʁiʒinal
this.M.SG	document(M).SG	fragile.M.SG	is	original.M.SG

'This fragile document is original.'

b.

Ces	documents	fragiles	**sont**	**originaux**.
se	dokymɑ̃	fʁaʒil	sɔ̃	oʁiʒino
these.M.PL	document(M).PL	fragile.M.PL	are	original.M.PL

'These fragile documents are original.'

The inflectional paradigm of adjectives in French is overall richer and more complex than that of nouns, due to involving a greater number of morphological features. The two basic morphosyntactic features which define the content paradigm of a French adjective are NUMBER (singular/plural) and GENDER (masculine/feminine). Furthermore, for at least some adjective lexemes, additional paradigm cells must be posited to account for inflectional contrasts dependent on syntactic position (see Bonami & Boyé 2005 and Bach accepted).

3.5 Liaison and uninflectability

A brief clarification regarding the phenomenon of VARIABLE LIAISON is also useful. Liaison is characterised as follows by Durand et al. (2011), in an article surveying the wide range of theoretical treatments which have been proposed:

> La liaison [...] conduit à ce qu'un certain nombre de mots apparaissent sous deux formes, une forme courte devant une initiale consonantique et, mais seulement parfois, une forme longue où une consonne finale de liaison s'ajoute devant un mot à initiale vocalique.
>
> (Durand et al. 2011: 105–106)
>
> 'Liaison [...] leads to a number of words appearing in two forms, a short form before a consonant–initial word, and, only in some instances, a long form in which a final liaison consonant is added before a vowel–initial word'.
>
> [my translation]

An example of potential contrast between realisations with and without liaison is shown in (4):

(4) a. without liaison
ces journaux originaux, ces documents originaux
se ʒuʁno oʁiʒino se dokymɑ̃ oʁiʒino

b. with liaison
ces journaux originaux, ces documents originaux
se ʒuʁnoz oʁiʒino se dokymɑ̃z oʁiʒino

Historically, liaison is a relic of the context-sensitive deletion of final consonants, which began in environments before a consonant-final word or a pause, and only later progressed to environments before a vowel-initial word (Pope 1934: 222–224; see also Section 3.6). In synchrony, it is traditionally treated as a phonological phenomenon of truncation, latency or epenthesis (see Durand et al. 2011 for discussion); more rarely, liaison has been integrated into morphological descriptions as a phenomenon of inflectional alternation (e.g. Bonami & Boyé 2005). Scholarly interest centres on the highly variable occurrence of liaison, which is sensitive to sociolinguistic and stylistic factors as well as syntactic cohesion and the frequency of a given construction (Durand et al. 2011, Durand & Lyche 2016). Variationist corpus analysis shows that, although liaison is licit in some contexts following plural nouns, such as the sequence of plural noun and vowel–initial adjective, occurrences of liaison in these contexts are vanishingly rare in spontaneous speech (Durand et al. 2011, Durand & Lyche 2016).

Under an analysis which integrates liaison forms into the inflectional paradigm of the word preceding the liaison context, French nouns such as *document* cannot be considered uninflectable for number since they show a marginal formal contrast mapping to a morphosyntactic contrast. However, the complex interaction of extramorphological factors governing liaison, together with the very low rate of occurrence for liaison following plural nouns in spontaneous speech, cast doubt on the empirical representativity of such an analysis, and thus this paper will not consider liaison forms to occupy a cell in the canonical inflectional paradigm of French nouns.

3.6 The historical source of uninflectability in French nouns

The loss of contrast between singular and plural realised forms of most French nouns is traceable to a single articulatory sound change, the deletion of final /s/. This change begins during the sixteenth century and is largely complete by the mid–seventeenth century (Pope 1934: 222–223).

The change itself does not automatically produce uninflectability. Instead, its inflectional consequences depend upon the wider context of the inflectional system. For nouns such as *mur* 'wall' < MURUM, in which number contrast hinged solely on the contrast between presence of final /s/ in the plural *murs*, and absence of /s/ in the singular *mur*, the deletion of final /s/ introduces syncretism between singular and plural forms. For other nouns, in which additional inflectional contrasts were correlated with number, the same sound change historically applied, but without causing syncretism. For example, for nouns such as *cheval* 'horse' < CABALLUM, singular and plural forms historically diverged due to the twelfth–century vocalisation of coda /l/ preceding a consonant, and the sixteenth–century monopthongisation of /aw/ to /o/; compare singular /tʃeval/ > /ʃeval/ > /ʃəval/ with plural /tʃevals/ > / tʃevaws/ > /ʃəvos/ > /ʃəvo/ (Pope 1934: 153–155, 199–200, 246–247).

Because the nominal paradigm only contains two cells, any changes producing syncretism between the forms which realise these two cells will also result in uninflectability. Adjective inflection is a useful comparator here. The loss of final /s/ equally affects adjectives, introducing syncretism between singular and plural forms in many lexemes (uninflectability for number). In adjectives such as *frêle* 'frail' < FRAGILEM which exhibited syncretism between masculine and feminine forms in either number value, the loss of final /s/ also produces overall uninflectability for the lexeme; whereas in adjectives such as *vif* 'lively' < UĪUUM which retained a contrast between the forms realising masculine and feminine values of gender in either number value, the lexeme continues to inflect for gender (Table 3).

Table 3: Inflection of *frêle* 'frail' and *vif* 'lively' in modern French.

	frêle 'frail'		vif 'lively'	
	SG	PL	SG	PL
M	frêle /fʁɛl/	frêles /fʁɛl/	vif /vif/	vifs /vif/
F	frêle /fʁɛl/	frêles /fʁɛl/	vive /viv/	vives /viv/

4 The feature CASE in the history of French

4.1 Loss of inflection and loss of uninflectability

The reduction and ultimate disappearance of case inflection in French has long formed the subject of detailed historical linguistic investigations (Schøsler 1984, 2013, van Reenen & Schøsler 2000; Sornicola 2011: 18–35; Smith 2011: 281–289; Ledgeway 2012, Buridant 2019, Sims-Williams & Baerman 2021). Beyond documenting the formal and functional changes in inflection, a key concern of many studies is the functional load of inflectional case distinctions in the context of word order and typological shift. This paper focuses instead on the formal analysis of the inflectional system.

As set out in Section 2.1, nouns in classical Latin have multiple values of the feature CASE, which are syntactically relevant and visible via agreement marking (Corbett 2006). Most scholars agree upon the existence of five distinct values (nominative, accusative, genitive, dative, ablative; Dragomirescu & Nicolae 2016); for a possible sixth category, the vocative, see Ashdowne (2016).

By contrast, in modern standard French, no case inflection is discernable on nouns or on the agreement targets which they control, as shown in example (5)[6] where the lexeme *mur* 'wall' appears in the same inflected form whether it is a subject (5a, 5d), a direct object (5b, 5e) or the complement of a preposition (5c, 5f); in the singular, the definite and indefinite articles have the forms *le* and *un*

[6]Examples taken from an online news article published 8 November 2019, https://www.franceinfo.fr/replay-radio/un-monde-d-avance/dans-le-monde-les-murs-se-multiplient_3673969.html, accessed 17 November 2023.

respectively, while in the plural, the definite article always has the form *les*; and the forms of adjectives and participles, although not fully exemplified here for reasons of space, likewise do not vary according to the case of the noun which they modify.

(5) a. Le mur de Berlin est tombé.
the.M.SG wall(M).SG of Berlin be.PRS.IND.3SG fall.PST.PTCP.M.SG
'The Berlin Wall fell.'

b. construire un mur
build.INF a.M.SG wall(M).SG
'build a wall'

c. avec le mur des Etats–Unis
with the.M.SG wall(M).SG of_the United States
'with the United States' wall'

d. Les murs se multiplient.
the.M.PL wall(M).PL REFL multiply.PRS.IND.3PL
'Walls are multiplying [i.e. becoming more numerous].'

e. pour contourner les murs existants
to circumvent.INF the.M.PL wall(M).PL existing.M.PL
'to circumvent existing walls'

f. comme les murs entre quartiers
like the.M.PL wall(M).PL between neighbourhood(M).PL
'like the walls between neighbourhoods'

Because the feature CASE is syntactically irrelevant for modern French nouns (i.e. French nouns do not inflect for case), this feature does not participate in the definition of the content paradigm. Thus, for modern French nouns, it would be incorrect to speak of uninflectability with respect to case. There is no expectation that any French noun should inflect for case; the feature is simply absent.

In existing literature, the historical trajectory from the five–case system of Latin to the absence of case on modern French nouns, via the well–attested intermediate two–case system of mediaeval French, is studied as an example of the gradual loss of inflected forms, of inflectional feature values and ultimately of the inflectional feature itself. Along the way, it is possible to observe both emergence and subsequent loss of uninflectability for case.

4.2 Development of the two–case system in mediaeval French

For detailed consideration of changes in noun inflection between Latin and mediaeval French, including the deletion of final consonants, the neutralisation of vowel length, assimilation of the minor declension classes IV and V to the more populous classes, and the shift of neuter plurals to the first declension, the reader is referred to the descriptions cited in Section 4.1. For the purposes of the present paper it will be sufficient to consider the three major declensional types illustrated in Table 1.

In classical Latin, all three major inflectional classes for nouns exhibit 7 to 8 distinct forms in the realised paradigm, distributed across 10 sets of feature values in the content paradigm (singular and plural NUMBER; nominative, accusative, genitive, dative and ablative CASE). Between this period and early French, syntactic changes and sound changes lead to a reduction in both the number of feature value sets in the content paradigm, and the number of distinct forms in the realised paradigm. The genitive and dative cases are largely supplanted by constructions comprising a preposition and the ablative or accusative, removing four cells from the content paradigm, while sound changes cause syncretism between accusative and ablative or accusative and nominative, and analogical changes replicate such syncretism. Many of these changes fall under Baerman's (2008) category of reassignment of case function.

Importantly, these changes affect not only the nouns which are the principal focus of this study, but also the agreement targets of nouns, principally adjectives, determiners and quantifiers. The loss of formal distinctions between accusative and ablative forms on other syntactic objects compromises the syntactic relevance of a distinction between the two case values altogether.

By mediaeval French of the twelfth century, only two case values are reliably distinguishable, traditionally termed NOMINATIVE (reflex of Latin nominative) and OBLIQUE (merger of Latin accusative and ablative).

4.3 Gender and inflectional class in the mediaeval French two–case system

By the twelfth century, the major inflectional classes for nouns have four-cell content paradigms with realised forms as shown in Table 4.

The *rose* class, continuing Latin first-declension nouns, has two distinct realised forms, with case syncretism in each number value. In the singular this is a direct consequence of regular sound change (see Section 2.2); in the plural, *roses* is the expected reflex of accusative ROSĀS, and replaces expected *rose < ROSAE in the nominative due to analogical changes in late Latin (see e.g. Schøsler 2013).

Table 4: Singular and plural forms of first–declension *rose* 'rose', second–declension *murs* 'wall', third–declension *flor* 'flower', *pedre* 'father' and *cons* 'count' in twelfth–century French.

	rose 'rose'	murs 'wall'	flor 'flower'	pedre 'father'	cons 'count'
NOM.SG	rose	murs	flor(s)	pedre(s)	cons
OBL.SG	rose	mur	flor	pedre	conte
NOM.PL	roses	mur	flors	pedre	conte
OBL.PL	roses	murs	flors	pedres	contes

The *murs* class, continuing Latin second-declension nouns, also has two distinct realised forms, distributed in a pattern which is entirely due to regular sound changes. Nominative singular MURUS and accusative plural MURŌS both develop to *murs*, while nominative plural MURĪ and accusative singular MURUM both yield *mur*.

Reflexes of third-declension nouns are more diverse. The majority develop according to regular sound change, but are then subject to analogy from the first and second declensions; feminine nouns such as *flors* 'flower' < FLORIS tend to gravitate towards the *rose* class, losing etymological final -*s* in the nominative singular, while masculine nouns such as *pedre* 'father' < PATER gravitate towards the *murs* class, gaining analogical final -*s* in the nominative singular and losing etymological final -*s* in the accusative plural (Sims-Williams & Baerman 2021). A small number of IMPARISYLLABIC nouns, most of which are of masculine gender, retain etymological stem alternations reflecting a difference of syllables between the nominative and all other forms in Latin, e.g. *cons* 'count.NOM.SG' < COMES 'companion.NOM.SG', *conte* 'count.OBL.SG' < COMITEM 'companion.ACC.SG'; thus, case inflection for imparisyllabic nouns involves stem contrasts in addition to the presence or absence of final /s/.

Note that traditional descriptions of mediaeval French declension, such as Buridant (2019), commonly conflate inflectional class and gender: for example, the *rose* class is termed *Type I feminine* while nouns such as *flor(s)* are termed *Type II feminine*. This classification is etymological in origin (Type I feminines continue the Latin first declension whereas Type II feminines are reflexes of third-declension nouns), and may also be motivated by the strong correlation between gender and inflectional class in the reflexes of Latin first- and second-declension nouns. Nevertheless, gender and inflectional class are qualitatively distinct: genders are agreement classes relating to morphosyntactic behaviour, while inflectional classes are morphomic classes relating purely to inflectional forms (see e.g. Corbett 2012). From the point of view of inflectional class, nouns such as *flor(s)* and *pedre(s)* must be classified depending on their realisation, as belonging to a distinctive class with final *-s* in all forms except oblique singular, belonging to the *rose* class, or belonging to the *murs* class.

4.4 Agreement, GENDER, and the morphosyntactic relevance of CASE

The discussion in Section 4.3 makes the canonical assumption that lexemes instantiating the same word class should have the same shape paradigm (see e.g. Corbett 2009). Under this assumption, mediaeval French nouns of the *rose* type have a content paradigm of four cells, but make no formal distinction between the realised forms associated with nominative and oblique case. Yet for the *rose* type to be classified as illustrating case syncretism or even uninflectability for case, the feature CASE must be demonstrably relevant to syntax.

This question is more interesting than it appears, because all nouns of the *rose* type are of feminine gender, and the morphosyntactic relevance of case in mediaeval French is dependent upon gender value. For masculine nouns, distinct values of the feature CASE are visible via agreement phenomena as well as mapping to inflectional alternations; but for feminine nouns, the vast majority of agreement targets do not make a formal distinction between nominative and oblique case (see Buridant 2019: 75–78, 139–142, 174–179, 287–298, for the inflection of nouns, adjectives and determiners). In the examples under (6), from the thirteenth-century text *La Châtelaine de Vergy*, the agreement targets of masculine *chevaliers* 'knight' vary with case: nominative singular *li* and oblique singular *le* contrast in the definite article, while adjectives typically present final *-s* in the nominative singular but not in the oblique singular. By contrast, the examples under (7), from the same text, show no discernable contrast between nominative and oblique agreement targets of feminine *dame* 'lady': the definite article is invariable for case, whether singular *la* or plural *les*, and adjectives have no distinctive inflection in either nominative or oblique.

(6) a. En un vergier li chevaliers toz jors vendroit.
in an orchard the.M.NOM.SG knight(M).NOM.SG all days would_come
'Every day the knight would come into an orchard.'

b. Li chevaliers fu biaus et cointes.
the.M.NOM.SG knight(M).NOM.SG was fine.M.NOM.SG and distinguished.M.NOM.SG
'The knight was handsome and distinguished.'

c. Il trova le chevalier.
he found the.M.OBL.SG knight(M).OBL.SG
'He found the knight.'

d. d'un chevalier preu et hardi
of=a knight(M).OBL.SG noble.M.OBL.SG and brave.M.OBL.SG
'of a brave and noble knight'

(7) a. Sui haute dame honoree.
I.am noble.F.NOM.SG lady(F).NOM.SG honoured.F.NOM.SG
'I am a lady of high degree.'

b. d'amer dame si souveraine
to=love lady(F).OBL.SG so sovereign.F.OBL.SG
'to love such a noble lady'

c. La dame li otria [...] s'amor.
the.F.NOM.SG lady(F).NOM.SG to_him granted [...] her=love(F).OBL.SG
'The lady granted him [...] her love.'

d. fors la dame
except the.F.OBL.SG lady(F).OBL.SG
'except the lady'

e. Les dames ont oï le conte.
the.NOM.PL lady(F).NOM.PL have heard the.M.OBL.SG tale(M).OBL.SG
'The ladies have heard the tale.'

f. querre toutes les dames de la terre
to_fetch all.F.OBL.PL the.F.OBL.PL lady(F).OBL.PL of the.F.OBL.SG land(F).OBL.SG
'to fetch all the ladies of the land'

Within this system, case is syntactically relevant for masculine nouns, but not for feminine nouns: in the terms of Baerman et al. (2005: 28–30), case is NEUTRALISED in the context of feminine gender.[7]

[7] Buridant (2019) notes a small number of adjectives such as *loiaus* 'loyal' < LEGALIS and *granz* 'big' < GRANDIS which for etymological reasons originate with a contrast between nominative (final /s/) and oblique (no final /s/) forms for both masculine and feminine gender; these are rapidly assimilated to the majority types with case distinctions only for masculine gender. In *La Châtelaine de Vergy*, for example, only *grant* < GRANDEM 'big.F.ACC.SG' occurs in the feminine singular, whether nominative or oblique is expected.

In this historical example, uninflectability emerges on a knife–edge: the processes causing case syncretism in the *rose* inflectional class (all members of which are of feminine gender) also cause case syncretism for feminine gender values of many adjectives and determiners. In the *rose* class, though not for other inflectional classes, uninflectability may be diagnosable for a brief period during which case remains visible on a few lexically specified agreement targets. However, as case syncretism in the context of feminine gender spreads to these items too, and as nouns of the *flor* type are assimilated to the *rose* class, the syntactic relevance of case in the context of feminine gender overall reduces, becoming entirely neutralised. Because these changes compromise the relevance of the feature CASE for syntax, and thus its presence within the inflectional paradigm of the *rose* class, it is more realistic to consider that the content paradigm of feminine nouns reduces to two cells defined only by the contrast of singular and plural NUMBER, rather than continuing to assume a four–cell paradigm exhibiting uninflectability for case.

This example illustrates a near–miss. By definition, uninflectability requires the absence of (realised) inflectional contrasts associated with different values of a given feature F within the paradigm of a given word class C, combined with the presence of (realised) inflectional contrasts associated with the same values of F for other word classes D, E which also inflect for F; contrasts in D, E are necessary to maintain the syntactic relevance of F. There is greater potential for uninflectability to arise if C has a small paradigm with minimal phonological contrast between realised forms. But stable uninflectability in diachrony is only possible where C loses contrasts while D, E retain contrasts. In the mediaeval French example, the phonological substance and morphological distribution patterns of inflected forms for feminine nouns' agreement targets are so similar to those for nouns of the *rose* class that the changes which cause loss of contrast in the *rose* class correspondingly undermine contrast in agreement targets. The result is neutralisation instead of uninflectability. From this example it can be inferred that stable uninflectability is more likely where inflectional expression of a given feature differs robustly across multiple word classes.

4.5 Loss of CASE as an inflectional feature

During the fourteenth century, oblique forms progressively supplant nominative forms for nouns of all inflectional classes and genders (for the occasional retention and refunctionalisation of nominative/oblique doublets, see Smith 2011), and for their agreement targets. This is a further example of reassignment of case function, since all functions previously associated with the nominative are assumed by the oblique. In the resulting system, case is no longer discernable in morphosyntax, and thus its presence as a feature in the content paradigm can no longer be justified. The content paradigm of nouns reduces to two cells, defined by the single feature NUMBER (Section 3.1).

5 Conclusions

This paper offers a first approach to the question of whether uninflectability can arise in diachrony by mechanisms other than language contact.

By means of comparison with the better–studied non–canonical phenomena of syncretism and defectiveness, the paper identifies loss of inflection as a process in the course of which uninflectability may potentially arise. Two case studies involving loss of (noun) inflection in the well–described history of French are then selected for more detailed investigation.

The first example concerns the feature NUMBER in the post–mediaeval period. For this feature, the range of values discernable in morphosyntax remains constant: the content paradigm of nouns in classical and modern French comprises two cells corresponding to singular and plural number. The inventory of forms in the realised paradigm, however, reduces: in the vast majority of lexemes, where the phonological contrast between singular and plural forms rested on the presence or absence of a single segment, the loss of this contrast due to a single sound change produces syncretism between both (i.e. all) realised forms of the lexeme, and thus uninflectability.

The second example concerns the feature CASE in early and mediaeval French. Between Latin and mediaeval French, the inventory of morphosyntactic feature combinations in the content paradigm reduces as the number of distinct values of case decreases, and the inventory of realised forms also reduces due to sound change and loss of paradigm cells. The familiar two–case system of mediaeval French is characterised by a small paradigm and by formal alternations which involve minimal phonological contrast. Yet although syncretism appears rife within this system, careful examination of agreement targets reveals neutralisation of case rather than uninflectability. From this example a more constrained principle can be inferred: stable uninflectability is more likely to develop in small paradigms where the inflectional forms of a given lexeme are minimally different, but where inflectional expression of a given feature differs robustly across multiple word classes.

Overall, this preliminary study indicates that uninflectability can indeed emerge via internal pathways, but only under very restrictive inflectional circumstances.

Internal routes to uninflectability require a specific combination of properties regarding (i) paradigm structure in the class of lexemes under investigation, (ii) the substance of inflectional alternations in this class of lexemes, (iii) paradigm structure in the classes of lexemes which are agreement targets for the class of lexemes under investigation, and (iv) the substance of inflectional alternations in the classes of lexemes which are agreement targets. The multiplicity of narrow conditions which must align suggests that emergence of *internal* uninflectability is likely to be a crosslinguistically rare eventuality.

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Chapter 7

French *voilà/voici*

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The French lexical items *voilà* and *voici* do not inflect, which is surprising given their syntactic behavior. They act syntactically as verbs by taking complements, assigning accusative case, and even appearing as the sole element in a relative clause. Despite these verbal characteristics, *voilà*/*voici* remain uninflected. Historically, unlike what is often stated, corpus research shows that *voilà*/*voici* do not originate from the imperative singular of *veoir* ('to see'). Instead, they originate from a syntagma combining *veez* (2PL/HON.IND/IMP of 'to see') and respectively the locative adverbs *ci* ('here') and *là* ('there'). From there, they are gradually evolving towards a multiword construction displaying what Spencer (2020: 140) calls "constructional uninflectedness", the intermediate form *vez* losing its specification for number in this process. By the 16th century, the forms *voilà*/*voici* are created, and this via reanalysis and alignment to the general sound change affecting the verbal paradigm (from *veoir* [vəwɛr] to *voir* [vwar]). The morpheme *ve-* is thus eventually replaced by *voi-*, which is the verbal root.

1 Introduction

The two French lexemes *voilà* and *voici* do not inflect. At the same time, for various reasons mentioned below, it is obvious that they do not fall under traditional Romance non-inflecting word classes such as prepositions or conjunctions. Hence, since scholars began to investigate them more closely in the 20th century, they are often classified under the vague term of "presentational particles" (Petit 2010: 152). Considering their syntactic distribution, however, this attribution to a group whose members are by definition not able to build phrases (cf. Bußmann 2008: 509) does not seem to hold, since the various configurations in which they appear show a different picture – namely them constantly taking over the

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verbal position and therefore behaving like members of an inflecting word class par excellence. In the terms of Spencer (2020: 142), we are thus not facing "uninflecting" items, that is words "that belong to non-inflecting classes", but well "uninflectable" ones, where "all forms of the paradigm are identical".

In recent years, presentative clauses in general began to receive attention in the linguistic literature (f. ex. Julia 2018; Zanuttini 2017 for Italian; Bonilla Carvajal 2020 for Latin; Wood & Zanuttini 2023 for English). Crosslinguistically, they all present characteristics that look exceptional at first glance. Analyzing the French presentative with a focus on one of its most striking particularities, the uninflectability of the presentative element, thus presents a promising approach to the phenomenon of such clauses. At the same time, a detailed case study of a specific lexical element becoming uninflectable in diachrony can contribute to the understanding of the dynamics of the phenomenon of uninflectability in general.

The fact that *voici* and *voilà* behave verb-like has already been observed by various scholars (first and foremost by Moignet 1974, Morin 1985). Also Zanuttini (2017: 241) attributes verbal features to the Italian *ecco* and analyzes it as "defective verb that starts out in v". For the French case, the examples below illustrate some of the evidence for treating *voilà* (and the same holds true for the analogous form *voici*) as verbs.

(1) Voilà Marie.
VOILA Marie
'There is/comes Mary.'

(2) a. Te=voilà.
you.2SG.ACC=VOILA
'There you are.'

b. * Tu=voilà.
you.2SG.NOM=VOILA

(3) La femme que voilà est mon épouse.
the woman that VOILA is my wife
'The woman that you see there (arriving) is my wife.'

The lexeme takes complements like in (1) and assigns accusative case to them, which becomes visible at the surface when replacing the noun phrase with a pronoun which displays the accusative form, clitic in nature (see 2a in contrast to 2b). In addition to that, the fact that the so-called particle suffices to constitute a full relative clause introduced by the complementizer *que* like in (3) suggests that it has to contain verbal properties. This impression is only reinforced by its ability to introduce subordinate clauses (4), pseudorelatives (5) and small clauses (6).

(4) Voilà pourquoi je voulais te parler.
VOILA why I wanted you.ACC talk
'This is why I wanted to talk to you.'

(5) Voilà mon frère qui pleure.
VOILA my brother who cries
'My brother is crying.'

(6) Voilà mon oncle content.
VOILA my uncle happy
'My uncle is happy now.'

Also when it comes to word formation, *voilà* in combination with the common verbal prefix *re-* is treated as a verbal element on the morphological level, as can be seen in example (7).

(7) Me re-voilà.
me.1SG.ACC again-VOILA
'Here I am again.'

At the same time, one could hypothesize that rather than as an uninflectable item, *voilà/voici* should be analyzed as a highly defective element where only one cell of the verbal paradigm is filled. However, although in those presentative sentences, the alleged subject of *voilà/voici* is never expressed overtly, cases like (8a–8b) suggest that there is no restriction of the addressee to a specific number feature. Since the focus of the paper lies on the morphosyntactic properties linked to uninflectability, the question, though no less intriguing, to know why sentences with *voilà/voici* lack an overt subject, especially in a non-pro-drop language like modern French, will be left aside here and needs to be investigated separately elsewhere.

(8) a. Addressee=2SG
Voilà un monde que tu n'as pas connu.
VOILA a world that you.2SG NEG have NEG known
'There you have a world that you did not know.'
[A. Jenny, L'Art Français de la Guerre, 2011, p. 241, via Frantext]

b. Addressee=2pl
Voilà pourquoi vous crèverez tous;
voila why you.2pl die.2pl.fut all
'This is why you all will die;'
[J. Roubaud, Poésie: Récit, 2000, p. 349, via Frantext]

In sum, what becomes clear considering the distribution of the uninflectable *voilà/voici* is that not only it behaves like a verb, but it also exhibits a high level of (morpho-)syntactic productivity. The predominant question that arises from the observation of this concrete phenomenon is thus the same as the one asked by Spencer (2020: 154) on a more general level: "How can an uninflected stem participate in syntactic relations that are normally the province of inflected word forms and the phrases they project?". To approach this question, we will concentrate on the part of the lexeme that is of verbal origin, namely the morpheme *voi-*, and on the development of its features over time. Whilst there is consensus about the assumption that *voici* and *voilà* etymologically arose from the combination of a form of the Old French verb *veoir* 'to see' and a locative adverb, respectively *ci* 'here' and *là* 'there', the exact nature of the former remains to be determined. One often propagated hypothesis in this regard is the one of a "frozen imperative singular" (Buridant 2000: 243; also stated f. ex. in Rabatel 2001: 141; FEW: vol. 14, 492), which needs to be tested and corrected, as will become apparent below. In the course of the paper, I will argue that what in Modern French spells out as *voi-* originates as 2pl/hon and undergoes a process of grammaticalization, involving contraction, univerbation, and reanalysis leading finally to root insertion.

In order to come to a fine-grained analysis, the following questions are addressed: How can the grammaticalization path of the lexemes *voilà* and *voici* be described accurately? Which mechanisms come into play during the transformation process into an uninflectable item? And finally, how does the feature configuration of the *voi*-morpheme develop until being solidified in its modern, uninflectable form?

By means of a corpus analysis for French language data from the 12^{th} to the 16^{th} century (see Section 2), the study traces back the diachronic development and the different stages of the particular lexical element under discussion. Section 3 discusses the origins of the source construction and its morphological features, Section 4 analyzes the intermediate stages *vez ci/la* and *veci/vela* specifically with regard to the emergence of uninflectability, and Section 5 explains the mechanisms that lead to the creation of the final forms *voici* and *voilà*.

2 Diachrony of *voilà/voici*: The data

2.1 Methodology

In order to explain the unusual (morpho-)syntactic behavior of the item under discussion, as a first step its etymology and diachronic evolution are traced back. Therefore, the design of a corresponding corpus research has to consider periods even before *voilà/voici* in its modern form is attested for the first time. Since in our corpus, the first occurrences are detected in the 15th century, and the rise of what can be seen as pre-forms of the particle (namely *vez la/ci*, see below) happens from the 13th century on, the detailed research starts in the 12th and extends to the 16th century. Employed were two already established digital corpora which cover an extensive range of texts and genres of their time and which are both annotated, tagged for parts of speech, and searchable via queries in Corpus Query Language (CQL): The *Base du Français Médiéval* (BFM) for the older language stages and the *Base textuelle Frantext* (Frantext) for data from the 14th century onwards. The total amount of data has been separated into subcorpora (as described in Table 1) and examined independently from each other. Of course, since the subcorpora vary considerably in size in terms of number of tokens, when comparing the results to one another to gain insights on developments over centuries, the results will have to be normalized.

Table 1: Quantitative description of the subcorpora.

corpus	century	no. texts	no. tokens
BFM 2019	12th	50	1,253,830
BFM 2022	13th	64	1,854,076
Frantext	14th	119	4,354,407
	15th	162	4,902,277
	16th	188	7,189,168

In order to compare and classify the results obtained by the corpus research, valid categories for comparison are needed that are eligible to picture the different stages of the construction before resulting in the presentational element *voilà/voici*. The first category comprises the regularly inflected Old/Middle French verb for ‘to see’ in combination with the locative adverb *ci* (‘here’) or *la* (‘there’), with its 2PL/HON form falling under category 1a and the form for 2SG under category 1b (see Table 2). The variance within this group is partially due to general language change taking place during the considered period, especially due to the morpho-phonological change on the verb itself from *veoir* to *voir*. However, since the research interest here focuses on the grammatical properties of the *voi*-morpheme of the univerbated particle, the pre-particle syntagmas have been classified in terms of verbal features and no distinction has been made regarding the actual realization of the verb. That way, a 12th century *veez ci* and a 16th century *voyez ci* fall in the same analyzing category (1a, see Table 2), which is only coherent in view of the research question.

Table 2: Categorization of presentative forms applied during the corpus study.

category		considered variants
1a	veoir.2PL/HON + ci/la	veez, vees, voiez, voyez, voiés, voyés
1b	veoir.2SG + ci/la	veis, veiz, vois, voiz
2	vez ci/la	vez, ves
3	ve(z)ci/ve(z)la	vezci, vezla, vesci, vesla, vezcy, vescy, veci, vela
4	voici/voila	voici, voicy, voila, voyci, voycy, voyla

During the first overall approaches to the corpus data, one form kept recurring that could not readily be classified and whose specific nature remains to be determined, namely occurrences of *vez ci/la*. The choice has therefore been made to analyze it separately (category 2 in Table 2), a choice that turns out to be crucial for the further analysis (see Section 4 below). The further categories 3 and 4 contain respectively the univerbated presentational items *ve(z)ci/ve(z)la* and *voici/voilà* still in use today. For the totality of the results, also variants due to graphical reasons were included by considering alternations between <v> and <u> as well as all possible combinations with diacritics; for instance, also a form like <voies> could have been classified under category 1a, but only after checking manually from the context that it does not stand for the plural of the noun *voie* 'way', but well for the 2PL of the verb 'to see'.

2.2 Results of the corpus study: the overall picture

Searching inside the corpus for the different categories of presentative forms specified above leads to the quantitative results summarized in Table 3. The relative frequencies given are calculated with a basis for normalization of one million tokens. In order to better recognize tendencies and possible dependencies, the results are visualized in a graph in Figure 1.

Table 3: Relative frequencies per million tokens, (absolute frequencies in brackets).

	1a (veoir.2PL/HON + ci/la)	1b (veoir.2SG + ci/la)	2 (vez ci/la)	3 (ve(z/s)ci / ve(z/s)la)	4 (voici / voila)
12th c.	36.69 (46)	5.48 (7)	10.37 (13)	0 (0)	0 (0)
13th c.	35.06 (65)	8.63 (16)	42.07 (78)	12.41 (23)	0 (0)
14th c.	33.07 (144)	9.88 (43)	71.19 (310)	98.52 (429)	0.46 (2)
15th c.	37.56 (161)	0.7 (3)	6.3 (27)	94.56 (405)	9.11 (39)
16th c.	10.29 (74)	4.59 (33)	0.14 (1)	23.79 (171)	438.16 (3150)

Since for one of the categories, namely category 4 (*voici/voila*), we observe an extremely steep rise during the 16th century in terms of frequencies, its values are represented on a separate scale, in order to zoom in onto the evolution of the remaining data points representing the earlier stages.

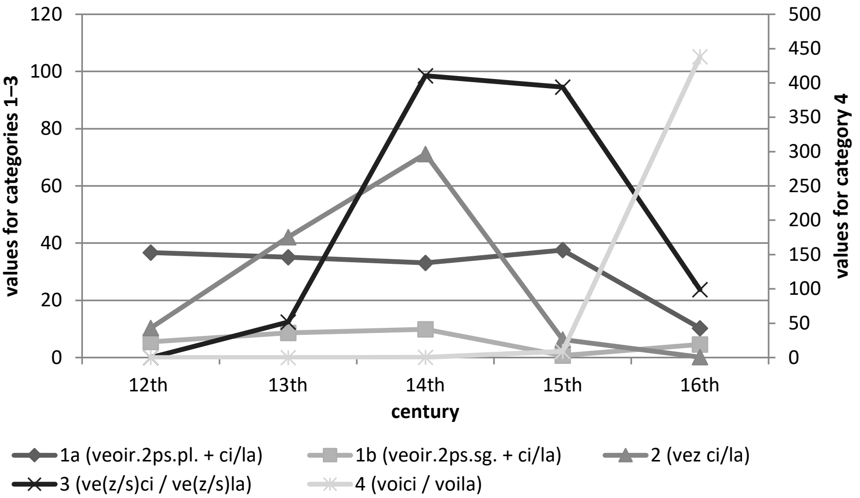

Figure 1: Relative frequencies of presentative categories per million tokens, right y-axis only valid for category 4.

A first observation that can be made is that throughout the 12th to the 15th century, within the syntagma consisting of a conjugated form of the verb ‘to see’ and the locative adverb, the plural form of the verb (category 1a in Figure 1) is significantly more frequent than the singular form (category 1b), until the occurrences of the former decline drastically in the 16th century. Furthermore, we see the numbers of the form *vez* in combination with the locative adverb rise steadily over the 13th and 14th century and fall drastically again in the 15th century (category 2). At the time when the *vez* based presentatives reach their highest peak, another univerbated form is prevailing (*vezci*, *veci*, etc., category 3), increasing sharply in the 14th century, directly after their first documentation in the corpus. Finally, in the 16th century, all frequencies of the formerly described presentative forms seem to collapse and reach their lowest point (with the exception of the singular syntagma 1b staying constantly low), whilst *voici/voilà* (category 4) increases drastically, with an extremely high rate of growth, from roughly 9 occurrences per million tokens in the 15th century to over 400 in the 16th century.

In what follows, we will analyze the results going step by step through the different diachronic stages of the presentative constructions, focusing specifically on the nature and features of the verbal morpheme involved that ends up developing into the uninflectable element still in use today.

3 Etymological origin

3.1 Fixed formula originating in 2PL/HON (*veez ci*/*la*)

As a first step, it is necessary to determine the still debated etymology of *voici* and *voilà*. The two lexemes are seen here as the result of a grammaticalization process, understanding grammaticalization as "the change whereby in certain linguistic contexts speakers use parts of a construction with a grammatical function. Over time the resulting grammatical item may become more grammatical by acquiring more grammatical functions and expanding its host-classes." (Brinton & Traugott 2005: 145). As will become apparent along the analysis in Sections 3–5, this is exactly what we witness for the French presentative element that starts out as a regular syntagma and develops into a verb. In order to determine the starting point of the process accurately, we will have to "focus on strings or constructions rather than lexical items alone" (Traugott 2003: 645), following the unanimously assumed hypothesis that *voici* and *voilà* in one way or the other involve the verb *voir* 'to see' and the locative adverb *ci* 'here' and *là* 'there' respectively.

Looking at the data from the 12th century, one can observe first, that within the considered constructions the combination of an inflected form of the verb 'to see' and the locative adverb *ci/la* presents indeed the most frequent option, and second, that the verb is far more often inflected for 2nd person plural than for 2nd person singular (compare categories 1a and 1b in Figure 1). This observation is crucial, since in order to assume a construction to be the source for further diachronic development in terms of grammaticalization, "we need to be able to show that the source constructions exist – and exist in sufficient numbers" (Brinton 2017: 284, here with regard to the development of pragmatic markers). The fact that the combination of singular forms with the respective locative adverbs does not yield significantly elevated numbers of frequencies that would go beyond the predictable cooccurrence of the two items in their regular function makes it indefensible to take the representants of category 1b for possible candidates for a source syntagma of the lexemes in category 3 and 4. On the contrary, for the 2nd person plural form of 'to see' in combination with the locatives *ci/la* (category 1a in Figure 1), we have such a case of outstanding frequency in comparison also to the other forms of the paradigm. Furthermore, their quantitative presence in language use seems to be strongly affected, like the other presentative constructions appearing at later stages, by the massive rise of the final presentative form *voici/voila*, which suggests that *veez ci/la* and variants are competing with the new presentative, and thus have to fulfill a similar pragmatic function. By contrast, the elements of category 1b are the only ones in Figure 1 not to be affected by the introduction of the novel presentative element, which in turn confirms the assumption that they are not directly involved in the evolutionary pathway of the French presentatives.

We can thus state that the source construction of the French presentative item is based on the Old French verb *veoir* displaying 2nd person plural morphology. It is important to note that, in Old French as in modern French, the verbal forms for the 2nd person plural are homophonous to those used to express the honorific form, singular and plural. In fact, in situations where the addressee of *veez ci* and variants can be unequivocally identified in the 12th century subcorpus, for example by means of directly preceding vocatives or other clear references from the context, the vast majority of the clauses turns out to be directed to a single person in a polite manner rather than actually addressing a group of people. For the latter case, only 4 clear examples can be identified, in contrast to 28 for the former. The blurred borders this homophony entails between polite singular and generic plural forms, already beginning to show here, will play a role again in the process analyzed in Section 4.2.

By the 12th century, not only the verbal element of the presentative construction is being solidified to a specific form, but also the locative element. Occasional variances in this period still attest to the fact that this stability had just been reached shortly before. For instance, expanded locative expressions like *la* *sus* 'up there' in (9a) or *ci desouz* 'below' in (9b) could still be encountered, although rarely, up to the 13th century, before not being attested any more in the 14th century. Up to this point, by gradually excluding alternatives like in (9a–9b), a clear restriction to the binary pair *ci* 'here' – *là* 'there' as possible candidates for the locative element in the presentative construction has been accomplished.

(9) a. Seingnors, fet il, [...] ces dames que **veez** **la** **sus**, ce
Sir.NOM does he [...] these ladies that see.2SG.HON there up this
sont les filles Adrastus, Deyphilé et Argÿa, [...].
are the daughters Adrastus.GEN D. and A.
'Sir, he says, [...] these ladies that you see up there are Adrastus' daughters, D. and A.'
[Anonyme, Roman de Thèbes 2, p. 127, ca. 1150, via BFM 2019]

b. Et ce pouez vous prouver par ces .iii. figures
and this can.2PL/HON you.2PL/HON prove by those three figures
que vous **veez** **ci** **desouz**.
that you.2PL/HON see.2PL/HON here under
'And you can prove this with those three figures that you see below.'
[Gossouin de Metz, Image du monde, ca. 1246, via BFM 2022]

All in all, this process involves a transformation from free collocation (in this case the combination of any conjugated form of *veoir* with a locative expression) into a fixed formula (displaying 2PL/HON morphology and reducing the options to two specific locative adverbs). The terms "free collocation" and "fixed formula" are borrowed from Brinton's (2001) analysis of English *look*-forms, which show clear parallels to the here studied phenomenon.

Furthermore, in both cases this last step not only entails morphological fixation, but also the involved words "become fixed syntactically" (Brinton 2001: 189). In the case of the French presentatives, a consolidation of the internal word order of the construction is witnessed: In general, in Old and Middle French, nothing precludes the placement of the locative adverb in front of the verb as shown in (10).

(10) Sire Willame, un petit m'=atendez. Ices couarz que
Sir W. a little me=await.2SG.HON those cowards that
vus **ici** **veez**, Ceste est ma torbe, [...].
you.2SG.HON here see.2SG.HON this is my troop
'Sir W., wait a bit. Those cowards that you see here, this is my troop, [...].'
[Anonyme, Chanson de Guillaume, p. 117, ca. 1140, via BFM 2019]

However, this constellation remains the exception in the examined presentative construction, also in later stages, where in general language use *ci* begins to be found mostly in proclitic position and is contrasted with *ici* in this regard (FEW: vol. 4, 425): In the 15th century subcorpus, only one instance of proclitic *ci* (*cy veez*) can be observed, and only six occurrences of *veez/voyez* with enclitic *ici*, compared to the 114 cases of the type *veez/voyez ci*. The fact that, in opposition to what is happening in the regular French language system, *ci* continues to be constantly used in enclitic position with *veez/voyez* and does not get replaced by *ici* confirms once more that the whole syntagma had by then evolved into a fixed expression.

3.2 Indicative or Imperative? Verbal mood of the source construction

Having identified the person and number features of the source syntagma as 2PL/HON, its grammatical mood remains to be determined. Given that the often-stated hypothesis of *voici/voilà* originating from a "frozen imperative singular" (see Section 1) already turns out not to be correct with regard to its second part, the question whether the construction involves an imperative still needs to be tested systematically. This undertaking turns out to be more complex than one might expect: First, unlike Modern French, Old French allowed for pro-drop, in imperatives as well as in declarative sentences (cf. Adams 1987). In a sentence like (11), which in this form is quite representative of a good number of presentative clauses in the 12th and 13th centuries, it is thus impossible to definitely tell based on pure grammatical arguments whether the sentence is meant to be a command or a declarative statement. To this is added that at this stage, the verb *veoir* just like its Latin etymon VIDERE does not exclude intentionality, unlike Modern French *voir* 'to see' in contrast to *regarder* 'to look', and could still trigger both semantic readings (Iliescu 2010: 212).

(11) Si dist li uns a l'autre: «**Veez** **la** le bon chevalier, [...]»
so said the one to the other see.2SG.HON there the good knight
'And one said to the other: «(You) see there the valiant knight, [...]»'
[Anonyme, Queste del saint Graal, ca. 1225/1230, p. 206d, via BFM 2022]

Also, the placement of clitics, which for the early presentatives happens systematically enclitically on the verb according to the pattern *veez le ci/la* cannot give a reliable indication about whether a declarative or an imperative is realized, since at this stage, unlike in Modern French, clitic positioning still follows similar rules for both clause types (cf. Hirschbühler & Labelle 2000). What can help to shed light onto the question is looking at instances of embedded presentative constructions. In general, imperatives cannot be embedded, or more precisely, only under certain conditions. One of those conditions is the obligatory presence of expressed subjects (Platzack 2007: 196), which we see is not met in Old French if we consider for instance example (12b) in contrast to (12a). We thus have good reason to assume the relative clauses in (12a–12b) to be embedded declaratives rather than imperatives.

(12) a. Donc valt mialz li uns de ces trois javeloz **que vos veez** ci, [...].
so be-worth.3SG better the one of these three darts that you.2PL see.2PL.IND here
'So it is better [to take] one of these three darts that you see here, [...].'
[Chrétien de Troyes, Conte du Graal (Perceval), 1181–1185, p. 361, v. 201, via BFM 2019]

b. Cez estencelemenz / **Que veëz** ci dedenz.
these sparkles / that see.2PL/HON.IND here inside
'These sparkles that you see in here.'
[Philippe de Thaon, Comput, 1113/1119, p. 10, v. 404, via BFM 2022]

On the other hand, there are other constellations where an interpretation of the verbal element as imperative seems more appropriate. In particular, this is the case when the presentative construction occurs in coordination with another imperative like in (13a) or right after an imperative forming with it a sequence of exclamatives as in (13b). In those cases, although once again from a morphological and syntactic point of view the presence of indicative mood cannot be excluded formally, the discourse context strongly suggests an imperative reading.

(13) a. [...] si comencierent a dire: «Tornez vos, veez ci la reïne.»
[...] so start.3PL to say turn.2PL.IMP you.REFL see.2PL.IMP here the queen
'[...] and they started to say: «Turn around, see here the queen.»'
[Anonyme, Queste del saint Graal, ca. 1225/1230, p. 162c, via BFM 2022]

b. Fuiés, fuiés! Veés ci venir le fol de
flee.2pl.imp flee.2pl.imp see.2pl.imp here come.inf the madman from
la fontainne!
the fountain
'Flee, flee! Watch out for the madman who is coming from the fountain!'
[Anonyme, Tristan en prose, 13th century, après 1240, p. 253, via BFM 2022]

Consequently, these observations result in the fact that the grammatical mood of the etymological source construction for the French presentative cannot be limited to one single clear-cut option. Instead, it originates, with regard to its modality, from a polysemic verbal form that is sometimes used with an indicative reading, sometimes with an imperative one, and where in between these two poles there is space for gradual interpretation depending on discourse contextual factors – and it might even be the case that precisely this polyvalence contributed to this particular syntagma being grammaticalized into a presentative item via the process described in the following section.

4 From syntagma to particle

4.1 Contraction (*vez ci/la*)

During the corpus research, the form *vez* manifests itself by showing a behavior that is not to be equated with the other forms in Figure 1. The presentative construction clearly evolves following a separate curve, rising gradually from the 12th to the 14th century and disappearing abruptly in the 15th century. The observation being made that constructions involving *vez* apparently constitute a distinct stage in the grammaticalization path of the presentative, its specific features need to be examined closely.

One could hypothesize that the form represents a mere variant of an already established verb form. This hypothesis has actually been stated frequently in the literature wherever an attempt to classify the element *vez* is undertaken: Buridant (2000: 311) treats it as "a monosyllabic, contracted form" (translation K.F.) of the imperative plural of the verb *veoir*, and so do Oppermann-Marsaux (2006: 80) and Danino et al. (2020: 48). The corpus data, especially the high frequency of the type *veez ci/la* (compare Figure 1) support the assumption of an underlying verb form inflected for 2pl/hon. This is aside from the fact that in view of the situation explained in 3.2. those forms expressing the imperative and the indicative mood form an opaque complex from where the grammaticalization process of the presentative is initiated. Speculations like the possibility of a newly created form based on Latin VIDE ECCE (FEW: vol. 14, 429) for which there is no evidence found in previous language development, are by this means also ruled out for the time being. Yet, the investigation should not stop at this point. By looking closer at the phenomenon, it becomes clear that one should not consider the changing lexeme in isolation, but rather take into account its frequent cooccurrence with the locative element and regard the fixed formula as described in 3.1. as a whole. In particular, what is special about *vez* is the fact that it is not found in free variation with the 2PL/HON form it is originating from, neither with the imperative nor with the indicative.

Table 4: Absolute frequencies of *vez* in the 14th century subcorpus.

vez ci	*vez la*	*vez* + other LOC. ADV.	*vez* without LOC. ADV.
283	70	15	11

Instead, we see in Table 4 that *vez* is in the overwhelming majority of cases (353 out of 379) found in combination with the locative adverb *ci* or *la*, that is within the fixed formula that has been developing until then. 15 other instances of the lexeme cooccur directly with further locatives, mostly with the adverb *ça* 'here' < Lat. HAC, which is formed on the basis of the ablative form of Lat. HIC that in turn also serves as etymon for fr. *ci* (FEW: vol. 4, 373). On account of this etymological and semantic comparability, also those instances can be treated as cases of the fixed presentative formula, or at least as strongly related forms. That leaves us with 97,1% (368) of the occurrences of *vez* being integrated into the specifically presentative source syntagma, versus with 2,9% (11) only a negligible part being used as variant of *veez*.2PL/HON.IMP/IND, most probably graphic in nature.

The fact that the contraction from *veez* to *vez* only takes place within the context of this specific construction suggests that the change is triggered for prosodic reasons. Once the fixed formula is established, the phrasal accent seems to be shifting from the ultima of the verb to the locative, which in turn enables the contraction of the verbal form in this specific constellation (*veéz ci* > *vez cí*). This process appears plausible also with regard to the "most important" general prosodic change from Old French to Middle French (Gess et al. 2020: 470). This change involves a shift from lexical stress, usually on the ultima of the word, to group stress, i.e., "stress assigned not at the level of the prosodic word, but at the right edge of a higher prosodic constituent" (Rainsford 2020: 63–64). At the same time, the fact that the stress has thus already been shifted towards the end of the construction paves the way for the univerbation process described in 4.3. Altogether, it becomes apparent that *vez* is indeed a form which originates from *veoir* inflected for 2PL/HON but which through contraction results into an item that is not congruent with its etymon with regard to its morphosyntactic features, as will be stated in the following.

4.2 Developing uninflectability

The consolidation of the fixed formula and the contraction of the included verbal element into *vez* entails wider consequences on the morphosyntactic level. Although *vez ci/la* in the 13th century subcorpus are still mostly found in contexts where the utterance is addressed to a person in polite manner (49 times) or to a group of people using the 2nd person plural (5 times), instances begin to arise where the contracted form appears surrounded by forms of address clearly marked for 2nd person singular. With 9 such occurrences in 5 out of the 64 texts within the 13th century subcorpus, although not extremely frequent, those do not seem to represent individual cases. Examples for the three different types of addressees described here can be seen in (14a–14c).

(14) a. Addressee=HON
Pere, fait Aucassins, ves ci **vostre** anemi [...].
Father makes A. VEZ here your.HON enemy
'Father, says A., here is your enemy [...].'
[Anonyme, Aucassin et Nicolette, beginning 13th century, p. 10, v. 19., via BFM 2022]

b. Addressee=2PL
Li rois le moustre **a ses compaignons** et **lour**
the king it shows to his.PL.M companions.PL.M and them.DAT.PL.M
dist: «Ves chi une nef [...].»
says VEZ here a boat
'The king shows it to his companions and says to them: «Here comes a boat [...].»'
[Anonyme, Suite du roman de Merlin, ca. 1230–1235, p. 314, via BFM 2022]

c. Addressee=2SG

Se	**tu**	voloies	mengier	pain,	/	vez=en	la	.I.	lez	cel
if	you.2SG	wanted	eat	bread	/	VEZ=PART	there	one	next-to	this

autel.
altar

'If you want to eat bread, there you have one next to this altar.'
[Anonyme, Roman de Renart, branche XI, beginning 13th century, p. 97, v. 12401, via BFM 2022]

Sentences like (14c) suggest that the contracted form *vez* can already be regarded as particle developing uninflectability, since their utterance is only conceivable under a certain condition: They can only be produced if what used to be an inflected verbal form, namely the 2PL/HON form of the verb 'to see', is not perceived any more as such by the authors in those specific cases.

Moreover, another observation in the diachronic data indicates that *vez* is not treated as a simple contracted form, but that a more fundamental change is happening under the surface: From the moment the *vez* particle first appears, overt subjects are immediately and consequently ruled out, even in embedded relative clauses containing *vez ci/la* at the position of the verb. This can be surprising at first glance, knowing that exactly during that time the French language system is losing its pro-drop properties which should normally cause an increase of subject pronouns. Instead, the incompatibility of *vez* with overt subjects indicates what already Bouchard has hypothesized for *voici/voilà*, i.e., that it has "no TNS/AGR" and hence that "nominative Case is not assigned" which in turn "prohibits the presence of a lexical NP" (Bouchard 1988: 93). Together with the evidence from cases like (14c), this reinforces the picture of the verbal part of the presentative construction changing at that point from a finite form to a form of non-finiteness.

Particularly interesting in this context is that the observed process of a plural form being applied to singular addressees fits into a mechanism that is attested crosslinguistically and that Bates & McKenzie (2021) call the "plural-to-singular reanalysis cycle". This process of number reanalysis is usually initiated when "a common ground disparity exists between the speaker and hearer" (ibid., 7). For example, what is likely to happen in our case, is that the hearer has to accommodate a sentence where it is not only unclear whether it is meant as an imperative or an indicative (see 3.2), but where also the number feature of the verb requires interpretation. Since *vez ci/la*, initially mostly found in reproduction of direct speech, is beginning in the 13th century to be used also in purely written forms of communication to address for instance the reader, this leaves room for ambiguity. As already mentioned above in Section 3.1, the verbal forms for the 2nd person plural are homophonous to those expressing the honorific forms in French. The verbal source morphology could either stand for a polite approach of one single addressee, a polite approach to a group of people, or for the 2pl used without conveying honorificity. The reader therefore finds him- or herself as the only addressee present in this communicative situation and has to deduce correctly the type of addressees the writer has in mind. The reader can then either opt for one of the morphologically encoded interpretations, or, in case he/she "considers the required accommodations implausible" (Eckardt 2012: 2688) and, in order to rescue the processing of the sentence, assume an "interpretative analysis" which triggers then the semantic reanalysis (Bates & McKenzie 2021: 7). When situations that trigger this reinterpretation do not happen only once by accident but occur redundantly amongst the language community, regular language change can apply so that "the new entry is permanently adopted" (Eckardt 2012: 2688). Since such instances of reinterpretation are first sparsely found in the corpus and then begin to increase, this seems to be what is happening to the French presentative *vez ci/la*.

Actually, it can already be observed during the 12th century that even the fixed formula *veez ci* that contains a verb inflected for 2pl/hon gets 2sg interpretation. The two cases found in the corpus displaying this behavior are specified in (15a–15b).

(15) a. **Veez** ci un val, **fai** les **tuens**
see.2pl/hon here a valley make.2sg.imp the yours.2sg.poss
assembler, E **pren** **tes** messages [...].
gather and get.2sg.imp your.2sg.poss messengers
'Here you have a valley [suitable for a battle], make your men gather, and get your messengers [...]'
[Anonyme, Chanson de Guillaume, ca. 1140, p. 10, v. 177, via BFM 2019]

b. Respundi li prestres [à David]: «**Vééz** ci la spéé
replied the priest [to David] see.2pl/hon here the sword
Golias le Philistien que **tu** **óceïs** [...].»
Golias.gen the Philistine whom you.2sg killed.2sg
'The priest replied to David: «Here you have the sword of Golias le Philistien whom you killed [...]»'
[Anonyme, Quatre Livres des Rois, ca. 1190, p. 43, via BFM 2019]

Already here, and more frequently once the inflected verb transformed via contraction (as exemplified in 14c), it becomes apparent that at this stage the French presentative construction is losing the specification for plural. On a more general level, this is also what is attested in other language phenomena subjected to plural-to-singular reanalysis: the plural feature of the construction under discussion is dropped and a form which originally carried plural marking becomes number-neutral. If then, which is clearly the case in our study, the number-neutral (former plural) form is more frequently used, the speakers would privilege the latter to resolve the semantic competition with the former singular form which by consequence disappears (Bates & McKenzie 2021: 10). This in turn can also explain why at a later stage of the grammaticalization path of the particle (see below, 4.3), there is only one single instance of a univerbated form based on the 2sg of the verb, *voizcy* in this case, and that aside from this one example, the singular no longer plays a role any more in the further development of the particle.

Concerning *vez ci* and *vez la* (and *voici* and *voilà* in the continuity), the plural-to-singular reanalysis cycle stops at this point with the presentative being applicable to plural as well as to singular contexts. In theory, the process could still go one step further. For a number of pronouns for example, it is observed that a new plural form emerges which in turn serves to disambiguate the plural-singular syncretism created in the course of the previous steps of the cycle, and which limits the former plural and then number neutral form finally to the singular (Bates & McKenzie 2021: 11–12). Such a continuation of the cycle is not attested for Standard French, where the forms *voici* and *voilà* are consolidated by the end of the 16th century and since then did not change their form. It could still be possible that in other dialects and under other conditions, the lexical item is evolving in a manner that reminds the disambiguation process decribed above, notably by redeveloping inflectional endings.[1] However, the central condition under which such a continuation of the cycle could happen, is that the proclitic object pronouns attached to *voilà* are reinterpreted as subject pronouns. Conversely, the fact that from the start, overt subjects are ruled out with *vez ci/la* hints at no syntactic subject position being projected and thus the verb being non-inflecting. Taken all those considerations together, we thus have clear indications for the assumption that the presentative formula at this stage is ultimately evolving towards constructional uninflectability.

[1]Morin (1985: 810) evokes the possibility to encounter inflected forms of *voilà* in the French of Québec on the basis of data he discovered in novels by Louis Fréchette written around 1900. Here, the presentative particle develops inflectional endings, for instance *nous v'lons* 'here we are'; *les v'lont* 'here they are' (examples (77e) and (77g) in Morin 1985: 811). However, this seems to be a marginal phenomenon that is rarely attested and which Morin classifies as "now obsolete" (Morin 1985: 810), a statement that would need to be targeted via a specific survey in order to be maintained properly.

4.3 Univerbation (*veci/vela*)

What has already begun via phonological and morphological contraction continues in the course of the further development of the presentative particle and finally manifests itself also in the graphic representation: *vez ci/la* is contracted to the extent that the two elements start forming one single item, not only on the prosodic level but from the 13th century onwards also in orthography. The merged writing soon overtakes the separated one in terms of frequencies in the 14th century, as can be seen in Figure 1 above. The fusion into one single item that this orthographic change is indicating is not unusual to happen in such grammaticalization processes, where "combinations of words and morphemes that occur together very frequently come to be stored and processed in one chunk" (Bybee 2003: 617).

At this stage of the evolution, one cannot ignore the role the previously established contracted particle *vez* is playing. Whilst the forms *veci* and *vela* are usually mentioned when discussing the origins of *voici* and *voilà*, little has been said about a far more frequent form of the 14th century univerbated presentative form, namely *vezci/vezla*. Figure 2 shows that during that period they represent the overwhelming majority of instances for univerbated presentatives (416 in total) and that, on the contrary, the writings omitting the intermediate consonant constitute with only 11 occurrences rather exceptional cases.

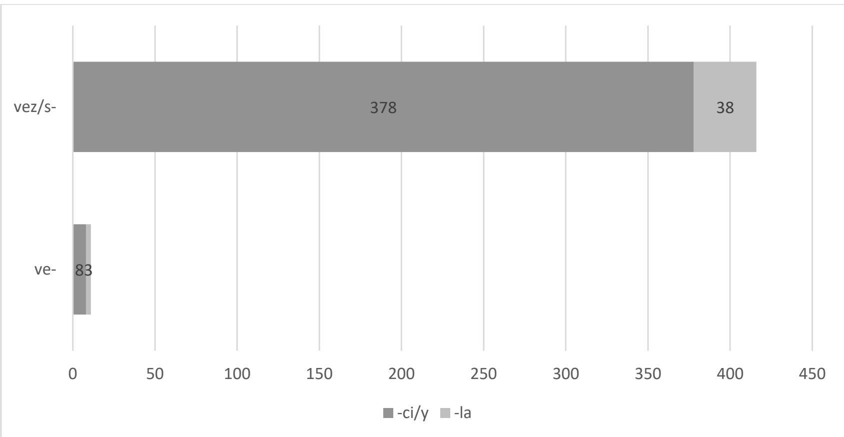

Figure 2: Absolute frequencies of *vez/sci/y* + *vez/sla* vs. of *veci* + *vela* in the 14th century subcorpus.

In view of the results depicted in Figure 2, one cannot but consider the fact that given the overrepresentation of forms including the full form of the particle *vez/s* described above in Sections 4.1–4.2, the univerbated form first of all originates from the contracted syntagma *vez ci/la* (or *ves ci/la*) which leads to the form *vezci/vezla* (or *vesci/vesla*) and that it is this construct which afterwards is further reduced into *veci/vela*. This means at the same time that at this stage, the morpheme *ve-* still is a relic of the reduced form of the 2PL/HON and does not represent, as it has been suggested, the verb stem ("base verbale", Oppermann-Marsaux 2007: 236) of the verb *veoir*, at least not from the start. A reanalysis as such is however possible and likely, as indicates the transformation described below in Section 5.

The merging of a former syntactic phrase into a single lexeme entails consequences amongst others with regard to the morphosyntactic behavior of the emerging element. In our case, a mechanism comes into play that is observed similarly for instance with noun-noun compounding: oftentimes lexemes that are integrated into a multiword expression lose their inflectability due to this combination. For example, within the English compound *bicycle lane*, it is impossible for the non-head noun to inflect for plural number (**bicycles lanes*). In this case, "the construction itself prevents an otherwise inflectable lexeme from inflecting", a phenomenon dubbed "constructional uninflectability" by Spencer (2020: 148). Even if we have seen that at this point of the evolution of the presentative element, *vez* has already lost, or at least weakened its formerly inherent number feature and is already developing towards uninflectability, it is important to bear in mind that this process only was engaged in combination with the locative adverb *ci* or *la* directly following. What we see for instance in the examples in (14) is to be considered an intermediate stage between the fully inflected *veez*.2PL on one side, and the compound *veci* on the other, with the morpheme *ve-* prevented from inflecting by a combination of two factors: first, by the plural-to-singular reanalysis which is responsible for preventing the creation of an additional, in theory conceivable particle implementing the 2SG form of the verb. And second, by the constructional constraints it finds itself due to the univerbation process. Given that in French, inflection normally occurs at the end of the verb, once the lexeme *veci/vela* was created, inflectional endings would have to adjoin to the part of the element that is still well recognizable as an adverb. The formerly inflecting morpheme thus finds itself blocked in a position where the possibility to redevelop inflectional endings in the future is extremely limited, if not impossible.

5 Reanalysis and root insertion (*voici/voilà*)

In the 16th century, we witness an extremely steep increase of the use of presentative particles of category 4, namely *voici* and *voilà* compared to the previous century (cf. Table 3 and Figure 1). They reach a relative frequency of 438.16 per million tokens, which is around 48 times as high as the century before where they emerged for the first time. The drastic collapse of all the other presentative constructions that this increase goes along with suggests strongly that the two events are linked causally and that it is *voici/voila* that is replacing them directly in the language use. The question thus arises first, by which mechanisms this ultimate form arises and second, which features the *voi-* morpheme bears. With regard to the second question, as we will see, there are arguments to assume that those differ in nature from the ones of the deverbal element in *veci/vela*.

Table 5 shows that the two forms did not develop in parallel. Whilst the type *veci/vela* has its peak in the 14th century and also continues in the following century to be the most employed presentative form, isolated occurrences of *voici/voila* only begin to be documented for the first time at this stage. Rather than assuming autonomous emergence of the two different forms *veci/vela* and *voici/voila*, the former turns out to be the basis from which the latter evolves in the course of time.

Table 5: Absolute frequencies in the 14th century subcorpus.

category 3		category 4	
ve(z/s)ci	ve(z/s)la	voici	voila
386	41	1	1

Taking into consideration the late Middle French language system in general shows a striking correlation with an ongoing phonological change at that time (also observed by Tacke 2022: 358). In fact, due to general sound change mechanisms, the verb ‘to see’ itself changes from *veoir* [vəwɛr] to *voir* [vwar] from the 12th onwards: first, [wɛ] opens to [wa] and by the beginning of the 14th century, the hiatus [ə] is eliminated (Andrieux-Reix 1995: 70). This change affected not only the infinitive, but all the forms of the inflectional paradigm, so that the diphthong usually represented by <oi> or <oy> is found in all of its cells. What is interesting here, is that the change from *veci/vela* to *voici/voila* in the 16th century takes place slightly after the process concerning the core paradigm of the verb already being accomplished. It seems thus as if the presentative is aligned to the changing paradigm not out of phonological necessity (which is not given since *veci/vela* would have been pronounceable also within the novel sound inventory), but through reanalysis by the speakers who still identify it as belonging to the verb *voir* and who effect the change on an analytical basis. That in turn suggests that *veci/vela* and subsequently *voici/voila* even after becoming uninflectable and losing the number feature still bear a verbal feature which is even actively recognized as such by the speakers.

In order to carry out the alignment of the presentative to the verbal paradigm of *voir*, *ve-* is replaced by *voi-*. This means that the form of *veoir* inflected for 2PL/HON which developed into the number neutral form *vez* and finds itself further contracted in *veci* is now replaced by the verb stem. Also here, we see a mechanism coming into play that is crosslinguistically attested: Like in other cases of constructional uninflectability, "we can suppose that the construction itself renders the lexeme locally uninflecting, so that the default reversion-to-the-root has to apply", assuming that the default realization of any lexeme is always the root form and that in case of uninflectability, this default is not replaced by any more specific statement (Spencer 2020: 156, 144). The case of *voici* supports this assumption. As a consequence of the univerbation into *veci* and the situation of constructional uninflectability of the verbal morpheme, it is naturally reinterpreted as verbal root and transformed accordingly, this reinterpretation becoming visible through the morpheme chosen in the process of analytical alignment.

Alternatively to root insertion, one could hypothesize that the speakers in the 16th century reanalyzed the verbal part of the construction as an imperative singular and replaced it accordingly. However, although of course orthography is not always a reliable factor especially in earlier language states, it is striking that not a single time one of the forms *vo[iy]sc[iy] / vo[iy]sla* or *vo[iy]zc[iy] / vo[iy]zla*, so forms containing the whole imperative singular of the verb *voir* are found in the 16th century subcorpus, compared to 959 occurrences with only *voi-* in front of *ci* and 2191 of *voi-* preceding *la*. Also, the placement of the clitics in preverbal position does not coincide with what would to be expected with imperatives at this stage.

Taking a closer look onto clitic positioning also helps determining the overall nature of the lexical entity. Given that the speakers are now dealing with a novel element that is composed of a former phrase with two different parts, a verb and a locative adverb, the question arises as to how they implement this new element in their grammar. We have already seen in the introduction that in Modern French, *voici* and *voilà* occupy the syntactic position of the verb. Investigating on clitic placement with the presentative construction in diachrony shows that this clear treatment as a verb is accomplished with the passage from *veci* to *voici* in the 16th century. At the time when the presentative construction still contained a clearly inflected form, object clitics were systematically placed enclitically on the verb and by consequence in front of the locative, as can be seen in examples (16a–16b).

(16) a. Or **Veez=mei** **ci** en vostre curt;
now see.HON=me.ACC here at your court
'Now here I am at your court;'
[Jordan Fantosme, Chronique concernant la guerre d'Écosse, ca. 1175, p. 26, v. 334, via BFM 2019]

b. Si li dïent: «**Veez=la** **la**.»
so him say.3PL see.HON=her.ACC there
'So they said to him: «There she is.»'
[Chrétien de Troyes, Chevalier au Lion ou Yvain, 1165–1200, p. 98, v. 4957, via BFM 2019]

This changes radically with the univerbation of the construction into *ve(z/s)ci* / *ve(z/s)la*: In the 14th century, using object clitics in combination with the univerbated form is for most of the authors not an option at all, this combination being found only three times in the entire subcorpus, among which two occurrences are produced by one and the same person. In the 15th century, object clitics with *ve(z/s)ci* / *ve(zs)la* in preverbal position become more common, 10 out of 162 texts in the subcorpus making use of them in combination with the presentative. This increase indicates that the univerbated construction as a whole bears the potential to be processed as a verb. However, examples like (17) show that this evolution does not come without hesitancy. Here, the authors place the clitic in between the two morphemes, maintaining the syntactic configuration of the earlier stage exemplified in (16), which shows that they still perceive the whole construction, although written together, as a verb phrase and not as a single lexical item. Yet overall, those examples are very rare, with only 3 cases being found in the 15th and 16th century subcorpora respectively. Instead, for the majority of the speakers, the transition towards treating the whole item as a verb seems to be fully assumed with the change from *veci/vela* to *voici/voila*. In the 16th century, 310 occurrences like the ones in (18) are to be noted, where proclitics of all conceivable person and number feature combinations are adjoined to the presentative element.

(17) a. L'EXECUTEUR / C'est fait. **Velela** bien pendu.
Executioner / this is done VE-him.ACC-there well hanged
'Executioner: It's done. Now he truly is hanged.'
[Anonyme, Mystère de la procession de Lille, III, 1485, p. 433, via Frantext]

b. Et tout incontinent **voilecy** qu'il se rameine
and all immediately VOI-him.ACC-here that he REFL bring-back
luymesme [...] à l'eglise.
himself [...] to the church
'And here he is going back [...] to the church immediately.'
[Bonaventure des Périers, Novelles récréations et joyeux devis, 1558, p. 199, via Frantext]

(18) a. THOMAS. / **Me=voicy**, Monsieur.
Thomas / me.ACC=VOICI Sir
'Thomas: Here I am, Sir.'
[Pierre de Larivey, Le Laquais : comédie, 1579, p. 194, via Frantext]

b. et maintenant **le=voila** abbatu sans espoir.
and now him.ACC=VOILA destroyed without hope
'and now there he is, destroyed and without hope.'
[Jacques Yver, Le Printemps, 1572, p. 1188, via Frantext]

The consistency of clitics being placed as in (18) indicates that the presentative element is not processed as a verb phrase anymore by the speakers, in contrast to the rare examples in (17) which are evidence for a corresponding transition phase. Instead, once the fixed formula univerbated into a single lexical item, the speakers start to interpret it as a verbal head. Apparently, in the case of *vezci* and *vesci,* and even more unequivocally after insertion of the verbal root *voi-*, no evidence from the input seems to speak against this option. The process thus results in the reanalysis represented in (19).

(19) $[_{\mathrm{VP}}\ [_{\mathrm{V}^\circ}\ \text{veez}]\ [_{\mathrm{ADV}}\ \text{ci}]] > [_{\mathrm{VP} > \mathrm{V}^\circ}\ \text{ve(z)ci}] > [_{\mathrm{V}^\circ}\ \text{voici}]$

This development follows a universally observed scheme. According to van Gelderen (2011: 13), "whenever possible, a word is seen as a head rather than a phrase", a syntactic mechanism that is referred to as "Head Preference Principle". Illustrating this principle on the basis of diachronic and synchronic observations, she states that it is preferable for an element to merge and move as a head rather than as a phrase for economy reasons (see also van Gelderen 2004). In the case of *voici/voilà* the treatment as verbal head might be coming along with similar advantages in view of language economy.

Taking all the effects of the mechanisms evolved together, *voici* and *voilà* thus end up being based on the root of the verb *voir* and processed as regular, although uninflectable, verbs. Recent developments however show that the evolutionary path of this specific presentative element has not come to an end yet. In Modern French, *voilà* tends to be used as discourse marker and shows similar behavior to related phenomena, for example to Italian verb-based particles analyzed by Cardinaletti (2015). This indicates that the verbal feature is about to disappear in those contexts and that the status as uninflectable verb may not constitute the final stage within the evolutionary path of the element under discussion.

6 Conclusions

The diachronic corpus study has shed light on the details of the grammaticalization path the lexical items *voici* and *voilà* are subjected to between the 12th and the 16th century. It can be summarized as depicted for *voici* in Figure 3, starting out with a regular syntagma that develops into a fixed formula and from there on undergoes a process of contraction and reanalysis.

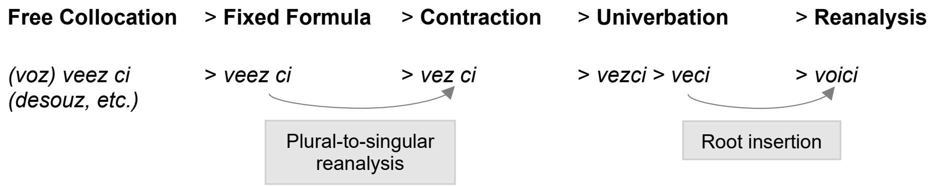

Figure 3: Grammaticalization Path of *voici*.

The mechanisms that come into play during this process are determined by an interplay of different factors: semantic reanalysis, constructional restrictions, and syntactic principles become visible at different stages as the grammaticalization process goes on. Finally, the results explain the unusual behavior the uninflectable *voici* and *voilà* display: despite their first morpheme having lost its number feature and due to reinterpretation as verbal root also displaying impoverished person features, the verbal feature itself still continues to be present and prevailing when it comes to the syntactic implementation of the item.

What is interesting to do as a next step is to extend the research and to pursue the current evolution of *voilà*, away from its verbal use and in direction of a discourse marker, to complete the picture of its cyclical change. To go beyond the case study of one single language, also the comparison with presentative constructions in other languages can be fruitful to find out about eventual crosslinguistical tendencies regarding the evolution of presentative clauses in general. For instance, the English particles *lookee* or *lookahere* investigated by Brinton

(2001), show striking similarities to the French case, and also Italian *ecco* continues to raise questions, less with regard to its etymology, but also with regard to its syntactic behavior. A question that emerges recursively when it comes to analyzing presentative clauses syntactically is centered around the nature of their subject, which is therefore to be approached in the first place in order to gain a larger picture of presentative clauses, regardless of whether they dispose or not of a separate presentative element like French.

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Chapter 8

Emerging uninflectedness in French clipped verbs

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This study documents the emergence of systematic verbal clipping in contemporary French, creating a novel source of morphological uninflectedness. While clipping is well-established for French nouns and adjectives (*introduction* → *intro*), verbal clipping represents a recent phenomenon primarily found in informal registers (*je me déconnecte* 'I disconnect' → *je me déco*). Crucially, these forms remain uninflected across paradigm cells despite being phonologically compatible with standard first conjugation patterns, suggesting conventionalized uninflectedness rather than grammatical impossibility. We devise a computational method for retrieving truncated verbs from the FrWaC corpus by identifying verbal tokens that are substrings of inflected verbs in the French lexicon. Distributional semantics determines the contextually optimal full form from multiple candidates, achieving 88% accuracy in human evaluation. A qualitative analysis of the extracted forms suggests that verbal truncation is productively used by speakers and can apply throughout the verbal paradigm. We follow this up with an experiment to demonstrate that clipped verbs are accepted by speakers in appropriate sociolinguistic contexts. The phenomenon displays interactions between lexeme formation and inflection that are relevant to morphological theory: while inflectional suffixes are fully lost in truncated verbs, stem allomorphy sometimes leads to the preservation of partial inflectional properties, and there is no biuniqueness requirement between full and clipped forms.

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1 Introduction

There are a variety of reasons that a lexeme can be uninflected, as witnessed in other papers in this volume. However a main source of uninflectedness is a situation where forms of a lexeme have a shape that is inappropriate for the application of usual inflectional regularities — in other words, the lexeme is uninflected because it is uninflectable. This happens often with borrowings. Consider the Czech undeclinable noun *iglú* 'igloo'. All established inflection classes of Czech entail that the NOM.SG, which is also the citation form, end in a consonant, or one of the vowels *-a*, *-o*, *-e*, *-í*, *-ý*, *á*, or *é*.[1] As the citation form *iglú* does not have one of the predictable shapes, a natural strategy to accomodate it in the system is not to inflect it at all.

In this paper we document a case where uninflectedness does not result from borrowing but from productive morphology. Many languages combine two interesting characteristics: having a productive truncation (or clipping) process in the word formation system, and relying on suffixal exponents in inflection. In such a situation, we may wonder whether truncation may lead to a situation of uninflectability for lack of an appropriate shape, parallel to the Czech example above. English is a language with these two characteristics, but the organization of its inflection system is such that the outcome just described does not arise: for instance, truncating the noun *rhinoceros* to *rhino* does not lead to uninflectability, as English inflection accomodates nouns with any vocalic ending in the singular. But while the coexistence of the two characteristics above does not deterministically yield uninflectability, there are nevertheless cases in which this outcome might be found. In this chapter, we document the emergence in contemporary French of productive clipping of verbs leading to uninflectedness, and consider whether uninflectedness is motivated by an impossibility of accomodating the clipped verbs in the inflection system. As it will turn out, the situation is more interesting: we will argue that French clipped verbs could have been inflected, but that they are not, probably to maintain the specific social meaning associated with clipping.

The structure of the chapter is as follows. Section 2 presents the relevant background facts on clipping and uninflectedness in French. In Section 3, we show that clipped verbs are readily found in corpora of contemporary informal French, and in Section 4, we present experimental evidence that French speakers find these forms to be fully acceptable in the appropriate contexts. Section 5 lays out the consequences of these results.

[1]The last three possibilities are usually not listed in reference grammars but are found with masculine, feminine and neuter nouns converted from adjectives.

2 Background on French morphology

2.1 Clipping in French

In this section we rely heavily on Antoine's (2000) dictionary of clipped forms, as well as the informal survey of the state of the art in Cerquiglini (2019).

Clipping in French has been the focus of attention of linguists and lexicographers as early as the 1920s (Kjellman 1920). It is attested since the 18th century, and is generally thought to have attained high productivity in the 20th century. Clipping is found with all major parts of speech, but with variable productivity, as is highlighted in Table 1, where we compare the proportion of lexemes in various parts of speech in Antoine's (2000) dictionary and in the general lexicon. As is immediately striking in the table, clipping is clearly overrepresented in nouns, and underrepresented in verbs.

(1) a. Nouns:
introduction > *intro* 'introduction',
vélocipède > *vélo* 'bicycle',
appartement > *appart* 'apartment', etc.

b. Adjectives:
dégueulasse > *dégueu* 'disgusting',
sympathique > *sympa* 'nice', etc.

c. Nouns or adjectives:[2]
anarchiste > *anar* 'anarchist',
catholique > *catho* 'catholic', etc.

d. Adverbs:
personnellement > *perso* 'personally',
directement > *directo* 'directly',[3] etc.

e. Verbs: *magnétoscoper* > *scoper* 'tape-record'

Although both apocope (clipping of the end of words) and apheresis (clipping of the beginning) are attested (1–2), only apocope is clearly productive, making up more than 95% of Antoine's (2000) data.

(2) *autobus* > *bus* 'bus',
capitaine > *pitaine* 'captain',
ricain > *américain* 'American'

[2]We group under 'nouns or adjectives' cases where the language uses homophonous items of related meanings in the two parts of speech, and truncation applies equally to both. We refrain from making any claim on the morphological relationship between these items.

[3]Refer to Footnote 4 for a discussion of -o suffixation in French clippings.

Table 1: Distribution of parts of speech in clippings as documented in Antoine (2000), compared to lexemes in the general lexicon, as documented in the Lexique 3 database New et al. (2007).

Part of speech	Clippings	General lexicon
Noun	79%	57%
Noun or adjective	9%	8%
Adjective	8%	17%
Adverb	3%	4%
Verb	<1%	13%
Preposition	<1%	<1%
Interjection	<1%	<1%
Total	1,152	47,359

There is evidence of various kinds that clipping is a true lexeme formation process, giving rise to the conventionalisation of new lexemes, rather than a mere online abbreviation mechanism. First, clipped forms sometimes supplant their nonclipped equivalent entirely. Striking examples of this are *vélo* 'bicycle', clipped from *vélocipède*, and *métro* 'subway', clipped from *train métropolitain*, literally 'metropolitan train'. In both cases, the nonclipped form has almost entirely fallen out of usage, to the point that ordinary speakers have no intuition that clipping took place.

Second, although the main import of clipping is to convey some form of social meaning related to informality, clipped forms sometimes have shifted or otherwise unpredictable meanings, a clear hallmark of lexical listedness. A most striking example is that of *beauf*, originally a clipped form of *beau-frère* 'brother in law', but mostly used since the 1960s as a derogatory term to designate an uncultivated person espousing reactionary politics; crucially, that meaning is unavailable for the unclipped form. Kerleroux (1999) discusses more systematic semantic effects of clipping in the particular case of nominalizations in *-ion*, a corner of the lexicon where clipping is particularly productive (*introduction* > *intro* 'introduction', *manifestation* > *manif* 'manifestation, demonstration', etc.). According to that study, clipping selects noneventive readings of these nouns, leading to contrasts such as that in (3).

(3) a. L'**introduction**/*l'**intro** du lynx dans le Vercors provoque des tollés.
'The **introduction** of the lynx in the Vercors causes outcry.'

b. L'**introduction**/l'**intro** de ta dissertation nous a bien fait rire.
'The **introduction** of your essay made us laugh.'

Although Anselme et al.'s (2021) more thorough empirical study does not confirm the categorical effect that Kerleroux hypothesized, it confirms the existence of differences in the statistical distribution of readings of clipped and nonclipped forms that could not be accounted for if they were a single lexical entity.

Third, clipping feeds all varieties of word formation. Table 2 documents cases where clipped forms are bases in either native or neoclassical compounding, while Table 3 provides some examples where they are bases for either affixal derivation or conversion. In none of these cases can the unclipped forms be substituted for the clipped form.

Table 2: Examples of clipping feeding compounding

Base	Clipped	Compound
vélocipède	vélo	vélodrome
'bicycle'		'bicycle racing arena'
cinéma	ciné	cinéphile
'cinema'		'cinema enthusiast'
automobile	auto	automitrailleuse
'automobile'		'machine guncarrying (armored) car'
homosexuel	homo	homophobe
'homosexual'		'homophobic'
microphone	micro	micro-cravate
'microphone'		'clip-on microphone'

Fourth, apocope is frequently accompanied by the suffixation of an extra morph redundantly conveying informality (4), most often but not exclusively *-o*.[4] To the extent that the use of a particular suffix in the context of a particular base is conventionalised, this is again a clear sign of lexical listedness of clipped forms.

(4) a. *prolétaire* > *prolo* 'proletarian',
intellectuel > *intello* 'intellectual',
régulier > *réglo* 'reliable',
socialiste > *socialo* 'socialist', etc.

[4]Suffixation occurs with 25% of apocopes in Antoine's data, and the suffix is *-o* in 36% of these cases. As discussed at length by Vihman (2019), this particular suffixation creates a larger family of clipped nouns fitting in a phonological template with a final *-o*, which is sometimes already present in the base, see e.g. *intro* from *introduction*, *ado* from *adolescent*, etc. Together, suffixed and unsuffixed *-o* forms make up 32% of Antoine's dataset. Note also that *-o* suffixation is often accompanied by allomorphy, see e.g. *crado* from *crasseux* 'filthy', *dirlo* from *directeur* 'director', *parigot* from *parisien* 'parisian'. See Kilani-Schoch & Dressler (1999) on the pragmatic value of *-o*.

Table 3: Examples of clipping feeding derivation

Base	Clipped	Derivative
vélocipède	vélo	vélociste
‘bicycle’		‘bicycle mechanic’
cinéma	ciné	cinéaste
‘cinema’		‘filmmaker’
mathématiques	math	matheux
‘mathematics’		‘mathematician (informal)’
camelote	came	(se) camer
‘junk’	‘drugs’	‘do drugs’
blaireau	blair	blairer
‘badger’	‘nose’	‘get along with someone’

b. *matériel* > *matos* ‘gear’,
cantine > *cantoche* ‘canteen’,
anglais > *angliche* ‘English’, etc.

Finally, clipping interacts in crucial ways with other nonconcatenative processes of French conveying related types of social meaning.[5] Most significantly, clipping often affects *verlan*, a word formation process whose output is characteristic of ingroup teenage speech,[6] and consisting in modifying the prosodic makeup of a word in a manner that obfuscates the sound-meaning association,

[5] Antoine (2000) describes examples such as those in (i) as cases of clipping feeding reduplication. Although these cases definitely have a family resemblance with those discussed in the main text in terms of their social meaning, and they do involve some form of truncation, the fact that they uniformly involve truncation to a single open syllable suggests that they follow a different pattern, widely attested in hypocoristics (ii), and documented in detail by Plénat (1984).

(i) *communiste* > *coco* ‘communist’, *crasseux* > *cracra* ‘filthy’, *nourrice* > *nounou* ‘nanny’, *mignon* > *mimi* ‘cute’, *général* > *gégène* ‘general’

(ii) *Sophie* > *Soso*, *Jean* > *Jeanjean*, *Émile* > *Mimile*

[6] The social history of verlan use is complex. Most sources agree that, although it has much deeper origins, it started gaining widespread popularity as a marker of ingroup status for youth of disinfranchised *banlieues* of the Paris area in the 1960s and 1970s. It then gained quick cultural prominence in the 1980s and 1990s, to become mostly a generational marker. Ironically, there are now multiple generations of *verlan* users attuned to its apropriateness only for younger speakers: many parents of contemporary teenagers used exactly the same forms as their children in the same context at the same age.

using a combination of vowel deletions and insertions, and, most compicuously, inversions of order (see Lefkowitz 1991, Plénat 1995 for detailed description). Importantly for our purposes, in many cases, verlan formations of two syllables are followed by clipping, resulting in a single closed syllable, as illustrated in Table 4. Again there is clear evidence of the convetionalization of clipped forms: for all the examples in Table 4, the nonclipped form is barely used, if at all. Likewise there are many other examples where verlan formation is not accompanied by clipping in the conventionalised form, although there is no reason that it could not, e.g. *pourri* > *ripou* 'rotten', *méchant* > *chanmé* 'mean', *tête* > *teuté* 'head'.

Table 4: Examples of verlan feeding clipping.

Base	Verlan	Clipped verlan	Translation
flic	keufli	keuf	'cop'
fête	teufé	teuf	'party'
femme	meufa	meuf	'woman'
père	reupé	reup	'father'
arabe	beura	beur	'arab'
mec	keumé	keum	'guy'
parent	rempa	remp	'parent'

From this quick overview of clipping in French, we conclude that clipping is clearly a productive word formation process integrated in a family of morphopragmatic means of conveying informality, but leading to institutionalized coinage of new lexemes. Hence it is a legitimate question to ask whether and how clipped lexemes inflect.

2.2 The potential for uninflectability

In this subsection we quickly review the structure of the French inflection system, and show that verbs are the only class of lexemes where uninflectability is likely to be observable.

Inflection is found in French in nouns, adjectives and verbs. However, inflectional variability is strongly impoverished in nouns and adjectives, and structured in such a way that uninflectability cannot arise.

Table 5 illustrates the main inflectional characteristics of French nouns. The vast majority of nouns pattern like *personne* 'person': the singular and plural form are identical in most contexts, although a final /z/ may optionally be realized in the plural in so-called *liaison* contexts, i.e., when followed by a vowel-initial attributive adjective.[7] This pattern accounts for 99.3% of the 31,002 nouns documented in the *Flexique* database (Bonami et al. 2014). The few remaining cases exhibit some alternation between singular and plural forms; /aj/~/o/ and /al/~/o/ are the only two alternations found with a sizable number of lexemes. Importantly for our purposes, there is no phonotactic requirement that a noun following the main pattern has to meet to be usable in either the singular or the plural: any phonotactically well-formed sequence will do. As a result, we expect any new noun to be able to inflect normally.

Table 5: Sample of inflectional patterns in French nouns.

Orthographic citation form	Inflected forms SG	PL	Translation
personne	/pɛʁsɔn/	/pɛʁsɔn(z)/	'person'
vitrail	/vitʁaj/	/vitʁo(z)/	'stained glass'
amiral	/amiʁal/	/amiro(z)/	'admiral'
œil	/œj/	/jø(z)/	'eye'

Moving on to adjectives, the situation is similar although a little more complex. A typical French adjective follows the pattern illustrated in Table 6 by *joli* 'pretty' and *net* 'clear': identical forms throughout, with the optional realization of a final /z/ in the plural in *liaison* contexts. This pattern is found with 39% of the adjectives in *Flexique*. Deviations of the pattern come in two main varieties. First, many adjectives such as *petit* 'small' exhibit an alternation between a final consonant in the feminine, and an optionally realized consonant in the masculine singular (realized, again, in *liaison* contexts). Second, a smaller set of adjectives have a final /o/ in the masculine plural alternating with a final /al/ in the other three cells of the paradigm.[8] The important point for our purposes is that there

[7]See Bonami & Delais-Roussarie (2021) for a recent overview of French *liaison*.

[8]There is a handful of other patterns of low type frequency that have been the focus of much theoretical attention but have no bearing on the discussion at hand. See Bonami & Boyé (2005) for details. Typical descriptions also include a set of fully uninflectable adjectives such as *marron* 'brown' or *gnangnan* 'soppy', which are written without a final <s>, and purportedly do not give rise to liaison, in the plural. To our knowledge this has never been verified empirically. A search in web corpora documents thousands of examples of the purportedly incorrect orthography in phrases such as *yeux marrons* 'brown eyes', and native speaker intuitions on liaison are far from clearcut. While more research would be necessary, this suggests that these adjectives may pattern with *joli* except in highly controlled formal speech and edited writing.

is at least one clearly productive pattern with high type frequency that places no phonotactic requirements on its members, namely the pattern found with *joli* and *net*: any phonotactic shape that is a legal French word can be used as an adjective in any of the four cells in the paradigm. As a result we reach the same conclusion that we reached for nouns: we expect any new adjective to be able to inflect normally.

Table 6: Sample of inflectional patterns in French adjectives.

Orthographic citation form	Inflected forms M.SG	M.PL	F.SG	F.PL	Translation
joli	/ʒoli/	/ʒoli(z)/	/ʒoli/	/ʒoli(z)/	'pretty'
net	/nɛt/	/nɛt(z)/	/nɛt/	/nɛt(z)/	'clean, clear'
petit	/pəti(t)/	/pəti(z)/	/pətit/	/pətit(z)/	'small'
global	/global/	/globo(z)/	/global/	/global(z)/	'global'

Things become more interesting when we turn to verbs. Unlike nouns and adjectives, French verbs make systematic use of affixal exponence. This is true in particular for verbs belonging to the first conjugation, which is the largest inflection class by far (87% of the verbs documented in *Flexique*[9] and the only one that is clearly productive, both in terms of hosting newly morphologically-formed verbs and borrowings. Table 7 shows the paradigm of one such verb. Here two important observations are in order. First, most inflectional affixes are vocalic. This is in particular the case for the (homophonous) most frequent suffixes of the infinitive, past participle, indicative present 2PL and imperative 2PL (note that other frequent cells, such as those in the present singular and 3PL, use a bare stem). Second and relatedly, there is strong phonotactic constraints on the shape of the basic stem of a first conjugation verb, as witnessed by the form used in the indicative present singular. Only 6.3% of the verbs in *Flexique* have a final vowel. That vowel is never a nasal nor /o/, /ɔ/, /ø/ or /œ/. The vowel /a/ is only attested after /w/ in verbs exhibiting a /wa/~/waj/ allomorphy (witness *il aboie* /ilabwa/ 'he barks' vs. *aboyer* /abwaje/ 'to bark'). /ɛ/ and /e/ are much rarer as final segments of verb stems than of stems for nouns and adjectives.

As a result of these observations, one might expect uninflectability in French verbs, if there is a source of new verbs ending in a vowel other than /u/, /i/, or /y/, as the phonological shape of these stems might make them poor candidates

[9]This is the size of the relevant macroclass according to Beniamine et al. (2017). There is some internal variability within the first conjugation, so that only 69% of verbs inflect *exactly* as illustrated in Table 7.

Table 7: The paradigm of the regular first conjugation noun LAVER ‘wash’

Finite forms

	1SG	2SG	3SG	1PL	2PL	3PL
IND.PRS	lav	lav	lav	lavɔ̃	lave	lav
IND.IMPERF	lavɛ	lavɛ	lavɛ	lavjɔ̃	lavje	lavɛ
IND.PST	lavɛ	lava	lava	lavam	lavat	lavɛʁ
IND.FUT	lavəʁɛ	lavəʁa	lavəʁa	lavəʁɔ̃	lavəʁe	lavəʁɔ̃
COND	lavəʁɛ	lavəʁɛ	lavəʁɛ	lavəʁjɔ̃	lavəʁje	lavəʁɛ
SBJV.PRS	lav	lav	lav	lavjɔ̃	lavje	lav
SBJV.IMPERF	lavas	lavas	lava	lavasjɔ̃	lavasje	lavas
IMP	—	lav	—	lavɔ̃	lave	—

Nonfinite forms

INF	PRS.PTCP	PST.PTCP			
		M.SG	F.SG	M.PL	F.PL
lave	lavɑ̃	lave	lave	lave	lave

for first conjugation inflection, the verbal inflectional class that neologisms and borrowings are assigned to.[10] In light of this, consider the dialogue in (5) between two Parisian teenagers, overheard by the last author of this paper in 2014:

(5) S1: Oh non, je me suis encore fait
Oh no 1SG.NOM 1SG.ACC be.PRS.1SG again make.PST.PTCP
déco!
disconnect.INF
‘Oh no, I was **disconnected** again!’

S2: Hein, il t’ a **déco**?
What 3SG.M.NOM 2SG.ACC have.PRS.3SG **disconnect.PST.PTCP**
‘What, he **disconnected** you?

[10]There are very few exceptions to this pattern. Neologisms that get assigned to conjugations other than the first have a clear analogical attractor in the target conjugation, such as *alunir* ‘to land on the Moon’, assigned to the second conjugation because formed in analogy to *atterrir* ‘to land (on Earth)’, which belongs to the second conjugation.

Eh ben **reco**-toi!
Well then **reconnect.IMP.2SG**-2SG.ACC
'Well then **reconnect**!'

[...]

S1: Je suis **redéco**!
1SG.NOM be.PRS.1SG **redisconnect.PST.PTCP**
'I am redisconnected!'

Here we clearly witness uninflectedness, both in the infinitive and the past participle, in three morphologically related clipped verbs whose stem ends in *-o* as a consequence of clipping. Hence we witness a good candidate for uninflectability for phonological reasons. What is striking, then, is that such cases are not documented in the literature. They do have a strong feeling of novelty, and tend to feel unacceptable to speakers over 40. In fact, apocope is usually said to be impossible with verbs, except in very specific cases discussed below, and such cases have not caught the attention of Cerquiglini (2019) in his recent overview of the phenomenon.

The goal of this paper is to document the existence of such cases, and to point out their theoretical importance.

2.3 Documented uninflected verbs

Although examples such as those in (5) are undocumented, there are some well known cases of uninflectedness in French verbs. One case that is important but orthogonal to our discussion is that of the lexical items *voici* and *voilà*, which are well-known to be best analyzed as subjectless presentative verbs, as they have a host of syntactic properties not found with other verbs (Morin 1985, Bouchard 1988). However these lexical items exhibit no inflectional variation, and are found only in root clauses conveying an assertion, with no co-occurrence with time adverbials; hence they seem to lack an inflectional paradigm altogether, rather than using forms lacking expected exponents in some or all paradigm cells.

(6) Voilà ton livre.
PRES 2SG.M.SG book
'Here is your book.'

More interesting are a series of verbs that alternate between overt inflection and uninflectedness, and are all used within the sphere of informal speech (see among other sources Antoine 2000, Cerquiglini 2019, Gadet & Wachs 2016). A prime but lonely example is the verb *fiche*, a euphemistic alternative to the verb *foutre*, originally meaning 'have sex', but which has almost entirely shifted to an argotic equivalent of 'do, make, give'. Importantly, *fiche* is found without a suffix in clearly infinitival contexts such as (7), in alternation with a suffixed form.[11]

(7)

J'	en	ai	rien	à	faire/	foutre/ fiche/ ficher.
1SG	PART	have.PRS.3SG	nothing	to	make.INF	do.INF

'I couldn't care less.'

A second relevant example is that of a small family of argotic verbs in *-ave* ultimately borrowed from Romani (*chourave* 'steal', *bicrave* 'sell', etc.) or formed by analogy to these (*bédave* 'smoke weed'). These are clearly attested in the infinitive and past participle, either uninflected or with normal first conjugation inflection. Easily accessible corpora contain many more inflected than uninflected forms, but this may be an effect of accessibility of the relevant varieties.

(8) a. T fou toi mec tu va **chourave** mon outil de travaille ?[12]
'Are you crazy, you would **steal** my working tool?'

b. Ben alors, Ford, on va **chouraver** des pneus à la concurrence maintenant ?[13]
'So what, Ford, you now **steal** pneumatics from the competition?

Table 8: Example of verlan verbs

Base	Verlan	Clipped verlan	Translation
énerver	vénère	—	'vex'
choper	pécho	—	'catch, seduce'
gonfler	flégon	flègue	'annoy'
niquer	kéni	ken	'fuck'
fumer	méfu	mef	'smoke'

[11] The rest of the paradigm of *fiche* is not well documented. The past participle seems to be uniformly *fichu*; there are documented uses of an uniflected form in the present 1PL or 2PL, although these are much less frequent than their regular alternatives *fichons* and *fichez*.

[12] https://uberzone.fr/threads/seine-et-marne-quatre-conducteurs-de-vtc-braques-en-trois-nuits.13833/page-2, consulted on October 2, 2023. Orthography as in the original.

[13] https://www.downshift.fr/ben-alors-ford-on-pique-des-pneus-a-mercedes, consulted on October 2, 2023.

A third example is that of verbs that have undergone *verlan*, sometimes accompanied with clipping of the final vowel. Table 8 presents some relevant examples. Notice that some of these verbs, such as *pécho*, do present a phonotactic shape unfit for inflection, but most don't. Be that as it may, we find again for these verbs alternation between uninflectedness and normal first conjugation inflection, as witnessed in (9–10).

(9) a. Dès qu'on te dit « bonjour », tu sens que tu vas **ken** ce soir. [14]

'As soon as they say 'hi', you understand that you will **have sex** tonight.

b. Les gens abuse [...] la fin est nul oui mais le film est parmis ceux ou y'a du sexe arrêtez de vous plaindre comme si vous **kenez** pas ![15]

'People exaggrate [...] Yes, the ending sucks, but this is a film where there is sex, stop complaining as if you never **had sex**!'

(10) a. Bon Marinette et Adrien ça fait 5 ans que j'attend que vous vous **pécho**.[16]

'Well, Marinette and Adrien, I've been waiting for 5 years for the two of you to **hook up**.'

b. Tu comptes la **péchoter** en oubliant son anniv ?[17]

You plan to **hook up** with her while forgetting her birthday?

This series of examples shows that, although it is a rare phenomenon, there is clear precedent in French for verbs to be uninflected, and that uninflectedness is associated with phenomena in the sphere of informal speech. This possibly set the stage for the emergence of a more systematic source of uninflectedness in clipped verbs, to which we now turn.

3 Clipped verb forms in corpora

In the previous section we have presented circumstantial evidence that younger speakers of French readily clip verbs, and that this leads to these verbs being uninflected. We now turn to a systematic exploration of the phenomenon. We begin by designing a method to retrieve clipped forms from corpora, and reconstructing the corresponding full form. The goal of this endeavour is to investigate the

[14]https://www.topito.com/top-preuves-tes-en-chien, consulted on October 2, 2023.
[15]https://twitter.com/NellyKoffi9/status/1519410504346836993, consulted on October 2, 2023. Orthography as in the original.
[16]https://twitter.com/if4irysieun/status/1317846593534558210, consulted on October 2, 2023.
[17]https://www.wattpad.com/amp/603513342, consulted on October 2, 2023.

scale of the phenomenon, and create a data set from which to extract generalisations and make inferences about verbal clipping.

Clipping consists of removing phonological material from a word form. A promising strategy therefore is the operationalisation of clipped verb forms in corpora as substrings of existing verb forms. This simple strategy would retrieve all clipped verb forms, but it would also yield a large number of false positives, as any word that happens to be a substring of another would be flagged as a clipped form (e.g. *disc* would be falsely flagged as a clipped form of *discuss*). Therefore the part of the method responsible for identifying potential clipped forms consists of finding substrings of existing verb forms and imposing filtering conditions to avoid including non-clipped forms that fulfil the main criterion. The second part of the method is responsible for retrieving the non-clipped version of the clipped form. This is done by means of distributional semantics and rule-based inference based on context surrounding the form.

3.1 Methodology

3.1.1 Finding candidate clipped forms

We build a reference lexicon containing non-clipped forms of the language. This will serve to check whether a word form is an existing word of French. As many commonly used lexicons of French such as the GLàFF (Hathout et al. 2014) or Wiktionary include clipped forms and filtering them out manually would be highly time consuming, we instead choose to build our own reference lexicon from corpora that do not contain clipped forms due to their formal register. We construct a lexicon of 331,332 words by combining tokens found in the Français-GUTenberg dictionary (Pythoud 1999), a resource created for automatic spelling correction, and in FREELANG (Vivier 2003), a list of French words that is commonly used for practical applications that need a normative filter, such as creating a word-based game. While the combination of these two resources does not cover the entirety of words in the French language, it does have good coverage of frequent words. This is sufficient for our purposes, since clipping in French is more likely to affect frequent words, and infrequent words tend to be longer than frequent ones (Zipf 1949, 1935), so it is likely that infrequent words will be superstrings of words already present in our lexicon, and therefore unhelpful for the identification of clipped forms. As the final method will be run on each token in the corpus in context, we organise the lexicon into a trie, as shown in Figure 1, so that searching whether a word is a subset of existing words and not an existing string in the dictionary is faster and more efficient than looking the string up in a dictionary as a whole.

A word in a corpus is considered a candidate clipped form if it is searched for in the trie, and it constitutes a valid path through it but its last letter does not constitute a leaf node.

We excluded from consideration words of one or two letters,[18] words in all caps (as these are overwhelmingly acronyms), and words starting with a capital letter but not located at the beginning of a sentence (as these are overwhelmingly proper nouns).

This part of the method returns a candidate clipped form with all the candidate full words it could correspond to.

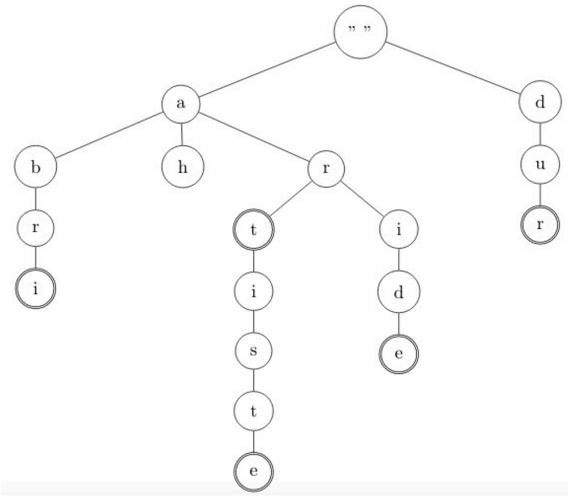

Figure 1: A subset of the lexicon containing the French words {*ah*, *abri*, *artiste*, *aride*, *dur*}, organised into a trie. Nodes with a thicker border represent potential end nodes - paths starting from the origin of the trie and stopping at an end node constitute words in the vocabulary

3.1.2 Reconstructing the correct form

The next step consists of assigning each candidate clipped form a full form. The idea behind the method is to choose between the candidate full forms identified by the previous step by relying on the sentential context surrounding the candidate clipped form: in English, *ref* in isolation could be the clipped form of both

[18]While clipped verbs of two letters may exist, candidate clipped words of less than three letters lead to too much indeterminacy when reconstructing their full form. While it would be conceivable to extend the method to two-letter forms, the data reported in this paper does not include them.

reference or *referee*, but knowing that the occurrence of *ref* was found in the sentence *'the ref sent off Zidane'*, one can infer with high certainty that it is a clipped form of the second word.

To capture lexical semantics, we rely on `word2vec` (Mikolov et al. 2013, `gensim` implementation), a model that derives word embeddings by training a neural network to predict a masked word. We specifically rely on the ability to query the model as to the probability of different candidate words in a context. We trained models on the Leipzig French News corpus (Goldhahn et al. 2012). The following hyperparameters were varied in training:

Table 9: Hyperparameter combinations attempted for training `word2vec` models.

Hyperparameter	Values
vector size	{10, 20, 50, 100, 150, 200}
number of epochs	0-200, in increments of 5
context window	{3, 5, 7}

Based on performance of clipped form detection and full form reconstruction on a specially created test corpus, we chose to integrate into the method the `word2vec` model with a size of 50, 125 training epochs and a context window of 3.

For every candidate token clipped form, we masked the clipped form and asked the model to predict how likely it was that each of the candidate full forms could fill the gap. The most likely form was outputed as the corresponding full form for the candidate clipped token.

3.2 Evaluation

We tested the method's performance on held out data containing 362 clipped forms. The method correctly identified 318 (87%) of these as candidate clipped forms. No false positives were attested, but the 44 remaining clipped forms were not identified – for 39 of them, this was because the clipped forms happened to already be in the lexicon despite our efforts to minimise this possiblility. The remaining 5 clipped forms were not identified because their corresponding full form was not included in the lexicon.

As for the method's performance in automatically identifying the full form, a form judged to be the correct one by human annotators was produced for 190 of the clipped forms (53% out of the 362 clipped forms, 60% of the 318 forms correctly identified as being clipped).

3.3 Discussion

While the performance of the full method is not sufficient to perform trustworthy large scale quantitative analyses of clipped forms, its output still provides a starting point for us to make qualitative generalisations about clipped forms, and their features in the verbal domain in French.

We set out to find clipped forms in the FrWaC corpus (Baroni et al. 2009), a large web-crawled corpus of French. By restricting our sample to clipped forms that have been given a verb as their full form and observing them in context, we observe several things.

Clipping of verb forms in informal French is pervasive, suggesting that we are indeed seeing the emergence of a new source of uninflectedness in the French language. Accounting for the low accuracy of our model and assuming that only 60% of the forms that have been flagged as clipped verbs have been correctly labeled (as indicated by the manual evaluation), we can conservatively estimate to have found 182,520 clipped verb tokens, in a wide variety of cells, suggesting that clipping is productive and possible throughout the verbal paradigm.

A broader discussion of what the clipped verbs extracted teach us about the phenomenon can be found in Section 5.

4 Clipped verb forms in language use

The previous section outlined a method to find clipped forms in corpora and discussed observations based on the sample. However, since the data is orthographic, one may question whether the observed clipped forms are in the best case a phenomenon limited to writing, or in the worst case simply typos. To address this possiblity, we perform an experiment to evaluate whether French speakers accept clipped verbs in speech.

4.1 Methodology

4.1.1 Task

The experiment is a modified acceptability judgement task. In each trial, participants heard a recording of two speakers, one asking a question and the other providing the answer. The task was to judge the acceptability of the answer along three dimensions: whether they could imagine overhearing this exchange between two third parties (testing for the acceptability of the clipped form), whether they thought the answer constituted correct French (this question was included to allow participants to express feelings about the infelicitousness of the items where the formality of the question was not matched to the formality of the answer, to prompt participants to separate their intuitions about register from their intuitions about grammaticality and usage) and whether they thought the answer was socially appropriate given the question. In addition, participants were asked to provide a rephrasing of the answer, to check whether they had understood it.

4.1.2 Items

Each item was composed of two recordings, one of a speaker asking a question, and the other of someone answering it. The question and the answer varied independently in formality, yielding four versions of each item. The choice to have the items vary in whether the question and answer match for formality, rather than presenting just formal and informal items as a means to distract participants from the purpose of the experiment, which is to establish the acceptability of verbal truncation conditional on register. The formality dimension contrasted an informal register with a neutral register (containing no markers of formality nor informality). To obtain the informal version of the sentence, we manipulated known markers of informal speech in French, such as lexical choice, the use of informal interjections, and *tutoiement* (the use of the informal second person singular pronoun *tu*, as opposed to the more formal *vous*). The informal version of each question or answer contained two of these markers, which were absent in the neutral version.

The answer portion of each item contained a verb or a noun that could appear in its truncated form. The informal version of the item contained the truncated form, as well as another marker of informality, while the neutral version contained the non-truncated form and no marker of informality.

A total of 90 items were created. One third of them contained a nominal crucial item in the answer while two thirds contained verbs. Each of the items could be presented in one of four versions, depending on the register of the question and the answer.

4.1.3 Participants

Participants were recruited by word of mouth and through links on social media. Participation was voluntary and they were not compensated for their time.

The experiment had a total of 25 participants, 18 women and 7 men. Their ages ranged between 12 and 74, with a median of 22. Three participants hadn't completed high school yet, 10 had a high school diploma, 1 had a bachelor's degree, 8 had a masters' degree or higher qualifications.

4.2 Results

Only items for which the question and the answer matched in formality were retained for analysis. Informal items contained truncated forms and formal items contained non-truncated forms of the same lexeme. This left 850 data points for modeling.

The three measures have a small amount of correlation (Table 10), suggesting that participants were treating the judgements as separate to a reasonable degree, as intended.

Table 10: Pearson correlation between the different measures collected from participants (Poss. = Possibility)

	Poss. of overhearing	Situation Fit	Correct
Poss. of overhearing	1.00	0.39	0.36
Situation Fit	0.39	1.00	0.33
Correct	0.36	0.33	1.00

The overall acceptability of items was extremely high, with 82% of datapoints having been given an acceptability above 0.9 out of 1 on an anchored but unmarked scale, a figure that diminished somewhat but remained comparable when looking at truncated verbs only, as shown in Table 11.

Table 11: The three judgements ($0 < x < 1$) the experiment required speakers to perform, by item condition and part of speech.

Part of speech	Formality	Poss. of overhearing	Situation fit	Correct
Noun	Neutral	0.96	0.90	0.85
Noun	Informal	0.92	0.89	0.63
Verb	Neutral	0.94	0.86	0.79
Verb	Informal	0.81	0.77	0.40

We fit a beta regression to responses to the acceptability question, which asked people whether they could imagine hearing the exchange on the street. The regression included random intercepts for item and participant, and age, formality, part of speech of the crucial item in the answer, and item order as fixed effects. The set of fixed effects employed was selected by stepping down a model which originally included other demographic information about the participants such as gender and education, and experimental variables such as item order, which were found to not contribute significantly to explaining the variance. Age and item order were standardised. Formality had neutral as a reference level, and the part of speech of the crucial word had noun as the reference level. The model summary is shown in Table 12.

Table 12: Coefficients for a generalised linear model predicting the acceptability of the answer

	Estimate	Std. Error	z value	p value
(Intercept)	2.23	0.21	10.68	<0.001
Formality (informal)	-0.18	0.11	-1.70	0.09
Part of speech (PoS) (verb)	0.01	0.11	0.05	0.96
Item order	0.00	0.00	0.41	0.68
Age	0.00	0.01	-0.84	0.40
Formality (Informal): PoS (verb)	-0.51	0.14	-3.58	<0.001
Formality (Informal): Age	0.00	0.00	-0.31	0.76

Age and item order do not have an effect, nor do formality or part of speech when considered in isolation. The only significant effect is that of the interaction between formality and part of speech: truncated verbs are rated lower than non-truncated verbs, and both truncated and non-truncated nouns.

4.3 Discussion

The overall high score given to truncated items is good evidence that both nominal and verbal truncations appear to be in use in the French language.

The model output validates the descriptive statistics, shows that compared to truncated nouns, truncated verbs have reduced acceptability. Nevertheless, this finding is to be interpreted in the context of overall very high acceptability for all items. The comparative novelty of verbal truncation likely explains its lower acceptability compared to nominal truncation, but data from the experiment suggests that they are both accepted in informal contexts.

While age did not appear to have an effect, the lack of older participants in our sample makes it hard to estimate the true effect of this demographic variable, though anecdotally verbal clippings are much less frequent and much less accepted in the speech of older French speakers. An additional caveat about the generalisability of our results comes from the fact that our sample was entirely composed of speakers of hexagonal French – other varieties of French might show different patterns of clipping, and different levels of acceptability for such formations. A survey of clipping in other varieties of French constitutes a potentially fruitful avenue for future research on the phenomenon.

5 General discussion

Combined, the two studies above show that speakers are familiar with truncated verbs, and that they are used in speech as well as in writing. Examining the outputs of the automatic truncation extraction method allows us to make inferences about the process of verbal truncation in French, its relation to uninflectability, and its place relative to other known truncation processes in the language.

When discussing uninflectedness, it is common to focus on the causal role of the phonology of the stem, a theme exemplified in the introduction by Czech uninflectability, a phenomenon often reconduced to the ill-suitedness of the phonological form of certain borrowed words to serve as the citation form. The case of French verbal clipping highlights the variety of motivations that can lead to a word being uninflected. In most of the examples discussed in this chapter, there is no factor in the grammar that motivates the lack of inflectional endings on these forms: any consonant-final stem word is a good candidate for serving as the stem of an inflected verb. For vowel-final truncations, such as clipped forms in *-o* like *déco*, *reco* (5), there is a credible analogical path for innovating inflected forms: conversion of vowel final lexemes to verbs typically involves epenthesis of a stem-final *t*, as in *noyau* 'core' vs. *noyauter* 'to core', *flou* 'blurry' vs. *flouter* 'to blur', *blabla* 'blather' vs. *blablater* 'to blather', etc.). Examples such as *péchoter* discussed above in (10), or *psychoter* 'worry disproportionately', from *psychose* 'psychosis', suggest that *t* epenthesis is easily mobilised as a repair strategy for adapting a stem that is a poor fit for verbal inflection. Yet we do not find *décoter*, *recoter* as inflected infinitives for clipped versions of *déconnecter, reconnecter*. It hence appears that French verbal uninflectedness is not motivated by a grammatical impossibility, and thus is conventionalized. We speculate that it may have started as language play introducing variation within the morphological system, persisting in the language thanks to the social meaning it has taken on, which would be a lot less visible with regular inflections.

In the sample extracted by our method, we observe a number of interesting phenomena about clipping. We note that there is no biuniqueness requirement for clipping, no constraint for a one-to-one correspondence between the clipped form and its full form. There are clipped forms that may correspond to different full forms depending on the context: we found *supp* as the clipped form of both *supposer* 'to suppose' and of *supporter* 'to give support to'. Conversely, the same verb may have more than one clipped form: *modifier* 'to modify' was found to be clipped as both *mod* and *modif*.

Perhaps the most interesting finding is the interaction between clipping and stem allomorphy. All verbal inflection is suffixal in French, and as clipping in French verbs operates on the right edge, most instances of clipping amount to uninflectedness. However, some French verbs take on a different stem depending on which paradigm cell they realise, so when they undergo clipping, they lose suffixal inflectional markers but not the information signaled by the stem change. Such an example is the verb VOULOIR 'to want', which has the stem *voul-* in the 1PL and 2PL of the present indicative, but *voudr-* in the future. In our corpus, we found clippings of VOULOIR in both the present and the future, with the forms *voul* and *voudr* respectively. In these cases, the process of clipping preserves information about TAM marking as well as about subject agreement. This data point is an interesting addition to the discussion about the domain of clipping. The forms above suggest that the grammar first generates the full inflected form, and clipping is a low-level process that happens at the interface between morphology and phonology. This situation contrasts with the nominal, adjectival, and adverbial clipping discussed in Section 2, showing that clipping can be taken as the input of word formation, and that it is therefore a more high-level process. Taken together, these observations suggest that clipping is not a single process in French, but instead a family of operations, each with their own domain of application.

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Contributions

The research reported here was conducted while Maria Copot was a PhD student, and Ninoh Agostinho Da Silva, Ahmed Beji and Arno Watiez were Master's students, at Université Paris Cité. Olivier Bonami's main contribution was to write Section 2, while the rest of the paper was written by Maria Copot and reports on joint work with the other authors under Maria's supervision.

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Chapter 9

Inflection dropping in the English-origin verbs of present-day French

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Fifty frequent English-origin verbs extracted from a corpus of 100 million French tweets from 2020–2022 were analyzed for uninflectedness in two composite verb forms – the *passé composé* construction and the periphrastic future construction. It was found that a majority of these verbs show a substantial preference for the nonstandard practice of dropping inflectional marking on the past participle (for the *passé composé*) and the infinitive (for the periphrastic future). This salient feature of written social media discourse is unexpected as it violates a presumably categorical norm but it is not truly exceptional in light of other underreported phenomena of verbal uninflectedness, especially when French is in contact with other languages. It is also suggested that unadapted forms are not only a marked choice flagging foreignness and peripherality, but also, in some cases, markers of in-group membership.

1 Introduction

The literature on the morphological adaptation of anglicisms in present-day French has so far largely focused on nouns and adjectives and Saugera (2012a,b) has convincingly argued that the specific classes of items that resist plural inflection marking generally do so in accordance with subrules that originate within the French morphological system. Verbs are generally less frequently borrowed than nouns (Tadmor 2009) and less attention has been paid to verbal anglicisms in the literature. Their adaptive behavior is generally appraised as follows:

Vincent Renner & Adam Renwick. 2026. Inflection dropping in the English-origin verbs of present-day French. In Enrique L. Palancar & Sebastian Fedden (eds.), *Voids in morphology: Exploring "uninflectedness"*, 271–287. Berlin: Language Science Press.
DOI: ??

while the adaptation of foreign-origin nouns and adjectives to the French inflectional system is variable, that of borrowed verbs is obligatory (Anastassiadis-Syméonidis & Nikolaou 2011, Poplack 2017) and contact with English has had no significant impact on the grammatical core of the language as French verbs continue to be conjugated (Saulière 2014). These general statements may, however, need to be partially reevaluated in light of the recent emergence of uninflected English-origin verbal forms, as illustrated by the following tweet excerpts, in (1–3), and news magazine article title, in (4):

(1) J'ai <u>crush</u> sur une meuf avec qui je travaillais.
'I had a crush on a chick I was working with.'
(the inflected past participle form *crushé* is expected)

(2) Je vais <u>spoil</u> donc lisez pas le thread si vous êtes pas à jour sur le manga.
'I'm going to spoil things so don't read the thread if you're not up to date on the manga.'
(the inflected infinitive form *spoiler* is expected)

(3) Je viens de <u>play</u> cette vidéo en salle de pause à fond.
'I've just played this video in the break room with the volume full on.'
(the inflected infinitive form *player* is expected)

(4) Esclavage : l'Écosse s'apprête-t-elle à "<u>cancel</u>" son patrimoine historique ?
'Slavery: Is Scotland about to "cancel" its historical heritage?'
(Marianne, 22 Dec. 2021)[1]
(the inflected infinitive form *canceler* is expected)

Such occurrences could be brushed aside as individual ungrammatical oddities but their recurring presence in random searches on the social network Twitter[2] acted as a spur to take this incipient phenomenon as seriously as it deserves and to attempt to measure the extent to which it is widespread in contemporary written social media discourse, in the broader context of keyboard-to-screen communication (KSC) (Jucker & Dürscheid 2012).

The remainder of this chapter is organized as follows: Section 2 describes the different steps taken to obtain a representative sample of 50 frequent English-origin verbs in a large corpus of French tweets. Section 3 presents the distribution of the inflected and uninflected forms in the *passé composé* and periphrastic

[1]https://www.marianne.net/monde/europe/esclavage-lecosse-sapprete-t-elle-a-cancel-son-patrimoine-historique (last accessed on 13 June 2024)

[2]This research was conducted before *Twitter* became known as *X*, which is why only *Twitter* is used in this chapter.

future constructions found in the corpus. Section 4 discusses the various findings in the light of other phenomena of verbal uninflectedness, particularly when French is in contact with other languages, and concludes.

2 Methods

Uninflected verbal forms in present-day French are hypothesized to be a morphological feature more typical of colloquial discourse and the difficulty of accessing vast corpora of spoken data led to relying on written social media data, which are known to epitomize register hybridity, mixing informal spoken features with more formal written ones, and to exhibit the same heterogeneity and patterns of linguistic change as non-KSC speech (Tagliamonte & Denis 2008, Bohmann 2020). Building an ad hoc corpus from Twitter data was opportunistic as the Academic Research Access API allowed for easy collection of millions of French tweets. Starting from the premise that verbal uninflectedness was an exceptional behavior and in order to keep the task of data processing manageable, only two constructions intuitively identified as more likely to contain instances of audible inflection dropping were selected for this exploratory study: the *passé composé* construction, which is formed with the auxiliary verb *avoir* ('to have') followed by a past participle and encodes either primary or secondary past tense (Mosegaard Hansen 2016), and the periphrastic future construction, which is formed with the auxiliary verb *aller* ('to go') followed by an infinitive and encodes secondary future tense (Mosegaard Hansen 2016). This choice was in part motivated by the fact that several highly frequent verbal inflectional markers are inaudible in French (by default, the three singular forms of the present indicative are, for instance, homophonous with the verbal root) and that spelling alone is not a reliable cue for inflectional (non-)marking in written social media discourse (in spontaneous written production, i.e. a situation in which spelling is not controlled, or standardized, it is not possible to determine whether inaudible deviations from the orthographic norm are deliberate, reflect the imposition of time or practical constraints on the act of typing, or indicate defective mastery of the standard written code).

Our corpus was assembled using the TAGS application (Hawksey 2016) to automate weekly queries for the two target constructions over a 15-month period, from November 2020 to February 2022, in pools of tweets automatically tagged as being in French (without any geographical specification) by Twitter. For each construction, the search pattern consisted of any string of characters containing a subject pronoun immediately followed by an inflected form of the target auxiliary (*avoir* or *aller*) in the present indicative. Inaudible uncontrolled spelling variants – e.g. *tu a* (/ty a/, you.SG have.3SG) instead of standard *tu as* (/ty a/, you.SG have.2SG) – were included in the queries. The collected data were then processed as follows:

(i) duplicates and retweets were automatically removed, which led to retaining approximately 220,000 tweets per day, for a total of approximately 100 million tweets over the sampling period;

(ii) the entire corpus was processed with TXM (Heiden 2010) to identify and rank in a decreasing order of frequency the variety of word tokens appearing immediately to the right of the searched patterns;

(iii) the top tokens were manually examined, non-verbs and non-English[3] verbs were discarded, and the review was stopped once 50 verb types were identified;

(iv) homophonous uncontrolled spelling variants for all 50 past participles and infinitives – e.g. *dropé* (/dʀɔpe/, give.up-PST.PTCP) in an infinitive context when standard *droper* (/dʀɔpe/, give.up-INF) is expected – were included;

(v) uncontrolled forms ending in a diacriticless <e> – e.g. *drope* – were considered ambiguous (deaccented graphemes are quite common in French KSC (Cougnon 2010) and a form like *drope* could correspond to either uninflected /dʀɔp/ or inflected /dʀɔpe/) and they were therefore excluded from the final token counts.

3 Results

Our sample of 50 frequent English-origin verbs of present-day French is presented in the list below in descending order of frequency (verbs are given in their uninflected forms, along with their primary senses in French and their total number of *passé composé* and periphrastic future tokens):

1. *dead (ça)* 'to kill it' (37,522)
2. *test* 'to put to test' (31,451)
3. *stream* 'to watch via streaming' (17,895)

[3]Cognate verbs for which it is unclear which lexical item (the French or the English one) appeared first and which one was borrowed – e.g. *contrôler* ('to control'; cf. *Oxford English dictionary, online edition*, s.v. *control*$_{\mathrm{V}}$) – were also discarded.

4. *flop* ‘to fail’ (13,328)
5. *bug* ‘to be baffled’ (12,283)
6. *drop* ‘to give up’ (11,339)
7. *follow* ‘to subscribe to someone’s tweets’ (10,999)
8. *check* ‘to verify’ (9,035)
9. *stop* ‘to cease’ (7,589)
10. *screen* ‘to screenshot’ (6,894)
11. *win* ‘to achieve victory’ (6,814)
12. *spam* ‘to post spam’ (4,970)
13. *boycott* ‘to refuse to deal with’ (4,719)
14. *spoil* ‘to reveal carelessly’ (4,680)
15. *pull* ‘to draw’ (4,422)
16. *stan* ‘to be an obsessive fan’ (3,894)
17. *leak* ‘to reveal inadvertently’ (3,791)
18. *skip* ‘to pass over’ (3,382)
19. *pop* ‘to appear’ (3,069)
20. *perform* ‘to give a performance’ (2,971)
21. *crush* ‘to have a crush’ (2,784)
22. *carry* ‘to lead a team’s success’ (2,662)
23. *switch* ‘to change’ (2,382)
24. *pack* ‘to obtain as part of a pack’ (2,344)
25. *tryhard* ‘to put a lot of effort’ (2,303)
26. *farm* ‘to repeat an in-game action’ (2,240)
27. *flex* ‘to show off’ (2,210)
28. *troll* ‘to act as a troll’ (1,850)
29. *play* ‘to entertain oneself’ (1,825)
30. *tag* ‘to add a tag’ (1,792)
31. *grind* ‘to repeat an in-game action’ (1,618)
32. *feat* ‘to appear on a song’ (1,578)
33. *boost* ‘to give a boost’ (1,541)
34. *rush* ‘to do quickly’ (1,513)
35. *tilt* ‘to get mad’ (1,304)
36. *clip* ‘to record videogame clips’ (1,298)
37. *shoot* ‘to take a picture’ (1,274)

38. *spawn* 'to appear (in a videogame)' (1,258)
39. *stonk* 'to gain' (1,244)
40. *cancel* 'to put an end' (1,215)
41. *ragequit* 'to quit in anger' (1,123)
42. *disband* 'to dissolve a group' (898)
43. *buy* 'to purchase' (888)
44. *pick* 'to choose' (831)
45. *kick* 'to remove forcibly' (777)
46. *stalk* 'to track online' (764)
47. *raid* 'to attack; to send viewers to another stream' (760)
48. *reset* 'to start again from zero' (742)
49. *bully* 'to harass' (664)
50. *back* 'to come/go back' (561).

The verbal meanings attested for French above are almost all attested in English. Two apparent exceptions are *screen*, a neologism presumably obtained by right-clipping the English verb *screenshot*, and *back*, which is apparently an output of deadverbial conversion. The French meanings often correspond to the general meaning of the English verb (e.g. *boost*, *check*, *test*), but there are also cases where borrowing leads to semantic reduction (Alexieva 2008, Winter-Froemel 2019) – only one domain-specific (and possibly marginal) meaning of the English verb is adopted in French (e.g. *clip*, *farm*, *pack*).

Beyond the wide attestation of these 50 verbs in French written social media discourse, the exact distribution of their various inflected and uninflected forms is of primary interest to assess the spread of uninflectedness. This distribution is presented separately for the two constructions in order to see if they exhibit different behaviors. Table 1 provides the percentages of uninflectedness for each individual verb.

In Figures 1 and 2, the percentages of uninflected tokens for each individual verb are clustered into deciles to gauge relative prevalence from an overall perspective. Looking at Figure 1, this means that for 27 of the 50 verbs, uninflected past participle forms account for at least 90% of all tokens. Looking at Figure 2, this means that for 26 of the 50 verbs, uninflected infinitive forms account for at least 90% of all tokens.

The overall distribution trends are strikingly similar in the two datasets: the extreme lopsidedness to the right is highly remarkable and means that not only do the overwhelming majority of English-origin verbs preferably appear in their

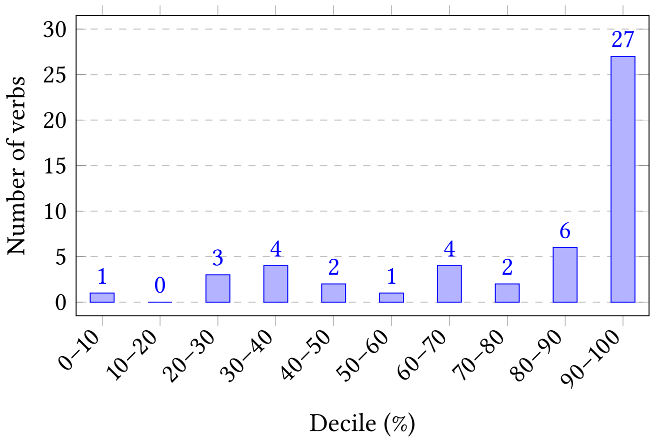

Figure 1: Decile distribution of the percentage of uninflected tokens of the past participle in the *passé composé* construction for the 50 verbs of the sample

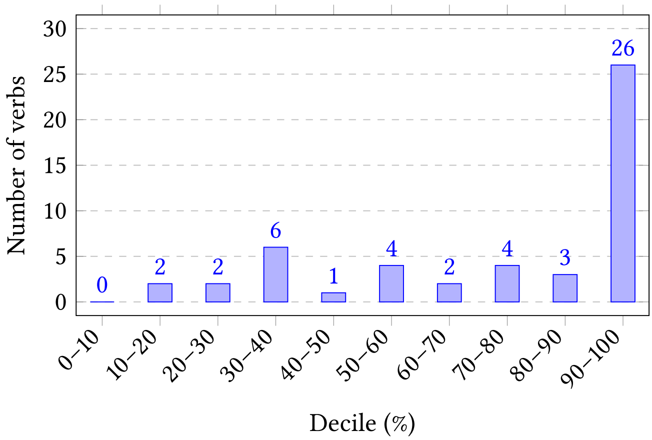

Figure 2: Decile distribution of the percentage of uninflected tokens of the infinitive in the periphrastic future construction for the 50 verbs of the sample

Table 1: Percentages of uninflected tokens in the *passé composé* (= PC) and periphrastic future (= PF) constructions of the corpus

Verb	PC	PF	Verb	PC	PF
dead (ça)	100%	99%	farm	64%	72%
test	8%	15%	flex	98%	97%
stream	82%	78%	troll	81%	41%
flop	77%	87%	play	99%	99%
bug	33%	20%	tag	40%	34%
drop	97%	97%	grind	97%	98%
follow	99%	99%	feat	94%	85%
check	59%	54%	boost	19%	12%
stop	33%	30%	rush	82%	85%
screen	95%	91%	tilt	31%	36%
win	99%	98%	clip	34%	30%
spam	85%	79%	shoot	20%	25%
boycott	25%	30%	spawn	99%	99%
spoil	76%	70%	stonk	98%	98%
pull	99%	99%	cancel	93%	92%
stan	100%	98%	ragequit	94%	95%
leak	94%	92%	disband	99%	99%
skip	96%	96%	buy	99%	99%
pop	96%	95%	pick	99%	99%
perform	42%	50%	kick	61%	38%
crush	91%	66%	stalk	64%	58%
carry	98%	100%	raid	99%	97%
switch	81%	66%	reset	99%	98%
pack	67%	54%	bully	100%	100%
tryhard	98%	99%	back	83%	95%

uninflected guise in the two constructions, but more than half of them massively favor uninflectedness to the point that their inflected occurrences are almost negligible. This is a rather unexpected result given the extreme paucity of uninflected verbal forms in controlled written French. At the individual level, almost all verbs are distributed in the same decile or in adjacent ones when comparing behavior in the two constructions, indicating that the ratio of uninflectedness is generally quite stable. The only exceptions are the verbs *crush(er)*, *switch(er)*, *troll(er)*, and *kick(er)*, and in all four cases it is the past participle form that is markedly less inflected, suggesting that the dominance of uninflectedness may be slightly more advanced for the *passé composé* construction.

The minority of verbs that are predominantly inflected are (in decreasing order of inflectedness): *test(er)*, *boost(er)*, *shoot(er)*, *bug(ger)*, *boycott(er)*, *clip(per)*, *stop(per)*, *tilt(er)*, *tag(uer)*, *perform(er)*, and *kick(er)*. The main linguistic parameter which is hypothesized to markedly affect the distribution of uninflectedness is the relative establishment of the sampled verbs in the French language. This parameter is operationalized here by the presence or absence of the English-origin verb in a large recent French dictionary of anglicisms, the *Dictionnaire étymologique et critique des anglicismes* (Weisman 2020). Examination of the data is partially conclusive as the two overall distribution trends in Figures 3 and 4 are in stark contrast to each other. The dispersion of the dictionary-sanctioned verbs is quite high, indicating that the presence of a verb in the dictionary of anglicisms is not a reliable predictor of the ratio of uninflectedness, while the distribution of neologisms is extremely lopsided to the right, suggesting that novelty is correlated to a fairly large extent with massive uninflectedness. Among dictionary-sanctioned anglicisms, relative anteriority is not found to be a significant factor. In the set of nine English-origin verbs that, according to Weisman (2020), have been attested in French for more than half a century, five — *test(er)*, *boost(er)*, *boycott(er)*, *shoot(er)*, *stop(per)* — are predominantly inflected, but four — *drop(er)*, *rush(er)*, *switch(er)*, *check(er)* — are predominantly uninflected.

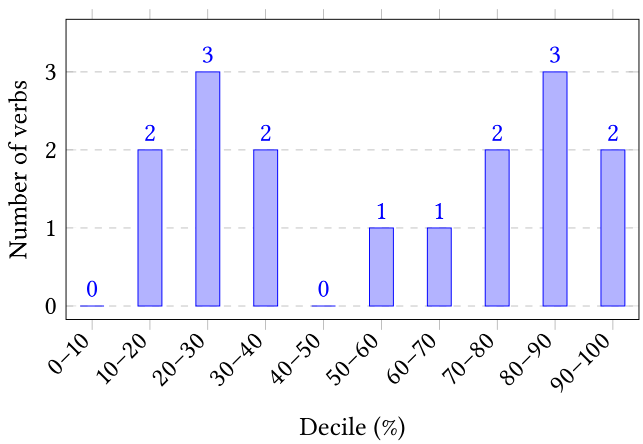

Figure 3: Decile distribution of the average percentage of uninflected tokens in the two constructions for the 16 dictionary-sanctioned verbs of the sample

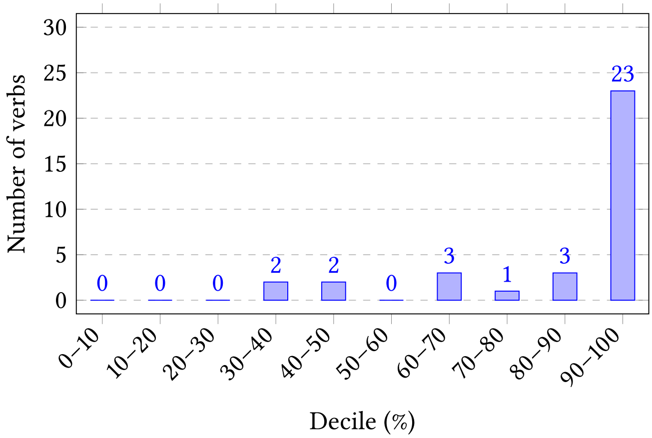

Figure 4: Decile distribution of the average percentage of uninflected tokens in the two constructions for the 34 neologisms of the sample

4 Discussion and conclusion

The main finding of this exploratory study is that two verbal constructions of present-day French massively display the nonstandard practice of dropping inflectional marking on the past participle and the infinitive in the context of written social media discourse and when the lexical verb is of English origin. The choice of examining the *passé composé* and the periphrastic future was initially guided by the intuition that inflection dropping seems to occur more frequently in these two constructions. This can be related to the fact that they correspond to doubly inflected composite verb forms (consisting of a lexical verb premodified by an auxiliary) while the other frequent finite forms of the French verb (the *présent*, the *imparfait*, and the *futur simple*) are simple verb forms which are singly inflected. Inflection dropping in simple verb forms is a less likely choice as it would lead to the loss of all morphological tense-aspect-mood (TAM) information. In contrast, in composite verb forms, dropping the inflectional marking on the lexical verb is not detrimental to the correct identification of the whole form and its TAM meaning. Inflection dropping should therefore be seen here as an unexpected and marked choice rather than an inconceivable one.

This view is also supported by the fact that verbal uninflectedness has been documented elsewhere in the periphery of the French language. In their analysis of the outstanding grammatical features of the *Multicultural Paris French* (MPF) corpus, Cappeau & Moreno (2017) point out that some groups of French verbs appear under only one — uninflected — form. This notably applies to verbs originating from *verlan* (a French argot based on syllable and phoneme reversal; cf. Méla 1997), as in (5–7), and from Romani, as in (8–9):

(5) J'ai envie d'sèpe!
'I need to piss!' (Tengour 2024)
(the inflected infinitive form *séper* is expected; *sèpe* is a verlanized form of *pisser*)

(6) J'ai vesqui l'service militaire.
'I've skirted military service.' (Tengour 2024)
(the inflected past participle form *vesquié* is expected; *vesqui* is a verlanized form of *esquiver*)

(7) C'est dead, Auxerre, ils m'ont tèj.
'It's over, Auxerre have thrown me out.' (Frantext)
(the inflected past participle form *téjé* is expected; *tèj* is a verlanized form of *jeter*)

(8) Comme dans les films qui collent des crampes au bide même quand on n'a rien bédave.
'Like in the movies that give you stomach cramps even if you didn't smoke anything.' (Frantext)
(the inflected past participle form *bédavé* is expected)

(9) Peur ! Peur ! Quand tu dois poucave ton pote…
'Fear! Fear! When you have to snitch on your buddy…' (Frantext)
(the inflected infinitive form *poucaver* is expected)

Verbal uninflectedness has also been documented to appear when French has been in intense contact with English. Dubois & Sankoff (1997) report that, in Louisiana French, English-origin verbs are typically adapted to the French morphological system, except for the past participle and infinitive forms, for which the bare stem is used, as illustrated in (10–11), and they conclude that such verbs can be seen as belonging to a distinct conjugation class:

(10) Tu crois que noncle Pierre a enjoy sa visite ?
'Do you think that Uncle Pierre enjoyed his visit?'
(the inflected past participle form *enjoyé* is expected)

(11) Je crois ici là on peut mieux discipline notre enfant.[4]
'I think that here we can better discipline our kid.'
(the inflected infinitive form *discipliner* is expected)

In an experimental study of the conditions influencing morphological (non-)integration, Root (2018) offers a nuanced and quantitatively driven examination of the phenomenon, positing the existence of a class of loan-verbs characterized by variable integration, for which some slots of the verbal paradigm are preferentially filled with bare forms and others with morphologically integrated ones. This hypothesis is partially supported by his data since inflected forms account for 50% of the tokens for the conditional, for 32% for the *imparfait*, for 17% for the past participle, and for 13% for the infinitive.

A similar situation is found in some French-based hybrid urban vernaculars of western Africa. In Nouchi, a hybrid language of Côte d'Ivoire mixing French with other European languages and several Niger-Congo languages (Sande 2015, Boutin 2021), French-origin verbs in *passé composé* and periphrastic future constructions are inflected in the same way as in Standard French while verbs of other origins (e.g. Dyula, Malinke, or English) remain uninflected (Ahua 2008, Atsé N'Cho 2014). Similarly, in Camfranglais, a hybrid language of Cameroon mixing French, English, Cameroonian Pidgin English, and a number of Cameroonian languages (Kießling 2022), verbs of non-French origin are uninflected in the *passé composé* and periphrastic future constructions, as in Bogni's (2018: 185) examples in (12–13):

(12) J'ai begin à play les cartes frons.
'I've been playing cards for a long time.'
(the inflected past participle form *beginé* and the inflected infinitive form *player* are expected)

(13) Nous allons back autour de 17 heures.
'We'll get back around 5 p.m.'
(the inflected infinitive form *backer* is expected)

Like Dubois and Sankoff for Louisiana French, Bogni (2018) concludes that verbs of non-French origin form a distinct ("fourth group") conjugation class in

[4]It is unclear why the form "*discipline*" (which is the only infinitive form given by the authors as an illustrative example) is considered to be an English-origin verb. In a form of circular reasoning, it may be that its uninflectedness is seen as a justification for its status as an anglicism. Other illustrative evidence can be adduced, such as *ça va flood cette section là* 'they're going to flood this section', or *je vas le back sûr* 'I'll definitely back him' (Root 2018: 19, 26).

Camfranglais. In line with Root's quantitative approach for Louisiana French, Nchare's (2010) analysis of a large dataset of online Camfranglais interactions provides a slightly nuanced picture of language in use and confirms a particularly pronounced tendency towards uninflectedness, with 98% of bare past participles, 91% of bare infinitives, and a morphological class of exceptions that reportedly comprises all denominal verbs.

This brief overview of similar phenomena in a variety of language-contact settings demonstrates that the massive Twitter-wide uninflectedness of many French verb forms of English origin, while unexpected because it violates a presumably categorical norm, may not, after informed consideration, be truly exceptional. It also suggests that unadapted forms are not only a marked choice flagging foreignness and peripherality, but also, in some cases, markers of in-group membership that may contribute to the characterization of a sociolect — here, that of (young) tweeters and, more generally, that of (young) KSC users.

An obvious limitation of this exploratory investigation is that the analysis is blind to the geographic and demographic distribution of the collected data. Regarding the former dimension, it can be hypothesized that European, North American, and African varieties of French accommodate uninflectedness in fairly similar ways, but this assumption will need to be empirically confirmed in future research. Regarding the latter dimension, Bouchard (2023) has found a highly significant correlation between age and the reported use of morphologically unintegrated English-origin verbs in Quebec French, with younger people much more likely to use such forms, and her findings are likely to generalize across varieties of present-day French. It also remains to be seen whether uninflectedness is becoming more prevalent, or will soon extend, in a quantitatively significant way beyond English-origin verbs, i.e. to the native lexicon. Cappeau & Moreno (2017: 79) and Gadet & Hambye (2018: 737) report various cases of inflection dropping in non-English and cognate verbs in the MPF corpus, as illustrated in (14–17):

(14) On s'est fait carotte en fait.
'We've actually been screwed.'
(the inflected infinitive form *carotter* is expected)

(15) Les gens quand ils sont asthmatiques ils leur donnent de l'air pas à graille.
'When people have asthma, they're given some air, not something to eat.'
(the inflected infinitive form *grailler* is expected)

(16) Je vais pas te mytho.
'I'm not gonna lie to you.'
(the inflected infinitive form *mythonner* is expected)

(17) Je me fais contrôle.
'I had my ticket checked.'
(the inflected infinitive form *contrôler* is expected)

It therefore cannot be ruled out that, in the course of a generation, uninflectedness in a number of verbal constructions will have become a hallmark feature of colloquial French.

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Chapter 10

Uninflectability in Italian nouns and adjectives

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This chapter illustrates the factors that render Italian nouns and adjectives uninflectable. Three phonological factors – ending in a stressed vowel, in a consonant or in /i/ in the citation form – are causes of uninflectability for both nouns and adjectives; while ending in a stressed vowel results in exceptionless uninflectability at all times, nouns and adjectives ending in a consonant were phonologically adapted by providing them with a final vowel, and subsequently integrated in an inflecting class, in older stages of the Italian language, but at present this does not happen anymore; a few nouns and adjectives ending in /i/ have occasionally developed an inflecting variant through backformation, when the form ending in /i/ has been interpreted as an inflected masculine plural form. Another factor that produces uninflectability of nouns is a mismatch between a noun's final vowel and its gender: neological masculine nouns in /a/ and feminine nouns in /o/, whose final vowels are typical of nouns bearing the opposite gender value, are treated as uninflectable in contemporary Italian. Even recently borrowed or recently coined nouns ending in the vowels most typical for their gender can be uninflectable; this is more common for nouns denoting inanimate entities, and for nouns bearing foreign characteristics in their spelling. There are also cases of constructional uninflectability, two of which are briefly discussed. The chapter also illustrates the inflectional classes of Italian nouns and adjectives and provides quantitative data about types and tokens in each inflectional class from several available databases.

1 Introduction

Spencer (2020) introduced a useful terminological distinction between *uninflecting* and *uninflectable* lexemes. He calls *uninflecting* lexemes and classes of lex-

Anna M. Thornton & Paolo D'Achille. 2026. Uninflectability in Italian nouns and adjectives. In Enrique L. Palancar & Sebastian Fedden (eds.), *Voids in morphology: Exploring "uninflectedness"*, 289–327. Berlin: Language Science Press. DOI: ??

emes that "are not associated with any inflectional paradigm", like English prepositions, and *uninflectable* "members of an otherwise inflecting class" that "unexpectedly fail to inflect", like the Russian noun KENGURU 'kangaroo' (Spencer 2020: 142).

In the Italian grammatical tradition, the same words, the adjective *invariabile* lit. 'invariable' and the noun *invariabilità* lit. 'invariability', are used for both cases. Traditional Italian grammars list nine parts of speech, of which five (articles, adjectives, nouns, pronouns and verbs) are said to be *variabili* 'inflecting' and four (adverbs, prepositions, conjunctions and interjections) *invariabili* 'uninflecting'. Within the inflecting parts of speech, specific nouns and adjectives are *invariabili* 'uninflectable', i.e. appear in the same phonological form in any syntactic context, as the noun *città* 'city, town' in (1) and the adjective *blu* 'blue' in (2):

(1) a. *l-a città vecchi-a*
def-f.sg city(f)[sg] old-f.sg
'the old town'

b. *l-e città italian-e*
def-f.pl city(f)[pl] Italian-f.pl
'the Italian cities'

(2) a. *l-a penn-a blu*
def-f.sg pen(f)-sg blue[f.sg]
'the blue pen'

b. *l-e penn-e blu*
def-f.pl pen(f)-pl blue[f.pl]
'the blue pens'

c. *il libr-o blu*
def.m.sg book(m)-sg blue[m.sg]
'the blue book'

d. *i libr-i blu*
def.m.pl book(m)-pl blue[m.pl]
'the blue books'

In previous work (Dressler & Thornton 1996, D'Achille & Thornton 2003, D'Achille 2005b, D'Achille & Thornton 2008) we have investigated qualitative and quantitative properties of Italian uninflectable nouns.

In this chapter, we illustrate uninflectability in Italian nouns, updating our previous findings on the basis of new data concerning usage in the first twenty years of the twenty-first century, and investigate adjectives with the same methodology used for nouns by D'Achille & Thornton (2003).[1]

Spencer (2020) also drew attention to *constructional non-inflectedness / uninflectedness / uninflectability*, a property exhibited in "grammatical constructions involving lexemes which cannot bear inflections [...] in that context", such as "the modifying non-head noun in an English noun-noun compound" (Spencer 2020: 148).[2] In this chapter, we will touch constructional uninflectability only briefly, in Section 4; a thorough investigation of this phenomenon in Italian must be left for further research.

The chapter is organized as follows: in Section 2 we present the inflectional classes of Italian nouns and adjectives, and quantitative data on types in each class from various available databases; in Section 3 we illustrate and discuss the factors that render Italian nouns and adjectives uninflectable, and their variation in diachrony, with a focus on the contemporary situation; in Section 4 we briefly discuss two cases of constructional uninflectability; Section 5 summarizes our conclusions.

2 Inflection and inflectional classes of Italian nouns and adjectives

Before delving into the investigation of Italian uninflectable nouns and adjectives, a brief sketch of Italian nominal inflection is in order.

Italian nouns inflect for NUMBER, which has two values: SINGULAR and PLURAL. They are usually assumed to have an inherent GENDER value, either MASCULINE or FEMININE. Nouns control agreement in Gender and Number on articles and adjectives (including possessives, demonstratives and some quantifiers) within NPs, and NPs control agreement in Gender and Number on predicates that agree

[1] Uninflectable Italian adjectives have been briefly discussed by Fedden (2019) in the context of his investigation of "sporadic agreement". To the best of our knowledge, they have not received specific attention in the context of Italian linguistics. Uninflectable verbs do not seem to exist in Italian.

[2] Spencer (2020) uses "constructional non-inflectedness" in the title of section 4, but then discusses the possibility that we should make a terminological distinction between constructional uninflectedness (*sic*) and constructional uninflectability; this second term is considered more appropriate for cases in which a lexeme which normally inflects fails to do so in specific constructions (Spencer 2020: 148–149). The Italian cases discussed in Section 4, then, are instances of constructional uninflectability.

(adjectives and past participles appearing in certain periphrastic verb forms, according to conditions described in Thornton 2019: 78), on most third person pronouns, and on periphrastic relative pronouns.

Table 1 illustrates the paradigm of the Italian definite article, a reliable exponent of agreement feature values in most contexts, which will often appear in our examples below. In the singular, final vowels are deleted before vowel-initial words (this deletion is represented orthographically by an apostrophe); the distribution of the m.sg shapes *il* and *lo* and the m.pl shapes *i* and *gli* is mostly phonologically conditioned, but there are some lexically conditioned oddities (e.g., *gli dei* 'the Gods' vs. *i denti* 'the teeth'; cfr. Maiden & Robustelli 2000: 62).[3]

Table 1: Forms and shapes of the Italian definite article

	masculine	feminine
singular	*il, lo, l'*	*la, l'*
plural	*i, gli*	*le*

Table 2 illustrates the inflectional classes of Italian nouns, according to the numbering proposed by D'Achille & Thornton (2003); class 7 is only attested in Old Italian, and is no longer used in contemporary Italian. Class 6 (shaded in grey) hosts uninflectable nouns.

The gender assignment system of Italian is mixed: most nouns denoting humans and animals that are economically important have semantic assignment (masculine is assigned to nouns denoting males, feminine to nouns denoting females), but there are a few epicene nouns denoting humans (e.g. persona (f) 'person', personaggio(m) 'character (in a novel, play, etc)') and many denoting animals (e.g. giraffa(f) 'giraffe', serpente(m) 'snake'). Inanimate and abstract nouns are assigned a gender value on the basis of phonological or morphological criteria, or sometimes by semantic "crazy rules" (Enger 2009; e.g., nouns denoting cars are feminine, cf. Thornton 2001).

As shown in Table 2, classes 1, 2 and 4, and the extremely non-canonical and inquorate class 5 (on which cf. Thornton 2019: 85–87) align with a gender value completely or with extremely few exceptions, while classes 3 and 6 host nouns of both genders. Table 3 shows the percentage of nouns of each gender in these classes.[4]

[3] For a recent survey of the various proposals on the distribution of the shapes of the Italian m.sg definite article, see Repetti (2020).

[4] Data in Table 3 are based on BDVDB (Thornton et al. 1997), a database containg about 4500 nouns belonging to the Italian Basic Vocabulary (De Mauro 1989).

Table 2: Inflectional classes of Italian nouns

Nouns' endings SG	PL	Class	Gender
-o	*-i*	1	M with one exception
-a	*-e*	2	F
-e	*-i*	3	M, F
-a	*-i*	4	M with two exceptions
-o	*-a*	5	SG M, PL F
stressed V	stressed V	6	M, F
C	C		
-i	*-i*		
...	...		
-o	*-ora*	7 (extinct)	SG M, PL F

Table 3: Percentage of masculine and feminine nouns in classes 3 and 6

	M	F
Class 3	50.7%	49.3%
Class 6	48.6%	51.4%

Table 4 illustrates the inflectional classes of Italian adjectives; for contemporary Italian we follow the classification proposed by Iacobini & Thornton (2016), which contains classes 1-5; we have added classes 6 and 7 which are only attested in earlier stages of the language; the rightmost column of Table 4 draws attention to correspondences between inflectional classes of nouns and adjectives. As in Table 2, classes which host adjectives completely or partially uninflectable are highlighted in grey. Classes 6 and 7 are only attested in Old Italian, and are no longer used in contemporary Italian.

In Tables 5 and 6 we present data on the percentage of lexemes in each class. The data are drawn from two different sources: BDVDB (Thornton et al. 1997), a database containing information of various kinds on the about 7,000 lexemes constituting the Italian Basic Vocabulary (De Mauro 1989), and DeGNI (De Martino et al. 2023), a database containing information on gender and inflectional class of the nouns contained in CoLFIS (Bertinetto et al. 2005), a frequency dictio-

Table 4: Inflectional classes of Italian adjectives

Adjectives' endings SG	PL	Gender	Class	Corresponding class for nouns
-o	*-i*	M	1	1
-a	*-e*	F		2
-e	*-i*	M=F	2	3
-a	*-i*	M	3	4
-a	*-e*	F		2
-e	*-i*	M	4	3
-a	*-e*	F		2
stressed V C *-i* …	stressed V C *-i*	M=F	5	6
-o	*-a*	M	6 (extinct)	5
-a	*-e*	F		2
-e	*-i*	M	7 (extinct)	3
-e	*-e*	F		6

nary of contemporary written Italian. BDVDB contains about 4,500 nouns and about 1,100 adjectives, i.e. those appearing among the top 5,000 most frequent lexemes in LIF (Bortolini et al. 1971, a frequency dictionary of written Italian based on a corpus of about 500,000 tokens, from texts published between 1947 and 1968), and those appearing among about 2,500 lexemes not present in the top frequency ranks in LIF, but deemed to be highly available for speakers, since they denote objects and situations commonly encountered in everyday life (e.g., *forchetta* 'fork', *ingorgo* 'traffic jam', *carino* 'cute', *pigro* 'lazy'); CoLFIS, on which DeGNI is based, contains over 20,000 nouns, from a corpus of written Italian of about 3.8M tokens, drawn from press and literature published between 1992 and 1994.[5] DeGNI contains only nouns; therefore, for adjectives we have only data from BDVDB (which contains 1,128 adjectives), presented in Table 6.

[5]The data in Table 5 are not completely comparable with respect to class 5, because DeGNI does not consider it a separate class, and merges data on nouns belonging to this class with data on other inflectionally odd nouns, such as various types of compounds, in a category "Other", which accounts for 4% of the nouns in the CoLFIS corpus on which DeGNI is based.

Table 5: Percentage of nouns in each inflectional class (types)

Class	percentage of types in BDVDB N = 4,583	percentage of types in CoLFIS / DeGNI N = 23,619
1	37.7%	31%
2	34.4%	23%
3	20.8%	20%
4	1.3%	1%
5	0.3%	??
6	5.4%	21%

Table 6: Percentage of adjectives in each inflectional class (types)

Class	percentage of types in BDVDB N = 1,128
1	65.2%
2	31.7%
3	0.9%
4	0.01%
5	2.0%

Clearly, inflectional classes of Italian nouns and adjectives do not constituite a canonical system, as defined by Corbett (2009; see also Thornton 2019: 79–87): membership in the various classes is not balanced, and uninflectable nouns and adjectives are few in BDVDB; however, in DeGNI, based on a corpus collected at the end of the twentieth century, class 6, containing uninflectable nouns, seems to be about as numerous as classes 2 and 3, which are inflecting. This seems to point to an increase of uninflectability of Italian nouns in recent times.

To investigate the evolution of inflectional class membership, we have created and manually annotated a small corpus, containing 500 tokens of nouns and 500 tokens of adjectives from Italian texts belonging to each one of eleven time periods, between 1211 and 2022, shown in Table 7.[6] We have as much as

[6]The data about nouns in the periods between 1211 and 2000 have been collected in 2000 and analyzed in D'Achille & Thornton (2003); for the present chapter, we have extended the noun corpus to cover the period 2001-2022, and have collected and analyzed data on adjectives.

possible avoided establishing the time periods mechanically, aligning them with centuries; whenever possible, we have chosen as initial and final dates for a given period years which are conventionally considered turning points in the linguistic history of Italian (such as Boccaccio's death in 1375, the publication of Bembo's *Prose della volgar lingua* in 1525, the publication of the first edition of the Italian dictionary by Accademia della Crusca in 1612, etc.). The twentieth century has been subdivided in three periods to have a better grasp of recent developments, and we have included texts from the twenty-first century up to 2022. For each time period, we have selected five prose texts belonging to different genres (such as novels and plays, scientific texts, essays, chronicles, ledgers, biographies and autobiographies, letters, legal texts, articles in newspapers and magazines) and have counted 100 tokens of nouns and 100 tokens of adjectives from a continuous portion of text, classifying each token in the inflectional class to which it belongs.

Table 7: Time periods used to partition the corpus analyzed

Time period	Dates
I	1211–1300
II	1301–1375
III	1376–1525
IV	1526–1612
V	1613–1691
VI	1692–1800
VII	1801–1900
VIII	1901–1945
IX	1946–1968
X	1969–2000
XI	2001–2022

3 Inflection and uninflectability of Italian nouns

3.1 Productivity of inflectional classes of Italian nouns

Previous qualitative and quantitative analyses (Dressler & Thornton 1996, D'Achille & Thornton 2003) have shown that the various inflectional classes of Italian nouns have different degrees of productivity, defined in terms of

Dressler's (1997, 2003) criteria of inflectional productivity, according to which an inflectional class is productive at a given time if it gains new members through integration of loanwords and other types of neologisms or through class shift of lexemes changing their inflectional class in diachrony.[7]

Class 5 is completely unproductive (while it was productive at least up to the fourteenth century, according to Gardani 2013), has been losing members to class 1 throughout the centuries (Santangelo 1981), and is now inquorate (it contains around two dozens of nouns, many of which are overabundant in their plural cell, with a plural of class 1 alongside one of class 5, cf. Thornton 2013).

Class 4 is losing members towards class 6, and is only fed by productive suffixation, most notably the formation of person-denoting nouns in *-ista* (such as *terrapiattista* 'someone who believes that the earth is flat' ←*terra piatta* 'flat earth').

Class 3 is losing members to class 6.

Classes 1 and 2 are productive according to several criteria; they were productive even according to the strongest criterion in Dressler's classification, integration of loanwords with unfitting phonological form, up to the nineteenth and even the early twentieth century (e.g., English *beefsteak*, "unfitting" because it ends in a consonant, was adapted as *bistecca*, attested since 1844, and is still inflected as a class 2 noun), but nowadays loanwords ending in consonants (i.e., with unfitting phonological form) are assigned to class 6. Nouns converted from verbs are assigned to class 1 (e.g., *scippo* 'bag-snatching' ←*scippare* 'snatch [a bag]') and some clipped nouns from deverbal suffixed action nouns are assigned to class 2 (e.g., *codifica* ←*codificazione* 'coding').

Class 6 is gaining members in several ways, mainly by hosting loanwords ending in a consonant or in /i/; even indigenous affixation feeds it, e.g. with quality nouns in *-ità* such as *padanità* 'quality of belonging to the Po valley territory [Val Padana]'.

Table 8 shows the increase of membership in class 6 of both tokens and types in our corpus, from the thirteenth to the end of the twentieth century; this increase has been shown to be statistically significant (D'Achille & Thornton 2003: 219).

D'Achille & Thornton (2003) concluded that the main system-defining property of Italian noun inflection is the alignment of back vowels (/a/ and /o/) with SINGULAR and of front vowels (/e/ and /i/) with PLURAL, and the alignment of the singular ending /a/ with FEMININE and the singular ending /o/ with MASCULINE; nouns not displaying such an alignment in their paradigm, such as class 3 nouns, that end in the front vowel /e/ in the singular and can belong to either gender (cf.

[7]See also Gardani (2013) for a critical discussion of Dressler's criteria.

Table 8: Percentage of types and tokens of nouns in class 6 in the corpus

	13^{th} century	end of 20^{th} century
tokens	2.4%	8.6%
types	2.7%	9.5%

Table 3), so that some plurals in /i/ are feminine, show an increasing tendency to become uninflectable, as do masculine nouns ending in /a/ (class 4) and feminine nouns ending in /o/ (investigated by D'Achille & Thornton 2008).

3.2 Properties of uninflectable Italian nouns

In this section we illustrate the properties that influence the uninflectability of nouns in Italian. These properties are mainly phonological, but the relation between a noun's gender and its ending in the singular also plays a role, as do orthographic and stratal properties.

Table 9 shows the progressive appearance of nouns of various kinds in our corpus. A ✓ in a cell in Table 9 means that nouns of the kind listed in the row are attested in the time period corresponding to the column. A continuous shading means that nouns of the kind listed in the row are unattested in the time periods corresponding to the columns, and are only attested later; the lighter shading in the row corresponding to nouns ending in a consonant is due to the fact that there is a doubtful occurrence of a noun ending in a consonant in period I (*dies* 'day', which, however, is a Latin word inserted in a vernacular context, a kind of code-switch), but then no more tokens till period VIII, that is till the twentieth century. An empty white cell means that nouns of the kind listed in the row are not attested in our corpus in the time period corresponding to the column, but are attested in the corpus in earlier periods.

Table 9: Appearance of various kinds of uninflectable nouns in the corpus

Time period Kind of noun	I 1211– 1300	II 1301– 1375	III 1376– 1525	IV 1526– 1612	V 1613– 1691	VI 1692– 1800	VII 1801– 1900	VIII 1901– 1945	XI 1946– 1968	X 1969– 2000	XI 2001– 2022
stressed final V	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MASCULINE in *-e*		✓								✓	
FEMININE in *-e*			✓	✓							
FEMININE in *-i*					✓			✓		✓	✓
MASCULINE in *-i*					✓					✓	
FEMININE in <*ie*>						✓	✓		✓		✓
final C	?							✓	✓	✓	✓
MASCULINE in *-a*									✓		
FEMININE in *-o*									✓	✓	

3.2.1 Phonological properties

The main property that makes a noun uninflectable for phonological reasons in Italian is ending in a stressed vowel.[8] Nouns ending in stressed vowels and treated as uninflectable are attested in all periods in our corpus (cf. Table 9), and have existed throughout the history of the Italian language. They have come about at first from Latin nouns, in various ways: some inherited nouns have come to end in a stressed vowel because a final consonant or consonant cluster has been lost (3a); a sizeable number of nouns end in a stressed vowel after loss by apocope of a final unstressed syllable (3b); another source of final stressed vowels is the deletion of a final /e/, which was etymological but was reanalyzed as epithetic and consequently deleted (3c).

(3) a. Lat. REX > *re* 'king'
 b. Lat. CIVITATE(M) > *cittate* / *cittade* > *città* 'city'
 Lat. VIRTUTE(M) > *virtute* / *virtude* > *virtù* 'virtue'
 c. Lat. DIE(M) > *die* > *dì* 'day'

Later the type has grown through loanwords, such as the ones in (4), encountered in our corpus:

(4) *falpalà* 'flounce', *caffè* 'coffee'

There has never been any attempt at integrating nouns ending in a stressed vowel in one of the inflecting classes. These are the nouns most truly uninflectable for phonological reasons.

A second group of nouns whose uninflectability correlates with a phonological property is represented by nouns ending in /i/.[9] These are attested in our corpus since period V, roughly corresponding to the seventeenth century. This type at first included only feminine nouns of Greek origin, such as the ones in (5):

(5) *ipotesi* 'hypothesis', *estasi* 'ecstasy', *crisi* 'crisis'…

Later, the type has grown through loanwords, most often assigned masculine gender, such as the ones in (6) from our corpus:

[8] In the standard Italian orthography, final stressed vowels are marked by an accent in polysyllabic native words and orthographically integrated loanwords, and in some but not all monosyllables, according to various normative rules which we will not illustrate here because they are orthogonal to the issue that concerns us. In the text we follow standard spelling, and provide phonological transcriptions when necessary.

[9] We do not address uninflectable nouns ending in unstressed /u/, which are very few and quite recent loanwords such as *guru, haiku, sudoku, tofu.*

(6) *baby* /ˈbɛbi/ 'kid',[10] *floppy* 'floppy disk'…

In masculine nouns frequently used in the plural, the final /i/ can be perceived as a M.PL ending of class 1, with consequent backformation of a singular in /o/; this happened in the case of *cachi* < Jap. *kaki* 'Diospyros kaki'.[11] Table 10 shows the frequency of the two options in the *ItTenTen20* corpus (a corpus which is part of the *TenTen* corpus family, and contains 12.4 billion tokens from texts downloaded from the Web in the years 2019-2020) in contexts in which the noun is used in the singular preceded by the indefinite article:

Table 10: Inflectional forms of singular *cachi* / *kaki* in *ItTenTen20*

uninflectable (class 6)	tokens	inflected in class 1	tokens
un cachi	106	*un caco*	203
un kaki	30	*un kako*	7

Inflection in class 1 through backformation clearly correlates with orthographic integration: in the standard spelling of contemporary Italian the letter <k> is only used in loanwords, and the data show that the most common option is inflection in class 1 for the orthographically integrated forms (with a ratio of 1.9 : 1 in favour of class 1), but uninflectability for the forms spelled with <k> (with a ratio of 4.3 : 1 in favour of class 6).

The last phonological property that makes a noun uninflectable in contemporary Italian is its ending in a consonant. In our corpus, this type appears only in the twentieth century, and is attested in all time periods since the VIII, with nouns such as the ones in (7), which are mostly loanwords (but a few, such as the brand name *domopak* 'plastic food wrap', are autochthonous formations):

(7) *lastex*, *rayon*, *sport*, *tweed*, *claque*, *refrain*, *soubrette*, *round*, *goal*, *supporter*, *assist*, *vermouth*, *Nord*, *Internet*, *computer*, *hardware*, *software*, *hard disk*, *mouse*, *monitor*, *modem*, *scanner*, *mail*, *domopak*, *hashish*, *babysitter*, *performance*, *governance*, *input*

[10] In the context in which it appears in our corpus, *i baby* refers to football players in a junior team, and could be glossed 'the colts'.

[11] In theory, a masculine plural form in -*i* could lead to backformation of a class 3 singular in -*e* or a class 4 singular in-*a*: however, such cases are not attested in the thorough investigation of backformation in Italian by D'Achille (2005a).

Previously, loan nouns ending in a consonant were adapted, appending a final unstressed vowel (/a/ for feminine nouns, /o/ for masculine ones) after gender assignment; only rarely, a final /e/ was supplied, either epenthetically or because the loanword was analyzed as containing a suffix ending in *-e*. These nouns were then inflected in the class appropriate to their gender and ending. This is shown by the data in (8), with examples of Arabic (from Mancini 1992) and English loanwords integrated before the twentieth century.

(8) a. Arabic loanwords
- Nouns denoting human males integrated in class 1:
  sulṭān > *sultano* 'sultan'
  šayx > *sceicco* 'sheikh'
- Nouns denoting inanimate entities:
  xaršūf > *carciofo* 'artichoke' — integrated in class 1
  līmūn > *limone* 'lemon' — integrated in class 3

b. English loanwords
- Nouns denoting humans:
  quaker > *quacchero* — masculine, integrated in class 1
  quaker > *quacchera* — feminine, integrated in class 2
- Nouns denoting inanimate entities:
  beef-steak > *bistecca* — integrated in class 2
  yard > *iarda* — integrated in class 2

Nouns denoting humans received gender by a semantic rule demanding congruence between gender and sex of the referent, and were subsequently provided with the vocalic endings of the most typical classes for each gender, i.e., class 1 for masculine and class 2 for feminine. Nouns denoting inanimate entities were either assigned masculine gender by default, or assigned feminine gender because they were associated with already existing nouns of similar meaning (see Thornton 2003: 72–73 for details on *bistecca* and *iarda*.) *Limone* was integrated in class 3 probably because it was analyzed as containing the suffix *-one*.[12]

In the twentieth century, morphological integration of loanwords ending in a consonant stopped. We assume that a final consonant was analyzed as a phonological factor preventing inflectability; however, Fedden (2019) claims that the reason for uninflectability of Italian adjectives ending in a consonant is "etymological" rather than phonological. It is hard to tear apart the two factors, since

[12] Gardani (2013: 43–45) draws attention to the role of native derivational suffixes in the integration of loanwords.

the overwhelming majority of Italian nouns and adjectives ending in a consonant are indeed loanwords, and probably awareness of their loan status was present in the speakers who first introduced these words into Italian. However, there are some nouns formed in Italian which end in a consonant and are uninflectable, such as *colf* 'domestic help' from *collaboratrice familiare* lit. 'family aide', and acronyms like *gip* from *giudice [delle] indagini preliminari* 'magistrate responsible for a preliminary investigation', *pec* from *posta elettronica certificata* 'registered e-mail', *Caf* from *centro di assistenza fiscale* 'tax service center', and others. This seems to point to the conclusion that it is the phonological property of ending in a consonant, rather than the etymological property of being a loanword, that makes a noun uninflectable in contemporary Italian.

3.2.2 Between class 3 and uninflectability: nouns in *-e*

The next group of uninflectable nouns we will discuss is that of feminine nouns in /e/. In this case there is no phonological impediment to inflectability, since nouns ending in /e/ can be inflected according to class 3.

However, feminine nouns in /e/ were treated as uninflectable in periods III and IV; they acquired a plural in *-e* by analogy with nouns of class 2 (feminines with singular in *-a* and plural in *-e*), but these analogical formations were not retained and nouns such as *meretrice* 'prostitute(F)', *cagione* 'cause(F)', *lode*(F) 'praise', attested as uninflectable in periods III and IV in our corpus, later reverted to inflecting according to class 3, as they still do.

A subgroup of feminine nouns in /e/ presents a problem that might be considered of the kind that Fedden (2019) calls "phonotactic": this is the case of nouns ending in /ie/ – orthographically <ie>, phonetically realized most often as [je] – such as *barbarie* 'barbarism, barbarity', attested in our corpus as uninflectable in period VI, and *serie* 'series', attested in our corpus as uninflectable from period VII. Plural forms of class 3 such as **barbarii* [barˈbarji] and **serii* [ˈsɛrji] are phonotactically illegal, since [ji] is not tolerated in Italian and is reduced to /i/ (Camilli 1965: 93, §64). A repair strategy producing plural forms such as **barbari* /barˈbari/ and **seri* /ˈsɛri/ would be possible: nouns of class 1 such as *occhio* /ˈɔkkjo/ 'eye', *operaio* /opeˈrajo/ 'worker' have the plural forms *occhi* /ˈɔkki/ and *operai* /opeˈrai/; however, for feminine nouns in /e/ this strategy is not attested. Later, even some nouns ending in orthographic <ie>, phonologically ending in a palatal affricate followed by /e/, such as *specie* /ˈspɛtʃe/ 'species', which had a regular plural of class 3 *speci*, started to be treated as uninflectable (Serianni 1988: §III.114–115).

More recently, a tendency to treat various kinds of nouns ending in /e/, both feminine and masculine, as uninflectable rather than as belonging to class 3 is gaining weight. Table 11 illustrates this tendency for a masculine loanword from Japanese, *kamikaze* 'kamikaze' and a feminine loanword from Greek, *stele* 'stele', presenting data on the frequency of the plural forms of these nouns preceded by the plural definite article in the *ItTenTen20* corpus.

Table 11: Plural forms of *kamikaze* and *stele*

class 6 (uninflectable)	frequency in *ItTenTen20*	class 3 (SG -*e* / PL -*i*)	frequency in *ItTenTen20*
i kamikaze	2036	*i kamikazi*	3
le stele	1164	*le steli*	324

Again, the lexeme whose foreign status can be perceived from the orthographic form has a stroger tendency to uninflectability.

3.2.3 Uninflectability due to mismatch between gender value and ending

3.2.3.1 Relation between gender values and endings in Italian nouns

Although the singular ending of inflectable Italian nouns cannot be considered an exponent of gender, since the same vowel appears in nouns of both genders (even minimal pairs, e.g., *fronte*(F) 'forehead' / *fronte*(M) 'front', *lama*(F) 'blade' / *lama*(M) 'lama; llama'), it cannot be denied that statistically the overwhelming majority of nouns ending in -*o* are masculine and the great majority of nouns ending in -*a* are feminine, as shown by the data in Table 12 (from Sgroi 2008: 109, with further elaboration by us; Sgroi based his counts on the 73644 nouns listed in the dictionary by De Mauro 2000; comparable results were obtained by De Martino et al. 2023: 126 on nouns in the CoLFIS corpus).[13]

Against this background, masculine nouns ending in -*a* and feminine nouns ending in -*o* are troublesome.

[13] Sgroi (2008) counted only once nouns that have the same citation form but can appear with both gender values; hence the column with data for "M & F" nouns. These are cases of masculine and feminine nouns referring to persons and used in both genders according to the sex of the referent (such as *cantante* 'singer' in class 3 and *giornalista* 'journalist', in class 4 when masculine and in class 2 when feminine), homonyms with different gender and different semantics (such as *fronte*(M) 'front' and *fronte*(F) 'forehead') and nouns whose gender oscillates in usage (such as *groviera* 'gruyère').

Table 12: Singular ending and gender value of Italian nouns

Final vowel in the SG	% M	% F	% M & F	Total number of types
-a	5.1%	86.8%	7.9%	24,126
-o	99%	0.4%	0.4%	25,614
-e	50.9%	40.6%	8.5%	16,540
-i	35.2%	54%	10.6%	1,566
-u	83.1%	5.3%	11.6%	95
Stressed V, C	51.6%	43.7%	4.7%	5,703
Total	52.4%	42.2%	5.3%	73,644

3.2.3.2 Masculine nouns in *-a*

Masculine nouns in *-a* have been thoroughly investigated by Migliorini (1957),[14] who provides an extremely detailed list of data. He observes that the earliest stratum of loanwords resulting in masculine nouns in *-a* is represented by Latin (9a) and particularly Greek learned loanwords (9b–9c):

(9) a. *scriba* ‘scribe’, etc.
  b. *papa* ‘pope’, *duca* ‘duke’, *collega* ‘colleague’, etc.
  c. *diadema* ‘tiara’, *diploma* ‘id.’, *diaframma* ‘diaphragm’, etc.

This type of nouns inflects according to class 4 (*scriba / scribi, papa / papi, diploma / diplomi,* etc.), and this schema has been productive, attracting agent nouns that had “a Greek flavour” (“di sapore greco”, Migliorini 1957: 59) even when their etymon contained /o/, as in the case of *stratega* ‘strategist’ (< Greek *stratēgós* ‘general’), *parassita* ‘parasite’ (< Greek *parásitos*), nouns in *-iatra* (< Greek *iatrós* ‘physician’) designating medical specialists, like *pediatra* ‘pediatrician’, *odontoiatra* ‘dentist’, and others.

A later stratum of masculine nouns in *-a*[15] is represented by nouns that Migliorini calls “exotic” loans, such as the ones in (10):

[14]The first edition of this article appeared in 1934; the article was republished in 1957 with many additions.

[15]It could be argued that here one should write “in /a/” rather than “in *-a*”, since uninflectable nouns do not have a segmentable ending, hence the hyphen may seem unwonted. We are aware of the issue, but here we keep to the common practice of inserting a hyphen before final vowels that could in theory be interpreted as inflectional endings in Italian.

(10) *lama* 'llama; Buddhist spiritual leader', *karma*, *alpaca*, *boa*, *cobra*, *gorilla*, *puma*, *anaconda*, *panda*[16]

Many of the nouns in (10) denote animals not autochthonous in Europe; they have been borrowed at the end of the nineteenth century, not directly from the "exotic"[17] languages in which they originate, nor through Spanish or Portuguese, in which they are feminine, but through French, in which they are masculine; they have received masculine gender in Italian because of their gender in French, and are treated as uninflectable.

It must be observed that the majority of these nouns do not betray their foreign origin through spelling and do not present any phonological or phonotactic oddness.[18] What are then the causes of their uninflectability?

A first possibility is that they are treated as uninflectable because of their foreign (or rather, exotic) origin, of which the first speakers who borrowed them were well aware. A second possibility is that they are treated as uninflectable because they exhibit a mismatch between gender and ending. Probably both factors play a role. The hypothesis that the gender / ending mismatch plays a role, notwithstanding the fact that it did not cause a problem for nouns of the first stratum (cf. (9)), is corroborated by the observation that there are other masculine nouns in *-a* that are treated as uninflectable, whose origin is not exotic. These are nouns like the ones in (11), which have been created by a variety of mechanisms of lexicon enrichment:

(11) a. Metonymy
boia 'executioner, hangsman, headsman' < Lat. *boia(m)* 'leather collar, thong'

b. Vossianic antonomasia
sosia 'look-alike' < *Sosia* in Plautus' and Molière's plays
balilla '8–14-year-old boy in Fascist youth organization' < *Balilla*, nickname of Giovan Battista Perasso

c. Deverbal conversion
tartaglia 'stutterer' < *tartagliare* 'stutter'

[16]There is no need to gloss the nouns that have been borrowed into English in the same form.

[17]For present purposes we define exotic loans as loanwords coming from countries that are faraway and little known (Mancini 1992: 7).

[18]An anonymous reviewer observes that a final sequence <oa>, found in *boa*, is not frequent in Italian; this is correct, but the fact remains that the majority of the nouns in (10) are not phonologically of phonotactically odd: compare, e.g., *karma* with native *arma* 'arm, weapon', *firma* 'signature', *forma* 'form', *Parma* toponym, *tarma* 'moth'...

d. Ellipsis of a masculine head or assignment of masculine gender because of a masculine hyperonym
avana 'Havana cigar' (*sigaro* 'cigar' is masculine)
barbera, *marsala* 'kinds of wine' (*vino* 'wine' is masculine)

e. «Nomi cartellino», lit. tag nouns (Migliorini 1975)[19]
vaglia 'money order' < 'be_worth.3SG.PRS.SBJV'
proclama 'edict, decree' < 'proclaim.3SG.PRS.IND'

Already Migliorini (1957: 104–108) attributes the uninflectability of masculine nouns in -*a* of the kinds in (10) and (11) to a gender / ending mismatch. He also lists a number of plural forms in -*a* of nouns that normally inflect according to class 4 that he observed in the press (such as *gli scriba, i fratricida, i due omicida, i regicida, i Belga, gli analfabeta, i despota, i monarca, i comma, i dogma, i panorama, i prisma*), noting that this type of deviation from standard inflection is far more frequent than simple typos, and interprets this as further evidence that at least for non-highly educated speakers masculine nouns in -*a* are uninflectable. Our own data confirm this tendency, which is not always limited to uneducated speakers: we encountered *gli scriba* (vs. *gli scribi*) in a text illustrating an exhibition in Venice's Palazzo Grassi, and heard *i sisma* 'the earthquakes' (vs. *i sismi*) on national TV news. Data from the *ItTenTen20* corpus on recently borrowed masculines in -*a*, shown in Table 13, confirm their uninflectability.

Table 13: Treatment of recently borrowed masculine nouns in -*a* in the *ItTenTen20* corpus

Noun	Plural NP queried Class 6	Frequency	Plural NP queried Class 4	Frequency
sherpa 'id.'	*gli sherpa*	620	*gli sherpi*	0
burqa / burka 'id.'	*i burqa*	77	*i burqi*	0
	i burka	45		
chakra 'id.'	*i chakra*	4,352	*i chakri*	0
parka 'id.'	*i parka*	287	*i parki*	0
			[*i parkas*	1]
mascara 'id.'	*i mascara*	1,004	*i mascari*	2

[19] Migliorini (1975) proposed to call "nomi cartellino", lit. tag nouns, those nouns resulting from the nominalization of verb forms which appeared as the first word of prayers (like *credo* 'I believe') or as words written with particular graphic prominence in legal documents (such as the examples in (11e)).

3.2.3.3 Feminine nouns in *-o*

Feminine nouns in *-o* have been thoroughly investigated by D’Achille & Thornton (2008). One of them, *mano* ‘hand’, is exceptionally inflected according to class 1, and is the only feminine noun in this class.[20] Any other feminine noun in *-o* is uninflectable; there are only a handful of such nouns in common usage, exemplified in (12) according to the way they have come into existence:

(12) Feminine nouns in *-o*

a. Loanwords (both from classical and modern languages)
consecutio temporum ‘sequence of tenses’
lectio magistralis ‘keynote lecture’
demo ‘demo’
macro ‘macro’
cabrio ‘convertible car’ (clipping from *cabriolet*)

b. Clippings (“Accorciamenti”)
auto ←*automobile* ‘car’
foto ←*fotografia* ‘snapshot’
moto ←*motocicletta* ‘motorbike’

c. Ellipsis of a feminine head
sdraio ←*sedia a sdraio* ‘deckchair’
lampo ←*chiusura lampo* ‘zipper’

d. Metaphors and metonymies
biro ‘ballpoint pen’ (from the name of its inventor László Biró)
caporetto ‘crushing defeat’ (from the toponym Caporetto, where Italy lost a battle in WWI)

Already Migliorini (1957: 106) considered the gender / ending mismatch as the reason why these nouns are uninflectable.

3.2.4 The catastrophe: masculine nouns in *-o* and feminine nouns in *-a* treated as uninflectable

There are cases of masculine nouns in *-o* and feminine nouns in *-a* treated as uninflectable, rather than as inflecting in classes 1 and 2 respectively. Some of these are instances of constructional uninflectability (cf. Section 4), but others

[20] *Sinodo* ‘synod’ and *parodo* ‘parodos’ are used both as masculine and as feminine nouns; when feminine, they are attested both as inflecting in class 1 and as uninflectable (D’Achille & Thornton 2008: 474).

are loanwords or recent coinages, such as *euro*, which could very easily fit in classes 1 and 2.

Table 14 shows data from the *ItTenTen20* corpus for feminine nouns in *-a*.

Table 14: Treatment of some borrowed feminine nouns in *-a* in the *ItTenTen20* corpus

Noun	plural NP queried Class 6	Frequency	plural NP queried Class 2	Frequency
geisha	*le geisha*	285	*le geishe*	366
/ˈɡɛiʃʃa/ ‘id.’	*le gheiscia*	0	*le gheisce*	3
iguana ‘id.’	*le iguana*	73	*le iguane*	692
keffiah	*le keffiah*	4	*le keffie*	1
/ˈkɛfja/ ‘id.’	*le chefia*	0	*le chefie*	1
casbah	*le casbah*	32	*le casbe*	1
/ˈkazba/ ‘id.’	*le casba*	1	*le casbe*	1
siesta ‘id.’	*le siesta*	0	*le sieste*	31
			[*le siestas*	1]

The data in Table 14 reveal that two factors are at play in treating borrowed feminine nouns in *-a* as uninflectable or as belonging to class 2. From the semantic point of view, uninflectability is stronger in inanimate nouns, and weaker in [+ human] and [+ animate] nouns, such as *geisha* and *iguana.* The second factor concerns orthographic integration: nouns that are orthographically integrated are usually also morphologically integrated and inflected according to class 2. The two factors intersect: for example, *geisha* is more commonly inflected in class 2 than uninflectable due to its being [+ human], but there is still a sizeable number of tokens of *le geisha*, while its orthographically integrated counterpart *gheiscia* (of very low frequency) is treated as a class 2 noun, and **le gheiscia* is not attested in the *ItTenTen20* corpus.

Turning to masculine nouns in *-o*, the most striking case is that of *euro*, overwhelmingly treated as uninflectable, notwithstanding the fact that its most immediately associated nouns, such as *dollaro* ‘dollar’, *marco* ‘mark’, *franco* ‘franc’, are inflected in class 1,[21] that it denotes an inanimate entity and that it has no foreign orthographic feature. Data about other borrowed masculine nouns in *-o* are presented in Table 15.

Table 15: Treatment of some borrowed masculine nouns in *-o* in the *ItTenTen20* corpus

Noun	plural NP queried Class 6	Frequency	plural NP queried Class 1	Frequency
bonzo 'bonze'	*i bonzo*	0	*i bonzi*	352
torero 'id.'	*i torero*	1	*i toreri*	250
			[*i toreros*	14]
kimono 'id.'	*i kimono*	374	*i kimoni*	33
	i chimono	0	*i chimoni*	0
eskimo 'hooded coat'	*gli eskimo*	17	*gli eskimi*	5
	gli eschimo	0	*gli eschimi*	3
poncho 'id.'	*i poncho*	115	*i ponchi*	17
			[*i ponchos*	30]
	i poncio	0	*i ponci*	1
gazebo 'id'	*i gazebo*	4,132	*i gazebi*	703
			[*i gazebos*	3]
avocado 'id.'	*gli avocado*	751	*gli avocadi*	73
			[*gli avocados*	36]

The same two factors at play with borrowed feminine nouns in *-a* appear to have an effect on the treatment of borrowed masculine nouns in *-o*. The two [+ human] nouns, *bonzo* 'bonze' and *torero* 'bullfighter', which happen to be orthographically undistinguishable from native nouns, are inflected according to

[21]The plural *euri* is a low variant, used jocularly or by uneducated speakers or in certain local varieties, mainly Tuscany and Rome. On the inflection or uninflectability of *euro* see Fanfani (2001), Gomez Gane (2003).

class 1; nouns denoting inanimate entities are preferably treated as uninflectable, particularly if their orthography bears foreign characteristics (<k> rather than <ch> in *kimono* and *eskimo*, <ch> rather than <ci> in *poncho*); nouns of Spanish origin can also appear in the Spanish plural form in *-s*.

A growing tendency to uninflectability is observable also in a lexical class of native nouns, those denoting months. Nouns denoting months are all masculine, and in the singular end either in *-o* (*gennaio, febbraio, marzo, maggio, giugno, luglio, agosto*) or in *-e* (*aprile, settembre, ottobre, dicembre*), inflecting according to class 1 or class 3; particularly the names of months ending in *-embre* are treated as uninflectable more often than as inflecting in the *ItTenTen20* corpus; names of months ending in *-o* on average are more commonly inflected according to class 1, but their treatment as uninflectable is attested too, as shown by the examples in (13), from the *ItTenTen20* corpus.

(13) a. *Come tutti i gennaio, sono iniziati i saldi.*
like all.M.PL DEF.M.PL January(M).[PL] AUX started.M.PL DEF.M.PL sales(M).PL
'Like every January, sales have started.'

b. *quel gennaio tra i gennai più freddi di sempre*
that.M.SG January(M).SG among DEF.M.PL January(M).PL more cold.M.PL of always
'that January among the coldest Januaries ever'

c. *come tutti i settembre*
like all.M.PL DEF.M.PL September(M).[PL]
'like every September'

d. *I settembri caldi si concentrano negli anni '80.*
DEF.M.PL September(M).PL hot.M.PL REFL concentrate.PRS.3PL in_DEF.M.PL year(M).PL '80
'Hot Septembers are concentrated in the 1980s.'

3.3 Conclusions on nouns

All the above data show that in contemporary Italian there is a strong tendency to treat various classes of nouns as uninflectable, even when their phonological properties, i.e., ending in the vowels /a/, /o/, /e/ in the singular, would allow them to be inflected according to classes 2, 1 and 3 respectively. Moreover, a mismatch between the gender value of a noun and the gender statistically more commonly associated with its final vowel is a categorical factor that makes a new noun uninflectable: class 4 (hosting masculine nouns ending in /a/, a prototypically feminine ending) does not acquire new members,[22] and some of its old members are becoming uninflectable (cf. Section 3.2.3.2); feminine nouns in /o/ are uninflectable (cf. Section 3.2.3.3).

Among the three main inflecting classes, class 3 is losing members to uninflectability more than class 1, and class 1 more than class 2. Nouns denoting inanimate entities have a stronger tendency to be treated as uninflectable than nouns denoting human or animate entities.

3.4 Uninflectability in Italian adjectives

3.4.1 Inflection of Italian adjectives

Italian adjectives inflect for gender and number; they are agreement targets, controlled by head nouns in NPs when used attributively and by subject NPs when used predicatively.

Inflectional classes of Italian adjectives have been presented in Table 4 (cf. Section 2).

In Table 16 we present data on the token frequency of adjectives in each class in the corpus used for the present article (cf. Section 2). Clearly, class 1 (which corresponds to classes 1 and 2 of nouns) has the majority of tokens in all periods; the only other class which has a sizeable frequency is class 2 (which corresponds to class 3 of nouns), whose token frequency appears to be growing since the second half of the twentieth century. All other classes have an extremely low token frequency in our corpus.

[22] Except (as already observed in Section 3.1) through derivation with the suffix *-ista* '-ist' or compounding with Latin or Greek combining forms such as *-cida* 'killer', *-iatra* 'doctor', etc., i.e., formatives that have been used in Italian lexeme formation for centuries.

Table 16: Percentage of tokens of adjectives in each class in the corpus

Period	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
Class	1211-1300	1301-1375	1376-1525	1526-1612	1613-1691	1692-1800	1801-1900	1901-1945	1946-1968	1969-2000	2001-2022
1	78.6	81.4	83.0	78.8	79.0	79.4	77.0	81.8	73.2	71.4	72.8
2	11.8	14.2	11.2	17.0	18.6	17.6	19.6	16.6	25.2	26.4	25.4
3	0	0.2	0	0	0	0	1.2	0.6	0.6	1.0	0.6
4	0	0	0	0	0	0	0	0	0	0.2	0
5	9.6	4.2	2.4	4.0	2.4	3.0	2.2	1.0	1.0	1.0	1.2
6	0	0	0.6	0.2	0	0	0	0	0	0	0
7	0	0	2.8	0	0	0	0	0	0	0	0
Total	100	100	100	100	100	100	100	100	100	100	100

3.4.2 Uninflectable adjectives in Italian

Class 5 hosts uninflectable adjectives. In tagging for membership in an inflectional class the adjectives in the corpus, we had to make choices about what to count as an adjective, and then as an *uninflectable adjective*. We adopted a very traditional definition of *adjective*, similar to the one used in descriptive grammars; therefore we included demonstratives, indefinites and possessives used as noun modifiers; basically, we included lexemes traditionally classified as adjectives in the Italian grammatical tradition and that can occur as modifiers of nouns bearing any combination of gender and number feature values. Therefore, we did not include numerals[23] and *ogni* 'each', which can occur only with singular controllers (albeit of either gender). We assigned to class 5 adjectives that do not appear in different forms, but that can occur in syntactic contexts which require agreement in both gender and number, such as *loro* 'their', which can occur with controllers displaying all possible combinations of gender and number values,[24] but has no overt inflectional exponent, in contrast with other possessives, as shown in (14), where *loro* is contrasted with the 1SG.POSS, which inflects according to class 1:

(14) a. *il* *mi-o* *libro*
DEF.M.SG 1SG.POSS-M.SG book(M).SG
'my book'

b. *il* *loro* *libro*
DEF.M.SG 3PL.POSS[M.SG] book(M).SG
'their book'

c. *le* *mi-e* *penne*
DEF.F.PL 1SG.POSS-F.PL pen(F).PL
'my pens'

d. *le* *loro* *penne*
DEF.F.PL 3PL.POSS[F.PL] pen(F).PL
'their pens'

Table 17 shows the full paradigms of Italian possessives. The uninflectability of *loro* is grounded in its diachronic origin: *loro* is the continuant of Latin (ĬL)LŌRŬ(M) 'DIST.DEM.M&N.PL.GEN', while the other possessives continue items that were adjectives with fully inflectable paradigms already in Latin. However, phonologically, nothing would have prevented the development of an analogical paradigm

[23] Numerals from '2' on do not agree in gender and only occur with plural nouns; the numeral '1' is homophonous with the indefinite article and is not considered an adjective.

[24] Italian possessives agree in gender and number with the noun designating the possessed item, and have inherent values for person and number of the possessor.

Table 17: Inflectional paradigms of Italian possessives

possessed / possessor	M.SG	F.SG	M.PL	F.PL
1SG	*mio*	*mia*	*miei*	*mie*
2SG	*tuo*	*tua*	*tuoi*	*tue*
3SG	*suo*	*sua*	*suoi*	*sue*
1PL	*nostro*	*nostra*	*nostri*	*nostre*
2PL	*vostro*	*vostra*	*vostri*	*vostre*
3PL			*loro*	

such as **loro / lora / lori / lore*, to align with all other possessives; but this did not happen.[25]

As shown in Table 18, *loro* is the only uninflectable adjective that occurs in all time periods in our corpus (and the high token-frequency of class 5 in period 1 is due to a text which contains an exceptionally high number of tokens of *loro*). Table 18 shows all the different types of uninflectable adjectives that we encountered in the corpus: a meagre harvest indeed, of only 14 types and 141 tokens (on a total of 5500 tokens of adjectives).

Most of the types have phonological shapes parallel to those of uninflectable nouns discussed in Section 3.2.1: four types end in a stressed vowel (15a), four end in /i/ (15b), and one ends in a consonant (15c); in (15) we list the contexts in which these types appear in the corpus and the time period in which they appear.

(15) a. Uninflectable adjectives ending in a stressed vowel
II ***più*** *cardinali* ‘several cardinals’
IV ***più*** *valore* ‘greater worth’
IV and V ***più*** *volte* ‘several times’
II ***che*** *novità* ‘what a piece of news’
III *in* ***che*** *modo* ‘in which way’

[25] A search in the OVI corpus (http://gattoweb.ovi.cnr.it/), which contains texts of Old Italian (from the first written attestations to the end of the fourteenth century), has yielded few attestations of *lori* used both as a third person plural masculine pronoun and as a possessive adjective, and even fewer cases of *lora* as a F.SG possessive adjective. There are also a few tokens of *lore*, used both as a third person plural pronoun and as a possessive adjective, in all possible combinations of gender and number values, mostly in non-Tuscan texts. It appears that *lore* has been used in some texts at some point instead of *loro*, but as an uninflectable form, just like *loro* in contemporary standard Italian, and not as part of a paradigm contrasting forms with different gender and number values.

Table 18: Presence of uninflectable adjectives in the corpus

Time period Type	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
loro '3PL.POSS'	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
più 'greater; several'		✓	✓	✓	✓	✓					
che 'what a'		✓	✓								
assai 'a lot of, many'			✓								
dabene 'honest'			✓								
altrui 'someone else's'						✓	✓				
pari 'equal, even'							✓		✓	✓	
grigio ferro 'iron gray'								✓			
pro-capite 'per capita'										✓	
qualsiasi 'ordinary'										✓	
così 'such, like this'										✓	
standard 'standard'										✓	✓
granata 'garnet red'										✓	
blu 'blue'											✓

X *un livido* ***così*** 'a bruise this big'
XI *bagliore* ***blu*** 'blue glow'

b. Uninflectable adjectives ending in /i/
III *tempo* ***assai***, ***assai*** *robe*, ***asai*** *disagi* 'a lot of time', 'many clothes', 'many hardships'
VI *cognizione dell'animo* ***altrui***, *le fortune* ***altrui*** 'knowledge of someone else's soul', 'other people's fortunes'
VII ***pari*** *ardore*, *insistenza* ***pari*** 'equal ardour', 'equal insistence'
X *altri acidi* ***qualsiasi*** 'other acids of whatever kind'

c. Uninflectable adjectives ending in a consonant
X *protocolli* ***standard***, *consumi* ***standard*** 'standard protocols', 'standard consumption'

A search of usage dictionaries yields a few other uninflectable adjectives with these phonological properties; some examples are given in (16):

(16) a. *agé, blasé, démodé, osé, retrò, rococò, tabù...*
  b. i. *impari* 'uneven', *dispari* 'odd (number)'
    ii. *mini, maxi*
    iii. *cachi* / *kaki* 'khaki', *cremisi* 'crimson', ...
    iv. *gay, baby, sexy...*
  c. *snob, chic, naïf, kitsch, folk, punk, halal, zen, mignon, pop, super, kasher, senior, junior, diesis, gratis...*

While a few adjectives ending in /i/ have been created by native processes, such as prefixation of a base already ending in /i/ (16b-i) or lexicalization of combining forms (16b-ii), others (16b-iii–16b-iv) are loanwords, as are practically all the adjectives ending in a stressed vowel (16a) or in a consonant (16c). The fact that borrowed adjectives of these types are fewer than nouns of the same type is in line with the fact that adjectives overall are the least borrowable type of content words. In the loanwords corpus analyzed by Tadmor et al. (2010), on 13446 loanwords, 79.7% are nouns, 14.4% are verbs, and only 5.9% are adjectives and/or adverbs (data elaborated by us on the basis of Table 1 in Tadmor et al. 2010: 231).

The phonological factors accounting for uninflectability of Italian adjectives are parallel to the ones already discussed for nouns. Adjectives ending in a stressed vowel are the most truly uninflectable, and repair strategies to make them inflectable are not attested. In theory, nothing would prevent adaptation of, e.g., *blu* 'blue' as **bluo*, with a full class 1 paradigm, as in (17a), since forms like the ones in (17b) are in common usage, and show that the starred forms in (17a) are phonotactically perfectly legal (as observed also by Fedden 2019: 308):

(17) a. **bluo* / **blua* / **blui* / **blue*
  b. *tuo* '2SG.POSS.M.SG', *suo* '3SG.POSS.M.SG', ...
  tua '2SG.POSS.F.SG', *sua* '3SG.POSS.F.SG', ...
  lui '3SG', *fui* 'BE.PST.PFV.1SG', ...
  tue '2SG.POSS.F.PL', *sue* '3SG.POSS.F.PL', ...

However, such adaptations are not attested. For adjectives ending in a consonant, instead, some cases of integration in an inflecting class are attested in older stages of the language, as in the case of *snello* 'slender' < Frankish **snel* (reported by Repetti 2012: 179); but in contemporary Italian this strategy is not employed anymore. Finally, adjectives ending in /i/ sometimes develop a full paradigm (or at least some inflected forms) of class 1 through back formation from a form interpreted as a masculine plural, as in the examples in (18):[26]

(18)	*choosy* /tʃusi/	>	*ciuso* /tʃuso/ 'choosy.M.SG' (jocular)
	tranqui[27]	>	M.SG *tranquo* / F.SG *tranqua* 'calm'
		>	M.PL *tranqui* / F.PL *tranque*

Uninflectable Italian adjectives have been discussed by Fedden (2019), who maintains that adjectives ending in a stressed vowel or in /i/ are uninflectable for phonological reasons, while those ending in a consonant are uninflectable for "etymological" reasons: "These adjectives [i.e., the ones ending in a consonant] are non-agreeing because they are loanwords" (Fedden 2019: 308). As already observed in the case of nouns, it is hard to tear the two factors apart, since almost all the adjectives ending in a consonant are also loanwords: however, again as in the case of nouns, there are a few examples of native adjectives ending in a consonant, like *doc* 'authentic' (from the acronym *DOC*, which stands for *denominazione di origine controllata* 'registered designation of origin'), which are uninflectable like the borrowed ones. This leads us to think that ending in a consonant is the real factor preventing inflectability, and that it is a phonological factor, not an etymological one. But it also true that ending in a consonant is a clue that makes non highly-educated speakers assume that the word having this property is a loanword, and specifically a loanword from English — so that there are anecdotical reports of speakers producing plural forms such as *kibbutzes*, *Führers*, *soviets* (Ratti et al. 1988: V), and even *i colfs e le colfs* 'the.M.PL domestic_help(M).PL and the.F.PL domestic_help(F).PL', jocularly used by Marilisa Marilì Lazzari in a post on OperaClick (https://www.operaclick.com/forum/viewtopic.php?t=1650, last access 19 May 2024), where the English plural exponent *-s* is applied to a native noun.

A possible case of uninflectability in adjectives that is not phonologically conditioned concerns color terms that are metonymically derived from nouns denoting objects that typically have that color, such as *rosa* 'pink' ⟵*rosa* 'rose', *viola* 'purple' ⟵*viola* 'violet'. These lexemes are uninflectable when used in typically adjectival functions, such as attributive modifier or nominal predicate, while the nouns from which they derive are regularly inflected in class 2. An intermediate

[26] However, the possessive *altrui* 'someone else's' (< Latin *ALTĔRŪI, a variant of ALTERI 'other.SG.DAT') did not develop a full paradigm through backformation.

[27] *Tranqui* is a clipped form from *tranquillo* 'calm'; the backformation of a full paradigm was already observed in Thornton (2007: 266) and D'Achille (2005a: 101).

situation concerns *marrone* 'brown' ←*marrone* 'chestnut': *marrone* 'brown' oscillates between inflection in class 2 and uninflectability, while the noun *marrone* 'chestnut' is regularly inflected in class 3. From the noun *arancio* 'orange (fruit)', two color terms have arisen: *arancio* 'orange (color)' is uninflectable, while *arancione* 'orange (color)' inflects in class 2. All other color terms derived metonymically from nouns of flowers, fruits and other entities typically bearing the color, such as *granata* 'garnet red' ←*granata* 'pomegranate' (present in our corpus), are completely uninflectable when used in adjectival function (Thornton 2004). Besides, all these color terms do not react positively to various tests of adjectivality, such as the creation of an elative form in -*issimo* or usage in prenominal position (while primary color terms allow both). Therefore, it can be debated whether *rosa, viola*, etc. are indeed adjectives, or rather nouns that can be used as attributive modifiers, as suggested by Thornton (2004: 530). If they are nouns, their uninflectability belongs to the realm of constructional uninflectability, addressed in Section 4; if they are adjectives, their uninflectability could be classified as due to "etymological" reasons (cf. Fedden 2019), since speakers are normally aware that the adjectival color term derives from a concrete noun.

4 Constructional uninflectability in Italian nouns and adjectives

Spencer (2020: 148) draws attention to constructional uninflectability, i.e., cases in which "it is the construction itself which prevents an otherwise inflectable lexeme from inflecting". We have not conducted a full investigation of constructional uninflectability, but we will document two cases of this phenomenon in Italian, one concerning nouns and one concerning adjectives.

A case of constructional uninflectability in nouns is represented by VN and some PN exocentric compounds. VN compounds denote instruments or agents, PN compounds with *fuori-* 'out' as their first member denote agents, events, or concrete objects. The nouns which are the second member of these compounds can appear both in their singular form (corresponding to the citation form) and in their plural form, and the gender of the second member noun can correspond to or differ from the gender of the whole compound. When the compound denotes a person, the gender of the compound aligns with the sex of the referent. A few examples of compounds denoting persons are given in (19):

(19) Compound — Noun base

a. *perditempo* 'idler', lit. lose_time — *tempo*(M).SG, class 1

b.	*strizzacervelli* 'shrink', lit. shrink_brains	*cervelli*(M).PL, class 1
c.	*portabandiera* 'standard-bearer', lit. carry_flag	*bandiera*(F).SG, class 2
d.	*portaborse* 'flunkey', lit. carry_bags	*borse*(F).PL, class 2
e.	*portavoce* 'spokesperson', lit. carry_voice	*voce*(F).SG, class 3
f.	*rubacuori* 'charmer', lit. steal_hearts	*cuori*(M).PL, class 3
g.	*fuoricorso* 'student who has not completed a program within the prescribed time', lit. out_course	*corso*(M).SG, class 1
h.	*fuoriclasse* 'champion, ace', lit. out_class	*classe*(F).SG, class 3

When the second member of these compounds is a plural noun, as in (19b, 19d, 19f) the whole compound is uninflectable, whatever its gender; when the second member is a singular noun, the possibility of having a distinct plural form depends on a number of factors. Pellegrini & Ricca (2019) have thoroughly investigated the issue of plural formation of VN compounds; data on their usage in corpora and on the prescriptions about plural formation in dictionaries are offered by Micheli (2016). We refer to Pellegrini & Ricca's study for an extremely detailed presentation of all the semantic factors that interact with the possibility and the probability of treating different kinds of VN compounds as inflectable; here we draw attention to the fact, already observed by Pellegrini & Ricca (2019: 110–112), that the correspondence or mismatch between the gender of the second member and that of the whole compound has a significant impact on the inflectability of the compound. Pellegrini & Ricca (2019: 110) report the data in (20) for *portavoce* 'spokesperson' (whose second member is a feminine singular noun, whose regular plural of class 3 is *voci*) when it refers to a man or a woman:

(20)	Plural forms and gender	Ratio uninflectable : inflected
	i portavoce / *i portavoci* (M)	24 : 1
	le portavoce / *le portavoci* (F)	3 : 1

Although VN compounds are frequently overabundant in the plural, with a plural of class 6 alongside one of the same class of their second member, inflectability of these compounds increases when the gender of the whole compound corresponds to that of its second member, and decreases when there is a gender mismatch between the two, in a statistically significant way (Pellegrini & Ricca 2019: 110–112). We interpret this finding as an instance of constructional uninflectability.

Our second example of constructional uninflectability in Italian concerns color adjectives. Several color adjectives are fully inflectable according to class 1, as shown for *rosso* 'red' in (21):[28]

(21) a. *vestito ross-o*
dress(M).SG red-M.SG
'red dress'

b. *gonna ross-a*
skirt(F).SG red-F.SG
'red skirt'

c. *calzini ross-i*
sock(M).PL red-M.PL
'red socks'

d. *scarpe ross-e*
shoe(F).PL red-F.PL
'red shoes'

However, when a color adjective is modified by a noun that specifies the color's shade "by comparing it with the quintessential color attributed to its referent (e.g., *capelli biondo-paglia* 'straw-blonde hair', *pantaloni blu notte* 'night blue pants') or, metonymically, with the color of an object intimately associated with, or typical of the referent" (Grossmann & D'Achille 2019: 71), color adjectives that are fully inflectable when unmodified become uninflectable, as shown in (22):

(22) a. *vestito rosso sangue*
dress(M).SG red[29] blood
'blood-red dress'

b. *gonna rosso sangue*
skirt(F).SG red blood
'blood-red skirt'

[28] *Verde* 'green' inflects according to class 2; *marrone* 'brown' and *arancione* 'orange' oscillate between uninflectability and inflection according to class 2. In the text we only illustrate the case of class 1 color adjectives, for which constructional uninflectability is most striking.

c. *calzini* *rosso sangue*
sock(M).PL red blood
'blood-red socks'

d. *scarpe* *rosso sangue*
shoe(F).PL red blood
'blood-red shoes'

In this construction, the color adjective appears in its citation form, which in the case of class 1 adjectives is a masculine singular form, and remains in this form even when there is a gender (and even a number!) mismatch with both the modified head noun and the color modifier, as shown in (23) with examples from the *ItTenTen20* corpus:

(23) a. *maglie* *grigio perla*
shirt(F).PL gray pearl(F).SG
'pearl-gray shirts'

b. *cappotti* *bianco panna*
overcoat(M).PL white cream(F).SG
'cream-white overcoats'

c. *bomboniere* *bianco nozze*
fancy_box(F).PL white wedding(F.PL)[30]
'wedding-white fancy boxes'

Grossmann & D'Achille (2019: 77), in a very thorough study of compound color terms in Italian, observe that these "A-N constructions are invariable. [...] rare instances of agreement on the head adjective [...] are definitely non-standard in modern Italian". This construction appears to be a very canonical case of constructional uninflectability, where a class 1 adjective, which shows the fullest set of contrasting inflectional forms when unmodified, appears in a "frozen" form (cf. Audring 2023) in a specific construction.[31]

[29] The issue arises of how to gloss the color adjectives appearing in this construction: their form is fully homonymous with the M.SG form, which is also the citation form, but this form does not carry MASCULINE and SINGULAR as contextual feature values, nor any other values, since it does not vary with different controllers. We have chosen to give only the lexical meaning in the gloss.

[30] *Nozze* 'wedding' is a *plurale tantum* noun, so it bears the value PLURAL inherently.

[31] The form in which the color adjective appears is its citation form, not its root, *contra* what is predicted by Spencer (2020). We do not address this issue here, but assume that Spencer's *reversion to the root* principle can be parametrized, and that languages which do not normally allow roots or bare stems as wordforms will specify which form from a lexeme's form paradigm is to be used in contexts of constructional uninflectability.

5 Conclusions

The survey presented in Sections 3–4 has shown that uninflectability is present in both Italian nouns and adjectives. However, in adjectives both its type frequency in the lexicon (Table 6) and its token frequency in our corpus (Table 16) is much lower than in nouns (cf. Table 5 and Table 8), and in contrast to nouns, it does not appear to have increased in diachrony. A possible reason for this different behaviour concerns the different proneness of the two word-classes to take in loanwords: many lexemes that are uninflectable for phonological reasons are loanwords, and since adjectives are much less borrowable than nouns, there are fewer uninflectable adjectives than nouns.

For both nouns and adjectives, certain phonological characteristics produce uninflectability: ending in a stressed vowel is the strongest impediment, for which no repair strategy has ever been applied; ending in a consonant is a strong impediment in contemporary Italian, while in the past repair strategies were employed, and loanwords ending in a consonant were provided with vocalic endings and inflected (in class 1 in the case of adjectives, in class 1 or class 2, depending on previous gender assignment, in the case of nouns, cf. (8)); ending in /i/ is also an impediment, very occasionally circumvented if the form ending in /i/ can be interpreted as a masculine plural, and a full paradigm of class 1 is created through backformation (cf. Table 10 and examples in (18)). These phonological characteristics are not the only causes of uninflectability. New masculine nouns ending in /a/ and feminine nouns ending in /o/ are uninflectable because their gender value and their ending clash, since /a/ and /o/ are typical endings for the opposite gender values (cf. Table 12). Finally, even some masculine nouns in /o/ and feminine nouns in /a/ are treated as uninflectable (although in many cases these lexemes have an overabundant plural cell, with a plural form of class 1 or class 2 alongside an uninflectable one). Most of these cases concern loanwords that bear some indication of their foreign origin in their spelling (such as <k> instead of <ch> or <q> not followed by <u> for /k/, <ch> instead of <ci> for /tʃ/, <sh> instead of <sc(i)> for /ʃ/, cf. Tables 13–15).[32] This orthographic factor begs the question whether speakers treat some lexemes as uninflectable for etymological reasons, since a foreign spelling makes them aware of a word's status as loanword; the same question may be asked in the case of uninflectable color terms that arise through metonymy from nouns denoting flowers, fruits or other entities typically exhibiting that color, since the speakers normally know both

[32]We have investigated data from written corpora; the treatment of these lexemes in spoken Italian remains a desideratum.

the base noun and the metonymically derived color term. The role of "etymological" knowledge must be considered also in the case of nouns and adjectives ending in a consonant: Fedden (2019) maintains that these are uninflectable not for phonological reasons, but because they are loanwords; we have shown that there are also native nouns and adjectives ending in a consonant, usually arising from acronyms; this may point to the generalization that consonantal ending rather than loanword status is the factor responsible for uninflectability, and this would be a phonological rather than an etymological factor; however, it may also be contended that speakers are aware that native uninflectable nouns and adjectives derive form acronyms, and this is a kind of etymological information; therefore, it is hard to tease apart the two factors in the context of Italian nominal and adjectival inflection. Finally, we have briefly discussed some cases of lexical or constructional uninflectability. We have shown a tendency to treat as uninflectable names of the months (cf. (13)), and D'Achille (2005b: 196) has reported a tendency to treat as uninflectable the noun *sabato* 'Saturday' (thus aligning it with the other masculine names of the days from Monday to Friday, which are uninflectable for phonological reasons, since they end in a stressed /i/). In Section 4 we have presented two cases of constructional uninflectability: nouns used as second members in exocentric compounds, and color adjectives used in a construction in which they are followed by a noun denoting an entity that prototypically bears that color, such as *rosso sangue* 'blood-red', where color adjectives that are fully inflectable when not accompanied by such modifiers (cf. (21)) appear in their citation form. Concerning constructional uninflectability in Italian, we believe that we have only scratched the surface of a phenomenon that deserves further investigation.

In conclusion, we have shown that uninflectability is quite widespread in Italian nouns and attested also in Italian adjectives, and that speakers appear to use a lot more information than just phonological information about the signifier of a lexeme when deciding whether to inflect it or not, since information about a noun's gender and orthographic, etymological, semantic, lexical and constructional factors also play a role.

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Chapter 11

Uninflectedness as a rule in Polish, an inflected language

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The chapter describes uninflected nouns referring to women in Polish. The absence of inflection on these nouns results from the lack of an inflectional paradigm that would suit the class in terms of both morphonology and semantics. In this case, uninflectedness, generally very peculiar in the language, has developed into a signal of feminine gender in its own right and given rise to a productive grammatical pattern. The lack of marking allows distinguishing the feminine words and the normally inflecting masculine counterparts. The text presents the functioning of the class that so poorly fits the Polish system, heavily reliant on inflection, competing patterns that do not involve uninflectedness as well as constraints on the use of the uninflected words. The last section gives a brief historical overview of the emergence of the category and shows that the process happened twice: first for surnames, then for personal common nouns. The article also addresses the ongoing debate among native speakers and recent developments which could potentially trigger the downfall of the discussed uninflected pattern.

1 Introduction

Polish is a language with rich inflection, involving nouns, pronouns, determiners, adjectives, participles, numerals and verbs. Yet, there are nouns (1% of all, cf. Stefańczyk 2007: 159) where the expected marking is lacking. This paper focuses on one type of uninflected nouns in Polish, originally masculine words used to refer to women (called U-FORMS here, cf. Section 3.1). Their uninflectedness, otherwise exceptional in the language, is essentially regular. Most of the words in question are borrowings, but we also find native ones in this group. Let us add that the

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said nouns are originally regularly inflecting words referring to men and having the masculine grammatical gender.[1]

The paper describes how this class of words functions in a language whose grammatical system it seems to fit so poorly, esp. considering the existence of rival morphological patterns. We will also sketch the historical development of U-forms, an open category at odds with basic Polish grammar, a regular pattern that as a whole is an exception.

Some aspects of the topic are socially very salient and constitute the subject of an ongoing public debate among native speakers. However, the bone of contention is not the uninflectedness of U-forms. It is the rival inflectable forms referring to women (which we call D-FORMS) whose increasingly common use (for the very same concepts) is controversial. U-forms surface in the debate, but as a peripheral issue. The roles are reversed here; our focus is on U-forms as a particularly interesting case of uninflectedness and the hotly debated D-forms are merely a point of reference. Thus, the paper is not meant to be a voice in the current debate in Poland; the aim is to analyse the usage without taking sides. We refer to the debate because it is a relevant part of the sociolinguistic reality.

Many examples come from the National Corpus of Polish (NKJP, Przepiórkowski et al. 2012), but the paper lacks an overall statistical analysis of the phenomena described. This is due to space limitations and particular difficulties with such a task. The tagging in the NKJP (or any corpus of Polish) logically relies on inflectional categories, which does not allow extracting reliable results for whole classes of uninflected nouns. However, the expressions referring to frequency used here are not mere opinions or guesses. First of all, they reflect the relative difficulty in finding examples. Furthermore, to compensate for the lack of the overall statistics, numbers for selected example phrases are included to support the points made at several points in the discussion. The related queries in the corpus were performed with the use of the PELCRA search engine (Pęzik 2012).

Section 2 of the paper briefly presents the Polish noun declension, then moves to factors conducive to uninflectedness of nouns. Section 3 is an analysis of the functioning of the uninflected words for women in the language at present

[1] Polish is said to have five gender categories: masculine personal (virile), masculine animate/animal, masculine inanimate, feminine and neuter. The plain label “masculine” will still appear in the paper. With words referring to people it means masculine personal. The three masculine genders are also syncretised in a number of grammatical positions. Let us stress that there is no grammatical context in Polish that would distinguish as many as five gender values. The five-way division proposed by Mańczak (1956) results from the superposition of the maximal distinctions made in the accusative singular and plural.

(21st century). Section 4 provides a historical overview of the development of uninflected and inflectable words for women, including current changes.

2 Declension and lack thereof

2.1 Declension in Polish

This section is meant to present Polish declension without delving into detail. For a comprehensive picture the readers are referred to grammars (Zagorska-Brooks 1975, Laskowski 1979, Bielec 1998, Swan 2002). Let us start with an example of declension in use. (1) is the opening sentence of the preface to one academic grammar of Polish, for clarity divided into clauses here.

(1) a. Książk-a t-a powstała na zamówieni-e.
book-NOM.SG this-NOM.SG.F arise.PST.F PREP commission-ACC.SG
'This book was commissioned.'

b. Nie odmawia się Wydawnictw-u,
NEG refuse.3SG IMPERS publishing_house-DAT.SG
'One does not refuse a publishing house,'

c. w któr-ym druk-Ø jest dla autor-a
in which-LOC.SG.N print-NOM.SG be.3SG for author-GEN.SG
honor-em.
honour-INS.SG
'to publish with which it is an honour for the author.'
(Nagórko 1997: 9)

(1) features a fairly representative selection of noun forms. We have a typical masculine nominative with a zero suffix, *druk* in (1c), a typical feminine one with the suffix *-a*, *książka* in (1a), a neuter accusative identical in form with the nominative, *zamówienie* in (1a), with a non-zero marker *-e*. There are distinct oblique case forms: a dative, *wydawnictwu* in (1b) and an instrumental, *honorem* in (1c), as well as a masculine personal genitive identical to the accusative, *autora* in (1c). The locative is exemplified by the relative pronoun *którym* in (1c). In sum, all Polish cases appear in (1), except the vocative, which is unlikely in academic prose. However, at the beginning of a preface one could easily imagine one of the vocative phrases in (2) preceding (1).

(2) a. drog-i Czytelnik-u
dear-VOC.SG.M reader-VOC.SG
'dear Reader'

b. drodz-y Czytelnic-y
dear-VOC.PL.VIR reader-VOC.PL
'dear Readers'

c. drogi-e Czytelnicz-k-i, drodz-y Czytelnic-y
dear-VOC.PL.NVIR reader-F-VOC.PL dear-VOC.PL.VIR reader-VOC.PL
'dear Readers [female], dear Readers [male]'

As for syntactic relations, the nominative forms in (1) are subjects of the clauses, the instrumental marks the subject complement after a copula, the accusative is governed by the preposition *na*, etc. Case marking fulfils the grammatical requirements of the words the nouns are structurally connected with. This phenomenon, syntactic government (in the traditional sense, cf. Matthews 1981: 246–250), relies on the category of case and is essential for signalling clause structure. Case (as well as number and gender) is also marked on words modifying nouns, as in the demonstrative *ta* in (1a) and the adjectives in (2). This kind of agreement is crucial for the cohesion of noun phrases. Agreement can also be seen on the verb *powstała* in (1a), which shows the number as well as gender of the subject of the clause. Gender is only marked on verb forms based on the past tense, historically a participle.

Polish nouns inflect for 7 cases and 2 numbers. This yields a 14-slot system, but because of case syncretism, mostly systematic, as in (1–2), no noun has more than eleven distinct forms and many have fewer. Adjectives, participles and determiners show the case and number of the noun they modify and additionally mark its gender, an inherent property of a given noun. The declension markers are solely suffixes, although inflection may be accompanied by sound alternations in the stem, most of them predictable, cf. the forms in (2). The suffixes are fusional; it is never possible to separate the marking of case and number (and gender in noun modifiers). Nouns fall into a number of declension classes which have different sets of markers for specific grammatical values. If this general description is reminiscent of classical Indo-European languages such as Latin or Sanskrit, the association is correct – the overall makeup of the Polish declension system is conservative.

Which declension class a given noun belongs to is the result of the interplay of its morphonetic shape and grammatical gender, though the two factors are correlated to a considerable extent. The nominative singular of a noun (its citation form) usually reveals the gender in the traditional three-way division. Most masculine nouns end in a consonant (the suffix being *-Ø*), as in (1c), most feminines end in *-a*, as in (1a), and most neuters end in *-o* or *-e*, as in (1a). This correlation is not absolute, but it delineates the major declension paradigms.[2] The three masculine genders are normally distinguished by different patterns of syncretism in the accusative forms (matched by different agreement markers, cf. fn. 1).

The defining parameter for gender is the distinctions of agreement markers on noun modifiers (Hockett 1958: 231 after Corbett 1991: 1, cf. also fn. 1). Thus, a Polish noun has a set value of this category, which manifests itself in the inflection of noun modifiers and verb forms based on the past tense. As previously mentioned, in practice gender is usually deducible from the form of the noun itself. The labels "masculine" and "feminine" imply a semantic basis for the gender system, not really strong in Polish (Kucała 1978: 175), but still perceivable. Gender is mostly a formal grammatical phenomenon apart from the opposition of masculine personal and feminine in nouns for people, which will prove crucial here.

To further complicate the assignment of declension class, the morphonetic shape concerns not only the very final sound of the basic form, but also the last sound of the stem (which is word-final if the suffix is zero). Stem-final consonants may be *hard*, i.e. non-palatalised, or *soft*, i.e. palatal(ised), there are also intermediate categories. Suffixes for some grammatical values depend on this division. Nouns with different types of stems are distributed across all genders. Table 1 exemplifies the major feminine paradigms and the hard-stem masculine personal paradigm for comparison.

Several minor paradigms include nouns falling outside of the scope of the correlation described above. There may be a "mismatch" of the nominative singular suffix and gender, as in masculine nouns ending in *-a*, e.g. *władca* 'ruler' and *mężczyzna* 'man' or feminine nouns ending in a consonant, e.g, *mysz* 'mouse' and *ciemność* 'darkness'. The nominative singular may end in a way not covered by the general rule, as in feminine nouns in *-i*, e.g. the honorific *pani* 'madam, lady, Mrs' and *władczyni* 'female ruler'. Still other nouns use the distinct suffixes of adjectives, e.g. *szeregowy* 'private (military rank), VIR', *królowa* 'queen, F', *znaleźne* 'finder's fee, N'. In general, minor declension classes have a clearer semantic profile than the major ones, cf. the discussion of (4) and fn. 10. Let us note that many of them are productive patterns due to association with specific derivational suffixes e.g. the agentive *-c(a)* in *władca*, feminative *-yn(i)*/*-in(i)* in *władczyni* or abstract *-ość* in *ciemność* above.

In sum, the main factors influencing noun declension in Polish are gender and morphonetics. Some (groups of) words show further idiosyncrasies. The re-

[2] They are "major" in terms of the number of nouns, but also the ease of acquisition. Smoczyńska (1985: 625–626) shows that nouns of minor classes are problematic for young speakers.

Table 1: Several major declension paradigms

	mama 'mum'	ciocia 'aunt'	córka 'daughter'	uczennica 'pupil, F'	doktor 'doctor, M'
	F hard stem	F soft stem	F velar stem	F historically soft stem	VIR hard stem
NOM SG	mam-a	cioci-a	córk-a	uczennic-a	doktor-Ø
GEN SG	mam-y	cioc-i	córk-i	uczennic-y	doktor-a
DAT SG	mami-e	cioc-i	córc-e	uczennic-y	doktor-owi
ACC SG	mam-ę	cioci-ę	córk-ę	uczennic-ę	doktor-a
INS SG	mam-ą	cioci-ą	córk-ą	uczennic-ą	doktor-em
LOC SG	mami-e	cioc-i	córc-e	uczennic-y	doktorz-e
VOC SG	mam-o	cioci-u	córk-o	uczennic-o	doktorz-e
NOM PL	mam-y	cioci-e	córk-i	uczennic-e	doktorz-y
GEN PL	mam-Ø	cioć-Ø	córek-Ø	uczennic-Ø	doktor-ów
DAT PL	mam-om	cioci-om	córk-om	uczennic-om	doktor-om
ACC PL	mam-y	cioci-e	córk-i	uczennic-e	doktor-ów
INS PL	mam-ami	cioci-ami	córk-ami	uczennic-ami	doktor-ami
LOC PL	mam-ach	cioci-ach	córk-ach	uczennic-ach	doktor-ach
VOC PL	mam-y	cioci-e	córk-i	uczennic-e	doktorz-y

sultant number of paradigms (exact sets of suffixes in the 14-slot grid) is considerable. However, there are numerous similarities across the paradigms. For example, the feminine classes in Table 1 share some suffixes in the singular. The suffixes for other cases differ, but all four patterns syncretise the dative and locative. In those represented by *ciocia* and *uczennica*, the genitive is also identical to these two cases. In the plural, the similarities are even more evident and some suffixes are shared with the masculine class (and others). Thus, the numerous exact sets of suffixes for the 14 slots are to a large extent variations on relatively few basic patterns.

2.2 Lack of declension in Polish

Let us now review the factors that make some Polish nouns resist inflection. Most of them are borrowings, but as we shall see, native words (and new coinages) can also be uninflected. The first factor is the incompatibility of morphonetic shape of the given word and available declension paradigms. Consider words ending in *-u*, which cannot be a nominative suffix, so the whole word is interpreted as the stem. This does not help much because such stems (with putative zero suffixes)[3] still cannot be associated with any pattern of declension. Consequently, loanwords like *malibu* in (3), *kakadu* 'cockatoo', *Peru*, *Nehru* or *Katmandu* remain uninflectable.

(3) butelk-a malibu
bottle-NOM.SG malibu[GEN.SG]
'a bottle of malibu'

Apart from the final *-u*, stressed final vowels are similarly problematic for the Polish system. These are found in loanwords that preserve the original oxytone pattern and in many acronyms (including native ones), e.g. *menu* [mɛˈni], *Delacroix*, *Calais*, *BBC* or *PKO* (name of a bank).

To make sense of (4), we must factor in semantics. Even though nouns with final consonants are commonplace in Polish and they normally inflect, they are masculine (with the suffix *-Ø*) and the corresponding paradigm is masculine, cf. (5).

(4) Czy nie przestaje Pan-i być rozsądn-ą
Q NEG stop.3SG Mrs-NOM.SG be.INF rational-INS.SG.F
bizneswoman?
businesswoman[INS.SG]
'Don't you stop being a rational businesswoman?'
(NKJP, Przepiórkowski et al. 2012)

(5) Czy nie przestaje Pan-Ø być rozsądn-ym
Q NEG stop.3SG Mr-NOM.SG be.INF rational-INS.SG.M
biznesmen-em?
businessman-INS.SG
'Don't you stop being a rational businessman?'

Because of its meaning, *bizneswoman* in (4) must be feminine and the gender assignment does not allow the noun to decline according to the masculine paradigm. This is a clear case of incompatibility: a potentially matching paradigm

[3] Some grammars of Polish posit zero suffixes for all paradigm slots of uninflectable nouns (e.g. Orzechowska 1998: 271). Here, subtler distinctions are necessary. In glosses for normally uninflectable nouns inflectional categories (retrievable from the context) are shown in brackets, cf. (3). In the case of normally inflectable nouns, zero suffixes (in opposition to non-zero suffixes on the same words) are used, cf. (5). In clear cases of contextual uninflectedness no morpheme segmentation is used and the category values given match the form of the noun and not the context, cf. (7a).

exists, but it is not applied because of the given noun's semantics, reflected in gender assignment.

A similar issue arises with words that have a final *-y* [ɨ], which is a nominative singular suffix in the masculine paradigm for adjectives. Nouns that inflect this way are typically masculine personal, cf. *szeregowy* 'private' in Section 2.1. Foreign male names that match the pattern, e.g. *Harry*, inflect, while semantically unfit borrowings, e.g. *brandy*, do not.

Neuter nouns in *-um*, e.g. *kuriozum* 'oddity' are a unique group in that they do not inflect in the singular, but do so in the plural. The pattern has arisen with Latin borrowings, but is productive, as many other minor ones. A very recent (and likely short-lived) example is *miraculum*, a name of type of magical object in *Miraculous: Tales of Ladybug & Cat Noir*, an animated television series from the 2010s.

Let us stress that when nouns are borrowed into Polish, the original form is taken to be the nominative singular. Numerous noun forms have a final *-u*, but it serves as a suffix of oblique cases. Borrowed foreign forms like *malibu* are never interpreted as having such a suffix.[4] The nominative singular functions as the basic form in the Polish grammatical tradition and is used as the entry form in dictionaries. The behaviour of borrowings shows that this convention is supported by the feeling of native speakers.

The second factor promoting lack of inflection can be called the unfamiliar character of the noun. Even though the place names *Vancouver* and *Hanower* 'Hannover' end in exactly the same sounds in Polish, only the latter regularly inflects. This factor is very elusive and cannot be boiled down to geographical distance or low frequency.[5] Names of many cities in Germany, a neighbour of Poland, do not inflect, e.g. *Essen* or *Monachium* 'Munich'. On the other hand, names of numerous very distant and clearly exotic places do inflect, e.g. *Osaka* or *Bangkok*.

Similar observations can be made about common nouns for concepts of foreign origin. Many such words are normally uninflectable, even though possibly matching paradigms exist, e.g. *judo*, *karate, reggae, kakao* 'cocoa', and *mango*. Other established borrowings do inflect, however. The following examples are

[4]Following this line of thought, (3) could be interpreted as having a genitive suffix *-u*, found in masculine inanimate paradigms, yielding a nominative form **malib*. In reality, borrowings are never treated this way and the very idea would seem absurd to native speakers. This fantasy scenario is inspired by the treatment of loanwords in Kambaata (Cushitic, Ethiopia), which are always inflected no matter what phonetic shape they have (Treis 2023).

[5]In the case of the two city names, this is completely mistaken. *Vancouver* is actually more frequent in the NKJP than *Hanower*, 4,503 vs. 3,331 attestations, respectively.

semantically similar to the previous set: *joga* 'yoga', *rap*, *cha-cha*, *kawa* 'coffee' and *papaja* 'papaya'.

To sum up, some foreign words function as quotes from other languages (Krzyżanowski 2013: 36, 40) and do not inflect, although which ones behave so is really a matter of language use, which is not uniform either (see below). According to Dunaj (2004: 84–85), the features promoting uninflectedness are low frequency of the potential paradigm and the given borrowing as such, as well as the learned character of the word.

We mentioned in passing the idea of using words as quotes. In fact, this is a typical environment for contextual suspension of inflection. Thus, in (6) the normally inflectable word *kwiat* does not adopt the form expected after the preposition *do.* The basic form is used for metalinguistic quoting.

(6) rym-Ø do "kwiat"
rhyme-NOM.SG PREP flower.NOM.SG
'a rhyme for [the word] flower' (NKJP)

Note that an inflected word form can be quoted as well, but then its overt categories are not tied to the actual context. However, the unchanged basic form of declinable words in Polish appears often in structures that are not straightforward quoting, namely APPOSITIONS, cf. (7).

(7) a. nad rzek-ą San
upon river-INS.SG San.NOM.SG

b. nad San-em
upon San-INS.SG
'on the river San'

The name in (7a) remains in the nominative, quoted, as it were, and the syntactic functions of the whole phrase are marked on the common noun *rzeka* 'river'. This happens despite the name being otherwise inflectable as in (7b). Appositional phrases where one noun (usually more general and coming first) is inflected while the other is given in the basic form, even though it is generally inflectable, constitute a common pattern in modern Polish, which easily incorporates uninflectable nouns. The phrase *strusi emu* in (8) follows the pattern of apposition observed in (7a). The more general noun is declined and thus shows the grammatical relations of the whole phrase to other words whereas the more specific noun bears no inflectional marking. The difference is that *emu*, like *malibu* in (3), is always uninflectable, unlike the river name in (7).

(8) przodk-owie dzisiejsz-ych strus-i emu
ancestor-NOM.PL today.ADJ-GEN.PL ostrich-GEN.PL emu[GEN.PL]
'the ancestors of modern emus' (NKJP)

For many of the nouns mentioned so far there is variation in usage. This is certainly true about the very fluid concept of unfamiliar character. If a given noun is normally not declined, but its morphonetic shape does not preclude assigning a declension paradigm, it is likely that some speakers will inflect it. For example, *Vancouver* may be uninflectable in standard usage, but some speakers do decline the name.[6]

A historical parallel is the usage of the noun *radio*. Initially, the word was uninflectable, presumably due to novelty of the invention, coupled with unfamiliar phonetics, the palatalised plosive [d^j] being rare in Polish. Today, however, the inflected forms have become dominant. Thus, some tendency to eliminate particular cases of uninflectedness can be noted. This process leads to variation in contemporary usage and changes of the properties of lexemes over longer stretches of time.

Changes such as that of the noun *radio* are rather expected; if nouns generally decline, it is natural that new "unfamiliar" nouns eventually comply with the regular pattern. However, Kurek (2019) documents an opposite tendency to avoid declension in present day Polish. One context in which normally inflectable nouns are used in their basic forms is official documents. This is linked to generating them on the basis of electronic databases. Various data, particularly surnames, are stored in nominative forms and then inserted directly into documents, whatever the actual linguistic context (Kurek 2019: 57). This kind of "officialese" may serve as a model for some speakers in other situations. Uncertainty about the actual forms of particular words also plays a role (Kurek 2019: 61–62). Faced with the risk of creating a wrong inflected form, speakers tend to prefer uninflectedness as the safest option, even though this may sometimes obscure the meaning or lead to misunderstandings (Kurek 2019: 157–159). Such usage has not been accepted as standard as yet, but is clearly present. The process is most advanced with surnames. These normally appear in apposition with given names, which resist the tendency much more and usually inflect.

[6]There are 2 attestations of *do Vancouveru* 'to Vancouver' in the NKJP and 132 of the standard phrase without inflection. There are 30 attestations of *w Vancouverze* 'in Vancouver' and 1210 of the version without inflection.

3 The uninflected words for women in present-day Polish

3.1 The array of constructions

The uninflected nouns referring to women represent a regular pattern, but are clearly a minority among feminine personal nouns in Polish. The examples below present both the uninflected forms and competing patterns. We shall start by defining the relevant word types. The focal group will be called U-forms (for ‘uninflected’), nouns referring to women that have no inflectional marking in syntactic positions where non-zero suffixes are expected. The major competing pattern are D-forms (for ‘derivation’), nouns referring to women that have clear morphologically related masculine counterparts (and so can be construed as derived from them), but inflect normally according to feminine paradigms. There are still M-forms (for ‘masculine’), masculine personal nouns with inflection according to masculine paradigms, but used with reference to women.[7] We should stress that there is actual competition among these word types, i.e. one lexical concept may be expressed by more than one type. (9–10) exemplify U-forms.

(9)

Z	psycholog	nie	chciała	gadać.
with	psychologist[INS.SG]	NEG	want.PST.F	talk.INF

‘With the psychologist she wouldn’t talk.’ (NKJP)[8]

(10)

Zainteresowałem	pomysł-em	pani-ą	notariusz
interest.PST.M.1SG	idea-INS.SG	Mrs-ACC.SG	notary[ACC.SG]

Agnieszk-ę	Sienkiewicz.
Agnieszka-ACC.SG	Sienkiewicz[ACC.SG]

‘I got the notary, Agnieszka Sienkiewicz, interested in the idea.’ (NKJP)

The common nouns *psycholog* in (9) and *notariusz* in (10) as well as the surname in (10) have no marking even though their syntactic positions require oblique cases that normally have non-zero suffixes. The three nouns are identical in form to the nominative singular of their masculine counterparts. With masculine referents they would inflect: *z psychologi-em*, *notariusz-a*, *Sienkiewicz-a*, cf.

[7] A non-negligible number of words referring to women (which we could label F-forms) fall beyond the scope of the division presented here. These are inflectable feminine nouns that are morphologically independent from their masculine counterparts, e.g. *siostra* ‘sister’ vs. *brat* ‘brother’.

[8] Examples like (9) should theoretically be retrievable from corpora because of the charateristic mismatch between the preposition (taking an oblique case) and the following U-form which looks like a nominative. Yet, such queries mostly yield noise, phrases with various forms wrongly tagged as nominatives.

the difference between (4) and (5). Applying the originally masculine nouns for feminine referents whilst keeping the unchanged basic form across the whole paradigm is the formation of *derived U-forms*, a productive pattern of word-formation. The common nouns involved are usually ultimately borrowings, as in (9–10), but many are old ones and not necessarily perceived as foreign. Among surnames a great number of U-forms are native. Apart from derived U-forms, there are also uninflectable loanwords referring to women, e.g. *bizneswoman* in (4). These are not identical to their masculine counterparts (or may not have them at all) and will be referred to as *independent U-forms*.

To explain the uninflectedness here, we need only generalise the analysis of (4). Words like *psycholog*, *notariusz*, etc. pose a problem for the Polish system if they are to refer to women. The word-final consonants in the basic form are characteristic of masculine paradigms, but the words are supposed to be grammatically feminine to match actual reference. As a result, the feminine declension cannot be applied because of the morphonetic structure of the words, and the masculine personal one is not used either because of semantics.[9]

Uninflectedness is only one possible resolution of the conflict of morphonetic shape and gender. Apart from inflection, Polish also has rich derivational morphology. If the morphonetic shape of a word is problematic for the given use, a derivational mechanism can rectify it. This is how D-forms come into existence, examples of which are (11–13).

(11)	Apetyczn-a	trzydziestolat-k-a	o	wyglądzi-e
	attractive-NOM.SG.F	thirty.year-NMLZ-NOM.SG	PREP	appearance-LOC.SG
	prowincjonaln-ej	nauczyciel-k-i.		
	provincial-GEN.SG.F	teacher-F-GEN.SG		

‘An attractive 30-year-old looking like a provincial teacher.’ (NKJP)

The word *nauczycielki* in (11) is clearly distinct from the U-forms above and can be regarded as a regular element of the Polish system; the inflection is present and it is in line with semantics. Note the feminative suffix *-k(a)*, also present in (2c), which turns the masculine noun *nauczyciel* ‘teacher’ into the feminine *nauczycielka*, which undergoes regular inflection.

[9] Section 2.1 mentions a paradigm for feminines ending in consonants. It might then seem that there is no conflict after all. Alas, this declension class only has nouns with (historically) soft word-final consonants, which is not the case for the majority of the words that appear as U-forms. It also contains mostly inanimate nouns and few for animals, e.g. *mysz* ‘mouse’. Any personal nouns following the pattern are long obsolete, e.g. *jątrew* ‘husband’s brother’s wife’.

Most D-forms are created by suffixation, but some are conversions (zero derivations).[10] A conspicuous (and controversial, cf. Section 4.4) example is *ministra* in (12), with exactly the same stem as the masculine *minister.* The word is formed by direct application of a major feminine paradigm (including *-a* for the nominative singular). Thus, even though the basic form *minister-Ø* has no matching feminine paradigm (as was the case with all the U-forms above), the stem *minist(e)r-* can be matched with no problem. This comes at the price of having a distinct basic form of the word, clearly a change noticeable for the speakers.

(12)

Now-a	ministr-a	kultur-y	Szwecj-i	ma
new-NOM.SG.F	minister-NOM.SG	culture-GEN.SG	Sweden-GEN.SG	have.3SG
dred-y.				
dreadlock-ACC.PL				

‘The new Swedish minister for culture has dreadlocks.’[11]

The noun *radna* ‘councillor, F’ in (13) shares the stem with the masculine *radny*, but these are nouns of the adjectival paradigm. Adjectives have gender as an inflectional category. Hence, whenever a reference to a woman needs to be made with concepts expressed by such nouns, the proper inflectional suffixes are directly available in the paradigm.

(13)

Odpowiedź-Ø	nie	usatysfakcjonowała	radn-ej
answer-NOM.SG	NEG	satisfy.PST.F	councillor-GEN.SG.F
Binder.			
Binder[GEN.SG]			

‘The answer did not satisfy councillor Binder.’ (NKJP)

Also the word *trzydziestolatka* ‘30-year-old, F’ in (11) has the same stem as the masculine *trzydziestolatek*. The two are parallel derivations from the phrase meaning ‘thirty years’ rather than a direct instance of conversion. Still, the morphological closeness of both words is absolutely clear.

M-forms are exemplified by the words *nauczycielem* in (14), *ministra* in (15) and *autora* in (1c). Even though the reference is clearly feminine, these are inflected forms with suffixes of masculine paradigms. The adjective in (14) has a masculine suffix, which shows that M-forms are grammatically masculine nouns, just with female referents.

[10]Derived U-forms are also conversions – from a major masculine personal paradigm to the “fully syncretic” class of uninflectable nouns.

[10]https://wyborcza.pl/7,75410,24399632,nowa-ministra-kultury-szwecji-ma-dredy-i-palila-trawke.html Accessed 24.06.2024.

(14) Pan-i Grażyn-a nie jest surow-ym nauczyciel-em.
Mrs-NOM.SG Grażyna-NOM.SG NEG be.3SG strict-INS.SG.M teacher-INS.SG
'[Mrs] Grażyna [given name] is not a strict teacher.' (NKJP)

(15) rocznic-a śmierc-i Izabel-i Jarug-i
anniversary-NOM.SG death-GEN.SG Izabela-GEN.SG Jaruga-GEN.SG
Nowack-iej, ministr-a polityk-i społeczn-ej
Nowacka-GEN.SG.F minister-GEN.SG policy-GEN.SG social-GEN.SG.F
'the anniversary of the death of Izabela Jaruga-Nowacka, the Minister for Social Policy'[12]

Let us add that (14) involves a noun used predicatively in a copular construction and this is one context in which M-forms appear regularly and U-forms are very rare. D-forms, by contrast, are fine; an alternative way to express the meaning of (14) would be with the phrase *surow-ą nauczyciel-k-ą*.

Table 2 summarises the properties of the three relevant word types. The distribution of the structures is partially determined by constructional, lexical and stylistic factors, discussed at length in Section 3.3. The restriction on U-forms being used predicatively has just been mentioned. As for the lexical factors, different concepts applied to women favour different structures, to the extent that only one of the three theoretically possible formations may be used. The role of register (and possibly ideology) can be noted with the words meaning 'minister' above. (12) uses a D-form, (15) an M-form, a U-form is also possible. (12) is part of a headline in a progressive daily *Gazeta Wyborcza*, which at some point decided to adopt the innovative D-form for female holders of ministerial offices, cf. Section 4.4. (15) comes from a government website and the M-form has a dry bureaucratic feel to it. In fact, the U-form *minister*, as in (24), is the most common way to refer to female ministers.

Thus, there is some form of "division of labour" between the word types, but at the same time there is also competition between them. Even though we can clearly distinguish three types of noun usage, the speakers seem to perceive a binary opposition: D-forms (showing feminine reference directly) vs. the other two. The same transpires from linguistic analyses: even if the problem of uninflectedness is acknowledged and discussed, it is subordinate to the main division: D-forms are typically labelled *feminine forms* while U-forms and M-forms are *masculine*.[13]

[12] https://www.gov.pl/web/rodzina/8-rocznica-smierci-izabeli-jarugi-nowackiej Accessed 24.06.2024.

[13] A notable exception is Krzyżanowski (2013: 21, 172ff.).

Table 2: Summary of the properties of the discussed word types

	U-forms	D-forms	M-forms	masculine nouns
reference	feminine	feminine	feminine	masculine
basic form (NOM SG)	same as M noun (and M-form)	distinctly F	same as M noun (and U-form)	distinctly M (and same as U-form and M-form)
inflectional paradigm(s)	no marking	feminine	masculine	masculine
morphological relation to M counterpart	identical to M NOM SG (conversion) or no direct relation	suffixation or (pseudo) conversion	full identity	n/a
examples in this section	*psycholog* in (9), *notariusz* in (10), *Sienkiewicz* in (10)	*panią* in (10), *trzydziesto-latka* in (11), *nauczycielki* in (11), *ministra* in (12), *radna* in (13), *Nowackiej* in (15)	*nauczycielem* in (14), *ministra* in (15)	n/a

Let us also consider the status of the three word types as lexical units. It is obvious that D-forms are separate lexemes from masculine nouns. Likewise, it is clear that M-forms are only special ways of using masculine nouns. Independent U-forms, like *bizneswoman*, must also be lexemes of their own. The question is more complex for derived U-forms. It is often assumed that they are twin lexemes of their masculine counterparts, i.e. that there is an inflectable masculine lexeme *minister*$_1$ and a separate uninflectable feminine one *minister*$_2$. The idea was apparently first suggested by Kucała (1978: 68) and is included in academic grammars (e.g. Grzegorczykowa & Puzynina 1998a: 368; Grzegorczykowa & Puzynina 1998b: 422). Some more recent works refer to this position as a given (Jadacka

2006: 126; RJP 2012; Andrejewicz 2019: 626), but it is not universally accepted. Łaziński (2006: 227) argues strongly against it while Bobrowski (2012: 229) considers it only as a possible option. The issue will resurface in Sections 3.3.2 and 4.5.

Table 3: Major inflectional types of Polish surnames

surname type	masculine form (further exs. in brackets)	feminine form	traditional feminine forms (as of early 20th cent. in standard language)
suffix *-sk(i)* and variants	Kowalski (Nowacki)	Kowalska	Kowalska
final *-a*	Duda (Jaruga)	Duda	Dudzina/Dudowa (married) Dudzianka/Dudówna (unmarried)
final consonant	Nowak (Sienkiewicz)	Nowak (U-form)	Nowakowa (married) Nowakówna (unmarried)

Women's surnames present a similar level of variation to common nouns. As shown in Table 3, surnames with the adjectival suffix *-sk(i)* have regular D-forms, while surnames with final consonants regularly have U-forms. The most interesting group, not fitting any of the word types above, are those with the final *-a*, which have identical inflectable forms for men and women, with exactly the same inflection in the singular, following the feminine paradigm. The grammatical gender of the surname is decided by semantics across all the types. Traditionally, all types had distinct D-forms, revealing not only gender, but also a given woman's marital status.[14]

[14]There used to be greater variation of D-forms than shown in Table 3. Women could also be referred to by D-forms based on their husbands or fathers' given names or positions. In the early modern times, similar derivative labels existed for young men, many of which developed into patronymic surnames.

3.2 U-forms in use

Let us explore the current usage of U-forms in detail. (16–17) are quite typical examples.

(16) Głos-Ø Wybrzeż-a ma now-ą redaktor
voice-NOM.SG coast-GEN.SG have.3SG new-ACC.SG.F editor[ACC.SG]
naczeln-ą.
main-ACC.SG.F
'[The newspaper] Głos Wybrzeża has a new editor-in-chief.' (NKJP)

(17) Lekarz powiedziała, że może bez problem-u
doctor[NOM.SG] say.PST.F that may.3SG without problem-GEN.SG
jeść.
eat.INF
'The doctor said that [the baby] may eat [this kind of food] without a problem.' (NKJP)

The suffixes on noun modifiers as in (16) and on past verb forms in (17) are standard signals of the feminine gender and the U-forms in the examples are clearly feminine. However, such elements are not always present, cf. (9), where there is no overt marking of gender, but the feminine reference is clear too. This proves that among personal nouns the lack of inflection functions as a signal of feminine gender in its own right, as noted by Obrębska-Jabłońska (1949).[15]

If a U-form appears in the subject position, it is identical to the corresponding masculine noun with a zero suffix. A past tense verb may show gender as in (17), but not all sentences are past. Those like (18) leave the gender of the referent underdetermined. This follows directly from the rules that establish the class of U-forms, but is untypical of Polish, esp. with specific referents (Łaziński 2006: 263), cf. fn. 17 and the discussion of (28) and (32).

(18) Jakby nie rozumiał, co doktor mówi.
as_if NEG understand.PST.M what.ACC doctor[NOM.SG] say.3SG
'As if he didn't understand what the doctor was saying.' (NKJP)[16]

[15] Curiously enough, even though she recognises uninflectedness as a gender signal, Obrębska-Jabłońska herself describes the use of U-forms as "masculinisation" (1949: 1), in line with the general tendency to block U-forms and M-forms together against D-forms, cf. Section 3.1.

[16] It is impossible to retrieve easily examples of U-forms like (18) from corpora. U-forms are tagged the same as masculine nouns and true examples of the latter dominate the results absolutely.

Misunderstanding sentences like (18) is probably relatively seldom a risk in real-life communication as the gender will have been established earlier in the discourse.[17] Without any such cues the default interpretation would likely be that the sentence is about a male doctor.[18] Thus, plain U-forms, even though attested and used, may feel somewhat inadequate.

In fact, U-forms are very often combined in apposition with other feminine nouns, typically inflectable ones. In (19–20) U-forms are preceded by more general inflectable nouns, which directly show the syntactic relation of the phrases to the surrounding context. The uninflectedness becomes much less problematic in apposition, the U-form functions like a label quoted in an unchanged basic form, as in (6), (7a) and (8).

(19) T-a spraw-a jest ważn-a dla pan-i
this-nom.sg.f matter-nom.sg be.3sg important-nom.sg.f for Mrs-gen.sg
prezes.
president[gen.sg]
'This matter is important for [you,] madame president [of the company].'
(NKJP)

(20) Węgr-y też mają kobiet-ę minister
Hungary-nom.pl also have.3pl woman-acc.sg minister[acc.sg]
spraw-Ø zagraniczn-ych.
affair-gen.pl foreign-gen.pl
'Hungary also has a female foreign minister.' (NKJP)

The use of the honorific *pani* in apposition with U-forms, as in (19) or (10) above, is a regular pattern in the language. Other words can be used similarly, but this rare usage always meets specific needs of a given context, e.g. in (20) the noun *kobietę* explicitly labels the gender of the referent. The honorific combines regularly not only with common nouns, but also with surnames of women, both inflectable and uninflected. Table 4 shows the importance of the appositional pattern in terms of frequency. With the phrase *dla pani prezes* from (19) as a starting point, the NKJP was queried for the preposition *dla* and the noun form *prezes* with one intervening word possible. The results show a strong dominance of the apposition with the honorific for the U-form. No such effect is visible in the data for corresponding masculine phrases. This suggests that *pani* has developed

[17] In the context of (18), we learn two sentences earlier that the doctor is a woman from her D-form surname.

[18] We can then see a point in regarding U-forms as "masculine".

into a kind of auxiliary noun with U-forms, a carrier of overt inflectional marking and analytic marker of the feminine gender, cf. the discussion of (32). Note also that in the query results there is no single example parallel to (20), i.e. with a common noun other than *pani* used in apposition with the noun *prezes*.

Table 4: Frequencies of various types of phrase structure in phrases *dla (...) prezes(a)* 'for the president (of a company)' in the NKJP.

phrase structure	no. of relevant examples	%
dla pani prezes (honorific + U-form)	43	80%
dla ADJ.F/DET.F *prezes* (F modifier + U-form)	4	7%
dla prezes (no intervening words)	7	13%
total F	54	
dla pana prezesa (honorific + M noun)	74	6%
dla ADJ.M/DET.M *prezesa* (modifier+ M noun)	137	11%
dla prezesa (no intervening words)	1,034	83%
total M	1,245	

As for women's surnames, they commonly appear in apposition, with the noun *pani* and with given names, most of which inflect.[19] In (10), such typical structures are joined into a single complex apposition: honorific – profession – given name – surname. Two nouns are inflected, two are U-forms.

Another natural combination is that of a given woman's function and her surname, as in (13). In fact, of the 7 examples with no intervening word before the U-form in Table 4, as many as 4 involve a surname in apposition, coming after the word *prezes*. In such phrases both nouns may well be U-forms, as in (21).

(21) polityczn-a burz-a po twitterow-ym wpisi-e
political-NOM.SG.F storm-NOM.SG after twitter.ADJ-LOC.SG.M post-LOC.SG
kurator Nowak
superintendent[GEN.SG] Nowak[GEN.SG]
'a political storm after a tweet by the school superintendent Nowak'[20]

[19]The vast majority of female given names in Polish end in *-a* in the basic form and inflect according to the major feminine paradigms. The few rather rare exceptions are U-forms.

[20]https://wiadomosci.dziennik.pl/polityka/artykuly/8578116,barbara-nowak-lex-czarnek-barbara-nowacka.html Accessed 26.06.2024.

As in (9), there are no direct signals of feminine gender in (21), but the feminine reference is perfectly clear because of the lack of expected inflection. A parallel phrase about a male superintendent by the same name would obviously need to inflect: *kurator-a Nowak-a*, cf. the discussion of (4–5) and (9–10).

The opposition between the inflectable male version of a surname and the uninflected female one extends to many foreign surnames. For example, the surname *Clinton* inflects when Bill Clinton is referred to, but does not when Hillary Clinton is, but cf. Section 4.2. Here the given names behave identically to the surname: *Bill* is regularly inflected as a masculine noun, while *Hillary*, not fitting any feminine paradigm, is a U-form.

(22), the title of a paper devoted to U-forms, exemplifies how surnames function as plain U-forms. As we can see, this is parallel to what we observed with common nouns.

(22) Baran mówi o Kowal.
Baran[NOM.SG] talk.3SG about Kowal[LOC.SG]
'Baran talks about [Ms.] Kowal.' (Pawłowski 1951)

The surname *Kowal* in (22) is in a non-subject position and the lack of overt inflection identifies it as a U-form and a woman's name, as in (9) and (21). However, for the surname *Baran*, which appears as the subject, it cannot be decided from the sentence alone if the person in question is a man or a woman, as in (18).

To sum up, U-forms are an established and productive pattern. Among them there are common nouns for professions, professional functions and titles as well as surnames (including numerous native ones). These words do not have any overt marking of the feminine gender, but trigger feminine agreement. However, the lack of inflection on nouns is a conspicuous absence in Polish[21] and so it has developed into a signal of the feminine gender by itself. This signalling fails, however, if the given noun is in the nominative singular, which for derived U-forms is identical to the masculine counterpart. There is no grammatical difference between common nouns and personal names. All the effects related to uninflectedness can be observed in both groups.

The feminine agreement triggered by U-forms is the reason Kucała (1978: 61, 68) concluded they are feminine nouns, thus excluding any mismatch of gender between the modifiers or past verb forms and the nouns of an apparently masculine shape.

[21] Krzyżanowski (2013: 35) vividly describes uninflected nouns, including U-forms, as "refraining from inflection".

Despite their established status in the system, U-forms are still a very odd class in Polish. This fuels the use of a rival pattern, D-forms. Taking into account the most general features of the language, one might expect a very strong systemic pressure for establishing D-forms across the whole personal noun lexicon. As can be inferred so far, this has not been the case, but see the discussion of recent developments in Section 4.4.

Another effect related to the "odd" character of U-forms is the ease with which they enter into appositional structures. In this way, they can "hide" behind inflectable nouns, which show syntactic relations through case marking and manifest gender explicitly. The main inflectable nouns entering into these structures are the honorific *pani* and given names. Apposition is thus a major strategy of integrating U-forms in the grammatical system heavily reliant on inflection.

3.3 Constraints on the use of U-forms

3.3.1 Overview

The use of U-forms is limited in several ways in present-day Polish. Some restrictions are grammatical. One is the virtual absence of U-forms from the copular construction where M-forms are used rather than U-forms, cf. the discussion of (14), which need not be expanded. U-forms are also less free in the plural, see the next section. Apart from that there are lexical and stylistic factors. Groups of lexemes show preferences for particular word types – this is the "division of labour" between U-forms, D-forms and M-forms. However, there is also substantial (and increasing) overlap and competition between the types. When forms of two or even three types co-exist for the same lexical concept, the difference between them is related to factors like register or ideology.

3.3.2 Plural number

The examples so far have all shown U-forms in the singular. Number is the most problematic category for U-forms. As has been shown in Section 3.2, case can usually be inferred from the context as its value must match the requirements of the word governing a given U-form syntactically, even though structurally ambiguous phrases appear in actual use cf. Szpyra-Kozłowska (2021: 340–341). Gender is understood to be feminine due to the lack of inflection. A U-form by itself has no way of differentiation left for number, which is by default understood to be singular.[22] A natural solution is to mark plurality around the noun: on

[22] An alternative would be for the number to be underspecified in such contexts. This is not the case, however, e.g. (9) refers to one female psychologist, a plural reading is highly unlikely if not downright impossible.

nouns in apposition, modifiers or verbs. In (23) the U-form surname is combined with the plural of the honorific *pani*. Yet again apposition makes the functioning of an uninflected noun in Polish syntax possible.

(23) Postanowiono pomóc pani-om Anflik w
decide.PST.IMPERS help.INF Mrs-DAT.PL Anflik[DAT.PL] in
inn-y sposób-Ø.
other-ACC.SG.M.INAN way-ACC.SG
'It was decided that the Anflik Ladies will be helped in a different way.'
(NKJP)

The plural honorific is used regularly, but other potential carriers of plurality are not unproblematic. Łaziński (2006: 227) argues that derived U-forms simply cannot be used as plural nouns in their own right, giving an example with the "evidently faulty" phrase **między dwiema minister* 'between two [female] ministers'. A phrase with an apposition, parallel to (23), *między paniami minister* would be acceptable, however. An independent U-form in plural is also fine: *między dwiema lady* 'between two ladies'.

Łaziński's conclusion is that derived U-forms are not separate uninflectable lexemes, but simply special cases of usage of masculine nouns (otherwise normally inflectable) for female referents. The lack of marking would then prove to be simply contextual uninflectedness – in contrast to the view taken by Kucała (1978), cf. Section 3.1. It follows that there are two separate ways in which masculine personal lexemes can be used to refer to women, U-forms and M-forms. The analysis runs into problems when we consider that U-forms trigger feminine agreement. Łaziński (2006: 231) explains the resultant gender mismatch (non-existent in Kucała's approach, cf. Section 3.2) as "agreement *ad sensum*", i.e. a grammatically masculine noun combines with feminine modifiers or past verb forms on the basis of actual feminine reference. This special kind of agreement would then be obligatory for U-forms, but never happens with M-forms, cf. (14). The firmness of Łaziński's conclusions is shaken by the fact that sentences with the supposedly unacceptable structures are in fact attested, e.g. (24).

(24) Słow-a t-e kierowały obi-e
word-ACC.PL this-ACC.PL.NVIR direct.PST.PL.NVIR both-NOM.PL.F
minister wyłącznie do osób-Ø, któr-ych (…)
minister[NOM.PL] only to person-GEN.PL who-GEN.PL
'These words were spoken by both [female] ministers only to people who (…)'
(NKJP)

In (24) plurality is marked on the modifier and the verb form. Ironically, the example includes the very same noun as Łaziński's "faulty" phrase. Yet, sentences like (24) are rather infrequent. To be fair, Łaziński gives a counterexample himself, quoting what he had heard said by a receptionist at the doctor's, given as (25).

(25) Doktor already przyszły.

Doktor	już	przyszły.
doctor[NOM.PL]	already	come.PST.PL.NVIR

'The doctors [all female] have come.' (Łaziński 2006: 227)

Plurality in (25) is shown on the verb only. Łaziński comments that this kind of usage "does not seem to be very common". This observation seems accurate, but such structures still do occur in real language. (26) is a parallel example with an independent U-form. Such sentences are easier to find than ones like (25) even though independent U-forms are generally less common than derived U-forms.

(26)

Dawniej	miss			były
early.COMP.ADV	beauty_queen[NOM.PL]			be.PST.PL.NVIR
zdan-e	tylko	na	siebie.	
rely.PTCP-NOM.PL.NVIR	only	on	oneself.ACC	

'In the earlier times, beauty queens were left on their own.'[23]

In the light of (24–26) it is impossible to accept Łaziński's categorical statements at face value, but the observations are not irrelevant; the counterexamples are infrequent. Table 5 sums up the use of U-forms in plural. The possible carriers of plurality form a hierarchy. Let us add that we are not aware of attestations of U-forms with clearly plural meaning and no formal marking of plurality. If such usage exists at all, it is rare, quite certainly substandard and strongly context-dependent. In contrast to U-forms, the other patterns: D-forms and M-forms as well as masculine personal nouns are used in the plural without restrictions.

In Table 5 we present the general picture of the usage of U-forms in the plural. Since it is extremely difficult to substantiate the frequency labels for the whole class, we resort to corpus data for specific example words, shown in Table 6. We selected three typical U-forms: *minister*, *doktor* 'medical doctor, Ph.D.' and *profesor* 'professor'. The data have been queried separately for different loci of plural marking. These were: the non-nominative plural forms of the honorific, non-nominative feminine forms of basic plural modifying words: *dwie* 'two', *obie*

[23] https://dziennikzachodni.pl/ewa-wojciechowska-dawniej-miss-byly-zdane-tylko-na-siebie-to-byl-blad/ar/486300 Accessed 26.06.2024.

Table 5: Occurrence of U-forms in plural contexts

locus of plural marking	occurrence	examples
pani in apposition	regular	(23)
noun modifiers	attested but not common	(24)
finite verb	rare, mostly with independent U-forms	(25–26)

'both', *wsystkie* 'all' and lastly top-frequency plural verb forms: *są* 'they are', *będą* 'they will (be)' and *mają* 'they have'. The first two are placed before the target U-forms in the query, the verb forms after them. The use of non-nominative forms in the first two excludes plural agreement on the verb. For comparison, Table 6 also includes figures for parallel plural phrases with a single D-form *nauczycielka* 'teacher, F'.

Table 6: Frequencies of various types of plural phrase structure for selected U-forms and D-forms in the NKJP

phrase structure	no. of query results	exs. with single locus of plurality	%
pani (PL, NNOM) + *minister*/*doktor*/*profesor*	23	19	95%
dwie/*obie*/*wszystkie* (PL, NNOM) + *minister*/*doktor*/*profesor*	1	1	5%
minister/*doktor*/*profesor* + *są*/*będą*/*mają*	3	0	0%
total U-forms		20	
pani (PL, NNOM) + *nauczycielki* (PL, NNOM)	83	81	47%
dwie/*obie*/*wszystkie* (PL, NNOM) + *nauczycielki* (PL, NNOM)	41	41	24%
nauczycielki + *są*/*będą*/*mają*	56	49	29%
total D-forms		171	

Since we are interested in comparing different loci of plural marking, examples with doubled signals of plurality (e.g. both on the modifier and the honorific) had to be removed from the final count. In the case of U-forms combining with the selected plural verb forms this meant excluding all examples. We have only a single example, cited as (24), of a plural modifier with a U-form even though three different words and three possible modifiers were used in the query. Table 6 shows both the initial figures and the examples with single locus of plurality after the recount. The data confirm the hierarchy of carriers of plural marking suggested in Table 5. The honorific is significantly more common relative to the other types in U-forms than in D-forms. The plural honorific dominates in the results with the U-forms. The distribution of the structures with the D-form is much more even.

3.3.3 Lexical range

Various areas of the personal noun lexicon of Polish show preferences for particular word types when feminine reference is made. In a way, each of the three types is restricted in its lexical range; default or at least common for some words, but rare or even non-existent for some others. We will briefly outline what kind of nouns prefer each of the morphological types, starting with words dispreferring U-forms.

Very many concepts are regularly expressed by D-forms when referring to women. (27) shows pairs of corresponding words for men and women, the feminine ones show relevant morphological division.

(27) a. Włoch ~ Włosz-k-a
 'Italian person'
 b. kuzyn ~ kuzyn-k-a
 'cousin'
 c. sąsiad ~ sąsiad-k-a
 'neighbour'
 d. artysta ~ artyst-k-a
 'artist'
 e. poseł ~ posł-ank-a
 'MP'
 f. sprzedawca ~ sprzedawcz-yn-i
 'shop assistant'
 g. szef ~ szef-ow-a
 'boss'

h. hrabia ~ hrab-in-a
'count(ess)'

i. diabeł ~ diabl-ic-a
'devil, evil person'

j. staruch ~ staruch-a
'old person (derog.)'

Word pairs such as those in (27) are most consistent with general rules of Polish and constitute the "basic" (Kucała 1978: 49) section among personal nouns. Feminine reference is made with D-forms and U-forms are rare or non-existent. Most of these concepts allow M-forms in copular constructions and masculine nouns in generic use (covering both men and women).[24]

As for semantics, there are names for nationalities in this group, cf. (27a), as well as family relations, cf. (27b). Apart from that, it is difficult to generalise, but we must note that some words for occupations and functions also appear here, cf. (27d–27f), as well as those for noble titles, cf. (27h).

The standard assumption in Polish linguistics is that D-forms are derived from their masculine counterparts (e.g. Grzegorczykowa & Puzynina 1998b: 422; Kucała 1978: 49). While this is right in many cases, Szpyra-Kozłowska (2021: 134–135) points out that it need not be a given. For example, nationality labels for both men and women, as in (27a), are most logically derived from country names, cf. also the discussion of *trzydziestolatka* in (11).

(27) shows the rich array of feminative suffixes in Polish: the most common *-k(a)* in (27a–27d), as well as (2c) and (11) above, *-ank(a)* in (27e), *-yn(i)/-in(i)* in (27f), *-ow(a)* in (27g), *-in(a)* in (27h), *-ic(a)* in (27i). Finally, some D-forms are conversions, cf. (27j) and (12).

The suffix *-k(a)* is the most common and can be considered default. Its disadvantage in the eyes of the speakers is that it is identical to the diminutive one, and the one forming names of objects from personal nouns, e.g *marynar-k-a* 'suit jacket, F' from *marynarz* '(navy) sailor, M', as well as the nominalising one, as in *trzydziestolatka* in (11).[25] The variety of suffixes itself may also be problematic when new D-forms are created as the speakers might not be sure which suffix to apply (cf. Szpyra-Kozłowska 2019, see also Sections 4.3 and 4.4).

[24](2a–2b) are examples of masculine nouns used generically. An alternative to generic masculines is splitting, as in (2c).

[25]The identity of the feminative *-k(a)* and suffixes with other meanings may lead to emergence of excessive homophones, which is a factor that supposedly blocks the creation of D-forms in some cases. Szpyra-Kozłowska (2021: 191–192, 262–265), however, points out that homophony of suffixes is common in Polish derivation.

As for inflection, most D-forms follow major feminine paradigms, those in *-yn(i)*/*-in(i)* belong to a minor class, while the suffix *-ow(a)* builds words of the adjectival declension. Note that even if the masculine noun has the typically feminine *-a* in the basic form, D-forms are usually distinct, cf. (27d), (27f) and (27h).

At the other pole, there are lexemes that can refer to both men and women, but have forms inflected according to masculine paradigms, i.e. they are regularly M-forms when the reference is feminine. D-forms and especially U-forms are rare for such words, but not completely non-existent, cf. (30) and Section 4.1. Examples include *szpieg* 'spy', *wróg* 'enemy', *przechodzień* 'passer-by', *tchórz* 'coward' or *gość* 'guest' as in (28).

(28) Musi przywitać nasz-ego gości-a osobiście.
must.3SG welcome.INF our-ACC.SG.M guest-ACC.SG personally
'He must welcome our guest in person.' (NKJP)

The noun *gościa* in (28) takes a suffix from a masculine paradigm and the masculine gender is evident on the modifier. Without context the sentence may well refer to a male guest. In fact, this would be the default interpretation. A female name appears several paragraphs earlier in the text (28) is taken from, thus making the actual reference clearly feminine.

Łaziński (2006: 267) claims that the nouns of this group are rarely used for specific reference, which is normally associated with clear marking of the referent's gender in Polish. The mismatch between feminine reference and ostensibly masculine form of the nouns is mitigated by the backgrounding of gender in their meaning and non-specific reference. For example, *tchórz* focuses on the feature of character, while *przechodzień* emphasises a momentary role in traffic. This line of reasoning fits some of the words less well, e.g. *gość*, which may often have specific reference, as it does in (28). Similarly, the noun *szpieg* could logically refer to specific individuals as well, the synonymous *agent* is apparently more likely in such contexts (Łaziński 2006: 268). The latter has a regular D-form *agentka* for women as referents.

Apart from the fully explicit marking of femininity by D-forms and its occlusion with M-forms, there is a group of concepts for which U-forms are used regularly. For some of these words, this is the dominant pattern. D-forms may exist as alternatives, but their prevalence and acceptability varies greatly depending on the word. The words appearing as derived U-forms so far in the text are mostly labels for professions and professional titles: *notariusz* 'notary', *psycholog* 'psychologist', *redaktor* 'editor', *lekarz* 'medical doctor', *doktor* 'doctor', *prezes* 'president (e.g. of a company)', *kurator* 'educational superintendent', *profesor* 'professor', *minister* ‘minister’. What is common for these words is some level of prestige.[26] Indeed, the more prestigious a concept is, the more likely it is to be expressed by a U-form. The model for such usage across various contexts is direct address, which takes the form of the apposition comprising the honorific *pani* and the given U-form, cf. (19). Łaziński (2006: 275–276) claims the use of a word as a regular title of address blocks the possibility of creating a D-form. The blockage is not complete, and has been loosening recently, cf. Section 4.4. However, the appositional structure for addressing women remains a stronghold of U-forms. (29) is an example of a creative use of the pattern. The noun *ekspert* is not normally used for direct address and has an established D-form *ekspertka*. However, in the given situation, the U-form was chosen as a way of semi-direct address.

(29) T-o dla pan-i ekspert.
this-NOM.SG for Mrs-GEN.SG expert[GEN.SG]
‘This is for [you,] expert.’ (NKJP)

(30–31) show words with relatively little prestige used as U-forms without the honorific. The examples are recent press headings which Szpyra-Kozłowska (2021) presents as instances of blatant overuse of the uninflected pattern.

(30) polsk-a szpieg uhonorowan-a w Londyni-e
Polish-NOM.SG.F spy[NOM.SG] honour.PTCP-NOM.SG.F in London-LOC.SG
‘A Polish [woman] spy [from WW2] honoured in London’
(Szpyra-Kozłowska 2021: 339)

(31) Piękn-a sołtys wyznaje.
beautiful-NOM.SG.F village_leader[NOM.SG] confess.3SG
‘The beautiful village leader confesses.’ (339)

(30) features the use of *szpieg* as a U-form, although the concept is usually expressed as an M-form. For *sołtys* in (31) a more or less established D-form exists. The examples are certainly not typical for this word type and may seem questionable for some speakers. On the other hand, such examples of U-forms prove the pattern is productive.

In sum, the nouns preferring D-forms are very much regular: their form and inflection are in line with semantics. The ones preferring M-forms also inflect,

[26] In the 20th century Polish linguists suggested various factors other than prestige as blocking the creation of D-forms, and promoting U-forms/M-forms in the lexicon, cf. Buttler et al. (1973: 108–110). Szpyra-Kozłowska (2021: 190–241) offers a convincing criticism of these factors, calling them “myths”. One of them is the identity of the feminative *-k(a)* and suffixes with other meanings, cf. fn. 25.

but ignoring the actual feminine reference. U-forms are exceptional because of their uninflectedness. As can be seen, they are not distributed freely in the lexicon, but concentrate in the domain of words for professions and professional titles. Despite these divisions in the lexicon, there is still a considerable field for competition between the three types.

3.3.4 Register

When they are in competition, U-forms are somewhat formal while the competing D-forms are informal and often colloquial. This matches the factor of prestige, cf. Section 3.3.3. Let us consider the word *dyrektor* 'school principal' as an example. Since many principals in Polish schools are women, it is naturally used as a U-form, typically with the honorific: *pani dyrektor*. This form is expected and used in contexts like teacher-student or teacher-parent communication, and esp. when addressing the principals. However, in less formal contexts, e.g. between students or between fellow teachers, the D-form *dyrektorka* is used as well. This pattern is found with many U-forms. It is generally easier for the U-forms to appear in less formal contexts, cf. (17) taken from an online forum, than for the D-forms to enter formal register.

As the official U-forms "trickle down" into less formal registers and contexts, this leads to partial semantic bleaching of the honorific *pani*. It may be retained in contexts where little or no marking of respect is needed or intended. The appositional structure with the word is simply the most certain way to assure clarity of the feminine reference. As a result, this polite word becomes an auxiliary noun, an analytic marker of the feminine gender and carrier of the inflectional marking, cf. the discussion of the data from Table 4.

(32) mentions a school principal, first giving the masculine noun (strictly speaking an M-form) and then correcting that the principal is actually a woman (with a U-form in apposition). The word *pani* is used to mark the gender rather than show respect. A plain U-form would feel inadequate, in line with the belief of the speakers that U-forms are "masculine", cf. Section 3.1.

(32) Mamy spoko dyrektor-a tzn. pani-ą dyrektor.
have.1PL ok principal-ACC.SG i.e. Mrs-ACC.SG principal[ACC.SG]
'We have a good principal, that is Ms principal.' (NKJP)

The same effect can be observed in metalinguistic comments. The title of the article by Szpyra-Kozłowska (2019) quoted above presents various options for labelling a female prime minister: a D-form created by conversion (*premier-a*), a

D-form created by suffixation (*premier-k-a*) and a U-form. The last one is tellingly presented in apposition with the honorific.

Similarly, (12) above is a newspaper headline. The use of the D-form *ministra* was criticised in the comments on the newspaper's website. One of them provided the "nicer" alternative: *pani minister*. The actual headline with the D-form does not need a honorific and neither would a similar headline about a man.

3.3.5 Ideology

In recent years the competition of U-forms and D-forms has gained ideological overtones in the public discourse in Poland. Stępień (2018, after Szpyra-Kozłowska 2021: 158) shows that D-forms are more common in liberal and left-wing media than in right-wing ones. Particularly well publicised instances of use of individual D-forms spark heated debates, cf. Section 4.4. The divide can be seen in the Sejm as well. On several occasions female MPs requested their plaques and official items like voting cards bear the D-form label of their position: *posłanka* rather than U-form/M-form *poseł*.[27] Only left-wing MPs made such requests.

Word type choices can lead to arguments. Let us recall one such event from the Sejm. It is customary for MPs to start their speeches by addressing the person presiding over the session. In 2019, the MP Krzysztof Mieszkowski started his speech with the phrase *Pani Marszałkini*, thus using a D-form of *marszałek* 'speaker of the parliament'. The presiding deputy speaker Małgorzata Gosiewska immediately protested with passion and requested that a U-form be used when she is addressed.[28] The said MP came from the then liberal opposition and the deputy speaker is a member of the then ruling conservative party.

4 Historical development of U-forms

4.1 Background

In earlier stages of Polish a novel usage of a concept with reference to a woman rather than a man resulted in feminative derivation. Thus, D-forms dominated, both among common nouns and surnames. Interesting proofs of the strength of the pattern are the nouns *papieżyca* 'popess' (limited range of application),

[27] https://oko.press/kancelaria-sejmu-nie-zmienila-tabliczek-poslanki-sa-poslami Accessed 27.06.2024.

[28] https://wiadomosci.onet.pl/kraj/spiecie-w-sejmie-marszalek-gosiewska-ja-sobie-nie-zycze/thz3bdx Accessed 23.06.2024.

szpiegini 'spy, F' or *gościni* 'guest, F' (both later supplanted by M-forms, cf. Section 3.3.3). However, M-forms (U-forms?)[29] were used too, not necessarily very seldom. Apparently, the oldest attested M-form in a text written in Polish comes from Mikołaj Rej, a 16th century writer (Nitsch 1951: 65).

Below we sketch the history of the development of U-forms and their competition with other types, but let us recall that changes affected only a relatively small part of the lexicon. Numerous common nouns have been D-forms for centuries, e.g. most of those in (27). Women's surnames in *-sk(a)* have also been stable as D-forms. Such words are hardly mentioned in the following sections. Yet, the reader should bear in mind that they were present in the system and co-existed with the changing forms we focus on.

4.2 Surnames

As for surnames, D-forms were originally the norm and involved a differentiation between married and unmarried women, cf. Table 3. Women were identified by the relationship to their fathers or husbands and the possessive flavour to these forms may be felt by some speakers even today. The masculine variant was considered basic, a surname being a reflection of patrilineal ancestry. Hence, occasional mentions of women with this "true" form can also be found. These were probably U-forms from the very beginning.

Kucała (1978: 63) lists examples of female surnames from a 17th century diary, most of them are D-forms, but there are several U-forms as well. The practice of simplifying women's surnames into U-forms gradually gained ground and allegedly was fashionable in the 18th century Warsaw (Nitsch 1951: 63). By the mid 19th century it became the subject of heated criticism by language purists, which is a proof of its spread. The criticism focused mostly on young women, esp. adolescent schoolgirls, who allegedly dared to use U-forms among themselves and with their female teachers. However, very similar complaints are attested from 1850s until 1900s (Pawłowski 1951: 50), which shows that U-forms and D-forms of surnames co-existed for decades in actual usage, most likely with register differences.

Another source of support for U-forms was foreign administration of partitioned Poland, which operated in German or Russian and did not use traditional Polish D-forms for surnames. The Austrian partition adopted Polish as the official language in the 1860s, but the use in documents in the following decades

[29] In the case of historical, and esp. metalinguistic evidence, it is often difficult to distinguish these two.

varied (Kowalska 2013). The region seems to have clung to D-form surnames more than the other two partitions (Nitsch 1951: 63; Pawłowski 1951: 56).

After the First World War, the newly independent Poland introduced mandatory use of traditional, "patriotic" D-form versions of female surnames in documents (Dacewicz 2021). Actual practice varied and D-forms proved difficult in use. With some surnames they involved sound alternations, which could lead to mistakes in recreating the male versions from the D-forms. Consequently, in 1930 the Ministry for Education relaxed the regulations – in school documentation D-forms remained mandatory for surnames in *-wicz* (Pawłowski 1951: 54), otherwise U-forms were allowed. Let us also recall that with most surnames, D-forms revealed the marital status, which was certainly felt as a flaw of this structure by many women.

After the Second World War, U-forms were once again introduced as the default versions of women's surnames in administration and official documents. Usage varied and the new standard was more eagerly used by younger people. In the early 1950s D-forms did not seem under threat in informal use (Nitsch 1951: 64), while actively used U-forms were typical of schools (Pawłowski 1951: 49). Pawłowski presents surveys conducted in secondary schools and among bank clerks. The data suggest differences in use based on gender and age. Young women (pupils, bank clerks, also teachers) generally preferred U-forms, while male pupils opted for traditional D-forms (Pawłowski 1951: 54–55). Interestingly, Pawłowski (1951: 49) suggests that women's surnames could have the form identical to male versions in the nominative (i.e. in the documents), but inflect as D-forms otherwise. It is unclear whether such hybrid usage was common at the time. It may well be just a compromise proposal by the scholar disapproving of uninflectedness, but aware of its spread.

With time, U-forms gradually dominated. Kucała (1978) recounts several surveys of surnames in the press, which clearly show that D-forms were used less and less. In the mid 1970s, traditional D-forms constituted 24% of women's surnames in press obituaries, 6% in a women's magazine and were absent from a youth magazine (Kucała 1978: 68) reflecting the steady decline across generations. The prevalence of D-forms in speech was probably higher, but the pattern was clearly losing ground. This tendency has never been reversed[30] and today D-forms for surnames are mostly spoken and informal. Yet, even within this limited range, the suffix *-ow(a)* remains productive, cf. the informal *Clintonowa*

[30]We should note a trend to use D-forms of surnames among women in artistic (maiden name versions) and scholarly (married versions) circles. The names of some authors cited here: Grzegorczykowa, Puzynina and Waszakowa represent the latter group. The fashion is now gone – the youngest followers of it were born in the 1960s.

'Mrs. Clinton' or *Merkelowa* 'Mrs. Merkel'. When using such forms many speakers today apply the suffix *-ow(a)* regardless of the marital status of the women involved.[31]

4.3 Common nouns

The "original" situation presented in Section 4.1 seems to have been the case for common nouns for a long time. D-forms dominated, appearing in the vocabulary when needed, but there were instances of M-forms, esp. in the copular construction (the earliest known M-form in Rej's writings, cf. Section 4.1, appears in such a context). Most likely there was some form of "division of labour" between the types (cf. Section 3.3.3) and a minority of personal nouns were typically M-forms.

The late 19$^{\text{th}}$ century saw an unprecented rise of women professionals as various occupations, positions and study programmes at universities were gradually opened to them. These social changes continued in the 20$^{\text{th}}$ century. The language needed labels for women doctors, lawyers, officials, etc. and these were often called by means of U-forms/M-forms, which spurred criticism by language purists roughly since the 1880s (Język Polski 1933: 28). The commentaries were emotional and used harsh language, e.g. accusing women with professional titles of cowardly and fraudulently hiding their femininity (Poradnik Językowy 1911: 118), cf. also fn. 32.

Such metalinguistic criticisms and later linguistic papers (notably Klemensiewicz 1957) give the impression that the progressive female professionals rejected D-forms, while conservative purists (mostly men) advocated this traditional structure. As the use of U-forms/M-forms was more and more common over decades, the protests allegedly became weaker as these structures gained some level of reluctant acceptance still in the interwar period (Klemensiewicz 1957: 112–113).

This narrative seems to present a partial picture only. First of all, it is not a given that the criticisms by individual language purists and conservatives, however vocal, are in any way representative of broader social perception of the structures, let alone the usage at the time. We should also note that some language authorities were tolerant towards M-forms (Język Polski 1931a: 30–31). We have surviving metalinguistic evidence from the critics of M-forms/U-forms, but very little from the other side. Woźniak (2014) cites copiously from feminist press of the interwar period, showing that D-forms and M-forms (she quotes no U-forms) were both used. In fact, it was normal to see a mix of the types in a single text.

[31] This practice is parallel to the standard usage of surnames in Czech, where most surnames have a single D-form in *-ová* (or *-ská*).

Individual women may have had their preferences, but the women's movement as a whole did not advocate any of the word types. Klemensiewicz (1957: 103) recounts a 1929 census of teachers in which D-forms and U-forms/M-forms for various positions in the educational system appear in comparable numbers. This suggests that differences in register between the types were relatively slight. We know of no official policy of the Polish state with reference to labelling women professionals.

An alternative scenario is that the early decades of the 20th century were a time of a constant rise in the number of women to whom the new words could realistically apply. It is then likely that this meant that both D-forms and U-forms/M-forms were used more and more often, but only the latter growth was noticed and discussed at length. Having been in use for several decades, the new formations naturally gained some acceptance. Furthermore, even though D-forms had been well established as a pattern, each new word of the type needed to be accepted by the speakers, esp. because of the variety of feminative suffixes (Woźniak 2014: 298–300; Obrębska-Jabłońska 1949: 1).

The concept that was at the forefront of the changes in the language and the society was the figure of a woman doctor. Female medics were an increasingly common sight, had contact with many people and the position had the prestige earlier reserved for men. Apparently, most of them preferred the U-form/M-form *doktor*, the argument being that this is the right translation of the Latin title they had been given (Poradnik Językowy 1911: 118). The purist critics disapproved of the plain U-form,[32] and even of the apposition with *pani*, today standard, well into the 1930s (Język Polski 1931b: 159).[33] Even though the vocal purists advocated the D-form alternative *doktorka*, which was in use, not all people concerned with language correctness agreed, thinking the form is either "depreciative" (Poradnik Językowy 1906: 55) or "not strictly grammatical" (Poradnik Językowy 1925a: 104).

We may conclude that U-forms/M-forms were becoming more common in the interwar period, but they did not gain acceptance of refined and traditional speakers. Alternative D-forms were not accepted straightforwardly either. The data

[32] The editors of Poradnik Językowy (1925b: 15) described a sign advertising a medical practice of a dermatologist *dr I Krzywy*. The acronym stands for the word *doktor* and the surname is not a D-form so it was perfectly possible to conclude that the said medic is a man, which was not the case. The comment criticises the sign in a dramatic tone: what a misunderstanding it would be if a male patient decided to consult Dr. Krzywy!

[33] Recounting his own stay in hospital, Benni (1933: 184) suggested a hybrid formation, the feminine honorific and an M-form: *z pani-ą doktor-em* 'with [Ms.] doctor'. Klemensiewicz (1957: 37) and Nitsch (1951: 65) present this structure as one in use in the interwar period and even later (cf. also Kupiszewski 1967: 374), but Benni's own mention of the surprise of the nurses and a response from Obrębska (1933) suggest it was not particularly frequent.

from Klemensiewicz (1957) and Woźniak (2014) quoted above suggest the register variation between the types was not very strong at the time. Yet, the example of female doctors suggests the modern opposition of formal/polite U-form with the honorific vs. the informal D-form, as described in Section 3.3.4, was forming already.

The communist rule installed in Poland in 1945 brought important changes to society, including massive employment of women. Consequently, there were even more real-life situations when they could be referred to by their professions or addressed by professional titles. The field of competition between D-forms and U-forms/M-forms grew immensely. The official ideological position that men and women are equal translated into the use of exactly the same labels for both sexes in official language. In other words, M-forms/U-forms were promoted against "backward" D-forms. Plain U-forms started to appear in written texts other than official records (Obrębska-Jabłońska 1949: 3–4) and gained momentum. The move away from D-forms might have been modelled on Russian (Woźniak 2014: 305), but some see the development in Polish as independent (Łaziński 2006: 250). The use of different formations for common nouns referring to women varied in the early years after the war and so did the views of the speakers. We have surviving mentions of disagreements between speakers (e.g. Doroszewski 1951) and criticisms of uninflectedness.

Varying opinions were expressed by linguists of the time. Pawłowski (1951: 60, 62) and Nitsch (1951: 65) accept the lack of inflection only in the appositional phrase *pani doktor*, condemning similar use of other nouns and plain U-forms. Yet, we must conclude that such usage (today standard) was clearly present in the language if it was criticised. Taking an opposite "progressive" stance, Klemensiewicz (1957) announces U-forms/M-forms to be finally victorious in their long competition against D-forms among words for (prestigious) professions and professional titles. As we shall see below, his conclusions were definitely too hasty. The supposed triumph was also limited, as a considerable portion of the feminine lexicon had always been served by D-forms, cf. Section 3.3.3, and this remained so. A decade later, Kupiszewski (1967) also sees the use of M-forms/U-forms as a strong and unstoppable tendency. Interestingly, his paper is inspired by a letter from a radio listener who protested againt the trend and insisted on using the "correct" D-forms.

Kucała's (1978: 51–61) survey of noun usage in press obituaries and announcements provides numerous examples of both D-forms and M-forms (more often than U-forms). Kucała (1978: 61) notes also the use of plain U-forms, but considers them to be colloquial. By the 1970s the use of U-forms was accepted by prescriptivist works – at least for those words where D-forms had not become established (Buttler et al. 1973: 111), although care was advised to avoid plain U-forms (Buttler et al. 1973: 116). Later prescriptivist works are still slightly more approving of U-forms (Jadacka 2006: 128–129).

4.4 Recent developments

After the overthrow of communism in 1989, the incipient feminist movement started advocating for D-forms instead of U-forms/M-forms among common nouns. For some time such usage was limited and perceived to a large extent as an ideological declaration (Łaziński 2006: 278). Gradually the "new" D-forms started to appear in public discourse and raise controversy among native speakers, by now used to the presence of U-forms. In recent years there were several occasions when the use of specific D-forms in public statements drew great attention and led to heated debates among native speakers. The particularly controversial words were e.g. *ministra* (vs. M *minister*), *powstanka* (vs. M *powstaniec* 'insurgent'), *gościni* (vs. M *gość* 'guest') and *posłanka* (vs. M *poseł* 'MP', cf. Section 3.3.5).

We will focus on the controversy with the first word. In the early 2012, the then Minister for Sport, Joanna Mucha, declared in a television interview that she prefers to be addressed *Pani Ministro*, i.e. with a vocative of the D-form *ministra*, as in (12). This provoked a flood of criticism, even from fellow ministers, but there were also voices of approval. Within days after the interview, a well-known linguist talked of Ms. Mucha "raping the language".[34] There are also at least two linguistic articles with Ms. Mucha's name in the title (Bobrowski 2012, Waszakowa 2014), much more moderate in tone. Most critics disapproved of any D-form, others would have accepted one created by suffixation: *ministerka*, cf. the analysis of the title of Szpyra-Kozłowska (2019) in Section 3.3.4. The Council for the Polish Language, an authoritative body on matters of language correctness, felt compelled to issue a statement. Their position (RJP 2012) was moderate, acknowledging the possibility of conversions like *ministra* but voicing scepticism about their chances of being generally accepted.[35] They also mentioned numerous messages they had received and asserted that most women were sympathetic to Ms. Mucha's D-form. One of the largest Polish dailies, *Gazeta Wyborcza*, uses

[34] https://www.wprost.pl/kraj/308115/ministro-mucho-nie-mozna-gwalcic-jezyka.html Accessed 28.06.2024

[35] Interestingly, the verdict was perceived as a victory for the D-form, since it was not fully condemned, cf. a title of a press report stating simply that "*ministra* is correct" https://www.newsweek.pl/polska/ministra-w-jezyku-polskim-jest-poprawna-rada-jezyka-polskiego/9b9q25g Accessed 28.06.2024.

the word *ministra*, cf. (12). It seems this practice started after Ms. Mucha's interview.

Despite fervent opposition from purists and conservatives,[36] D-forms keep spreading in the public sphere. Occasionally, D-forms appear in official contexts as well, cf. the title of the dean of one of the faculties at Adam Mickiewicz University in Poznań: *dziekana* (vs. standard M *dziekan*),[37] or the title of the deputy mayor of Sopot: *wiceprezydentka* (vs. M *wiceprezydent*).[38]

As can be seen, the pace of change is quick and the more often D-forms are used, the more acceptance they eventually gain. Paulina Mikuła, a youtuber specialising in language and language correctness, made two videos concerning D-forms. In the later one (made in 2019), she is more approving and admits that she has changed her attitude, she "simply got used to" D-forms.[39] The earlier video was created only 5 years prior.[40]

We must note that the current debate about D-forms completely ignores surnames, here the use of U-forms is not contested and no change in usage is detectable. Interestingly, both the dean from Poznań and the deputy mayor from Sopot mentioned above have surnames that are U-forms according to the current norm, *Pawelczyk* and *Jachim* respectively, and it is these forms that appear officially next to the innovative D-forms for their positions.

4.5 U-forms: the history of a category

U-forms appeared in Polish as an innovative pattern in those cases where the general tendency to create D-forms for feminine referents was overcome by the need to preserve the "true" form of the words in question, i.e. that identical to the masculine version. The resultant conflict of morphonology and semantics made most such formations uninflected, but not all: M-forms are attested today and were very likely more common than U-forms at early stages of the diversion from D-forms. As Sections 4.2 and 4.3 show, the development happened twice, first for some types of surnames (in the period from the 18th century until the second half of the 20th century) and then for common nouns denoting prestigious

[36] Ironically, at least until the 1960s a conservative approach was to use and support D-forms rather than U-forms, cf. Section 4.3.

[37] https://anglistyka.amu.edu.pl/strona-glowna/wladze-wydzialu Accessed 28.06.2024.

[38] https://trojmiasto.wyborcza.pl/trojmiasto/7,35612,28938409,magdalena-czarzynska-jachim-zastepczyni-prezydenta-sopotu.html Accessed 28.06.2024.

[39] https://www.youtube.com/watch?v=TjJZ8THmtXQ Accessed 28.06.2024.

[40] https://www.youtube.com/watch?v=VJZm2vAh_Oc Accessed 28.06.2024.

professions, positions and professional titles (in the period from the late 19th century until the second half of the 20th century). In the latter case an intermediate stage of M-forms is detectable in the first half of the 20th century.

The category of U-forms expanded in three ways: among speakers, in the lexicon and across linguistic contexts. The gradual spread in the population is presented in Sections 4.2 and 4.3. As for the lexical range of U-forms, it has always been limited, D-forms being the majority among lexical concepts with possible feminine reference. However, U-forms successfully dominated their niches in the lexicon and were eventually accepted by the speakers for these types of words (this happened roughly in the 1960s and 1970s), to the point that the newly (re)created D-forms are rejected by more conservative speakers at present. The acceptance for U-forms is still weak beyond their typical domains, cf. Szpyra-Kozłowska's (2021: 336–341) criticism of (30–31). Yet, attestations of such spread in the lexicon are a proof of the strength and continuing productivity of the pattern. As for the syntactic contexts, the appositional construction has always been a stronghold of U-forms. It is very likely that it was their original context (with given names for surnames and the honorific *pani* for common nouns). The construction remains a typical environment for U-forms, cf. Table 4.

Today U-forms are used freely in the singular (except for the copular construction), but are still limited in the plural. It is likely that the hierarchy of contexts for the use in the plural shown in Table 5 is also a historical development path of U-forms in general: from the "quoted" basic form in apposition to the fully independent use where the lack of inflection may be the sole marker of the feminine gender.[41] This perspective allows us to look again at the status of U-forms in the lexicon. We may reconcile Łaziński's (2006: 227) argumentation against U-forms as separate lexemes and the counterexamples, cf. Section 3.3.2. The latter represent a more mature stage of the development which has not been reached (fully) by too many speakers yet. In the wake of the recent developments and the rebound of D-forms, U-forms may well retreat and never fully mature.

It is rather sure that U-form surnames and common nouns are facing different fates in the nearest future. The former are not threatened in any way while the latter might lose territory to D-forms. This is due to different attitudes and choices of the speakers regarding the two groups.

As for common nouns, we cannot tell whether the current increased use of D-forms will eventually overcome the practice of using U-forms or prove a tem-

[41] In the early 1950s, the use of the U-form *doktor* was accepted by conservative linguists (Pawłowski 1951: 65) in apposition with the honorific, the first context in Łaziński's hierarchy, but the use of a U-form with a feminine possessive, the next one, was novel and contestable (Doroszewski 1951: 32).

porary fashion. Most likely the coexistence of the types will remain for long, but it might be demarcated differently. To oust U-forms from their current contexts of use, D-forms would need to lose their ideological flavour so that they feel neutral (enough) for the majority of speakers.

5 Conclusions

The paper has presented how a productive uninflected pattern functions in Polish and how it arose in spite of the omnipresent inflection. Current usage shows symptoms of the difficulty posed by uninflected forms and repair mechanisms, chiefly apposition. The case of U-forms provides an interesting insight about the concept of systemic pressure. U-forms are exceptional in Polish grammar, but they were able to emerge as a class, spread and establish their position. The pressure of the system appears to have lain dormant for several decades as they became more and more common. The current (partial) retreat of U-forms is linked first of all to the awareness of speakers and their conscious choices.

A notable side effect of the existence of U-forms is the new function of the feminine honorific *pani*, which is regularly used in contexts where marking respect is not needed or intended. The word is then an auxiliary noun, carrying the inflection lacking on the U-form and marking the gender explicitly.

Our analysis also shows that Polish is rather tolerant to the coexistence of different patterns and subclasses of words. The situation established by the late 20^{th} century involves standard use of U-forms for some surnames and some common words, while surnames in *-ska* and most common words referring to women remain D-forms. This kind of tolerance was also a necessary part of the emergence of U-forms. The above is linked to another obsevation: Polish easily allows productivity of minor inflectional patterns, which we have noted several times in Section 2. U-forms are only one example of this.

Another feature of Polish that surfaces in the analysis is the notion of the "basic form" of a noun. This status of the nominative singular is more than a convention applied by linguists and lexicographers; it seems to match the perception the speakers have. The desire to keep this "true" form of a word across all contexts (which may be triggered by a number of factors, cf. Section 2.2) is instrumental for uninflectedness, including the case of U-forms.

We may note the role of individual words in paving the way for whole categories in historical changes. It seems that the first successful and fully accepted U-form was the word *doktor*, cf. Section 4.3. Similarly, publicised controversies concerning few words have played a significant role in bringing D-forms back in recent years, cf. Section 4.4. We can also assume that in the final decades of the 19th century, the presence of U-form surnames in the language helped the rise of uninflected common words referring to women.

Abbreviations

Less typical abbreviations used in the paper are:

IMPERS	impersonal	NVIR	non-virile
INAN	inanimate	PREP	preposition
NNOM	non-nominative	VIR	virile (masculine personal)

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Chapter 12

The uninflecting word class *rentaishi* in Modern Japanese

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In this paper, I focus on the Japanese word class *rentaishi* ('noun-modifying words'), which is often overlooked in Western linguistics but has typological significance in the debate on uninflectedness. While Japanese verbs and adjectives exhibit rich inflectional morphology for categories like tense, aspect, and modality, *rentaishi* are restricted to attributive position, and lack inflection completely. Furthermore, they display morphological diversity, contrary to the major Japanese word classes (e.g., *aru* 'a certain', *rei-no* 'mere'). Historically, the word class *rentaishi* emerged in the early 20th century when Japanese linguists recognized that Japanese adjectives inflect for tense and can appear predicatively independently, unlike adjectives in Western languages that require a copula, both to express tense and to appear predicatively, and must agree with the head noun in case, number and gender in attributive position. To account for non-predicative, attributive modifiers that do not inflect, Japanese linguists introduced the word class *rentaishi*, grouping syntactically restricted and morphologically deficient lexemes. As a future consideration, I explore whether *rentaishi* should remain a distinct word class in Japanese or if relevant lexemes should be reclassified as defective members of other word classes or even as part of the adjective group they were originally separated from, presenting arguments for both approaches.

1 Introduction

In this paper, I discuss the not universally accepted word class of so-called *rentaishi* in Modern Japanese. *Rentaishi* are a group of lexemes that are syntactically restricted to their use as attributive modifiers. From a morphological perspective,

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this syntactic restriction translates to absence of inflection. Different instances of lexemes that have been called *rentaishi* are given in (1) below, as part of a DP and highlighted in boldface (cf. Sekine 1937, Kai 1980, Kim 2006, Matsubara 2009).

(1) a. **aru** gakusei
certain student
'a certain student'

b. **taishita** sa
considerable difference
'a considerable difference'

c. **shutaru** gen'in
main reason
'the main reason'

d. **ōkina** ie
big house
'big house'

As visible above, every relevant lexeme serves as a modifier of a noun, and that is the only syntactic environment such lexemes appear in. Importantly, they all exhibit different morphological endings. This morphological heterogeneity differentiates them from the major inflecting word classes in Japanese, namely verbs, which end with the vowel *-u*, and adjectives, which either consistently end with the element *-i* (*i*-adjectives) or with the element *-na* in attributive and *-da* in predicative position (*na*-adjectives). This is due to the fact that many lexemes that have been classified as *rentaishi* are derived from other word classes and have become frozen or fossilized.

In this paper, *rentaishi* will be discussed from a perspective of inflection, or inflectability. Concretely, I will show that, historically, the emergence of the word class *rentaishi* is tied to the criterion of uninflectedness and, in fact, a direct result of the greater awareness exhibited by Japanese linguists at the beginning of the 20th century concerning the differences between Japanese and the languages of the West, especially English and Dutch. The need to fill the slot of dependent attributive modifiers in Japanese as equivalents to the Western adjectives, which they perceived as defective contrary to the independent and inflecting Japanese adjectives (*keiyōshi*), lead to the creation of the word class *rentaishi*. This creation marks a linguistic turn at the beginning of the 20th century.

However, there is more to be discussed about the role of these modifiers in the Japanese word class system from a modern perspective and in a cross-linguistic context in order to evaluate whether they deserve a status as an independent word class. For example, Muraki (1996 et seq.) has put forth a new analysis of *rentaishi* as non-predicative adjectives, which marks relevant lexemes as *uninflectable* rather than *uninflecting* in the terms of Spencer (2020). Towards the end of this paper, I will explore a possible re-analysis of relevant lexemes as uninflectable and defective, rather than uninflecting.

This paper is structured as follows. First, the relevant terminology will be discussed in Section 2. In Section 3, the notion of *inflection* will be discussed from the perspective of Japanese linguistics, and the category of *rentaishi* will be introduced. The main analysis is carried out in Section 4 with special reference to the historical dimension of this group and its relevance for the debate on *uninflectedness*. Section 5 then discusses this group from a modern perspective and Section 6 concludes the paper.[1]

2 Terminology

Uninflectedness refers to the inability of a given lexeme or a group of lexemes to inflect. A prerequisite for adopting this term is that the language under investigation, in principle, allows for inflection and that inflection has a syntactic dimension. Baerman et al. (2005: 33) define uninflectedness as follows:

(2) (i) There is, in certain lexemes only, a loss of all values of a particular feature F found elsewhere in the language. This loss may depend on the presence of a particular combination of values of one or more other features (the context).

(ii) Other syntactic objects distinguish values of feature F, either generally or in the given context, and feature F is therefore syntactically relevant.

What this quote emphasizes is that uninflectedness is a morphological phenomenon, the syntax is autonomous, as will become clearer below.

Furthermore, the notion *uninflectedness*, in the first place, denotes the general inability of a given lexeme to inflect, or a general lack of inflection, but in fact,

[1]For all transliteration, I adopt the Hepburn-Romanization System for Japanese, even when citing from other sources. Japanese names are given with the last name first written in capital letters. I am very grateful to Ken Hiraiwa for judgments on the Modern Japanese data as well as to two anonymous editors for very useful comments and especially for pointing my attention towards the recently published PhD thesis of Nespoli (2023). All remaining errors are, of course, my own.

it is multidimensional. A first important distinction can be drawn between *uninflecting* and *uninflectable* (groups of) lexemes. In this paper, I will follow Spencer (2020: 142) and Baerman et al. (2005, 2017) and with the terminology *uninflecting* refer to parts of speech and elements belonging to these parts of speech that never, under no circumstances, inflect in a given language. Depending on the language, those can be lexical or functional parts of speech. A typical example of an uninflecting word class are conjunctions as those very rarely inflect across the world's languages. Examples are Russian *li* 'or', German *und* 'and' or the equivalents in English. On the other hand, the lexical class of nouns shows inflection in some languages, in this case makes up an *inflecting* word class, but not in others, where nouns then are uninflecting.

On the contrary, the notion *uninflectable* refers to specific elements that belong to an otherwise inflecting, lexical or functional, part of speech in a given language, but which for some reason or other do not inflect, that is do not show the relevant endings and inflectional paradigms of other elements belonging to the same part of speech in the relevant syntactic environments. Crucially, uninflectable lexemes belong to an inflecting word class, hence are expected to inflect (Spencer 2020: 142). An important prerequisite for analyzing a lexeme as *uninflectable* is therefore that it appears in all the relevant syntactic environments other lexemes from the same part of speech appear in (Spencer 2020: 142–144).

An often-quoted example of an uninflectable lexeme is the Russian noun *pol'to* 'coat'. Nouns in Russian inflect for case and number, therefore compose an inflecting word class. However, the noun *pol'to* simply does not show any specific inflectional ending(s) for those categories, all forms being syncretic. In other words, this noun displays the same inflectional form across all cells in its inflectional paradigm (Baerman et al. 2005: 30). Nevertheless, importantly, *pol'to* can appear in all relevant syntactic environments in which nouns in Russian typically appear, for example as the subject or object of a sentence. Its uninflectable nature becomes visible when a controller appears, such as the preposition *v* 'in'. As visible in (3b) below, when this preposition takes a DP containing a noun and an adjective as its complement, the adjective shows obligatory concord within the PP and is inflected for prepositional case. However, the noun *pol'to* is unaltered and appears in the same morphological form as in nominative contexts, exemplified in (3a).

(3) a. bel-oe pol'to
white-NOM.SG.N coat.Ø
'white coat'

b. v bel-om pol'to
in white-PREP.SG.N coat.Ø
'in a white coat'

Lexemes can also be *partially uninflectable*. This means they belong to an inflecting part of speech, but show a restricted inventory of inflectional forms across their paradigm. In other words, they show inflection in some contexts, but not in others. Staying in the nominal domain, *context* refers to case, gender and number. For example, the Polish noun *muzeum* 'museum' only shows distinct inflectional forms for case in its plural cells, but in singular all entries are syncretic (Baerman et al. 2005: 30; Spencer 2020: 143).

Finally, *contextual uninflectability* means that certain lexical items do not inflect in specific syntactic environments, but otherwise do (Spencer 2020: 149–150). This is for example the case for German (and Dutch) adjectives, which do inflect, but never in predicative position as shown below (cf. Roehrs 2008, Zeijlstra 2020 among others).

(4) a. das rot-e Buch
the red-NOM.SG.N book.NOM.SG.N
'the red book'

b. Das Buch ist rot.
the book.NOM.SG.N is red.Ø
'The book is red.'

Importantly, however, all the lexemes depicted above do appear in the relevant syntactic environments expected from their part of speech. If a lexeme fails to appear in the syntactic environments it is supposed to appear in in the first place, it can be called syntactically *defective* (Spencer 2020: 143). In this sense, this notion can refer to adjectives which cannot appear predicatively for syntactic reasons. This is the case for so-called *non-intersective* or *intensional* adjectives such as *former* or *alleged* and their cross-linguistic counterparts (Bolinger 1967, Kamp & Partee 1995, Larson 1998), or so-called relational adjectives (Levi 1978, Fábregas 2007, Bortolotto 2016 a.o.). For instance, although adjectives in Russian, different to German, inflect in attributive as well as predicative position, the group of relational adjectives, such as *knižnyj* 'pertaining to books' derived from *kniga* 'book', is simply illicit in predicative position as shown in (5) (Mezhevich 2002, Bailyn 2002).

(5) a. kniž-n-yj magazin
book-RA-NOM.SG.M store.NOM.SG.M
'bookstore' (lit. 'store pertaining to books')

b. * Magazin byl kniž-n-yj.
store.NOM.SG.M was book-RA-NOM.SG.M
Intend.: 'The store was booky/bookish.' (Mezhevich 2002: 100)

The distinction between morphologically and syntactically restricted modifiers will become relevant for our discussion of *rentaishi* and Japanese adjectives in Section 5 of this paper.

3 Japanese

This section gives an overview over what inflection denotes from a Japanese perspective, illustrating with different parts of speech. After this, the category of *rentaishi* will be outlined with reference to its relevance to the present discussion.

3.1 Overview

Japanese is a strictly head-final SOV language in which the verb cannot be moved from sentence-final position (Kuno 1973, Martin 1975, Iwasaki 2013 a.o.). Another key feature is its agglutinative character which means that grammatical information is encoded via affixes, predominantly suffixes. Concerning word classes, since the Edo-period (1603–1868), a classical divide has been drawn between *inflecting* and *uninflecting*, as well as *dependent* and *independent* word classes. *Inflecting* word classes can be further differentiated by their distinct morphology.

3.2 Inflection from a Japanese perspective

While established linguistic notions such as *inflection* are often based on European languages – usually Latin or Ancient Greek – the preceding outline has shown that *inflection* has been highly relevant for differentiating word classes in Japanese. In fact, a similar notion was established by linguists in Japan during the Edo-period, namely the term *katsuyō* 活用. The linguistic notion of *katsuyō* presumably goes back to MOTOORI Norinaga (1730–1801), one of the most important national linguists (*kokugakusha*) of the Edo-period, and was included as a criterion to categorize Japanese parts of speech in the *school grammar* by HASHIMOTO Shinkichi (1882–1945) (Hashimoto 1948 [1934]), so-called because it is to date the grammar taught in Japanese school curricula.

3.2.1 Inflection of verbs

It was the goal of the *kokugakusha* to analyze the Japanese language independently of other languages. And while *katsuyō* is in English often translated as *inflection*, it does not mean exactly the same thing. But then what is *katsuyō*? As mentioned above, an important difference is drawn between inflecting and uninflecting parts of speech in Japanese. The Japanese terms are *katsuyōgo*, meaning *katsuyō*-words or *inflectional words*, and *hi-katsuyō-go* (also *mu-katsuyō-go*), meaning non-*katsuyō*-words or *non-inflectional words*. *Katsuyōgo* are verbs (*dōshi* 動詞), adjectives (*keiyōshi* 形容詞) and auxiliary verbs (traditionally called verbal suffixes, *jodōshi* 助動詞). On the other hand, non-inflectional words are nouns (*meishi* 名詞) and adverbials (*fukushi* 副詞), as well as most functional word classes, most importantly particles (*joshi* 助詞).

Importantly, the term *katsuyō* differentiates verbal from nominal categories. And while *katusyō* is the key characteristics of verbal categories, nouns are seen as categorically uninflecting. This suggests that the term *katsuyō* in fact denotes *conjugation*, rather than *inflection*, and we can ask ourselves whether this might not be a more fitting translation.[2]

A second important aspect concerns the exact dimension of *inflection/katsuyō*. From an Indo-European perspective, we expect inflection to denote different morphological forms dependent on number and person as well as various TAM (temporal-aspectual-modal) categories. For example, in German, a verb such as *sprechen* 'to speak' shows different forms for person and number (*ich spreche* 'I speak.1SG', *du sprichst* 'you speak.2SG', *wir sprechen* 'we speak.1PL'), tense (*du sprachst* 'you spoke.2SG') and mood (*du sprächest* 'you would speak.2SG.SUBJ'). However, Japanese lacks the categories number and person and by extension different inflectional forms for those.

Arguably, tense, aspect and modality are not expressed on the verb itself, in the first place, but via auxiliary verbs (same as, for example, German expresses deontic modality with the auxiliary verb *müssen* 'must').[3]

Rather than morphological alternation, *katsuyō* denotes the conjugation of verbal categories in different inflectional forms, or *katsuyōkei* (活用形). For Modern Japanese, different authors have put forth different inventories of inflectional

[2]This is in fact the translation adopted in Martin (1975) for *katsuyō*. On the other hand, most generative linguists use the term *inflection* which is arguably the more established term. It is also found in most grammars of Japanese (see the discussion in Uehara 1998: 58–59). This discussion is reflected in Uehara (1998), who uses the term '*katsuyō*-inflection' as a compromise.

[3]With respect to tense, this was definitely the case in Classical Japanese, whereas in Modern Japanese the expression of past tense on verbs could be seen as almost fusional as it involves specific phonological processes.

forms, but a standard paradigm, that was established by linguists in the Edo-period on the basis of Classical Japanese, and is nowadays still used with regard to Classical, but also Modern Japanese, contains the following six inflectional forms (see for example Shibatani 1990: 222).[4]

(6)	I	未然形	*mizenkei*	(negative form/irrealis)
	II	連用形	*renyōkei*	(nominalization/adverbial form)
	III	終止形	*shūshikei*	(sentence-final form, also conclusive)
	IV	連体形	*rentaikei*	(attributive form)
	V	已然形	*izenkei*	(concessive/conditional form)
	VI	命令形	*meireikei*	(imperative form)

Some of these inflectional forms express specific functions independently, but more importantly, specific functional and lexical elements are suffixed to specific inflectional forms. For example, the negative form (*mizenkei*) does not express negation by itself.[5] Rather, it is the inflectional form an inflecting word class needs to inflect to in order to host those suffixes which negate the proposition expressed by the relevant member of this word class.

Verbs in Modern Japanese can be classified into three inflectional classes (compared to nine in Classical Japanese). Besides a group containing only two irregular verbs, which will therefore not be addressed here, inflection is especially visible for verbs belonging to the *consonant inflectional paradigm*, also called *consonant verbs* or *consonant-stem verbs*. Those verbs end in a consonant which is followed by a vowel. It is this vowel where inflection becomes visible, as it differs across inflectional forms. There are various analyses on how to treat this vowel (see Kuno 1973; Shibatani 1990: 221–235 for an overview). Shibatani (1990: 224) treats it as an inflectional ending, which together with the consonant root forms the stem. Iwasaki (2013: ch. 5) entertains a more modern analysis of this vowel as a *stem forming suffix*.

In any case, it is this vowel which shows inflection as exemplified below for the verb *nom* 'to drink' where typical suffixes for each inflection form are added and the alternation of the inflectional/stem forming vowel is highlighted in boldface and italics[6].

[4]When speaking of *Classical Japanese* (or *bungo*) I refer to the premodern written stage of the language based on the literary style of the Heian-period (794–1185) and in use till the language reform carried out during the Meiji-period (1868–1912).

[5]This is the case neither in Modern, nor in Classical Japanese.

[6]I only gloss the vocalic elements as separate morphological elements if they allow the verb to fulfill an independent function.

(7) Watashi-ga ocha-wo nom***a***-na-i.
I-NOM tea-ACC drink-NEG-PRS
'I do not drink tea.' (negative form)

(8) Watashi-ga ocha-wo nom***i***-ta-i.
I-NOM tea-ACC drink-VOL-PRS
'I want to drink tea.' (nominalized form)

(9) Watashi-ga ocha-wo nom-***u*** koto-ga suki-da.
I-NOM tea-ACC drink-PRS COMP-NOM like-COP
'I like drinking tea.' (sentence-final form)

(10) Watashi-ga ocha-wo nom***e***-ba i-i.
I-NOM tea-ACC drink-if good-PRS
'If I drank tea, it would be good.' (concessive)

(11) Ocha-wo nom-***e***!
tea-ACC drink-IMP
'Drink (the) tea!' (imperative)

Exemplifying with (7), *nom* 'to drink' is inflected for the negative form (*mizenkei*), which is expressed via the vowel *-a*, yielding *nom-a*, to which then the negative auxiliary verb *-na* is suffixed. This auxiliary verb is itself inflecting and shares its paradigm with *i*-adjectives (see 3.2.2), therefore it ends in *-i* in non-past. The result is *nom**a**-na-i* drink-NEG-PRS 'do not drink'.

In contrast, the sentence-final and the attributive forms, which have become unified in Modern Japanese for all word classes but *na*-adjectives – and thus for verbs can no longer be differentiated (Kuno 1973, Frellesvig 2010), which is why they are not given separately – fulfill independent functions as does the imperative form.

This classification in six inflection forms is not shared by all authors (see Shibatani 1990: 229). For example, Kuno (1973: 27–28) has proposed different inflectional paradigms on the basis of Modern Japanese, illustrated below for *hanas* 'to speak'.

Table 1: Inflectional forms of Modern Japanese according to Kuno (1973)

Inflection Form	Verb	Translation
Present/Non-Past	hanas-u	'speak'
Past	hanash-ita	'spoke'
Imperative	hanas-e	'speak!'
Cohortative	hanas-ō	'let's speak'
Continuative	hanash-i	'speaking'
Gerundive	hanash-ite	'speak-and'
Conditional	hanas-eba	'if…speak'
Perfect Conditional	hanash-itara	'if…spoken'
Perfect Suppositional	hanash-itarō	'I suppose…spoke'

On the other hand, inflection is harder to see for so-called *vowel-stem verbs*, or *vowel verbs*. This is because their stem already ends in a vowel, i.e., according to Iwasaki (2013: 80), their stem forming suffix is Ø. Nevertheless, they are typically still counted as an inflecting category (Shibatani 1990; Uehara 1998; Iwasaki 2013 among others). Their paradigm is shown below for the vowel-stem verb *tabe* 'to eat'.

(12) Watashi-ga gohan-wo tabe-na-i.
I-NOM rice-ACC eat-NEG-PRS
'I do not eat rice.' (negative form)

(13) Watashi-ga gohan-wo tabe-ta-i.
I-NOM rice-ACC eat-VOL-PRS
'I want to eat rice.' (nominalized form)

(14) Watashi-ga gohan-wo tabe-ru koto-ga suki-da.
I-NOM rice-ACC eat-PRS COMP-NOM like-COP
'I like eating rice.' (sentence-final form)

(15) Watashi-ga gohan-wo tabe-reba i-i.
I-NOM rice-ACC eat-if good-PRS
'If I ate rice, it would be good.' (concessive)

(16) Gohan-wo tabe-ro!
rice-ACC eat-IMP
'Eat (the) rice!' (imperative)

3.2.2 Inflection of adjectives

Japanese has two canonical adjective groups, one referred to here as *i*-adjectives, the other as *na*-adjectives. Whether *na*-adjectives actually inflect is subject to debate. In fact, this debate is tied to the very understanding of *katsuyō* (cf. Uehara 1998: 59–64; Kishimoto & Uehara 2016). The reason why *i*-adjectives are usually perceived as an inflecting category is illustrated in (17) through (20) below in which different inflection forms and, if applicable, typical suffixes are given for the *i*-adjective *aka-i* 'red'.

(17) Kono hana-wa aka-***i***.
this flower-TOP red-PRS
'This flower is red.' (sentence-final form)

(18) Kono hana-wa aka***ku***-na-i.
this flower-TOP red-NEG-PRS
'This flower is not red.' (negative form)

(19) hana-wa aka***ke***-reba
flower-TOP red-if
'if the flower was red' (concessive)

(20) Kono hana-wa aka***k***-atta.
this flower-TOP red-PST
'This flower was red.' (past)

At first glance, *na*-adjectives have to co-occur with a copula which is itself inflecting. The respective patterns are given below for the *na*-adjective *nigiyaka-na* 'lively'.[7]

(21) Kono machi-wa nigiyaka-da.
this city-TOP lively-COP.PRS
'This city is lively.' (sentence-final form)

(22) Kono machi-wa nigiyaka-de-wa-na-i.
this city-TOP lively-COP-TOP-NEG-PRS
'The city is not lively.' (negative form)

[7]For reasons of space, the question of which adjective groups in Japanese inflect will not be discussed here. Readers are referred to Uehara (1998), Nishiyama (1999), Yamakido (2005) among others.

(23) machi-wa nigiyaka-de-areba
city-TOP lively-COP-if
'if the city was lively' (concessive)

(24) Kono machi-wa nigiyaka-datta.
this city-TOP lively-COP.PST
'This city was lively.' (past)

3.2.3 Non-inflection of nouns

As mentioned above, the other major lexical word class, the nominal class, is uninflecting. Japanese lacks number also in the nominal domain, but it does have case. However, case is not expressed via inflection but via case particles which are usually treated as (themselves noninflecting) clitics (Uehara 1998, Iwasaki 2013). This is exemplified below.

(25) Sake-***ga*** suki-da.
alcohol-NOM fond-COP.PRS
'I like alcohol.' (nominative)

(26) Watashi-ga sake-***wo*** nom-u koto-ga suki-da.
I-NOM alcohol-ACC drink-PRS COMP-NOM fond-COP.PRS
'I like drinking alcohol.' (accusative)

(27) sake-***ni*** mizu-wo ire-ru
alcohol-DAT water-ACC put.in-PRS
'to put water in alcohol' (dative/locative)

This discussion emphasizes that the existence of *uninflectable* lexemes is not expected in Japanese in the system presented here. There should be no lexeme that belongs to an inflecting word class, but fails to show the relevant morphology associated with this word class.[8]

3.3 *Rentaishi*

3.3.1 Their development

The question is then how *rentaishi* fit into the picture. This word class literally translates to 'adnominal-noun-word' and is in English usually referred to as

[8]Potentially, this might be related to the fact that uninflectable lexemes might be more often found in the realm of nominals than verbals in general (Spencer 2020: 143). Nevertheless, we do find uninflectable verbs in some languages as demonstrated by Spencer (this volume).

adnoun or *prenoun*, German equivalents are *Adnomen*, sometimes *Adnominalie* (Rickmeyer 1995). An alternative name is *fukutaishi*, which means something like 'added word'. *Rentaishi* can be described as a word class with very few lexemes that have all been derived from other word classes and have become fossilized. Syntactically, they are restricted to their role as attributive modifiers. Morphologically, they can be defined as being rather heterogeneous which also sets them apart from the word classes of verbs and adjectives in Japanese which, as seen above, are normally characterized via their distinct morphology (cf. Nitta et al. 2000).

The origin of *rentaishi* can be traced back to the Meiji-period (1869–1912), thus they entered the Japanese word class system later than the other major word classes (Frellesvig 2010). The first mention of this word class can be attributed to MITSUYA Shigematsu in his lectures given in 1915/16, but published in 1932. Mitsuya describes *rentaishi* as "attributives that are entirely uninflecting", that are "adjective-like without being adjectives" (Mitsuya 1932: 238). The important linguists of the time, especially MATSUSHITA Daizaburō (1878–1935), realized that there are several lexemes in Japanese that are restricted to attributive use, do not inflect and whose sole function is to modify nouns. In his *Revised Standard Grammar* (*Kaisen hyōjun Nihon bunpō*), Matsushita (1976 [1928]), who calls them *fukutaishi*, writes.

(28) Fukutaishi (Adjective) are lexemes that express an attributive concept which is in turn dependent on another concept, that modify the meaning of other lexemes, and that do not have a predicative character.
(Matsushita 1976 [1928]: 204, [bracket in original])

This highlights the following four characteristics of *rentaishi.*

(29) Characteristics of *rentaishi* according to Matsushita (1976 [1928]):
1. lexical status
2. dependency
3. modifiers
4. no predicative use

Due to the influence of Matsushita and their incorporation into Hashimoto's Standard Grammar (1948 [1934]), in which Hashimoto highlights their lexemic status as well and puts special emphasis on their role as attributive modifiers of nominals and their inability to appear as subjects of a sentence (Hashimoto 1948 [1934]: 63–65), *rentaishi* had become a recognized member of the Japanese word class system.

Although neither author emphasizes the characteristic *uninflectability*, we will see that the perception of *rentaishi* as an independent category is very much tied to this notion.

3.3.2 Syntactic characteristics and morphological subgroups

Potentially, the most obvious cases of *rentaishi* are lexemes that are derived from verbs and have become fossilized in attributive form. As briefly mentioned above, the sentence-final form (*shūshikei*) and the attributive form (*rentaikei*), cannot be differentiated for verbs in Modern Japanese, but used to be different in Classical Japanese.[9] While the sentence-final form of verbs belonging to the *ri*-inflectional-paradigm ended in *-ri*, the attributive form ended in *-ru*.[10] Verbal derivates belonging to the group of *rentaishi* are solely derived from verbs of the *ri*-paradigm, thus end in *-ru*. Examples include the following.[11]

(30) *aru* '(a) certain'
saru 'previous'
akuru 'following'
kitaru 'ahead'

All those lexemes are impossible in predicative position, illustrated with *akuru*.

(31) a. akuru toshi
following year
'the following year'

b. * Toshi-wa akuru.
year-TOP following
Intend.: 'The year is the following.'

There are other verbal derivates in which a lexemic verb is first combined with the suffix *-yu* (or *-iu)*, which was used to express passive and spontaneity in Classical Japanese. This suffix is then itself inflected for attributive form by means

[9]This unification potentially happened in Late Middle Japanese (12–16 AD; Lewin 2003, Frellesvig 2010).

[10]In Iwasaki's terms (2013), this would mean that the stem forming suffix of the *shūshikei* was *-i*, that of the *rentaikei -u*.

[11]Note that *aru* '(a) certain' is homophonous to the existential verb/copula *aru* 'to exist' which is why some authors treat them as the same element (Lehmann & Nishina 2015 a.o.). However, the existential verb *aru* is restricted to inanimate entities and with these entities can express predication relations, which is why these two elements are kept apart here since the *rentaishi aru* never appears in predicative position.

of the suffix *-ru*. Representative examples are *iwayuru* 'so-called' and *arayuru* 'all kinds of'. Exemplifying with *iwayuru*, this word can be divided into the verb *iu* 'to say', which is inflected for *mizenkei*, (negative form) and the marker for spontaneity/passive *-yu*. This marker is in turn inflected for the attributive form. Again, the *rentaishi iwayuru* is not used in Modern Japanese, except for attributive contexts.

(32) a. iwayuru gakusei
so-called student
'a so-called student'

b. * Kono gakusei-wa iwayuru.
this student-TOP so-called
Intend.: 'This student is so-called.'

There are also members of the word class *rentaishi* which are are derived from adjectives. Besides those ending in the element *-na* to be discussed in Section 5, there are those which co-occur with the attributive form of the premodern copula *nari* or *tari*.[12] Again, the relevant lexemes in Modern Japanese are restricted to mono-attributive use.

(33) a. tannaru machigai
mere misunderstanding
'a mere misunderstanding'

b. * Machigai-wa tannaru.
misunderstanding-TOP mere
Intend.: 'The misunderstanding is mere.'

Finally, there are *rentaishi* derived from nouns. An example is *waga* 'my, our' which is derived from *wa*, the noun used in Classical Japanese to refer to the 1st person singular, and *-ga*, which is the nominative particle, but was also used as the genitive particle in Old, and partly in Classical Japanese (Frellesvig 2010). *Waga* is basically used only in some fixed expression in Modern Japanese, such as in (34) given below. The same restrictions apply.

(34) a. waga kuni
our country
'our country'

[12] *tari* can be seen as an allomorph which was restricted to Sino-Japanese lexemes in Classical Japanese.

b. * Kono kuni-wa waga.
this country-TOP our
Intend.: 'This country is ours.'

There are other lexemes which can be classified as nominal *rentaishi*. Those end in *no*, the particle used to mark genitive and a variation of other adnominal functions in Modern Japanese (for example Saito et al. 2008). In Classical Japanese, this particle was used also to mark the subject, although some uses are attested where it attaches to a nominal modifier within the DP. Examples include *reino* (or potentially *rei-no*) 'ordinary, aforementioned' and *honno* (*hon-no*) 'mere'. Neither do these lexemes themselves nor the part preceding *no* occur predicatively. We will get back to this group in Section 5.

(35) a. honno kodomo
mere kid
'mere kid' (Nagano 2016: 122)

b. * Kono kodomo-wa honno (da).
this kid-TOP mere (COP)
Intend.: 'This kid is mere.'

Table 2 illustrates all relevant subgroups of *rentaishi* (cf. Köhlich 2020: 181–182).

4 The historical development of *rentaishi*

As emphasized by Corbett (this volume, p. 43) "loss of inflection can reflect a change of part of speech". Having outlined the syntactic restrictions and morphological heterogeneity of *rentaishi*, we are in a position to analyze the relevance of this word class for the debate on uninflectedness. This will be done by taking the historic circumstances of its development into consideration, specifically in connection to the Meiji-period, the field of *Dutch Studies* and their connection to adjectives.

4.1 The Meiji-period

The word class *rentaishi* was established and subsequently became the object of interest towards the end of the Meiji-period, thus around the beginning of the 19^{th} century. Historically, this period is characterized by a stronger interest of Japan in the West. After decades of very limited contact, the intellectuals of the Meiji-period sought out the Western cultural and political achievements

Table 2: Different morphological subgroups of *rentaishi*

Group	Subgroups	Examples
verbal *rentaishi*	attributive form of verbs in non-past	*aru* '(a) certain'; *saru* 'previous'; *akuru* 'following'; *kitaru* 'ahead'
	attributive form of verbs with morpheme *-iu*	*iwayuru* 'so-called'; *arayuru* 'all kinds of '
	attributive form of verbs in past	*taishita* 'considerable'; *tonda* 'remarkable'
adjectival *rentaishi*	lexemes ending in attributive forms of premodern copula *-naru/taru*	*tannaru* 'mere'; *shutaru* 'main'
	three lexemes in combination with element *na*	*ōkina* 'big'; *chiisana* 'small'; *okashina* 'strange'
nominal *rentaishi*	lexemes ending in *-ga*, often in particular collocations	*waga* 'my/our'; *onoga* 'my'
	lexemes ending in *-no*	*reino* 'ordinary, aforementioned'; *honno* 'only, mere'

and started comparing them to those of their own country. This trend also pertained to the field of linguistics (cf. Heinrich 2002 among others). Several linguists started turning their eyes towards Western linguistics, comparing the languages of the West with the Japanese language as well as linguistic traditions in all areas, including grammar, but also word classes (cf. Sugimoto 1991: 47–54).[13] In turn, this also led to a better understanding of the Japanese language.

Another important trend of the Meiji-Period was the beginning unification of the Japanese language, known as *genbun'itchi.* Residues of diglossia between the written language, which mostly followed the norms of Classical Japanese, and the spoken language had hindered a unification of the Japanese language for decades, but from around 1885 onward, as a result of the *genbun'itchi* movement,

[13]Important linguists were TANAKA Yoshikado (1841–1879), NAKANE Kiyoshi (NAKANE Kōtei, 1839–1913) and NAKAGANE Masahira (1824–1884).

a standardized unified language was gradually achieved. Notably, this trend was tied to the fact that Japanese linguists started to realize that this lack of unification was not observable in other countries (Gottlieb 1991).

4.2 Adjectives in *Dutch Studies*

The two languages of interest during the Meiji-period were English, which was identified as the most important language of the West, and Dutch. Linguists were interested in Dutch since it had been the most important language of trade during the Edo-period when trade with outsiders was formally forbidden and was solely carried out with the Netherlands in the artificial island of Dejima, close to Nagasaki (Sugimoto 1967, 1991, Nespoli 2023). The study of Dutch had in fact already started during the Edo-period and resulted in the education of several translators and interpreters between the two languages as well as several translations of important Dutch works in various areas, including linguistics. This study of the Dutch language in Japan is referred to as *Dutch Studies* or *rangaku* 蘭学 (ibid.).[14]

The emergence of *rentaishi* as a word class, perceived independently by Japanese linguists, is tied closely to *rangaku*. This is because the linguists of the Meiji-period came to realize that the Japanese adjectives, or *keiyōshi*, are an inflecting category, and more importantly a category that inflects *independently*. In their view, this contrasted them sharply from the adjectival category in Dutch, and by extension all European languages as they thought, which only inflects in accordance with the head noun and never appears in predicative position without the aid of the copula (Kim 2006).

In this respect, note that also in the European tradition, the adjective had for a long time been regarded as *nominem adjectivum* ‘added noun’, following the first terminology of Thomas of Erfurt (12$^{\text{th}}$ century), whose purpose it is to follow the *nominem subjectivum* and to acquire its features (Hajek 2004). Dutch linguists of the 18$^{\text{th}}$ century, noteworthy all those relevant to the present article as they were perceived by contemporary Japanese linguists,[15] referred to the Dutch adjective as *bijvoeglijk naamwoord* or *toevoeglijk naamword*, both approximately translatable as ‘added noun’ (Sugimoto 1991: 624; Kim 2006: 128; Nespoli 2023).[16] What

[14] As argued in Nespoli (2023: 48), it might be more precise to refer to the dedicated studies of the Dutch language as *rangogaku*. However, *rangaku* was the preferred term used by scholars of the time as also evidenced by the titles of books and studies concerning the Dutch language.

[15] This concerns essentially works of Willem Séwel (1653 – 1720), François Halma (1653 – 1722) and Pierre Marin (1667/8 – 1718) (cf. Nespoli 2023).

[16] Compare also the German terms *beystendiges Nennwort* and *Beynahmen*, close to ‘accompanying noun‘, used by German linguists in the 18$^{\text{th}}$ century (Spitzl-Dupic 2011: 1–5).

is reflected in these terms is that adjectives do not occur independently, but only together with an *independent noun*, called among others *zelfstandig naamwoord* (ibid.). Crucially, adjectives are dependent because they need to inflect according to the *phi*-features exhibited by the head noun they modify.[17]

The Japanese linguists working on Dutch and writing their own Dutch grammars created various Japanese translations, or rather equivalents, for *bijvoeglijk* (*toevoegelijk*) *naamwoord*. Some of these terms were neologisms (Kim 2006, Sugimoto 1991), but others were calques taken either from the tradition of *kanbunten* ('Grammar of Chinese', cf. Sugimoto 1991: 44–45) or earlier works in the tradition of *kokugogaku* (for example from OGYŪ Sorai (1666–1728), Nespoli 2023: 395–399). A (inconclusive) list of relevant terms and in which works they are given is presented below (cf. Köhlich 2020: 188; especially the appendix in Sugimoto 1991).

All authors characterize the adjective as a lexeme, via the suffixes *-shi* or *-gen*. Some authors are more explicit and refer to them as *nouns* (*meigen* and *meishi*), while others do not. What all these terms reflect is the *added* and *dependent* character of the adjective. This is evident in the prefixes *fuzoku-* ('affiliation', 'membership'), *setsu-* ('connection') and *bai-* ('addition'). The morpheme *sei* 'stative' differentiates adjectives from *dynamic* (*dō*) verbs.

4.3 Adjectives and *rentaishi*

At the same time, the linguists of the Meiji-period realized that the Japanese adjectives, or *keiyōshi* 'quality words', can inflect independently for tense, in attributive as well as in predicative position, and do not show agreement with their head noun. That is, they do not simply just mirror the (phi-)features of nouns and therefore are not *added* or *accompanying*. In accordance with the recent popularization of linguistic comparisons of Japanese and Western languages, adjectives were stylized as *independent* against the Western adjectives, *dependent* and *added*. This view, and the historical development depicted on the preceding pages, is reflected in the works of ŌTSUKI Fumihiko (1847–1928), the grandson of ŌTSUKI Gentaku (1757–1827), one of the most important experts in *Dutch Studies*, and

[17]This only concerns their use in attributive position, whereas they do need a copula in order to appear in predicative position and show no inflection. See next section.

[18]Presumably, this is a revised version of *Oranda-go shihinkō* 和蘭語詞品考 (Thoughts on the parts of speech of Dutch) by Nakano Ryūho. The manuscript of this title cannot be identified, although some authors believed that this work is the same as *Ryūho Nakano sensei bunpō*. The fact that the terms for the Dutch adjective are different in the two works, for example, does suggest otherwise. Cf. Sugimoto (1991) and Nespoli (2023: 394) for discussion.

Table 3: Different terms for adjectives in *Dutch* grammars in Japan

Author	Book	Adjective
Nakano Ryūho (Shizuki Tadao, Wilgen Akker, 1760–1806)	*Ryūho Nakano sensei bunpō* (Grammar of Master Nakano Ryūho, around 1798)	*bōki-myōgo* 傍寄名語 (lit. 'noun depending on what is nearby')
Baba Sadayoshi (Baba Sajūrō, 1787–1822)	*Teisei rango kyūhin-shū* (Collection of the nine parts of speech of Dutch, revised, 1814)[18]	*kyoseishi* 虚静詞 (*kyo-* = empty, *-sei-* = stative, *-shi* = word)
Yoshio Shunzō (1787–1843)	*Rokkaku zenpen* (On the six cases, first edition, 1814)	*baimeishi* 陪名詞 (lit. 'added/dependent noun')
Fujibayashi Taisuke (Fujibayashi Fuzan, 1781–1836)	*Oranda gohō kai* (Analysis and grammar of Dutch, 1815)	*fuzokumeigen* 附屬名言 ('added noun')
Ōtsuki Genkan (Ōtsuki Banri, 1785–1837)	*Rangaku-han* (Dutch colloquial speech, 1816, 1824)	*setsumeishi* 接名詞 ('connected noun')
Ōba Sessai (1805–1873)	*Yaku Oranda bungo* (Dutch written language, 1855)	*baiji* 陪辭 ('added/dependent word')

nephew of ŌTSUKI Genkan. Ōtsuki Fumihiko wrote two of the first relevant grammars of Japanese as it approached its modern stage: *Genkai* (Sea of Words, 1891) and *Kō-Nihon bunten* (Detailed Japanese Grammar, 1897). In *Genkai*, Ōtsuki writes the following.

(36) The English *Adjective* [sic!] merely precedes the noun and bestows it with qualitative features. The Japanese adjective [keiyōshi, V.K.] also adds qualitative features to the noun, however in terms of morphological build-up, it is entirely different. The fact that it shows inflection in word-final position and different inflection forms is a feature it shares with verbs. Moreover, placed after the noun, it marks the end of a sentence.

(Ōtsuki 1891: 28)

In this definition, it becomes obvious that Ōtsuki sees the Japanese adjective more powerful than the English, the Western, adjective. This is illustrated below with the examples repeated from above where special emphasis should be placed on the English translation. Although this is not the precise state of the language during Ōtsuki's time, the examples show that the Japanese adjective occurs independently, whereas the English adjective is always accompanied by a copula when it appears in predicative position.[19]

(37) a. Hana-wa aka-i.
flower-TOP red-PRS
'The flower is red.'

b. Hana-wa akaku-nai.
flower-TOP red-NEG
'The flower is not red.'

c. Hana-wa aka-katta.
flower-TOP red-PST
'The flower was red.'

YAMADA Yoshio (1873–1958), the linguistic figure following Ōtsuki, argues that in Japanese, both verbs and adjectives, have the "ability of predication" (*"chinjutsu-no nōryoku"*), an ability which should only apply to inflecting word classes (Yamada 1908: 861). In contrast, he describes English adjectives as follows.

(38) The English *keiyōshi*, in other words the *adjective* […], should be considered a *baiji*, but not a *keiyōshi*.
(Yamada 1908: 869)

Initially, Yamada uses the term *keiyōshi* with respect to both languages, English and Japanese. Afterwards, he decides to refer to the English adjective as *baiji* or 'added word', a direct relict of the *rangaku* terms *baiji* used by ŌBA Sessai and *baimeishi* used by YOSHIO Shunzo. Yamada's use of the term *baiji* can be interpreted as pejorative and possibly even politically motivated, highlighting the superiority of the Japanese *keiyōshi* (Kim 2006).

Having established the superiority and the more independent character of the Japanese, contra the Western, adjective, Japanese linguists also became aware

[19]Note that I give an example of an *i*-adjective here, abstracting away from the question in how far *na*-adjectives show inflection. In fact, all adjectives, also those inflected for past or bestowed with the negative suffix, are possible in attributive position without additional elements such as relative clause markers.

that there is a group of attributive modifiers in Japanese which do not inflect, thus do not appear independently at the end of a sentence and express tense. One such element is *aru* 'a certain'. They realized that such elements could hardly be *keiyōshi* as they are not independent. Furthermore, a lexeme such as *aru,* which looks verbal on the morphological level, does not share any features with verbs as it lacks all inflectional forms of the verbal paradigm. It therefore could also hardly be classified as a verb.

Building on the grammatical descriptions of Ōtsuki (1891) and Yamada (1908), Matsushita (1976 [1938]) introduced *rentaishi* (*fukutaishi*) in his grammar and saw several successors (Sekine 1937, Kieda 1938 among others). Those linguists recognized that many lexemes in Japanese simply do not have any other function than to modify nouns, and that most of these lexemes are fossilized derivates of other word classes and frozen in their morphological form and usage domain.

Keeping apart the English *adjective* from the Japanese *keiyōshi*, I argue (and see Kim 2006 for an analysis similar in spirit) that the category of *rentaishi* was created subsequently in order to fill the slot of dependent attributive modifiers in Japanese. All those attributive modifiers derived from other word classes and unable to inflect were sorted into this newly formed word class. Thus, Japanese linguists avoided the existence of *uninflectable* lexemes, maintaining the variable *inflection*, or *katsuyō*, as the main difference between verbal and nominal categories in Japanese and predicting that word classes are either entirely inflecting or entirely uninflecting. This is why authors such as Mitsuya describe *rentaishi* as "adjective-like without being adjectives" (Mitsuya 1932: 238). In a nutshell, the integration of the word class *rentaishi* marked an important turning point in linguistics in Japan, highlighting the criterion *inflection* and setting Japanese word classes in relation to other languages.

From the perspective of (un-)inflectedness, the ability to inflect is clearly connected to word class status. Uninflectable lexemes were not part of the Japanese word class system during the Meiji-period, and presumably until the middle of the 20th century. Rather than accepting the existence of uninflectable lexemes, Japanese linguists established a new word class which is characterized via the very feature [+uninflecting].

Finally, consider what Baerman et al. (2005) call *neutralization* and define as follows:

(39) (i) In the presence of a particular combination of values of one or more other features (the context), there is a general loss of all values of a particular feature F found elsewhere in the language.

(ii) No syntactic objects distinguish any values of feature F in the given context, and feature F is therefore syntactically irrelevant in that context.

(Baerman et al. 2005: 30)

Neutralization basically accounts for cases in which a feature becomes syntactically irrelevant in certain contexts. The example given by the authors is *gender* in plural contexts in Russian (Baerman et al. 2005: 28) as there is simply no morphological distinction on nouns according to this feature in this context. This means that, different to uninflectedness, neutralization presupposes the existence of a specific feature that is relevant in syntax.

But that is not the case for *rentaishi*, at least not as depicted here, since no inflection is observable in the first place. A case of *neutralization* can indeed be put forth for in-between stages especially of verbal *rentaishi* in which they were still perceived as verbs but only used in particular contexts, that is, in attributive position. Kieda (1938: 356–359) still recognized an in-between stage for lexemes such as *aru* '(a) certain' which is why he decides not to classify them as *rentaishi* (Kieda 1938: 359). This means we manage to depict an evolution, at least according to the view of this author, from a neutralized lexeme to a member of an uninflecting word class.

5 Discussion: The modern dimension of *rentaishi*

Over the years, the understanding and acceptance of *rentaishi* as a word class, predominantly in Japan, but also abroad, has shifted. Komatsu (1973: 53) documents a rise in critical studies and surveys of this category and its inclusion into the Japanese word class system, following their establishment towards the middle of the 20^{th} century. These critical studies include Tokuda (1934) and Suzuki (1959) and prominently Watanabe (1971) and were followed by a gradual marginalization of this category. Nowadays, *rentaishi* are absent from most Standard Grammars (Hinds 1988, Tsujimura 2014, Hasegawa 2015 among others).

While some authors have performed dedicated analyses of *rentaishi*,[20] most authors who explicitly argue against the word class *rentaishi* instead sort relevant lexemes into the groups they morphologically belong to (Mio 1942, 1958, Suzuki 1959, Lehmann & Nishina 2015). That means a lexeme such as *aru* '(a) certain' is

[20]For example, Watanabe (1971) treats *rentaishi* as *adnominal auxiliaries* (*rentai-fukushi*), Martin (1975: 742–753) treats them as defective nouns and Katō (2003: 27) as a subgroup of dependent nominals.

classified as a verb, a lexeme such as *waga* 'my, our' as a noun and so on. These authors then enable the possibility of *uninflectable* entries inside the group of otherwise inflecting parts of speech in Japanese, an analysis which opens up room for discussion.

5.1 Possible *na*-adjective *rentaishi*

This discussion will be lead here, as it is most eminent, with regard to *rentaishi* which are arguably derived from adjectives and are thus argued by the above-mentioned authors to belong to this group. First and foremost, this concerns the three lexemes *ōki-na* 'big', *chiisa-na* 'small' and *okashi-na* 'strange' (or potentially *ōkina, chiisana* and *okashina*), which all end in *-na*, thus look like *na*-adjectives. However, they do not show inflection, in other words lack predicative use with the copula *da* and adverbial use with the element *-ni*. This is illustrated for *ōki-na* 'big' below.[21]

(40) a. ōki-na ie
big-NA house
'big house'

b. * Kono ie-wa ōki-da.
this house-TOP big-COP
Intend.: 'This house is big.'

c. * Ie-wa ōki-ni nat-ta.
house-TOP big-ADV become-PST
Intend.: 'The house became big.'

However, there exists *i*-adjective variants of these lexemes that are fully productive and do not adhere to these restrictions. Crucially, the meaning of the attributive modifier remains the same. This is shown below.

(41) a. ōki-i ie
big-I house
'big house'

b. Kono ie-wa ōki-i.
this house-TOP big-PRS
'This house is big.'

[21]To illustrate potential inflection, I will transliterate the ending *-na* with a hyphen.

c. Ie-wa ōki-ku nat-ta.
house-TOP big-ADV become-PST
'The house became big.'

There are three possible ways to analyze these lexemes in terms of word class status. We can either classify them as *rentaishi*, thus as an uninflecting category, presupposing that they somehow have become fossilized. This explanation seems reasonable, although it is hard to determine at what point of the language this fossilization occurred. The first result for *ōki-na* in the Corpus of Historical Japanese (CHJ[22]) stems from 1642, thus from Middle Japanese. Furthermore, it is unclear if they are derived from their *i*-adjective counterparts or potentially derived from forms such as *ōki-naru* in which a lexical adjective is combined with the historical copula *nari*. The first result for the form *ōki-naru* is attested on the CHJ in 1001.

Before we discuss their incorporation into the *na*-adjective category, another prominent option needs to be discussed, namely the analysis as *i*-adjectives with a secondary attributive form -*na*, which seems to be the preferred option in the research literature (Suzuki 1972, Rickmeyer 1995, Muraki 2012, Lehmann & Nishina 2015 among others). One piece of evidence presented by authors such as Suzuki (1972) and Lehmann & Nishina (2015) in favor of this analysis is that there is no semantic difference between the respective forms. Therefore, there should be no need for positing two separate lexical entries in two different word classes.[23] Although the existence of uninflectable lexemes, and alternative inflectional forms for that matter, is not foreseen according to the more traditional analyses of Japanese word classes depicted above, this option is justified, considering the semantic closeness of the two forms. In fact, whether we allow the existence of uninflectable lexemes in Japanese is close to an ideological debate and especially with regard to these three lexemes, nothing hinges on this distinction (see also Oshima & Hayashi 2021a: Fn. 15).

This brings us to the third alternative, namely to classify these lexemes as *uninflectable* or *defective na*-adjectives. This approach will be discussed below.

[22]https://clrd.ninjal.ac.jp/chj/overview-en.html

[23]Lehmann & Nishina (2015) also count lexemes such as *atataka-na* 'warm' as alternative forms of existing *i*-adjectives. However, such lexemes are not syntactically restricted and should thus be treated as true *na*-adjectives. Japanese then possibly simply has two separate lexical entries across two word classes in such cases.

5.2 *Rentaishi* as defective adjectives

The most contemporary understanding of *rentaishi* is equal to 'adjectives without predicative use'. This view emerged in the past 20 years and is found most prominently in the works of Muraki Shinjirō (for example 1996, 2009, 2012, 2015, see also Oshima et al. 2019 and Oshima & Hayashi 2021a,b). Muraki, in his role as important adversary of the school grammar, uses *rentaishi* synonymously to 'mono-attributive adjectives' (*kitei-keiyōshi*), that is adjectives which can only be used attributively. He argues in favor of three morphologically diverse adjective groups, ending in *-i*, *-na* and *-no* respectively, and emphasizes that all adjectives should in principle be able to occur attributively, predicatively and adverbially Muraki (2012: 36–38). If an adjectival lexeme lacks one of these uses it is described as "defective", or literally "incomplete" (*fukanzen*, Muraki 2012: 37; Muraki 2015: 26). One important group of incomplete adjectives are those that are mono-attributive, and those mono-attributive adjectives are what he calls *rentaishi*. Under this view, the three lexemes discussed in the previous section are *na-rentaishi*.

Importantly, however, most mono-attributive (non-predicative) modifiers in Japanese can be found among the group of modifiers co-occurring with the element *-no*, named *no*-modifiers in Köhlich (2023), but analyzed as *no*-adjectives in Muraki (2012), as evidenced in recent quantitative surveys on Japanese modifiers performed on the Balanced Corpus of Contemporary Written Japanese (BCCWJ[24]) (Abe et al. 2022, Köhlich 2023). In the corpus analysis of Abe et al. (2022), all modifiers given in Muraki (2012) are analyzed for the variables *attributive* and *predicative use* (among others). The authors determine 12 lexemes that completely lack predicative use, which they classify accordingly as *rentaishi*.[25]

Oshima et al. (2019) and Oshima & Hayashi (2021a,b) go in the same direction, although for them *rentaishi* are not necessarily adjectives. If a lexeme appears attributively with *-no* but cannot be used predicatively they call it '*no*-Type Adnominal', or '*no-gata rentaishi*', lit. '*no*-type *rentaishi*'. They give the following definition.

(42) Those lexemes that form a noun-modifying clause being accompanied by the attributive copula form *no*, but cannot be accompanied by other copula forms such as *da* and *de*.

(Oshima & Hayashi 2021b: 520)

[24] https://clrd.ninjal.ac.jp/bccwj/en/

[25] More lexemes, around 300, are classified as non-predicative in Köhlich (2023) which is due to the facts that more lexemes are analyzed in general and that also lexemes between one and five occurrences, or a ratio of 1:25 towards attributive use are classified as non-predicative modifiers.

The authors analyze *rentaishi* as compositional, as a combination of a lexeme and the element *-no*, which they define as the copula.[26]

To summarize, for the authors mentioned above, the term '*rentaishi*' does not refer to an uninflecting word class, but is perceived as a feature, namely that of being non-predicative or [-predicative]. Under this view, the lexemes in question are *defective* in the sense of Spencer (2020) as they simply do not occur in those syntactic environments they are expected to appear in if they are treated as adjectives. They are thus on a par with English *former* or Russian *knižnyj* 'pertaining to books', as mentioned in Section 2.

This potentially opens a new avenue for future research, as the connection of uninflectedness and uninflectability respectively to restricted uses of adjectives to the attributive or predicative domain has, to my knowledge, not been extensively discussed in the research literature.[27] Another cross-linguistic case where such a discussion could be relevant are mono-predicative adjectives in German. While German adjectives, in general, only show inflection in attributive, never in predicative, position there are some adjectival lexemes such as *barfuß* 'barefoot' or *pleite* 'bankrupt, broke', which are strongly biased toward predicative use and show highly restricted or marginal attributive distribution (Trost 2006). Treating the possibility of inflection as a criterion for adjectivehood, those lexemes are not adjectives, but could be perceived as some reversed counterpart to Japanese *rentaishi*. On the other hand, their uninflectability is connected to their syntactic behavior. Such lexemes should be studied with regard to the debate at hand.

While the view of the term *rentaishi* as mono-attributive adjectives in Japanese is appealing, I would like to conclude this section with two remaining problems this view entails. First, we are in danger of losing the historical dimension of the notion *rentaishi*. Lexemes classified as *rentaishi* in earlier approaches can be found at very early stages of Japanese. For example, *reino* 'ordinary, aforementioned' can be traced back at least to the Heian-period (794–1185). The first result on the CHJ dates to the year 900. This is different for other non-predicative *no*-modifiers, such *as bōgai-no* 'unexpected' which is classified as a *rentaishi* in Abe et al. (2022) but for which the first entry on the CHJ dates back only to 1887. From a historical point of view, namely following an analysis of the word class

[26]One needs to abstract away from the fact that the authors understand *no* as the copula when they talk about non-predicative lexemes. This analysis is of course not tenable as non-predicative elements should not be able to co-occur with the copula.

[27]I express my gratitude to an anonymous reviewer who pointed out this issue to me and lead me to rethink my, formerly rigid, opinion on this matter.

rentaishi as fossilized derivatives, it might therefore be difficult to argue for membership of lexemes such as *bōgai-no* to the word class *rentaishi*. This would thus be one problem one has to live with under this more modern approach.

A bigger problem specific to the work of Muraki (2012 among others) regards the fact that this author does not recognize verbally derived mono-attributive lexemes such as *aru* '(a) certain' as *rentaishi*, as he only discusses adjectival lexemes. However, I do not see a reason why only adjectives should be treated as having a possible restriction to attributive or predicative position in the first place.

I propose to maintain the historical dimension of *rentaishi* as an uninflecting word class whose members only serve as adnominal modifiers. On the other hand, I think it is appropriate to extend the notion *rentaishi* to refer to lexemes from other word classes that lack predicative use even if they do not belong to this word class from a historical point of view. One consequence of this is that the group of *rentaishi* becomes an open word class, although linguists in the Meiji-period probably did not perceive the word class *rentaishi* as open. Another crucial consequence of this is that the morphological heterogeneity of the word class *rentaishi* falls out nicely as different morphological subgroups are not only allowed, but predicted.

6 Summary

This paper looked at Japanese with regard to the debate on *uninflectedness*. Although the notion of *inflection*, in Japanese *katsuyō*, is slightly different from what we might expect from our knowledge of Indo-European languages, this notion has been used as the basis of word class classification in Japan roughly since the 17th century. Verbs and adjectives are classified as categorically inflecting, contra nouns as categorically uninflecting parts of speech.

It was shown that from the beginning of the 20th century, linguists in Japan became aware of the differences between their adjectives and those of Western languages, mostly English and Dutch. They realized that adjectives in those languages cannot inflect independently, but only in accordance with their head noun. Having realized this difference, they defined the Japanese adjective, the *keiyōshi*, as more powerful.

At the same time, they realized that there are some lexemes in Japanese whose sole purpose is attributive modification. In order to fill the slot of dependent and defective attributive modifiers, they literally created the word class of *rentaishi* as the Japanese counterparts to the Western adjective and sorted all those attributive modifiers into this group. This was famously done in the grammar of

Matsushita Daizaburō (1976 [1928]) and the school grammar of Hashimoto Shinkichi (1948 [1934]) allowing for the emergence of *rentaishi* as an independent word class.

In a nutshell, *rentaishi* present a case of an *uninflecting* word class established on historical grounds. It was exactly the characteristic *uninflectedness* which served as the motivation for word class status of relevant lexemes. From a modern perspective, this word class, as it was perceived at the beginning of the 20^{th} century, arguably poses an example of a closed category of only a few lexemes, defined via the characteristic *uninflectedness*.

Towards the end of this paper, it was outlined that starting around the middle of the 20^{th} century, several authors performed a re-analysis of *rentaishi* and, instead, sorted relevant lexemes into the word classes they morphologically belong to. Nowadays, the notion *rentaishi* is mostly treated as a syntactic feature, synonymous to 'mono-attributive' or [-predicative]. I argued that, if we are able to maintain the historical definition of *rentaishi*, this view can open a new avenue of research.

For future research, it would thus be interesting to tie the debate on uninflectedness to the discussion of adjectives without attributive use and to extend this approach to those without predicative use. Furthermore, it is desirable to test the characteristics inflecting/uninflecting or inflectable/uninflectable against the notion of word classes in other languages and to compare them, alongside any historical relevant developments, to the Japanese case.

Abbreviations

ACC	accusative
COM	comitative
COMP	complementizer
COP	dummy copula
IMP	imperative
N	neuter
NEG	negative
NOM	nominative
PRED	predictive copula
PREP	prepositional case
PRS	non-past
PST	past
PST.PTCP	past participle
RA	relational adjective
RC	relative clause
SG	singular
SUBJ	subjunctive
TOP	topic
VOL	volitional

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Chapter 13

The diachronic stability of uninflectability in Berber

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Most varieties within Berber, an Afroasiatic language family of Northern Africa, feature a robust system where nouns inflect for *state*. This typically involves two forms: the citation form (*free state*) and a marked form (misleadingly called the *construct state*), used mainly for postverbal nominatives and objects of prepositions. The system has been analyzed as either a marked-nominative case system or a typologically unique phenomenon. However, state inflection only applies to nouns with a gender-marked prefix, leaving a substantial minority of nouns uninflected. This includes many borrowed Arabic or Romance nouns, as well as some inherited terms like kinship words, which can be traced back to proto-Berber. While there are instances of state marking extending to previously uninflected nouns, it has never been fully generalized to all nouns. Comparative data suggests that selective uninflectedness has remained stable over a period of two millennia.

1 Introduction

If a lexeme is not marked for a morphological feature for which other members of its class are marked, it is *uninflectable* (Spencer 2020), or *uninflected* (Baerman et al. 2005: 30). The former term will be adopted here to avoid ambiguity, as the latter is used by Spencer in a different sense. Thus, for instance, English nouns inflect for number, but *sheep* is uninflectable.

In most of the Berber (Amazigh) languages of North Africa, most nouns are inflected for a feature conventionally termed *state* (arguably case; explained below), as well as for number. For example, 'man' in Kabyle is *argaz* in the free state, but

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wərgaz in the annexed state. However, a significant proportion of nouns are uninflectable for state; Kabyle 'thirst', for instance, is *fad* in both free and annexed state. In this article, it is shown that uninflectability in Berber is both synchronically widespread and diachronically stable over a millennium or more.

Such a result is surprising a priori. It appears reasonable to expect some kind of relationship between the frequency of uninflectability for a feature and the likelihood of the morphology continuing to mark that feature. After all, at the extremes, a feature for which all words are uninflectable is not a feature relevant to inflectional morphology at all. One for which all potentially inflectable items are inflectable can potentially carry a high functional load; one for which, say, half are uninflectable must perforce have a rather low one, making it much easier to dispense with. Such arguments tend to suggest that a high lexical frequency of uninflectability for a given feature should be diachronically unstable. Berber thus provides an important data point for the study of how uninflectability relates to linguistic change.

1.1 Background

Berber, or Amazigh, constitutes a family of closely related languages within the Afroasiatic phylum, spoken by tens of millions of people across North Africa. The family is primarily concentrated in Morocco and Algeria, but extends from Mauritania to western Egypt, and as far south into the Sahel as northern Burkina Faso. Some of its best known members include Tashlhiyt (Shilha) in southern Morocco, Kabyle in northern Algeria, and Tamajeq (Tuareg) in Niger.

In all Berber varieties, the prototypical noun (including most high-frequency nouns and almost all inherited nouns) starts with a prefix explicitly marked for masculine or feminine gender and for singular or plural number; values other than masculine singular are usually additionally marked by a suffix or by stem-internal alternations. Both features are relevant to agreement – of the verb with the subject, of pronouns (bound or free) with their referents, of adjectives with nouns… In most varieties, the nominal prefix is also marked for a non-agreement feature traditionally termed *state* (*état* in French, *addad* in Kabyle), which can take two values: *free* or *absolute* (*libre, ilelli*; henceforth glossed FS) and *annexed* or *dependent* (*d'annexion, amaruz*; glossed AS). The latter is often misleadingly called *construct state*, a term best reserved for head marking rather than dependent marking (Creissels 2009a). To a rough first approximation, these terms correspond to the absolutive and nominative respectively in a marked-nominative case system; however, such an analysis is problematic for certain Berber varieties and obscures some of the functions of these states, as discussed in more detail in Section 1.3 below. The paradigms given in Table 1 are representative.

Table 1: Paradigm of 'old person' in Kabyle and Tamasheq

Kabyle	free	annexed
M.SG	a-mɣar	wə-mɣaṛ
M.PL	i-mɣaṛ-ən	yə-mɣaṛ-ən
F.SG	ta-mɣaṛ-t	tə-mɣaṛ-t
F.PL	ti-mɣaṛ-in	tə-mɣaṛ-in

Tamasheq	free	annexed
M.SG	a-mɣar	ă-mɣar
M.PL	i-mɣar-ăn	ə-mɣar-ăn
F.SG	ta-mɣar-t	tă-mɣar-t
F.PL	ti-mɣar-en	tə-mɣar-en

Every Berber variety that retains state marking has a substantial number of nouns which are uninflectable for state. A rough estimate of their type frequency may be gathered by picking a set of pages at random from a Berber dictionary ordered by roots and counting nouns. For Tamazight (Taïfi 1991), such a procedure applied to ten pages suggests that about 39% (37/95) of nouns are uninflectable for state. For Tamajeq (Prasse et al. 1998), applied to five pages, it yielded 29% (34/116). By token frequency, a quick count suggests that uninflectable nouns (excluding personal pronouns and numerals) account for about 37% of the first two pages of the Kabyle folk tale "The Orphan's Cow" (Allioui 2017: 79–80); in more loanword-friendly genres, the figure would likely be higher. Figures like these are a far cry from the marginality of uninflectable nouns in languages like Russian or Polish. To understand the high frequency of state-uninflectable nouns across Berber, we must examine the factors conditioning state-uninflectability and their history. Before doing so, some background on Berber subclassification and on the nature of *state* is necessary.

1.2 Berber-internal relationships

To interpret the predominantly synchronic Berber data from a historical perspective, it is necessary to consider its family tree. Berber fits no better into a pure tree model than, say, Romance; the influence of Berber languages on other Berber languages remained substantial until a few centuries ago, and continues in some areas (Souag 2017). Nevertheless, some splits within the family are evidently much deeper than others. The varieties that likely split off earliest – Zenaga and Tetserret in the southwest, Awjili and to some extent Ghadamsi in the far east – have lost state marking, and will therefore not concern us here. Most of the remaining varieties fall naturally into four subgroups:

1. Tuareg across the central Sahara and Sahel: Tamahaq in Algeria and Libya, Tamasheq in Mali, Tamajeq in Niger

2. Zenati from eastern Morocco to southern Tunisia, notably including Tarifiyt (eastern Morocco) and Chaoui (eastern Algeria), along with numerous smaller varieties such as Figuig (eastern Morocco), Bissa (west-central Algeria), Tumzabt (south-central Algeria)

3. Kabyle in north-central Algeria

4. Atlas varieties in most of Morocco, especially Tamazight (southeast) and Tashlhiyt (southwest), but also a couple of smaller varieties, such as Senhaja, in the north.

It remains debatable whether any of these four can be subgrouped together more closely, but 2–4 share some phonological innovations, and can be classed together areally as “northern Berber”.

1.3 *State* in Berber

Most Berber varieties continue to mark nouns for state, although some peripheral varieties, such as Siwi (Souag 2013), Awjili (van Putten 2014), or Tetserret (Lux 2013), have lost the annexed state, and hence no longer have state marking. The formal marking of *state* is relatively uniform across Berber, with the exception of Tuareg. As illustrated above in Table 1, in Tamasheq, and Tuareg more generally, the *annexed state* (AS) can be given a unitary morphophonological analysis as the result of vowel reduction from the *free state* (FS), taking peripheral *a, e, i* to central *ă, ă, ə* with subsequent lower-level phonological adjustments (Heath 2005a: 147). For other Berber varieties, in which **ă > ə*, such an analysis might appear tempting for the feminine forms; it is, however, ruled out, at least synchronically, by the semivowels that appear in the masculine forms. These semivowels, moreover, are clearly original rather than innovative; some Tuareg varieties preserve relict traces of them in poetry and in placenames (Brugnatelli 1997), as does Zenaga in placenames (Taine-Cheikh 2005).

The function of *state* differs in detail across Berber varieties. However, a few generalizations appear to be exceptionless across those varieties that have retained this category. State is never an agreement feature: when a noun is followed by a nominal adjective[1] or by another noun in apposition to it, the latter is always in the free state irrespective of the state of the preceding noun. The free state is always the citation form of the noun, used in contexts that minimally include preposed topics (including preverbal subjects) and in situ direct objects. The annexed state is always conditioned by material preceding (usually directly preceding) the noun on which it appears. In every variety that has retained state marking at all, the annexed state appears in at least the following three contexts:

A) complements of some prepositions, such as genitive *n*, dative *i*, instrumental *s*, etc. - but not of all prepositions (the specific list varies from language to language); contrast (1) with (2).

(1) Kabyle (Mettouchi & Frajzyngier 2013: 24)

a. ɣər **wə**-drar
at MSG.AS-mountain
'at the mountain'

b. s **a**-drar
to MSG.FS-mountain
'to the mountain'

B) the dependent noun in tight compounds following certain heads, as in (2).

(2) Tashlhiyt (El Moujahid 1997: 116)
ayt **u**-zaɣar
sons.of MSG.AS-plain
'plainsmen'

C) a noun governed by a preceding numeral or similar quantifier, as in (3–4). (Not all numerals do this in any one language; on the variation in numeral-noun syntax across different varieties, see (Galand 2002: 212).)

[1] Like nouns (of which they constitute a subclass), nominal adjectives usually start with a nominal prefix. They can be used as the head of a noun phrase without any further marking; thus for instance *a-zəggʷaɣ* 'red (M.SG.)' can also be '(the/a) red one (M.SG.)'. In such a case, they are marked for state like any other noun.

(3) Bissa (Genevois 1973: 67)
yiğ **wə**-ryaz
one.M **MSG.AS**-man
'one man'

(4) Tumzabt (Abdessalam & Abdessalam 1996: 95)
səmms-ət **t**-sədnan
one.M **FPL.AS**-women
'five women'

In all these contexts, the state assigner must directly precede the noun to which it assigns the annexed state; Heath (2005a: 148) notes that, in Tamasheq, even a pause after the preceding preposition is enough to force its complement to appear in the free state.

In some peripheral varieties, such as Senhaja (Gutova 2021: 362, 407) or Ouargla (Delheure & Ochoa 2019: 79, 113), A-C are apparently the only contexts where annexed state appears. Across most varieties, however, e.g. Tashlhiyt (El Moujahid 1997: 114–116), Tamasheq (Heath 2005a: 147, 247), Tamahaq (Cortade 1969: 25), and Figuig (Kossmann 1997: 88–89; Kossmann 2016), the annexed state is also assigned to:

D) a postverbal subject – but never a preverbal one, nor a right dislocated one; contrast (5a) to (5b).

(5) Chaoui (Djarallah 1993: 135)
   a. H-ənna=as **h**-gəʕʕuṣ-t.
   3FSG-say.PFV=3SG.DAT **FSG.AS**-goat-FSG
   'The goat told him.'
   b. **Ha**-gəʕʕuṣ-t a=y t-uš ta-žəɣɣim-t.
   FSG.FS-goat-FSG IRR=1SG.DAT 3FSG.give FSG.FS-gulp-FSG
   'The goat will give me some milk.'

Unlike A-C, condition D does not typically require strict adjacency. Heath (2005a: 573) reports that Tamasheq requires the subject to be immediately postverbal unless fronted, but this does not hold true even in other Tuareg varieties.

While "state" is not traditionally identified with case, these four categories strongly suggest an analysis of "state" as a marked-nominative case system (Sasse 1984, König 2008, Belkadi 2024). Blake (1994: 1) defines case as "marking dependent nouns for the type of relationship they bear to their heads". Conditions 1–3 fit unproblematically into such a definition. Since most Berber varieties have VSO basic word order, Condition 4 can be understood as the application of case marking to in situ subjects; preverbal subjects are rather to be analysed as topics, assigned default case. The analysis of state as case has accordingly also been widely adopted in generativist work on Berber (Guerssel 1992, Felice 2020, Belkadi 2024). On this analysis, "free state" should properly be called the absolutive, and "annexed state" the nominative.

To these four conditions, however, Kabyle (Mettouchi & Frajzyngier 2013), Tumzabt (Abdessalam & Abdessalam 1996: 96), and at least to some extent Tarifiyt (Lafkioui 2000; Mourigh & Kossmann 2020: 39, 132) add a fifth – apparently not applicable in other varieties examined, such as Tashlhiyt (Mettouchi & Fleisch 2010) – which complicates such an analysis:

E) a right dislocated noun other than the subject of a verb: i.e. a subject following the predicate of a non-verbal sentence, or a non-subject co-referential with a preceding pronominal clitic. Both are illustrated in (6).

(6) Tumzabt (Abdessalam & Abdessalam 1996: 96)

a. D a-ssṛa n w-urəɣ **tə**-yẓiwt.
   COP MSG.FS-necklace GEN **MSG.AS**-gold FSG.AS-girl.
   She's (as beautiful as) a golden necklace [lit. a necklace of gold], the girl.

b. Ǧği-n=as ta-yətti **u**-bəṛhuš y-rəwl=asən.
   do.PFV-3MPL=3SG.DAT FSG.FS-care **MSG.AS**-dog 3MSG-flee.PFV=3PL.DAT
   'They took care of him, the ungrateful dog, and he ran away from them.'

The analysis of state in such varieties – especially in Kabyle – is accordingly controversial. Building upon the work of Galand (1969), Mettouchi & Frajzyngier (2013) reject a case analysis, observing that any of subjects, objects, and objects of prepositions can occur in either state, and consider state as a previously unrecognised typological category "provid[ing] the value (in the logical sense) for the variable of the function grammaticalized in a preceding constituent" (p. 13), i.e., for a function such as subject, object, or number, specifying the noun phrase fulfilling or being quantified by that function. Arkadiev (2015), on the other hand, argues that Kabyle "state" fits well into the cross-linguistic typology of marked-nominative case systems, and is best analysed as such, as in Creissels (2009b).

Under any analysis, Kabyle "state" marking cannot be accounted for in terms of argument structure alone, but interacts strikingly with information structure.

The question of where state marking fits into broader cross-linguistic comparative categories, however, is not our primary concern here. State marking is evidently a morphological feature of the noun; Table 2 illustrates that it shows allomorphy that is not synchronically predictable from the stem alone nor from the free state alone. For example, stems sharing the shape CiCəC differ in their free state marker (*a-* vs. *i-*) – likewise stems sharing the shape CaC (Ø vs. *a-* vs. *l-*) – and nouns sharing the same shape aCaC in the free state show different forms in the annexed state (uCaC vs. waCaC). State marking is equally clearly conditioned by syntax. Nouns which fail to mark it are therefore, by definition, uninflectable for state: such nouns are "insensitive to distinctions which are still syntactically relevant" (Baerman et al. 2005: 30).

Table 2: Examples of allomorphy in state marking in Kabyle (cf. Dallet 1982)

	free	annexed
'warming oneself'	a-ẓiẓən	u-ẓiẓən
'rope'	i-zikər	i-zikər
'hunger'	fad	fad
'finger'	a-ḍad	u-ḍad
'land'	akal	w-akal
'omen'	l-fal	l-fal

2 Uninflectability for state

All Berber varieties have a substantial number of nouns which are uninflectable for state. These arise for several reasons: phonological, morphological, and to a lesser extent even semantic.

2.1 Phonologically motivated

2.1.1 Vowel-initial stems

As implied by the segmentation in Table 2, contrasts like *aḍad – uḍad* 'finger' vs. *akal – wakal* 'land' are most conveniently analysed as reflecting a difference in the form of the stem – consonant-initial vs. vowel-initial – rather than of the prefix (Prasse 1964, Penchoen 1973). While consonant-initial noun stems are rather more numerous, all Berber varieties with state marking also have a certain number of vowel-initial ones. Such stems can often – arguably always (Prasse 1969, 2011) – be reconstructed as resulting from the loss of a former initial laryngeal, but that is of no synchronic relevance. No Berber variety allows a surface sequence of two successive vowels word-internally; in this context, the vowel of the nominal prefix is regularly deleted (*akal* 'land' rather than **aakal*; Tamasheq *taɣma* 'thigh', uninflectable for state, rather than free **taaɣma* and annexed **tăaɣma*).

This results in systematic uninflectability in situations where the vowel of the nominal prefix would otherwise be the only distinctive marker of state. As discussed in Section 1, the annexed state is formed by vowel reduction for feminine nouns in northern Berber and across the board in Tuareg. As a result, feminine vowel-initial stems across northern Berber, and all vowel-initial stems in Tuareg, are systematically uninflectable, as illustrated in Table 3.

Table 3: Vowel-initial stems contrasted across Kabyle and Tamasheq

	SG free	SG annexed	PL free	PL annexed
Kabyle (Dallet 1982)				
'wind'	aḍu	w-aḍu	-	-
'heart'	ul	w-ul	ul-awən	w-ul-awən
'lung'	t-ur-ətt^{s}	t-ur-ətt	t-ur-in	t-ur-in
'thigh'	t-aɣma	t-aɣma	t-aɣm-iwin	t-aɣm-iwin
'sweat'	t-idi	t-idi	t-id-iwin	t-id-iwin
Tamasheq (Heath 2006)				
'wind'	aḍu	aḍu	aḍu-tăn	aḍu-tăn
'heart'	ulh	ulh	ulh-awăn	ulh-awăn
'lung'	t-orr	t-orr	t-orr-awen	t-orr-awen
'thigh'	t-aɣma	t-aɣma	t-aɣm-iwen	t-aɣm-iwen
'sweat'	t-ide	t-ide	t-ad-iwen	t-ad-iwen

2.1.2 i-/y- neutralisation

In proto-Berber, the prefix vowel *a- regularly shifted to *e- in certain contexts, notably before *ă (van Putten 2016, 2018). With the merger of *ă > *ə* and *e > *i* in almost all of Northern Berber, the conditioning factor was lost, and this allomorphy went from being phonetically conditioned to being lexically conditioned. Almost all Northern Berber varieties thus have a minority class of nouns taking the prefix vowel *i* in the singular, alongside the more numerous ones in *a*.

For feminine nouns, this merger neutralised number marking within the prefix but poses no difficulties for state marking: free *te- > *ti-*, whereas annexed *tă- > *tə-*. For masculine nouns, free *e- > *i-* as expected; for the annexed state, however, we find *yə-* rather than **wə-, posing some difficulties for reconstruction (cf. Prasse 1974: 15) and also for inflectability. Northern Berber (with minor exceptions in Tunisia) never allows schwa to occur in an open syllable (cf. Kossmann 1995), and does not contrast *i-* with *y-* before a consonant: *yə* > *i* / _CV, as independently motivated for example by the behaviour of the 3m.sg. subject marker. As a result, CV-initial masculine stems which lexically happen to take *i-* are also systematically uninflectable for state, as seen in Table 4. In some varieties this is taken further. Notably, in Tamazight and Tashlhiyt, schwa is no longer phonemic, irrespective of syllable structure (Dell & Elmedlaoui 1985, Ridouane 2008). Accordingly, *i*-prefix nouns are normally uninflectable for state there even before CC-initial stems (El Moujahid 1982: 51).

Table 4: *i*-prefixed stems contrasted with *a*-prefixed ones in Kabyle and Tamazight

	sg free	sg annexed	pl free	pl annexed
Kabyle (Dallet 1982)				
‘share’	a-mur	u-mur	i-mur-ən	i-mur-ən
‘eagle’	i-gidər	i-gidər	i-gudar	i-gudar
‘cave’	i-fri	yə-fri	i-fr-an	yə-fr-an
Tamazight (Taïfi 1991)				
‘share’	a-mur	u-mur	i-mur-n	i-mur-n
‘eagle’	i-gidr	i-gidr	i-gidr-n	i-gidr-n
‘cave’	i-fri	i-fri	i-fr-an	i-fr-an

2.1.3 Prefix vowel reduction

Early in its history, both Tuareg and Zenati underwent a sound change reducing the vowel of the singular free state prefix to *ă preceding certain stems of the form CV… (in particular CVC, Ca…, or Cu…a…), where V represents a full vowel rather than a reduced one (Prasse 1974: 19–21; Kossmann 1999: 31; Brugnatelli 2006: 60). This *ă was retained as such in Tuareg (or in some contexts changed by vowel harmony to ə), but ended up as Ø in most of Zenati, since (with some exceptions in Tunisia) the latter does not allow reduced vowels in open syllables. Subsequent changes and loanwords have since made the distribution of reduced free state prefixes more opaque in Zenati.

In contexts where state is marked only by vowel reduction – as it is for all state-marked nouns in Tuareg and for all feminine ones in Zenati – words with reduced free state prefixes are accordingly systematically uninflectable for state in the singular. In the rare cases where they also happen to be unpluralisable, this results in full uninflectability for state, as in Table 5.

Table 5: Nouns uninflectable due to originally phonologically conditioned prefix vowel reduction

	free	annexed
Tamasheq (Heath 2006)		
‘pubic hair’	ă-šinšăḍ	ă-šinšăḍ
‘health’	tă-ssoxe-t	tă-ssoxe-t
Beni Snous (Destaing 1914)		
‘butter’	t-lussi	t-lussi
‘sun’	t-fuy-t	t-fuy-t
Cf. Tashlhiyt (Destaing 1920)		
‘butter’	ta-lussi	t-lussi

At least one Zenati variety, Ouargli (Delheure & Ochoa 2019: 79), has lost the prefix even in the construct state of words with reduced free state prefixes, making words like *fus* ‘hand’ uninflectable. This change, however, cannot be accounted for through regular sound change alone; it rather constitutes a reanalysis of such words as prefixless nominals (see Section 2.2.1 below), generalising the principle that nouns with no prefix vowel in the free state are uninflectable.

2.2 Morphologically motivated

As seen in the introduction, canonical nouns across Berber start with a nominal prefix capable of being marked for gender, number, and state. Many nouns, however, are not canonical in this respect. Some lack a prefix altogether. Others start with an invariable prefix, for reasons that are purely historical and lexical rather than being explained by phonological developments. In either case, state marking is unavailable.

2.2.1 Nominals with no prefix

The presence of a nominal prefix is a necessary – though not sufficient – condition for a noun to be marked for state. But, across the entire family, all personal pronouns, most numerals, and a variable set of core kinship terms take no nominal prefix. Certain types of verbal nouns are also normally prefixless, especially *fad* 'thirst' and *laẓ* 'hunger', along with some baby talk forms and loanwords, and certain types of compound noun (Brugnatelli 2006: 58). Many such forms are also uninflectable, or only suppletively inflectable, for number. Table 6 illustrates the range of such forms in Kabyle.

Table 6: Some prefixless nominals in Kabyle (Dallet 1982)

	SG free	SG annexed	PL free	PL annexed
'daughter'	yəlli	yəlli	yəssi	yəssi
'one (M)'	yiwən	yiwən	-	-
'two (M)'	-	-	sin	sin
'he'	nətt^{s}a	nətt^{s}a	nutni	nutni
'you (M)'	kəčč	kəčč	kunwi	kunwi
'thirst'	fad	fad	-	-
'owie'	diddi	diddi	-	-
'pennyroyal'	fləggu	fləggu	-	-

Within the inherited vocabulary, such forms are few and limited in productivity despite their high frequency. However, loanwords from languages such as Hausa, Songhay, and Bambara (cf. Heath 2005a: 164), or Spanish (Kossmann 2009), are regularly borrowed without a prefix, making this an unambiguously open class in languages like Tamasheq or Tarifiyt, as illustrated in Table 7.

Not all loanwords from these languages are borrowed without the prefix; contrast, for instance, Tarifiyt *a-kwaḏrun* 'doorpost', *ṯ-kuppa-ṯ* 'cup' (< Spanish *cuadro*

Table 7: Some prefixless loanwords in Tarifiyt (< Spanish) and Tamasheq (< Songhay)

	SG free	SG annexed	PL free	PL annexed
Tarifiyt (Mourigh & Kossmann 2020)				
‘cheese’	kisu	kisu	-	-
‘mechanic’	mikaniku	mikaniku	mikaniku-ṯ	mikaniku-ṯ
‘seagull’	ɣaḇyuṭ-a	ɣaḇyuṭ-a	ɣaḇyuṭ-aṯ	ɣaḇyuṭ-aṯ
Tamasheq (Heath 2006)				
‘stalk’	koɣəri	koɣəri	koɣəri-t-ăn	koɣəri-t-ăn
‘frog sp.’	mbăla	mbăla	-	-
‘ladle’	zăɣto	zăɣto	zăɣto-t-ăn	zăɣto-t-ăn

‘frame’, *copa*), Tamasheq *ă-karfu* ‘rope’, *tă-kăŋkani-t-t* ‘watermelon’ (< Timbuktu Songhay *karfo*, *kaŋkani*). In Tarifiyt, however, Mourigh and Kossmann’s data indicate that almost all Spanish nouns are borrowed without the prefix, suggesting that this has become the default strategy. The factors determining whether or not a loanword takes the prefix in such cases have yet to be studied.

2.2.2 Nouns with inherently uninflectable prefixes

All Berber languages have borrowed extensively from Arabic. Many Arabic nouns are given Berber prefixes, e.g. Kabyle *a-bənnay* ‘builder’ < *bannāy* (Classical *bannāʔ*). However, a very common strategy is to borrow them together with the Arabic definite article *l-/əl-/ăl-*, whose *l-* assimilates to a following coronal consonant (cf. Kossmann 2013: 208). This is then treated as an invariant prefix with no semantic interpretation (glossed here as N), although both its extremely high frequency and its behaviour in morphological derivations indicate that it is nevertheless treated as segmentable. Very occasionally this prefix even gets extended to an inherited Berber noun, as in Siwi *l-əgʷrazən* ‘dogs’ < *a-gʷərzni* ‘dog’ (Souag 2013: 76). As a result, a rather large proportion of nouns in any given Berber language start with an inherently uninflectable prefix *(ə)l-* (Tuareg *ăl-*). Table 8 gives a small sample.

This issue does not just arise with loanwords; some Berber varieties also use an inherited uninflectable masculine prefix *wa-* with a small set of nouns, typically names of flora and fauna (Brugnatelli 1998). The prefixal status of this *wa-*

Table 8: Some loans with the Arabic article

	SG free	SG annexed	PL free	PL annexed
Kabyle				
'calculation'	l-ḥisab	l-ḥisab	l-ḥisab-at	l-ḥisab-at
'carrion'	l-ğif-a	l-ğif-a	l-ğif-at	l-ğif-a
Tamasheq				
'family'	ăl-ʕăyal	ăl-ʕăyal	ăl-ʕăyal-ăn	ăl-ʕăyal-ăn
'usefulness'	ăl-fayd-a	ăl-fayd-a	ăl-fayd-at-ăn	ăl-fayd-at-ăn

is sometimes debatable, but is often confirmed by other members of the same word family: thus Kabyle *wa-zduz* 'broomrape (*Orobanche* sp.)' is presumably so-called for its shoots' resemblance to a pestle (*a-zduz*). Language-internal variation leads to similar conclusions; for 'goosefoot (*Chenopodium*)', Kabyle variously has *wa-ktun* or *a-ktun* (Zidat 2016, Dallet 1982). The form of this prefix recalls the M.SG. element *wa* used in the formation of demonstrative pronouns. Its history is poorly understood, but it has been argued to be a relict from a previous stage of the nominal prefix system in Berber (Brugnatelli 1997). Like *l-*, this prefix is inherently uninflectable; in most cases it also seems to be unpluralisable. A few derivational prefixes with more specific meanings are also uninflectable for state: privative *war-* (M.) / *tar-* (F.), evaluative prefixes like Kabyle *yir-* 'bad', etc.

Some demonstrative/relative/support pronouns are marked for case, as highlighted by Brugnatelli (2006), though most are not. Both types can be analysed as starting with (or consisting of) an element comparable to the nominal prefix, as illustrated in Table 9.

2.3 Semantically motivated

Semantically motivated variation in state marking in Berber, and indeed language-internal variation in state marking more generally, has yet to be seriously explored. However, state marking can be affected by sematic factors in at least one context.

Proper nouns, like common nouns, can be inflectable for state if they start with an appropriate prefix; thus in Tamasheq, for instance, the annexed state

Table 9: Selected demonstrative/relative/support pronouns in Kabyle (Dallet 1982)

	SG free	SG annexed	PL free	PL annexed
'this, already'	ay-a	w-ay-a	-	-
'this, what, whatever'	ay-ən	w-ay-ən	-	-
'this (M)'	wa-gi	wa-gi	wi-gi	wi-gi
'this (F)'	ta-gi	ta-gi	ti-gi	ti-gi

of the place name *a-măjon* "Amajon" is *ă-măjon*, and occurs where one would expect it to (Heath 2005b: 40). It has recently been observed by Nacira Abrous (2023), however, that in Kabyle, at least, proper nouns may variably be treated as uninflectable even when the prefix is present, appearing in the expected free state in contexts which normally require the annexed state. Examination of Kabyle written texts confirms this observation in seemingly identical syntactic contexts within the same document, as in (7a–7d).

(7) Kabyle (Sadi & Landri 2020)

a. Yə-qqim **A**-rəzqi di l-ḥəbs.
3MSG-stay.PFV **MSG.FS**-Rezki in N-jail
'Arezki stayed in jail.' (p. 26)
(cf. condition D)

b. Yə-sla **U**-rəzqi bəlli…
3MSG-hear.PFV **MSG.AS**-Rezki COMP
'Arezki heard that…' (p. 27)
(cf. condition D)

c. L-yaqut tə-nna=as i **A**-rəzqi.
N-Yakout 3FSG-say.PFV=3SG.DAT to **MSG.FS**-Rezki
'Yakout said to Arezki.' (p. 27)
(prepositions *n* and *i* assign the annexed state – condition A)

d. l-əhduṛ n **U**-rəzqi
N-speech GEN **MSG.AS**-Rezki
'Arezki's speech' (p. 25)
(cf. condition A)

Comparable examples can be found with feminine proper nouns, as in (8a–8b).

(8) Kabyle (Hamadouche 2007: 18)

a. I d=bədd-ən **Ta**-saʕdi-t d
REL CENTRIP=stand.PFV-3MPL **FSG.FS**-fortunate-FSG with
tə-rbaʕ-t=is.
FSG.AS-group-FSG=3SG.GEN
'Where Tasadit and her group stood.'
(cf. condition D)

b. yiw-ət n tə-lwəs-t n **T**-saʕdi-t
one-F GEN FSG.AS-sister.in.law-FSG GEN **FSG.AS**-fortunate-FSG
'a sister-in-law of Tasadit's'
(cf. condition A)

Personal names that can be marked for state at all are strikingly rare in Kabyle – in a list of some two hundred traditional personal names, Dallet (1982: 1027–1035) includes only nine, all either identical with or transparently connected to common nouns or adjectives: *akli* [lit. 'slave'], *a-məq^{w}ṛan* [lit. 'big'], *a-məẓyan* [lit. 'little'], *a-ṛəzqi* [based on *əṛṛəzq* 'lot, property'], *ta-saʕdi-t* [lit. 'fortunate'], *ta-wəṛduš-t* [a hypochoristic based on *ta-wəṛdi-t* 'pink'], *a-zwaw* [lit. 'Zwawi, Kabyle'], *aʕṛab* [lit. 'Arab'], *ta-məʕzuz-t* [lit. 'darling']. It is thus not surprising that this variation has so far received little attention. Much further work, ideally examining large text corpora, will be required before it can be determined whether this situation is representative or is just a peculiarity of (21st century?) Kabyle. For the moment, it constitutes the only clear evidence for semantically motivated variation in state marking in Berber.

3 The diachrony of state uninflectability

In contemporary Berber varieties, state uninflectability is a prominent phenomenon affecting much of the lexicon. Does this reflect an unstable intermediate situation on the way towards the loss of state inflection – an outcome attested in some peripheral varieties? To test this hypothesis, it is necessary to examine its history. Premodern records of Berber are rather sparse and often difficult to interpret, but span a period of more than two millennia. The internal diversity of Berber is sufficient to allow considerable progress to be made in reconstruction, despite the persistent difficulty of distinguishing common inheritance from family-internal contact. Combining both sources of data makes it possible to estimate roughly how long different categories of nouns have been uninflectable.

3.1 Written attestations

The earliest attestations of Berber – Libyco-Berber inscriptions from the pre-Roman period – do not transcribe vowels, except to some extent word-finally. Even allowing for that limitation, they provide no evidence for state marking through nominal prefixes. In the few cases of genitive constructions with a masculine singular possessor where a modern speaker would expect the annexed state to be marked by *w-* even in a consonantal text, no such marker is found: thus the bilingual text RIL 1 (Chabot 1940: 1–2), dating probably to the 2nd century BC, has *nbbn n-š?rh* "cutters of wood" and *nb[.]n n-zlh* "founders of iron" (cf. Kabyle *a-sɣaṛ* a.s. *wə-sɣaṛ* 'wood', *uzzal* a.s. *w-uzzal* 'iron'). The orthographic *-h* in such cases appears to be a suffix, fitting into a broader spectrum of evidence that suffixation could mark a wider range of functions on Libyco-Berber nouns than it does in any modern Berber variety (van Putten 2017). It is possible that *-h* was used to mark the genitive in some circumstances, but the evidence is too limited to allow any firm conclusions about case or state marking – nor, *a fortiori*, about uninflectability for either.

Similarly, common nouns of Berber origin recorded in African Roman texts usually do not display the characteristic nominal prefixes found throughout Berber today. As late as the 5th century, we find Latin *gemio(n-)* in the Albertini Tablets corresponding to modern *a-gəmmun* 'irrigated square of land', and *gelela(m)* in Cassius Felix for modern *ta-găll-ăt* (etc.) 'colocynth' (Múrcia Sànchez 2011). If the nominal prefix were already an obligatory part of the citation form of such nouns, one would expect them to remain so when borrowed into Latin, which has no phonological constraints on word-initial vowels. It thus seems likely that the modern pan-Berber system of state marking had not yet reached Roman Africa or Numidia, though it might already have taken shape somewhere outside the empire's borders. Even as late as the 9th century, Ibn Quraysh's lexical comparisons between a west Algerian Zenati variety and Hebrew include unexpectedly prefixless forms such as *kəmmus* 'parcel' or *šəfay* 'milk' alongside regularly prefixed ones such as *a-səkkum* 'asparagus' (Cohen 1972); cf. Tarifiyt *a-ḵəmmus, a-šəffay, a-səkkum* (Abarrou 2023). A possible relic of this stage today could be the very sporadic attestations in Kabyle proverbs and riddles of prefixless versions of M.SG. nouns that would normally require the prefix (Brugnatelli 2006: 58).

The earliest unambiguous direct evidence for obligatory prefixal state marking in Berber dates to the second millennium. By the 11th century (Chaker 1981; van den Boogert 1997: 113), even sources without vowel marking attest to M.SG. free *a-* (the citation form) vs. construct *w-* (attested primarily in genitives), alongside M. PL. free *i-* vs. construct *y-*. By the 12th century, the Arabic-Berber glossary of Ibn Tunart confirms a state system basically similar to today's (although the evidence must be interpreted carefully, since surviving copies are late); more importantly, it provides examples relevant to uninflectability. As in modern Berber, a significant minority of nouns are prefixless, e.g. *ḍukkuk* '(the star) Alcor', *miḍrus* 'cadaver', *mṭṭiḍ-ulli* 'Schneider's skink' (lit. 'sucker(of)-goats'), while others start with *wa-*, e.g. *wa-biba* 'mosquito', *wa-yləllu* 'henbane' (van den Boogert 1997: 106, 119; Amennou 2021). In most such cases, the noun is only attested in citation form. However, in *tibi n Mṣr* 'mallow of Egypt', the prefixless proper noun *Mṣr*, from Arabic *Miṣr*, appears in a context where the annexed state is expected, suggesting that prefixless forms were indeed uninflectable then as now.

The uninflectability of vowel-initial feminine stems is confirmed for the 12th century by Al-Bayḏaq, who records both *<ā tūmart īnaw> a t-umər-t inəw* 'oh my joy!', with *t-umər-t* in the free state, and *Ibn Tūmart* "Ibn Tumart" (named after the former exclamation), where *t-umər-t* must be in the annexed state (condition B), judging by comparable forms in the same source such as *Ibn Wə-mɣar* 'son of the chief' (van den Boogert 1997: 109–112; Marcy 1932). Further examples are to be found in Al-Kutāmī's 13th century commentary on Dioscorides' *Pharmacopoea*, which provides a number of plant names written plene (with vowels indicated by letters), including many genitive compounds showing the expected state marking. Among these, we find instances of vowel-initial stems in the annexed state (condition A), showing apparently uninflectable forms: thus *t-aɣaṭ* 'goat', *t-ili* 'ewe' in *ta-mar-t ən t-aɣaṭ* 'goat's beard', *ta-məzzuɣ-t ən t-ili* 'ewe's ear' (van den Boogert 1997: 121). If such forms reflect the loss of an original stem-initial consonant, a suggestion discussed above, then it appears that this consonant had already been lost 900 years ago.

The so-called Leiden fragment, tentatively dated to the 14th century,[2] attests a high concentration of Arabic nouns borrowed with *əl-*, as illustrated in example 9.

(9) Medieval Moroccan Berber
<dāyās yatārā rabba ’lʕalamyn alājr nālššahyd>

*da-yas	yə-tara	ṛəbb-əl-ʕalamin	əl-ajr	n	əš-šahid
PROG-3SG.DAT	3MSG-write.IMPF	lord-N-worlds	N-reward	GEN	N-martyr

'The Lord of the Worlds decrees for him the reward of a martyr.' (Leiden Ms Or. 23.306v)

[2] I am grateful to Evgeniya Gutova for allowing me to view a scan of this manuscript.

The first letter – always written as *alif* – is normally left entirely unvocalised when preceded by an orthographically separate word (including prepositions such as <kaɣ> *gəɣ* 'in'); sentence-initially, however, it is written as <'a>, and when preceded by an orthographically proclitic preposition (such as genitive <n-> *n*) appears as <ā>, due to the preservation of the letter *alif.* This pattern of variation is evidently not conditioned by state marking, and can be explained as purely orthographic. It thus appears that the modern Berber pattern of making Arabic loanword in *əl-* uninflectable for state already applied 700 years ago.

So far, our medieval examples have derived from Moroccan sources, reflecting an Atlas variety relatively close to modern Tashlhiyt. Further east, however, scattered Ibadi sources provide data from a similarly early period, despite some difficulties in dating. A late copy of a chronicle probably dating to the 12th century shows the expected inflection patterns for inflectable nouns, e.g. *am-an* 'water' but (cf. condition A) *əs w-am-an* 'with water' (Lewicki 1934: 288–289). At the same time, it confirms the uninflectability of pronouns, with examples like *ɣas šək* 'only you' (where free state is expected) alongside (with a preposition expected to assign annexed state, cf. condition A) *am šək* 'like you' (Lewicki 1934: 280–281). It also attests to the existence of prefixless *fad* 'thirst' (*ibid*:283), *middən* 'people' (*ibid*:295) as well as *əl*-prefixed Arabic loans like *əl-məruwwət* 'bravery', *əd-din* 'religion' (*ibid:*286). The unprefixed noun *Yuš* 'God' occurs as the object of prepositions (cf. condition A) in forms like *tamazyida ən Yuš* 'mosque of God' (*ibid:*288), *ɣəf Yuš* 'on God' (*ibid:*282), with no sign of state marking. Ongoing work on a book of religious law suspected to date from the same period paints a similar picture, though also revealing relics of a third unprefixed *state* used only in numeral+noun combinations (Brugnatelli 2011). In this work (Calassanti-Motylinski 1905: 75–77), we find the prefixless noun *šra* '(any)thing' both in the free state context of a postverbal direct object (*wəl yə-li* ***šra*** 'he does not have anything') and in the annexed state context of a postverbal subject (*i-d yə-qqim* ***šra*** 'if anything remains', cf. condition D), providing direct confirmation that it was uninflectable as expected.

In modern Berber, while canonical nouns are inflectable for state, a large number of nouns are uninflectable for state due to having no prefix, the wrong prefix, or a stem whose beginning is phonologically incompatible with the relevant prefix distinctions. A priori, this might appear to be an unstable state, in which a child acquiring the language might readily be tempted to overgeneralise one class or the other, and in which an overgeneralisation of uninflectability can hardly lead to any semantic misunderstandings. Yet all the available written evidence, as examined here, suggests that this situation has remained stable in most varieties for about a millennium.

3.2 Reconstruction

A noun prefix system with state marking can reliably be reconstructed for proto-Berber based on comparison of existing systems. A recent proposal which appears convincing is given in Table 10 (van Putten 2015).

Table 10: Proto-Berber nominal prefixes

	free	annexed
M.SG	*a-	*wă-
M.PL	*i-	*yə-
F.SG	*ta-	*tă-
F.PL	*tə- (> *ti-)	*tə-

The reconstruction of F.PL. free *tə-, however, depends crucially on the evidence from eastern Berber varieties which have lost state marking. All varieties that have actually retained state marking reflect a proto-system in which the F.PL. free state prefix was *ti-, not *tə-, apparently reflecting analogy with the masculine forms; such a system should probably be reconstructed for a lower-level proto-language dating to a period after some eastern Berber varieties had already split off. The *y in the M.PL. annexed forms may plausibly be derived from pre-proto-Berber *w, as in Prasse (1974: 14–16); doing so allows greater symmetry between M.SG. and M.PL.

Most proposals for the internal reconstruction of the prefix system draw upon its striking similarities to various demonstrative/relative/support pronoun series, e.g. Kabyle definite M.SG. *wa*, F.SG. *ta*, M.PL. *wi*, F.PL. *ti* 'this'. The best-known cross-linguistic grammaticalisation process that could account for these similarities is what Greenberg (1978) baptised the Demonstrative Cycle: demonstrative > definite article > non-generic article > class affix. Brugnatelli (1997, 2006), following Vycichl (1957), argues for this model's applicability to Berber, and explains the state distinctions as having emerged secondarily through stress shifts and the loss of initial semivowels in certain contexts (though neither change appears regular). Prasse (1974: 12) and Prasse (2002), on the other hand, sees the free state forms as deriving from a different pronoun series than the annexed state ones, comparing the former to Tamahaq neutral relative heads like *a* and the latter to definite ones like *wa*. In his scenario, the intermediate stage would have reflected a usage of such heads to mark predication and possession/attribution respectively.

Neither of these models has yet produced a fully satisfactory account for the proto-Berber forms or the syntactic distribution of state marking. Nevertheless, the Demonstrative Cycle hypothesis has important implications for the history of uninflectability in Berber. There is considerable cross-linguistic variation in which nominal elements require or permit articles and under which circumstances; in English, for instance, we note that pronouns and demonstratives can never take them (**the you*, **the that*), while generic nouns and certain kinship terms do not require them (*hunger*, *thirst*, *Mom*, *Grandpa*), and toponyms typically exclude them (*London*/**The London*) but occasionally require them (*The Hague*/**Hague*). The Definiteness Hierarchy (Aissen 2003) is useful in plotting much of this variation:

> Personal pronoun > Proper name > Definite > Indefinite specific > Non-specific

Personal pronouns and proper names (and by extension inalienable kinship terms) are inherently definite, and as such typically require no article; as expected assuming the Demonstrative Cycle, they obligatorily or typically take no nominal prefix throughout Berber. As Greenberg (1978) demonstrates, generic (or non-specific) nominals (including typical uses of numerals) are frequently left unmarked even in languages that require definites and specific indefinites to be marked overtly by an article. Nominals which usually or always occur at the extreme ends of the Definiteness Hierarchy are accordingly more likely than others to escape the effects of the Demonstrative Cycle, and thus to end up without a class prefix.

As seen in Section 2.2.1, all Berber varieties have a significant class of prefixless nouns, and show no nominal prefixes in personal pronouns and most numerals and demonstratives. This situation can be reconstructed for proto-Berber, as illustrated in Table 11a (which continues below as 11b). (For the reconstruction of prefixless kinship terms, see also Souag 2022; on the development of 'thing' into an indefinite pronoun, see Souag 2018.) If the nominal prefix system originated through the Demonstrative Cycle as generally assumed, then this situation must primarily reflect retention from pre-proto-Berber, rather than any kind of prefix loss.

Other types of morphologically motivated state uninflectability are also likely to go back to proto-Berber, but are supported by a narrower and less secure range of attestations. The uninflectable prefix *wa- was probably present in proto-Berber, as argued by Brugnatelli (1998), but it seems impossible to securely reconstruct any specific noun as having had this prefix, if only because so many

Table 11a: Some nouns which are reconstructible for proto-Berber with no prefix [Table continues in 11b]

	Zenaga (Taine-Cheikh 2008)	Tashlhiyt (Destaing 1920)	Figuig (Kossmann 1997)
'hunger'	-	ḷaẓ	ḷaẓ
'thirst'	fäđ	fad	fad
'sister'	yäđmäh	ultma	wətna
– PL	t^yšäđmäh	istma	yəssma
'daughter'	-	illi	yəlli
– PL	-	isti	yəssi
'(some)thing'	kārāh	kra	šṛa

Table 11b: Some nouns which are reconstructible for proto-Berber with no prefix [Table begins in 11a]

	Kabyle (Dallet 1982)	Tamahaq (Prasse 2010)	Awjili (van Putten 2014)	proto-Berber
'hunger'	laẓ	laẓ	tə-laz-at	*laẓ
'thirst'	fad	fad	tə-fad-at	*fad
'sister'	wəltma	wălătma	wərtna	*wălăt-ma
– PL	yəssətma	šetma	sətma	*ysăt-ma
'daughter'	yəlli	yăll	wəlli	*yăwle
– PL	yəssi	ešš	-	*yăste
'(some)thing'	kra	(hărăt)	kəra	*kăra

varieties have lost it; it likely had a more specific function than *a- did. Among uninflectable derivational prefixes, a pejorative is widely attested – Kabyle *yir-*, Tamasheq *erk-*, Zenaga *ä-gär-* – but the sound correspondences appear irregular, making its reconstruction challenging.

Nevertheless, proto-Berber clearly had a narrower range of uninflectable nouns than its modern descendants. Most of the phonologically motivated instances of state uninflectability discussed above can be reconstructed as originally inflectable, as made clear in Section 2.1. It is likely that proto-Berber already had some vowel-initial noun stems, but even for this category Prasse plausibly reconstructs an unattested laryngeal *h_1 in most instances (Prasse 1969, 2011). Semantically motivated state uninflectability is so far attested only in one variety, and therefore cannot be reconstructed back at any depth, although this might change as more data becomes available. Even among morphologically motivated instances, it is implausible a priori that proto-Berber, presumably spoken before contact with Arabic began, could have had a nominal prefix *ăl-, uninflectable or otherwise – and Arabic loanwords are now numerous even in basic vocabulary. The Proto-Berber lexicon would thus have had a rather lower proportion of uninflectable nouns than any major modern Berber variety.

4 Conclusions

State uninflectability in Berber is synchronically mostly predictable from the citation form of a noun alone, with a few exceptions; it is conditioned primarily by morphological and secondarily by phonological factors. Semantic factors seem to play only a very marginal role based on available data, although further work is needed.

State uninflectability can be reconstructed as diachronically stable in Berber over a time span that must amount to millennia for a significant number of core prefixless terms, including kin terms and some verbal nouns as well as pronouns and numerals. It has observably remained stable for at least a millennium or so in such cases, while coming to be extended to a wider range of phonologically and morphologically motivated categories of uninflectability nouns. This illustrates that partial uninflectability need neither be a transitional stage, as in Old French, nor a phenomenon limited to the margins of the lexicon, as in Russian or Polish. Any challenges it may pose for learnability are evidently far from insuperable.

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Name index

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Voids in morphology

If we were to imagine an inflectional morphological system as a galactic universe with its matter and governing laws, “uninflectedness” would mark the outer boundaries of this system, the “voids”, zones where the gravitational pull of inflection weakens and different structural principles come into play. Uninflectedness helps trace the limits of inflectional organization and offers valuable insights into how inflection operates and interacts with its periphery. Uninflectedness is thus a complex phenomenon that this book examines in depth for the first time, assembling a collection of 13 edited chapters that explore the phenomenon both theoretically and descriptively with the purpose of building a better typologically informed theory of inflection.